SECTION 01 11 00 - SUMMARY OF WORK

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 WORK INCLUDED IN THE CONTRACT

A. Work Included in the Contract: All construction and services required for a [_*BRIEF_PROJECT_DESCRIPTION*_], California Building Code (CBC) Type [_____] construction, of approximately [_____] sf floor area, including:

THE FOLLOWING ARE EXAMPLES ONLY. EDIT TO SUIT PROJECT REQUIREMENTS.

1. Site preparation.
2. Site utilities.
3. Site paving.
4. Landscape irrigation system and landscape planting.
5. Site fencing and site appurtenances.
6. New [_*FACILITY_DESCRIPTION*_].
7. Plumbing and heating, ventilating and air conditioning systems.
8. Wet-pipe fire suppression (sprinkler) system, to be provided on a design/build basis, with deferred approval by Code authority having jurisdiction, to suit the requirements of the facility and conforming applicable Codes, ordinances and standards of authorities having jurisdiction.
   a. Private fire service main shall be provided as indicated on Civil Drawings.
   b. Details of connections to private fire service main shall be included according to approved design/build wet-pipe fire suppression system.
9. Electrical power, lighting and signal systems.
10. Coordination of work being performed by others under separate contracts with University, described in Article below titled "CONCURRENT WORK UNDER SEPARATE CONTRACTS."
11. Additional general information concerning the Project is provided on the Architectural Drawings.

1.3 CONCURRENT WORK UNDER SEPARATE CONTRACTS

A. Work Under Separate Contracts: University may award separate design and construction contracts concurrent with this Contract and in the future, as determined by the University, for work listed below and for other work as University may determine. Such work under separate contracts may be indicated on the Drawings and in the Specifications as "Not in Contract", "NIC", "Future" or "Under Separate Contract".

1. [_*DESCRIPTION – Typically list furniture, emergency generators, utility work by campus, or other work that will take place on site that is not the responsibility of the contractor.*_].
2. [_*DESCRIPTION*_].
3. [_*DESCRIPTION*_].

B. Relationship to Work Under the Contract: Work under the Contract shall include all provisions necessary to make such concurrent work under separate contracts complete in every respect and fully functional, including field finishing. Provide necessary backing, supports, piping, conduit, conductors and other such provisions from point of service to point of connection, as shown on Drawings and specified herein. See [Section 01 31 00 - Project Management and Coordination] for additional requirements.

C. Documents for Work Under Separate Contracts: University's Representative will make available, in a timely manner, drawings and specifications of work under separate contracts for coordination and further description of that work.
   1. If available, such information will include drawings, specifications, product data, lists and construction schedules for such work.
   2. Information concerning work under separate contracts or directly by University will be provided for convenience only and shall not to be considered Contract Documents.

D. Permits, Notices and Fees for Work under Separate Contracts: Notices required by and approvals required of, authorities having jurisdiction over work under separate contracts and related fees, will be solely the responsibility of University.

1.4 PROTECT THE WORK FROM VANDALISM

A. During Work Hours. Protect the Work from theft, vandalism, and unauthorized entry. The Contractor shall have the sole responsibility for job site security. Refer to Specification 01 54 01 – SECURITY for additional information

B. During Off-Work Hours. During all hours that Work is not being prosecuted and at the Contractor's discretion, furnish such watchman's services as Contractor may consider necessary to safeguard materials and equipment in storage on the Project site, including Work in place and in process of fabrication, against theft, acts of malicious mischief, vandalism, and other losses or damages.

INCLUDE THIS ARTICLE FOR REMODELING AND RENOVATION PROJECTS. EDIT TO SUIT PROJECT REQUIREMENTS.

1.5 ALTERATIONS WORK DESCRIPTION

A. Alterations Work Description: [*Remodel*] [*Renovate*] the following areas, complete including operational mechanical and electrical Work:
   1. [_*DESCRIPTION*_].
   2. [_*DESCRIPTION*_].
   3. [_*DESCRIPTION*_].
   4. [_*DESCRIPTION*_].

B. Refinishing: Refinish all surface areas of the following, as specified:
   1. [_*DESCRIPTION*_].
   2. [_*DESCRIPTION*_].
   3. [_*DESCRIPTION*_].
   4. [_*DESCRIPTION*_].

C. In addition to specified replacement of equipment and fixtures restore existing plumbing, heating, ventilation, air conditioning, electrical, and [_____] systems to full operational condition.

1.5 OWNER-FURNISHED/CONTRACTOR-INSTALLED PRODUCTS

A. Owner-Furnished/Contractor-Installed (OFCI) Products: University will furnish, for installation by Contractor, products which are identified on the Drawings and in the Specifications as "OFCI (Owner-Furnished/Contractor-Installed)", "installed by General Contractor," or similar terminology. See Drawings for identification of such products. Refer to Section 01 64 00 - Owner-Furnished Products.

B. Relationship to Work under the Contract: Work under the Contract shall include all provisions necessary to fully incorporate such products into the Work, including, as necessary, fasteners, backing, supports, piping, conduit, conductors and other such provisions from point of service to point of connection, and field finishing, as shown on Drawings and specified herein. See Section 01 64 00 - Owner-Furnished Products for additional requirements.

1.6 PERMITS, LICENSES AND FEES

A. Permits, Licenses and Fees, General: Refer to Contract General Conditions, Article 4.11.

B. Licenses: Contractor shall obtain and pay all licenses associated with construction activities, such as business licenses, contractors' licenses and vehicle and equipment licenses. All costs for licenses shall be included in the Contract Amount.

C. Parking Fees: Contractor shall obtain and pay for all parking permits and fees for all vehicles parked on University property. Refer to Section 01 55 00, Vehicular Access and Parking for additional parking requirements.

1.7 PARTNERING

A. The Trustees intend to encourage the foundation of a cohesive partnership with the Contractor and its Subcontractors, the Architect and its consultants, and the Trustees. This partnership will be structured to draw on the strengths of each organization to identify and achieve reciprocal goals. The objectives are effective and efficient Contractor performance, intended to achieve completion within budget, on schedule, and in accordance with the Contract Drawings and Specifications.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

Not Applicable to this Section.

END OF SECTION 01 11 00

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SECTION 01 14 00 - WORK RESTRICTIONS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 CONTRACTOR'S USE OF PREMISES AND SITE, GENERAL

A. Contractor's Use of Premises and Site, General: Refer to Contract General Conditions, Article 4.00.

1. Contractor shall at all times perform Work so as to impose no hardship on the Trustees or others engaged in the Trustees' work nor cause unreasonable delays or hindrance thereto.
2. Construction activities shall be scheduled to minimize disruption to the University and to Campus users.
3. Contractor may not interrupt any Campus utilities without prior written permission from the Trustees.
4. Contractor shall leave the site clean and neat each day. See 01 74 00 CLEANING REQUIREMENTS for additional information.

1.3 USE OF PREMISES

A. Use of Site [*and Existing Building*]: Limit use of premises to work in areas indicated. Do not disturb portions of site beyond areas in which the Work is indicated.

1. Limits: Confine constructions operations to Project Area indicated on the Drawings. Use of other areas shall be only with the approval of University's Representative. Confine constructions operations to [*Description of areas where Work is permitted*].
2. University Occupancy: Where existing buildings and site areas are indicated for continued use by University, make provisions to continued use by scheduling and sequencing of Work under the Contract. Make provisions for temporary barriers, enclosures, covers, directional signage and other construction facilities and temporary controls to enable continuing use.

1.4 CONTRACTOR'S USE OF PROJECT AREA

A. Location of Work: The Work shall be accomplished within areas indicated on Drawings as Project Area or, if not indicated, to areas as directed by University's Representative. Use of other areas, including parking areas, shall be subject to approval by University's Representative. Refer to Section 01 52 05 - Construction Staging Areas and Section 01 55 00 - Vehicular Access and Parking for additional requirements.

1. Contractor shall not unreasonably encumber the site with materials or equipment.
2. Contractor shall assume full responsibility for protection and safekeeping of products stored on the premises.

3. Contractor shall move any stored products which interfere with operations of University or contractors performing work under separate contracts for University.
4. Temporary closures or restrictions of use of public thoroughfares, necessary to accomplish the Work, shall be made only as approved in advance by public safety and parking authorities having jurisdiction, as directed in writing by the University's Representative.
5. Additional off-site laydown, staging or parking area may be available for Contractor use. If available, the area may be rented from the University for a fee and shall not be back charged to the University. Contractor shall coordinate with the University’s Representative if additional off site space is needed.

B. Contractor's Use of the Project Area: Unless otherwise specified or indicated on the Drawings, during the construction period the Contractor shall have full use of the designated Project Area for construction operations, including use of the site. Contractor's use of Project Area shall be limited only by University's right to perform construction operations with its own forces or to employ separate contractors on portions of the Project in accordance with the Contract General Conditions.

EDIT PARAGRAPHS BELOW TO SUIT PROJECT REQUIREMENTS.

C. [*FOR RENOVATION PROJECTS*] Continued Use of Existing Building: Maintain existing building in a weather tight condition throughout construction period. Repair damage caused by construction operations. Protect building and its occupants during construction period.

D. Protection of Existing Improvements and Facilities: Contractor shall protect property adjacent to the Project Area and all existing improvements and facilities within the Project Area, including paving and landscaping indicated to remain.

1. All existing improvements and facilities, except those specifically indicated for removal or reconstruction, shall be protected with temporary barriers, enclosures and passageways. Refer to additional requirements specified in Section 01 56 00 - Temporary Barriers and Enclosures.
2. After completion of Work, existing improvements and facilities shall be restored to original condition and location. Project Area shall be cleaned and restored to presentable condition, equivalent to or better than the condition prior to start of Work.
3. Should existing improvements and facilities be damaged or soiled beyond renovation or repair, new products shall be provided by Contractor equivalent to existing products, as directed by University's Representative.

E. Project Area Access: Limit access to the site to routes and access points as indicated. If routes and access points are not indicated, access shall be as approved and as directed by University's Representative. Do not restrict access to adjacent facilities and do not restrict access for those performing work under separate contracts for University.

1. Access to and egress from Project Area shall be in strict conformance to prearranged routes approved by University's Representative, with the

understanding that curtailment of construction traffic or revision of access routes may be required on short notice if University's operations mandate such changes because of excessive noise or problems of safety, service or supply.

2. Driveways and Entrances: Keep driveways and entrances serving premises clear and available to service and emergency vehicles at all times. Do not use these areas for parking or storage of materials.
   a. Schedule deliveries to minimize use of driveways and entrances. Deliveries shall be restricted to certain times through each work day. Refer to section 1.5 below.
   b. Schedule deliveries to minimize space and time requirements for storage of materials and equipment on-site.

F. Emergency Access: Provide pathways, drives, gates, directional signage and other provisions as required by authorities having jurisdiction for emergency access to Project Area and adjoining campus facilities.

G. Emergency Egress: Maintain all pathways, drives, gates, and other means of egress during construction as required by public safety authorities having jurisdiction.

1.5 TIME RESTRICTIONS

A. Contractor's Work Hours: Work shall be limited to Monday through Friday, except University-observed holidays and periods when classes are not in session, during hours of 7:00am to 5:00pm.
   1. Work on other days and at other hours shall be permitted only with written approval of University's Representative.
   2. Prior to start of construction: Obtain a calendar from the Trustees indicating major campus events, study and examination periods, holidays and quarter breaks.
   3. Work during final exam periods at ends of class sessions shall be restricted to minimize noise, vibrations and other distracting and inhibiting activities.
   4. The Contractor may be asked to suspend work during the following or similar University events:

THE FOLLOWING IS AN EXAMPLE ONLY. EDIT TO SUIT PROJECT REQUIREMENTS. WORK IN CLOSE PROXIMETY TO HOUSING REQUIRES ALTERNATE WORK HOURS, TYPICALLY NOT ALLOWED TO START UNTIL 10AM. CONFIRM WITH HOUSING.

Commencement: (NO WORK)	Generally the third Saturday in June Generally the third Saturday in December
Open House: (NO deliveries allowed)	Generally the third Friday and Saturday in April
Residence Hall Move-in: (NO Deliveries allowed)	Generally weekend before Fall Quarter
Finals Weeks: (noise restrictions in place)	Generally the third week in March The week before June commencement

	The week before December commencement

5. If it becomes necessary to perform Work on weekends and holidays, in order to meet milestone and final completion dates, Work shall be performed at no change in Contract Amount unless authorized by written Change Order or Field Instruction.

B. Utility Outages and Shutdown: Schedule utility outages and shutdowns to nights, weekends, school holidays or times and dates acceptable to and approved by University's Representative.
1. Time and duration of outages and shutdowns shall not hinder normal campus activities except as authorized in writing by University's Representative.
2. Provide seven (7) calendar days notice in writing to University's Representative of all utility outages and shutdowns. Describe Work to be performed, which utilities will be interrupted and time and duration of interruption.
3. Contractor shall provide temporary utilities to occupied facilities and adjacent properties when utilities must be interrupted for more than two hours, unless otherwise directed by University's Representative.

THE FOLLOWING IS AN EXAMPLE ONLY. DELETE IF NOT APPLICABLE. IF INCLUDED, EDIT TO SUIT PROJECT REQUIREMENTS. PUT IN BID PROPOSAL IF APPLICABLE.

4. Power interruptions beyond the authorized time shall be subject to liquidated damages in the amount of $1,000 per day.
5. Refer also to requirements for temporary utilities specified in Section 01510, Temporary Utilities.

THE FOLLOWING IS AN EXAMPLE ONLY. REVIEW REQUIREMENTS AGAINST THE CONTRACT GENERAL CONDITIONS INCLUDED IN THIS CONTRACT. DELETE IF NOT APPLICABLE. IF INCLUDED, EDIT TO SUIT PROJECT REQUIREMENTS.

1.6 NOISE AND VIBRATION RESTRICTIONS

A. Noise Restrictions: These requirements are in addition to Article 35.03 of the Contract General Conditions. Minimize noise from construction activities. Limit loud construction activities to times when classes are not in session in adjacent [_*LIST FACILITIES*_] [_LIST SPACES, ROADS OR OTHER SIMILAR AREAS_].
1. Maximum noise levels within 1,000 feet of classroom, laboratory, residence, business, adjacent buildings, or other populated area for:
   a. Trenchers, pavers, graders and trucks: 90 dBA maximum at 50 feet as measured under the noisiest operating conditions.
   b. Other equipment: 85 dBA at 50 feet as measured under the noisiest operating conditions.
2. Equipment:
   a. Jackhammers: Equip with exhaust mufflers and steel muffling sleeves.

b. Air compressors: Quiet type such as a “whisperized” compressor. Keep hoods closed while equipment is in operation.
c. Portable Noise barriers: Provide around jack hammering or similar construction; 3/4-inch plywood lined with 1-inch thick fiberglass on the work side at a sufficient height and width to reduce noise to acceptable limits.

3. Operations:
   a. Keep noisy equipment as far as possible from noise-sensitive site boundaries.
   b. Do not leave machines idling.
   c. Use electric power in lieu of internal combustion engine power wherever possible.
   d. Maintain equipment properly to reduce noise from excessive vibration, faulty mufflers, or other sources.
   e. Engines shall have properly functioning mufflers.
4. Scheduling:
   a. Schedule noisy operations to minimize their duration, and disruption to the adjoining users.
   b. Notify the Trustees Representative of seven (7) calendar days minimum in advance of performing work creating unusual noise.
   c. Schedule work at mutually agreeable times.
5. Do not play radios, tape recorders, televisions, and similar items at construction site.
6. When work occurs in or near occupied buildings, keep noise associated with construction activities to a minimum. Noisy operations that may disrupt academic or residential activities shall be scheduled after normal work hours.
7. All noisy work within the area of residence halls and other campus residences shall be restricted between the hours of 10:00 am to 10:00 pm seven (7) days per week, throughout the year. No work will be allowed in residence halls on campus residences during finals week.
8. Trustees reserve the right to stop construction work, including but not limited to noisy work, during the following events: Commencement, Open House, Finals Week, residence hall move-in, Week-of-Welcome, or at other times that may be identified by the Trustees. Trustees reserve the right to stop noisy work when said work disrupts classes or residential areas. Refer to Section 01 11 00 - Summary of Work.

B. Vibration Restrictions: Do not perform activities that cause vibrations in adjacent occupied spaces, including spaces above and below location where Work is performed. If vibrations transmit through structure, perform Work at times when University activities are not being conducted. Work may proceed with written authorization from University representative.

1.7 UNIVERSITY RESTRICTIONS

A. Grand Avenue, Perimeter Road, Highland Drive, or California Boulevard: No large or slow-moving vehicles between the hours of 7:30 a.m. and 8:30 a.m., Monday through Friday, when school is in session.

a. Clearance is restricted to 12 feet-6 inches under train trestle on Highland Drive.
b. Construction Traffic is restricted to exit campus on Grande Ave. only.

B. The use of the surrounding residential streets for any contractor use is not allowed.
a. Observe all traffic laws on and off campus at all times.

C. Temporary traffic control and temporary traffic signs:
a. Follow Caltrans guidelines and regulations.
b. Large equipment working in and around pedestrian areas shall require a spotter when backing up. Back up beepers are not sufficient.
c. Drivers of the equipment or back-up alarms are not considered “spotters.”
d. Large equipment includes, but is not limited to, backhoes, dump trucks, concrete trucks and delivery trucks.
e. Flag Persons: Provide trained and equipped flag persons to regulate traffic when construction operations or traffic encroach on public traffic lanes.
f. Flares and Lights: Use flares and/or lights during hours of low visibility to delineate traffic lanes and to guide traffic.

1.8 UNIVERSITY'S USE OF SITE AND PREMISES

A. University's Use of Site and Premises: University reserves the right to occupy and to place and install equipment in completed or partially completed areas of buildings and site. Such placing of equipment and partial occupancy shall not constitute acceptance of the total Work.

1. Full University Occupancy: University will occupy site [_*and existing building*_] during entire construction period. Cooperate with University during construction operations to minimize conflicts and facilitate University usage. Perform the Work so as not to interfere with University's operations.
2. Partial University Occupancy: University reserves the right to occupy and to place and install equipment in completed areas of building provided such occupancy does not interfere with completion of the Work. Such placement of equipment and partial occupancy shall not constitute acceptance of the total Work.
3. Before partial University occupancy, mechanical and electrical systems shall be fully operational, and required tests and inspections shall be successfully completed. Unless otherwise agreed, University will provide operation and maintenance of mechanical and electrical systems in portions of the building used by University. Unless otherwise agreed in writing by the University, warrantee periods shall not begin until date established by Notice of Completion filed at Contract closeout.
4. Upon occupancy, University will assume responsibility for maintenance and custodial service for occupied portions of building.

PART 2 - PRODUCTS

Not Applicable to this Section

PART 3 - EXECUTION

Not Applicable to this Section

END OF SECTION 01 14 00

SECTION 01 15 01 - CONSTRUCTION AND DEMOLITION MATERIALS RECYCLING REQUIREMENTS

PART 1 GENERAL

1.1 SUMMARY

A. Section includes: Requirements and procedures for ensuring optimal diversion of construction and demolition (C&D) waste materials generated by the Work from landfill disposal within the limits of the Construction Schedule and Contract Sum.

1. California State law (Assembly Bill 75), requires the California State University to develop source reduction, re-use, recycling, and composting programs, to reduce the tonnage of solid waste disposed in landfills 50% by the year 2004. Construction waste materials generated by the Work are targeted to achieve these diversion rates.
2. The Work of this Contract requires that a minimum of 85% by weight of the construction and demolition materials generated in the Work is diverted from landfill disposal through a combination of re-use and recycling activities.
3. For LEED® projects, requirements for submittal of LEED documentation in compliance with Materials and Resources Credit 2.1 and Materials or Resources Credit 2.2, Construction Waste Management.
3. Requirements for submittal of Contractor’s Construction Waste and Recycling Plan prior to the commencement of the Work.
4. Contractor’s quantitative reports for construction waste materials as a condition of approval of the third progress payment. For contracts of shorter duration, this requirement will be enforced as determined by University Representative.

1.2 DEFINITIONS

A. Class III Landfill: A landfill that accepts non-hazardous resources such as household, commercial, and industrial waste, resulting from construction, remodeling, repair, and demolition operations. A Class III landfill must have a solid waste facilities permit from the California Integrated Waste Management Board (CIWMB) and is regulated by the Enforcement Agency (EA).

B. Construction and Demolition Debris: Building materials and solid waste resulting from construction, remodeling, repair, cleanup, or demolition operations that are not hazardous as defined in California Code of Regulations, Title 22, Section 66261.3 et seq. This term includes, but is not limited to, asphalt concrete, Portland cement concrete, brick, lumber, gypsum wallboard, cardboard and other associated packaging, roofing material, ceramic tile, carpeting, plastic pipe, and steel. The debris may be commingled with rock, soil, tree stumps, and other vegetative matter resulting from land clearing and landscaping for construction or land development projects.

C. C&D Recycling Center. A facility that receives only C&D material that has been separated for reuse prior to receipt, in which the residual (disposed) amount of waste in the material is less than 10% of the amount separated for reuse by weight.

D. Disposal. Final deposition of construction and demolition or inert debris into land, including stockpiling onto land of construction and demolition debris that has not been sorted for further processing or resale, if such stockpiling is for a period of time greater than 30 days; and construction and demolition debris that has been sorted for further processing or resale, if such stockpiling is for a period of time greater than one year, or stockpiling onto land of inert debris that is for a period of time greater than one year.

E. Enforcement Agency. Enforcement agency as defined [i.e. in Public Resources Code 40130].

F. Inert Disposal Facility or Inert Waste Landfill: A disposal facility that accepts only inert waste such as soil and rock, fully cured asphalt paving, uncontaminated concrete (including fiberglass or steel reinforcing rods embedded in the concrete), brick, glass, and ceramics, for land disposal.

G. Mixed Debris: Loads that include commingled recyclable and non-recyclable materials generated at the construction site.

H. Mixed Debris Recycling Facility: A processing facility that accepts loads of commingled construction and demolition debris for the purpose of recovering re-usable and recyclable materials and disposing the non-recyclable residual materials.

I. Recycling: The process of sorting, cleansing, treating and reconstituting materials for the purpose of using the altered form in the manufacture of a new product. Recycling does not include burning, incinerating or thermally destroying solid waste.

J. Reuse. The use, in the same or similar form as it was produced, of a material which might otherwise be discarded.

K. Separated for Reuse. Materials, including commingled recyclables, that have been separated or kept separate from the solid waste stream for the purpose of additional sorting or processing those materials for reuse or recycling in order to return them to the economic mainstream in the form of raw material for new, reused, or reconstituted products which meet the quality standards necessary to be used in the marketplace, and includes materials that have been “source separated.”

L. Solid Waste: All putrescible and nonputrescible solid, semisolid, and liquid wastes, including garbage, trash, refuse, paper, rubbish, ashes, industrial wastes, demolition and construction wastes, abandoned vehicles and parts thereof, discarded home and industrial appliances, dewatered, treated, or chemically fixed sewage sludge which is not hazardous waste, manure, vegetable or animal solid and semisolid wastes, and other discarded solid and semisolid wastes. "Solid waste" does not include hazardous waste, radioactive waste, or medical waste as defined or regulated by State law.

M. Source-Separated: Materials, including commingled recyclables, that have been separated or kept separate from the solid waste stream at the point of generation for the purpose of additional sorting or processing of those materials for reuse or recycling in order to return them to the economic mainstream in the form of raw materials for new, reused, or reconstituted products which meet the quality standards necessary to be used in the marketplace.

N. Waste Hauler: A company that possesses a valid permit from the local waste management authority to collect and transport solid wastes from individuals or businesses for the purpose of recycling or disposal in the locality.

1.3 SUBMITTALS

A. Contractor’s Construction Waste and Recycling Plan

1. Review Contract Documents and estimate the types and quantities of materials under the Work that are anticipated to be feasible for on-site processing, source separation for re-use or recycling. Indicate the procedures that will be implemented in this program to effect jobsite source separation, such as, identifying a convenient location where dumpsters would be located, putting signage to identify materials to be placed in dumpsters, etc.

2. Prior to commencing the Work and within ten (10) calendar days from the date of the Notice to Proceed, submit Contractor’s Construction Waste and Recycling Plan. Submit in format provided (Section 01 15 01A). The Plan must include, but is not limited to the following:
   a. Contractor’s name and project identification information;
   b. Procedures to be used;
   c. Materials to be re-used and recycled;
   d. Estimated quantities of materials;
   e. Names and locations of re-use and recycling facilities/sites;
   f. Tonnage calculations that demonstrate that Contractor will re-use and recycle a minimum 85% by weight of the construction waste materials generated in the Work.

3. Contractor’s Construction Waste and Recycling Plan must be approved by the Construction Administrator prior to the start of Work.

4. Contractor’s Construction Waste and Recycling Plan will not otherwise relieve the Contractor of responsibility for adequate and continuing control of pollutants and other environmental protection measures.

B. Contractor’s Reuse, Recycling, and Disposal Report
Submit Contractor’s Reuse, Recycling, and Disposal Report on the form provided (Section 01 15 01B) with each application for progress payment. Failure to submit the form and its supporting documentation will render the application for progress payment incomplete and delay progress payments. If applicable, include manifests, weight tickets, receipts, and invoices specifically identifying the Project for re-used and recycled materials:

1. Reuse of building materials or salvage items on site (i.e. crushed base or red clay brick).

2. Salvaging building materials or salvage items at an off-site salvage or reuse center (i.e. lighting, fixtures).
3. Recycling source separated materials on site (i.e. crushing asphalt/ concrete for base course, or grinding for mulch).
4. Recycling source separated material at an off-site recycling center (i.e. scrap metal or green materials).
5. Use of material as Alternative Daily Cover (ADC) at landfills.
6. Delivery of soils or mixed inerts to an inerta landfill for disposal (inert fill).
7. Disposal at a landfill or transfer station (where no recycling takes place).
8. Other (describe).

Contractor's Reuse, Recycling, and Disposal Report must quantify all materials generated in the Work, disposed in [Class III] landfills, or diverted from disposal through recycling. Indicate zero (0) if there is no quantity to report for a type of material.

As indicated on the form:
1. Report disposal or recycling either in tons or in cubic yards: if scales are available at disposal or recycling facility, report in tons; otherwise, report in cubic yards. Report in units for salvage items when no tonnage or cubic yard measurement is feasible.
2. Indicate locations to which materials are delivered for reuse, salvage, recycling, accepted as daily cover, inert backfill, or disposal in landfills or transfer stations.
3. Provide legible copies of weigh tickets, receipts, or invoices that specifically identify the project generating the material. Said documents must be from recyclers and/or disposal site operators that can legally accept the materials for the purpose of re-use, recycling, or disposal.

Indicate project title, project number, progress payment number, name of the company completing the Contractor's Report and compiling backup documentation, the printed name, signature, and daytime phone number of the person completing the form, the beginning and ending dates of the period covered on the Contractor's Report, and the date that the Contractor's Report is completed.

C. For LEED Projects, LEED Letter Template: Materials and Resources Credit [2.1 or 2.2] Construction Waste Management

Complete and sign LEED Letter Template in format provided under the U.S. Green Building Council's Leadership in Energy and Environmental Design (LEED) program. Prepare Letter Template on company letterhead.

1. Certify that the project has completed a waste management plan and diverted construction, demolition, and land clearing waste to uses other than landfill.
2. Provide quantities of diverted materials and means of diversion in the table provided in the LEED Letter Template.
3. Indicate how and where waste was diverted.
4. Indicate quantities of waste diverted in tons [or cubic yards].
5. Letter Template will calculate: Total quantity of diverted waste, total quantity of waste, and the percentage of waste diverted.

6. For projects where 50% of waste is diverted, one LEED credit will be achieved; where 75% is diverted, two LEED credits will be achieved.
7. Include name, organization, role in project, provide signature and date completed.

PART 2 PRODUCTS

2.1 GUIDE TO LOCAL COMPANIES

A. Guide to construction and demolition recycling and disposal - the following list is not an exhaustive list and it is the contractors responsibility to verify the information:

CONSTRUCTION AND DEMOLITION RECYCLING GUIDE

			Cardboard	Drywall	Plastic	Asphalt Roofing	Scrap Metals	Wood & Pallets	Green Waste	Scrap Metals	Concrete &
RECYCLING COMPANIES THAT ACCEPT C & D MATERIALS											
A-1 Metals & Salvage	238-3545	Paso Robles					●				
Bedford Metals	922-4977	Santa Maria	●				●				
Heilman Salvage	466-4893	Atascadero					●				
Paso Robles Recycling	238-4678	Paso Robles	●		●						
Zanker Landfill	408/ 263-2384	Gilroy		●		●					

CONSTRUCTION AND DEMOLITION RECYCLING GUIDE

			Cardboard	Drywall	Plastic	Asphalt Roofing	Scrap Metals	Wood & Pallets	Green Waste	Scrap Metals	Concrete &
ROLL-OFF COMPANIES ★ Note: These companies accept mixed boxes for recycling											
American Equip Svc.	489-9521	Arroyo Grande	●				●	●			
API Roll-Off Services★	928-8689	Santa Maria	●				●	●			
Coastal Roll-Off★	543-0473	San Luis Obispo	●				●	●			
Have Bins	466-3636	Atascader o	●				●	●			
Mid-State Solid Waste	434-9112	Templeto n	●				●	●			
Paso Robles Roll-off	238-2385	Paso Robles	●				●	●			
R & R Roll-Off★	929-8000 528-8440	Nipomo	●				●	●			
San Miguel Roll-Off	239-1266	San Miguel	●				●	●			
LAND FILLS & TRANSFER STATIONS											
Chicago Grade Landfill	466-2985	Templeto n	●				●	●	●	●	●
Cold Canyon Landfill	549-8332	San Luis Obispo	●				●	●			
Paso Robles Landfill	238-2028	Paso Robles	●				●	●			
Nipomo Transfer Station	922-9255	Nipomo	●				●	●			

PART 3 EXECUTION

3.1 SALVAGE, RE-USE, RECYCLING AND PROCEDURES

A. Identify re-use, salvage, and recycling facilities.

B. Develop and implement procedures to re-use, salvage, and recycle new construction and excavation materials, based on the Contract Documents, the Contractor's

Construction Waste and Recycling Plan, estimated quantities of available materials, and availability of recycling facilities. Procedures may include on-site recycling, source separated recycling, and/or mixed debris recycling efforts.

1. Identify materials that are feasible for salvage, determine requirements for site storage, and transportation of materials to a salvage facility.
2. Source separate new construction, excavation and demolition materials including, but not limited to the following types:
   f. Asphalt.
   g. Concrete, concrete block, slump stone (decorative concrete block), and rocks.
   c. Drywall.
   d. Green materials (i.e. tree trimmings and land clearing debris).
   e. Metal (ferrous and non-ferrous).
   f. Miscellaneous Construction Debris.
   g. Paper or cardboard.
   h. Red Clay Brick.
   i. Reuse or Salvage Materials
   j. Soils.
   k. Wire and Cable.
   l. Wood.
   m. Other (describe)

3. Miscellaneous Construction Debris: Develop and implement a program to transport loads of mixed (commingled) new construction materials that cannot be feasibly source separated to a mixed materials recycling facility.

3.2 DISPOSAL OPERATIONS AND WASTE HAULING

A. Legally transport and dispose of materials that cannot be delivered to a source separated or mixed recycling facility to a transfer station or disposal facility that can legally accept the materials for the purpose of disposal.

B. Use a permitted waste hauler or Contractor’s trucking services and personnel. To confirm valid permitted status of waste haulers, contact the local solid waste authority.

C. Become familiar with the conditions for acceptance of new construction, excavation and demolition materials at recycling facilities, prior to delivering materials.

D. Deliver to facilities that can legally accept new construction, excavation and demolition materials for purpose of re-use, recycling, composting, or disposal.

E. Do not burn, bury or otherwise dispose of waste on the project job-site.

3.3 RE-USE AND DONATION OPTIONS

A. Implement a re-use program to the greatest extent feasible. Options may include:

1. California Materials Exchange (CAL-MAX) Program is sponsored by the California Integrated Waste Management Board. CAL-MAX is a free service provided by the California Integrated Waste Management Board, designed to help businesses find markets for materials that traditionally would be discarded. The premise of the CAL-MAX Program is that material discarded by one business may be a resource for another business. To obtain a current Materials Listings Catalog, call CAL-MAX/California Integrated Waste Management Board at (916) 255-2369 or send a FAX to (916) 255-2200. The CALMAX Catalog is available through the Internet Site at http://www.ciwmb/ca.gov/calmax.

3.4 REVENUE

A. Revenues or other savings obtained from recycled, re-used, or salvaged materials shall accrue to Contractor unless otherwise noted in the Contract Documents.

END OF SECTION

SECTION 01 15 01A
CONTRACTOR'S CONSTRUCTION WASTE AND RECYCLING PLAN

(Submit After Award of Contract and Prior to Start of Work)

Project Title:		
Contract or Work Order No.:		
Contractor's Name:		
Street Address:		
City:	State:	Zip:
Phone: ( )	Fax: ( )	
E-Mail Address:		
Prepared by: (Print Name)		

Date Submitted:		
Project Period:	From:	TO:

Reuse, Recycling or Disposal Processes To Be Used

Describe the types of recycling processes or disposal activities that will be used for material generated in the project. Indicate the type of process or activity by number, types of materials, and estimated quantities that will be recycled or disposed in the sections below:

01 - Reuse of building materials or salvage items on site (i.e. crushed base or red clay brick)
02 - Salvaging building materials or salvage items at an off site salvage or re-use center (i.e. lighting, fixtures)
03 - Recycling source separated materials on site (i.e. crushing asphalt/concrete for reuse or grinding for mulch)
04 - Recycling source separated materials at an off site recycling center (i.e. scrap metal or green matls)
05 - Recycling commingled loads of C&D matls at an off site mixed debris recycling center or transfer station
06 - Recycling material as Alternative Daily Cover at landfills
07 - Delivery of soils or mixed inerts to an inert landfill for disposal (inert fill).
08 - Disposal at a landfill or transfer station.
09 - Other (please describe) ____

Types of Material To Be Generated

Use these codes to indicate the types of material that will be generated on the project

A = Asphalt	C = Concrete	M = Metals	I = Mixed Inert	G = Green Matls
D = Drywall	P/C=Paper/Cardboard	W/C = Wire/Cable	S= Soils (Non Hazardous)	
M/C = Miscellaneous Construction Debris		R = Reuse/Salvage	W = Wood	O = Other (describe)

Facilities Used: Provide Name of Facility and Location (City)
Total Truck Loads: Provide Number of Trucks Hauled from Site During Reporting Period
Total Quantities: If scales are available at sites, report in tons. If not, quantify by cubic yards. For salvage/reuse items, quantify by estimated weight (or units).

SECTION I - RE-USED/RECYCLED MATERIALS

Include all recycling activities for source separated or mixed material recycling centers where recycling will occur.

Type of Material	Type of Activity	Facility to be Used, Location	Total Truck Loads	Total Quantities Tons	Cubic YD	Other Wt.
(ex.) M	04	ABC Metals, Los Angeles	24	355		
a. Total Diversion			-	-	-	-

SECTION 011511A

CONTRACTOR'S CONSTRUCTION WASTE AND RECYCLING PLAN

Continued

SECTION II - DISPOSED MATERIALS						
Include all disposal activities for landfills, transfer stations, or inert landfills where no recycling will occur.						
Type of Material	Type of Activity	Facility to be Used, Location	Total Truck Loads	Total Quantities Tons	Cubic YD	Other Wt.
(ex.) D	08	DEF Landfill, Los Angeles	2	35		
b. Total Disposal				-	-	-

SECTION III - TOTAL MATERIALS GENERATED						
This section calculates the total materials to be generated during the project period (Reuse/Recycle + Disposal = Generation						
				Tons	Cubic YD	Other Wt.
a. Total Reused/Recycled				-	-	-
b. Total Disposed				-	-	-
c. Total Generated				-	-	-

SECTION IV - CONTRACTOR'S LANDFILL DIVERSION RATE CALCULATION			
Add totals from Section I + Section II			
	Tons	Cubic Yards	Other Wt.
a. Materials Re-Used and Recycled	-		
b. Materials Disposed	-		
c. Total Materials Generated (a. + b. = c.)	-	-	-
d. Landfill Diversion Rate (Tons Only)*	#DIV/0!		

** Use tons only to calculate recycling percentages: Tons Reused/Recycled/Tons Generated = % Recycled*

Contractor's Comments (Provide any additional information pertinent to planned reuse, recycling, or disposal activities):

Notes:

1. Section 01 15 01A is a Division 01 General Requirement.

2. Suggested Conversion Factors: From Cubic Yards to Tons (Use when scales are not available)
   - Asphalt: .61 (ex. 1000 CY Asphalt = 610 tons. Applies to broken chunks of asphalt)
   - Concrete: .93 (ex. 1000 CY Concrete = 930 tons. Applies to broken chunks of concrete)
   - Ferrous Metals: .22 (ex. 1000 CY Ferrous Metal = 220 tons)
   - Non-Ferrous Metals: .10 (ex. 1000 CY Non-Ferrous Metals = 100 tons)
   - Drywall Scrap: .20
   - Wood Scrap: .16

SECTION 01 15 01B

CONTRACTOR'S REUSE, RECYCLING, AND DISPOSAL REPORT

(Submit With Each Progress Payment)

Project Title:		
Contract or Work Order No.:		
Contractor's Name:		
Street Address:		
City:	State:	Zip:
Phone: ( )	Fax: ( )	
E-Mail Address:		
Prepared by: (Print Name)		

Date Submitted:		
Period Covered:	From:	To:

Reuse, Recycling or Disposal Processes Used

Describe the types of recycling processes or disposal activities used for material generated in the project. Indicate the type of process or activity by number, types of materials, and quantities that were recycled or disposed in the sections below:

01 - Reuse of building materials or salvage items on site (i.e. crushed base or red clay brick)
02 - Salvaging building materials or salvage items at an off site salvage or re-use center (i.e. lighting, fixtures)
03 - Recycling source separated materials on site (i.e. crushing asphalt/concrete for reuse or grinding for mulch)
04 - Recycling source separated materials at an off site recycling center (i.e. scrap metal or green matls)
05 - Recycling commingled loads of C&D matls at an off site mixed debris recycling center or transfer station
06 - Recycling material as Alternative Daily Cover at landfills
07 - Delivery of soils or mixed inerts to an inert landfill for disposal (inert fill).
08 - Disposal at a landfill or transfer station.
09 - Other (please describe) ____

Types of Material Generated

Use these codes to indicate the types of material that were generated on the project

A = Asphalt	C = Concrete	M = Metals	I = Mixed Inert	G = Green Matls
D = Drywall	P/C=Paper/Cardboard	W/C = Wire/Cable	S= Soils (Non Hazardous)	
M/C = Miscellaneous Construction Debris		R = Reuse/Salvage	W = Wood	O = Other (describe)

Facilities Used: Provide Name of Facility and Location (City)
Total Truck Loads: Provide Number of Trucks Hauled from Site During Reporting Period
Total Quantities: If scales are available at sites, report in tons. If not, quantify by cubic yards. For salvage/reuse items, quantify by estimated weight (or units).

SECTION I - RE-USED/RECYCLED MATERIALS

Include all recycling activities for source separated or mixed material recycling centers where recycling occurred.

Type of Material	Type of Activity	Facilities Used, Location	Total Truck Loads	Total Quantities Tons	Cubic YD	Other Wt.
(ex.) M	04	ABC Metals, Los Angeles	24	355		
a. Total Diversion			-	-	-	-

SECTION 011511B

CONTRACTOR'S REUSE, RECYCLING, AND DISPOSAL REPORT

Continued

SECTION II - DISPOSED MATERIALS						
Include all disposal activities for landfills, transfer stations, or inert landfills where no recycling occurred.						
Type of Material	Type of Activity	Facilities Used, Location	Total Truck Loads	Total Quantities Tons	Cubic YD	Other Wt.
(ex.) D	08	DEF Landfill, Los Angeles	2	35		
b. Total Disposal				-	-	-

SECTION III - TOTAL MATERIALS GENERATED						
This section calculates the total materials generated during the project period (Reuse/Recycle + Disposal = Generation						
				Tons	Cubic YD	Other Wt.
a. Total Reused/Recycled				-	-	-
b. Total Disposed				-	-	-
c. Total Generated				-	-	-

SECTION IV - CONTRACTOR'S LANDFILL DIVERSION RATE CALCULATION					
Add totals from Section I + Section II					
	Tons	Cubic Yards	Other Wt.		
a. Materials Re-Used and Recycled	-				
b. Materials Disposed	-				
c. Total Materials Generated (a. + b. = c.)	-	-	-		
d. Landfill Diversion Rate (Tons Only)*	#DIV/0!				

** Use tons only to calculate recycling percentages: Tons Reused/Recycled/Tons Generated = % Recycled*

Contractor's Comments (*Provide any additional information pertinent to planned reuse, recycling, or disposal activities*):

Notes:

1. Section 01 15 01A is a Division 01 General Requirement.

2. Suggested Conversion Factors: From Cubic Yards to Tons (Use when scales are not available)

Asphalt: .61 (ex. 1000 CY Asphalt = 610 tons. Applies to broken chunks of asphalt)

Concrete: .93 (ex. 1000 CY Concrete = 930 tons. Applies to broken chunks of concrete)

Ferrous Metals: .22 (ex. 1000 CY Ferrous Metal = 220 tons) Drywall Scrap: .20

Non-Ferrous Metals: .10 (ex. 1000 CY Non-Ferrous Metals = 100 tons) Wood Scrap: .16

SECTION 01 21 00 - ALLOWANCE PROCEDURES

OMIT THIS SECTION IF ALLOWANCES ARE NOT INCLUDED IN THE CONTRACT. IF ALLOWANCES ARE INCLUDED IN THE PROJECT, THE BID PROPOSAL FORM MUST INDICATE THESE AMOUNTS.

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Allowances indicated in the Bid Proposal Form to be included in Contract Amount.

1. Selected materials and equipment, and in some cases, their installation, are shown and specified in the Contract Documents by allowances. Allowances have been established in lieu of additional requirements and to defer selection of actual materials and equipment to a later date when additional information is available for evaluation. Additional requirements, if necessary, will be issued by change order.
2. Allowances may be used in lieu of metering for temporary construction site utility services or to reimburse project related work performed by University forces, for example, keying.

1.3 RELATED SECTIONS

A. Section 01 51 00 - Temporary Utilities: Coordination with Allowance for temporary power.

B. Refer to product Specifications Sections identified in Allowance description.

1.4 GENERAL REQUIREMENTS FOR ALLOWANCES

A. Contractor shall submit cost data and other descriptive data to establish basis used by Contractor for determining costs in Contract Amount attributable to each Allowance.

B. Any amount not fully consumed shall be adjusted by change order.

1. The Contractor will be credited for his actual cost of labor, materials, and other actual costs WITHOUT mark-up.
2. Any unused allowances shall be returned to the Trustees using a credit change order for the full amount of the value unused plus six (6) percent.
3. Should the Contractor's actual costs exceed the specified allowance, the Contractor's Contract Amount will be adjusted by change order in accordance with Contract General Conditions, Article 6.00.

1.5 ALLOWANCE COSTS FOR CONTRACTOR-PROVIDED PRODUCTS

DELETE THIS ARTICLE IF NOT APPLICABLE

A. Contractor-Provided Products: Amount for each Allowance, for procurement of products to be selected by University's Representative or Architect after execution of the Agreement, shall include:

1. Net cost of product(s) to Contractor. Trade discounts and rebates shall be included.
2. Delivery to site.
3. Labor, equipment and related consumable products required for application, installation and finishing of product when Allowance is indicated to include costs for incorporation into completed construction.
4. Applicable taxes, permits and fees.

B. Costs Included in Contract Amount: In addition to amount identified for each Allowance, include in Contract Amount all costs for:

1. Handling and storage at site, including unloading, uncrating, and protective measures.
2. Protection from weather, soiling and physical damage.
3. Labor, equipment and related consumable products necessary for application, installation or finishing, except when Allowance is indicated to include costs for incorporation into completed construction.
4. Contractor's and all subcontractor's field and home office overhead expenses, bonds, insurance and profit.
5. All other costs attributable to incorporation of Allowance into completed construction, such as design fees and reworking of adjoining construction.

1.6 ALLOWANCE COSTS FOR EXECUTION

DELETE THIS ARTICLE IF NOT APPLICABLE

A. Owner-Furnished/Contractor-Installed (OFCI) Products: Amount for each Allowance, for application, installation and finishing of products provided by University (Owner-Furnished/ Contractor-Installed products), shall include:

1. Delivery to site, unless specifically noted otherwise.
2. Applicable taxes, permits and fees.
3. Handling and storage at site, including unloading, uncrating, and protective measures.
4. Protection from weather, soiling and physical damage.
5. Labor, equipment and related consumable products required for application, installation and finishing of product when Allowance is indicated to include costs for incorporation into completed construction.
6. Contractor's and all subcontractor's field and home office overhead expenses, bonds, insurance and profit.
7. All other costs attributable to incorporation of Allowance into completed construction, such as design fees and reworking of adjoining construction.

PART 2 - PRODUCTS

2.1 LUMP SUM ALLOWANCES

A. Allowance No. 1 - Temporary Power: Allow sum of [_AMOUNT_WORDS_] ($[##.##] for charges for serving utility for temporary power consumed during construction. Utility shall be metered by the University.

B. Allowance No. 2 – Construction Water - Allow sum of [_AMOUNT_WORDS_] ($[##.##] for charges for serving utility for water consumed during construction. Utility shall be metered by the University.

PART 3 - EXECUTION

3.1 SELECTION OF PRODUCTS

A. University's Representative and Architect will:
  1. Consult with Contractor for considerations to be given in selection of products, suppliers and qualified installers.
  2. Make selection in consultation with University staff. Obtain written direction by University's Representative designating:
     a. Product, color, design and finish.
     b. Accessories and attachments.
     c. Suppliers and qualified installers, as applicable.
     d. Allowance amount to be included in Contract Amount.
     e. Construction Contract warranty and manufacturer's guarantee provisions.

B. Contractor shall:
  1. Assist University's Representative and Architect in determining qualified suppliers or installers.
  2. Obtain proposals from suppliers and installers when directed by University's Representative.
  3. Make cost and constructability recommendations to University's Representative and Architect for consideration in product, supplier and qualified installer selections.
  4. Notify University's Representative and Architect promptly of:
     a. Reasonable objections Contractor may have against any supplier or party under consideration for installation.
     b. Effects on Construction Schedule anticipated by selections under consideration.

3.2 CONTRACTOR'S RESPONSIBILITIES

A. Upon notification of selection, Contractor shall execute purchase agreement with designated supplier and enter into contract with designated qualified installer, as applicable. Should a purchase agreement already exist between University and supplier, Contractor shall assume the purchase agreement for the University.

B. Contractor shall make all arrangements for and submit shop drawings, product data and samples as required.

C. Contractor shall make all arrangements for pick-up, delivery, handling and storage of products.

D. Upon delivery, Contractor shall promptly inspect products for damage or defects. Should damage or defects be found, Contractor shall effect return, replacement or repair of products, as appropriate, and process claims for transportation damage.

E. Contractor shall apply, install and finish products in compliance with requirements of applicable Sections of Specifications.

3.3 ADJUSTMENT COSTS

A. Should the net cost of the Allowance be more or less than the amount included in the Contract Amount, the Contract Amount shall be adjusted in accordance with provisions of the Contract General Conditions and a Change Order shall be executed.

B. Adjustment shall be made only for:
1. Increase or decrease in handling costs at site, labor, installation costs, overhead, profit, and other expenses resulting from final selection under Allowance.
2. Increase or decrease in product cost resulting from final selection under Allowance.
3. Increase or decrease in product cost from data provided by University's Representative or Architect and used to determine Allowance product cost.
4. Increase or decrease in product, application, installation and finishing costs resulting from change in quantity stated in Allowance.

C. Contractor shall submit claim and supporting documentation for cost increase or decrease within ten (10) days of execution of Construction Change Directive. Failure to submit documentation within designated time shall constitute a waiver of claims for additional costs.

END OF SECTION

SECTION 01 23 00 - BID ALTERNATIVE PROCEEDURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Requirements and descriptions for products and scopes of Work identified as Bid Alternative in the Drawings and Specifications and listed as "Bid Alternative" on the Bid Proposal Form.

1.3 RELATED DOCUMENTS AND SECTIONS

A. Division 01 through Division 49: Refer to product Specification Sections indicated in Bid Alternative descriptions and as may be affected by alternate products and scope descriptions.

1.4 GENERAL REQUIREMENTS FOR ALTERNATIVES

A. To enable University to compare total costs where alternative materials and methods might be used or where scope of Work might be altered, Bid Alternative Work items have been established as described in this Section.

1. Unless otherwise specifically provided, the work described in Alternatives shall be completed with no increase in Contract Time.
2. Alternatives will be accepted in ***any*** order listed until the Construction Budget is reached. Additive and deductive alternates will be selected at the discretion of the University.

B. Contract Amount included in Base Bid and as stated in executed Agreement shall include all costs for Work described in Contract Documents.

C. Contract Amount shall include all necessary provisions for Work described in alternatives, whether or not Alternatives are accepted. Base Bid specifications shall govern Work of alternatives unless otherwise specified.

D. Bid Proposal Form or other means prescribed for submission of proposed cost of Work shall include line items for each Alternative described in this Section. No Alternatives other than as described in this Section shall be submitted, except in accordance with product options and substitutions provisions specified in Section 01 61 00, Basic Product Requirements.

E. Each Alternative is identified herein by number. This identification shall be used whenever referring to Work described in Alternative and when submitting cost proposals and payment requests.

F. Alternative construction described in Alternatives and revised scopes of Work shall be performed only when such Alternative is made a part of the Work by specific provision in the University-Contractor Agreement, if selected by University prior to execution of the Agreement, or by Change Order or Change Directive if selected subsequent to execution of the Agreement.

G. Costs for Alternatives shall be valid for no less than sixty (60) calendar days from date of Agreement, and University may select any or all Alternatives during that time. Once an Alternative is selected and the Contract modified for Work as described in the Alternative, changes to return to original scope of Work will be made only by Change Order or Change Directive in accordance with provisions of the Contract General Conditions for changes.

1. If this time has expired, the bid price also expires for the listed Alternate, unless extended voluntarily by the contractor.
2. If the time limit expires and pricing is not extended voluntarily by the contractor, the cost for the listed Alternate will be re-calculated according to the Change Order process. Refer to Article 6 of the Contract General Conditions for additional information on the Change Order Process.

1.5 PRODUCTS AND EXECUTION

A. If University elects to proceed on the basis of one or more of the described Alternatives, Contractor shall make all modifications to Work as required to provide products complete, in place and fully functional, including all labor, equipment, services and incidental consumables necessary to apply, install and finish Work described in Alternative in accordance with requirements specified in related product Sections of these Specifications.

B. Cost for Alternatives shall be complete and include all net increases and decreases in Contract Amount for Work described in Alternative and for all changes in related Work. No claims for additional costs to University will be honored other than as stated in cost proposal for each Alternative.

1.6 ALTERNATIVES [_*Titles and descriptions should be developed by the Design Professional in Charge*_]

A. [Additive or Deductive] Alternative Bid No. 1 - [_*Title*_].
   1. Base Bid condition: [_*Description*_].
   2. Alternative Bid condition: [_*Description*_].

B. [Additive or Deductive] Alternative Bid No. 2 - [_*Title*_].
   1. Base Bid condition: [_*Description*_].
   2. Alternative Bid condition: [_*Description*_].

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

Not Applicable to this Section.

END OF SECTION 01 23 00

SECTION 01 27 00 - UNIT PRICES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Administrative and procedural requirements for unit prices.

COORDINATE UNIT PRICES IDENTIFIED IN THIS SECTION WITH UNIT PRICES STATED ON BID PROPOSAL FORM.

1.3 RELATED SECTIONS

1. *VERIFY THAT SECTION TITLES REFERENCED IN THIS SECTION ARE CORRECT FOR THIS PROJECT'S SPECIFICATION; SECTION TITLES MAY HAVE CHANGED.*
2. *FIRST DIVISION 1 SECTION BELOW CONTAINS REQUIREMENTS THAT RELATE DIRECTLY TO UNIT PRICES.*

1. Section 01 21 00 - Allowance Procedures: Procedures for using unit prices to adjust quantity allowances.

RETAIN SUBPARAGRAPH 2. BELOW ONLY IF UNIT PRICES MAY BE EMPLOYED AS A RESULT OF SPECIAL TESTING AND INSPECTING PROCEDURES REQUIRED BY LOCAL CODES AND SPECIFIED IN

2. Section 01 45 00 - Quality Control: General testing and inspecting requirements, including those tests and inspections based on unit prices.

EDIT PARAGRAPH BELOW AND ADDITIONAL PARAGRAPHS TO IDENTIFY SECTIONS IN SPECIFICATIONS WHERE UNIT PRICES ARE USED.

3. Section [_*Number*_] - [_*Title*_]: Procedures for measurement and payment for [_*Description_of_Work*_].

REPEAT PARAGRAPH ABOVE AS NECESSARY TO INCLUDE ALL UNIT PRICES REQUIRED FOR PROJECT.

1.4 DEFINITIONS

DEFINITION BELOW ASSUMES THAT CONTRACT DOCUMENTS INDICATE OR PROVIDE ESTIMATED QUANTITIES AND WHERE BIDDERS STATE ON THE BID PROPOSAL FORM UNIT-PRICE AMOUNTS THEY PROPOSE. REVISE IF BIDDERS ARE REQUIRED TO ESTIMATE QUANTITIES AS WELL AND INCLUDE THEM WITH THE BID.

A. Unit Price: An amount proposed by Bidder and stated on the Bid Proposal as a price per unit of measurement for materials or services that will be added to or deducted from the Contract Sum by Change Order in the event the estimated quantities of Work required by the Contract Documents are increased or decreased.

1. Unit prices quoted in the Bid Proposal are for additions or deletions of approved items of Work.

2. All unit prices quoted shall be for the items completely installed, furnished, and operable in accordance with the Contract Documents, and shall include profit, overhead, taxes, cost of coordinating the unit price work with adjacent work, compensation for risk of loss or damage to the work regardless of cause, all expenses due to delays in performance, so they are the complete price to the University.

3. Unit prices shall not apply to work the Contractor elects to do for its own convenience or to correct errors committed by the Contractor.

4. All unit prices shall remain in effect for the full term of the Contract.

5. Quantities listed in the Contract Documents are approximate only. Contract Amount shall be adjusted by change order using unit prices listed for actual quantities of Work performed.

1.5 PROCEDURES

USUALLY RETAIN PARAGRAPH BELOW. IT WILL SUFFICE FOR ALL BUT THE MOST COMPLEX PROJECTS. REVISE IF MANY UNIT PRICES ARE ANTICIPATED AND IF METHODS FOR MEASURING WORK-IN-PLACE ARE COMPLEX. A SPECIAL ARTICLE OR ADDITIONAL PARAGRAPHS OUTLINING PROCEDURES FOR MEASUREMENT AND PAYMENT MIGHT BE NEEDED TO DEFINE RESPONSIBILITIES FOR COMPLEX SITUATIONS.

A. Measurement and Payment Procedures: As stated in General Conditions of the Contract. Refer to individual product Specification Sections for Work that requires establishment of unit prices. Basis of each unit price is specified in those Sections.

1. Measure: University reserves the right to reject Contractor's measurement of work-in-place that involves use of established unit prices and to have this Work measured, at University's expense, by an independent surveyor.

RETAIN BELOW AND REVISE TO SUIT SPECIAL PROJECT REQUIREMENTS.

B. List of Unit Prices: A list of unit prices is included at the end of this Section. Specification Sections referenced in the schedule contain requirements for materials described under each unit price.

C. The Trustees are not obligated to use the unit prices.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

3.1 LIST OF UNIT PRICES

IN PARAGRAPH AND SUBPARAGRAPHS BELOW, REMOVE TEXT ENCLOSED IN ANGLE BRACKETS AND INSERT TEXT TO SUIT PROJECT.

A. Unit Price No. [_*Number*_]: [_*Title*_].
  1. Description: [_*Unit_Price_Description*_], according to Section [_*Number*_] - [_*Title*_].
  2. Unit of Measurement: [_*Description*_].

REPEAT ABOVE AS OFTEN AS NECESSARY TO INCLUDE ALL UNIT PRICES REQUIRED FOR PROJECT.

END OF SECTION 01 27 00

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SECTION 01 31 00 - COORDINATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Coordination of Work under Contract.

1.3 RELATED SECTIONS

A. Section 01 11 00 - Summary of the Work: Various types of Work to be coordinated, including Owner-Furnished/Contractor-Installed products and work under separate Contracts.

B. Section 01 31 19 – Project Meetings: Schedule appropriate meetings with Sub trades and University Representative for proper coordination.

C. Section 01 61 00 - Basic Product Requirements: Coordination of products, especially general requirements for system completeness and product substitutions.

1.4 COORDINATION

A. Coordination, General:

1. Coordinate the Work according to provisions stated in Contract General Conditions. Do not delegate responsibility for coordination to any subcontractor.
    a. Anticipate the interrelationship of all subcontractors and their relationship with the total work.
    b. Resolve differences or disputes between subcontractors and materials suppliers concerning coordination, interference, or extent of work between sections. The Contractor's decisions, if consistent with the Contract Documents, shall be final. The Architect and/or University is not required to coordinate work between sections and will not do so.
    c. Coordinate the work of subcontractors and material suppliers, so that their work is performed in a manner to minimize interference with, and to facilitate the progress of the work.

2. Coordinate Work under the Contract with work under separate contracts by University.

3. Coordinate utility and building services shut-downs and closures of vehicular and pedestrian thoroughfares, including access to buildings and parking areas, to minimize disruption of University activities. Notify the University of all scheduled disruptions a minimum of fourteen (14) calendar days. For disruptions that may affect other campus buildings or services, notify the University a minimum of twenty one (21) calendar days in advance of the outage.

4. Contractor shall be responsible for providing anchorage, blocking, joining and other detailing as required to provide complete project.

5. Do not obstruct spaces required by Code in front of electrical equipment, access doors, etc.

6. Do not cover any piping, wiring, ducts, etc., until properly inspected and approved.

7. Remove and replace any and all Work under any Section which is not in accordance with the Contract Documents with other materials and Work which is in conformance with the Contract Documents. Repair or replace all other Work damaged by these operations at no additional cost to the University. No additional time will be granted.

8. This work shall be coordinated with all associated Work in a manner that will insure that all work will be accomplished as rapidly as the progress of the project will permit and so that no work will be delayed for want of associated work.

B. Coordination of OFCI Products: Contractor shall cooperate with University and others as directed by University's Representative in scheduling and sequencing the incorporation into the Work of Owner Furnished/Contractor Installed (OFCI) products identified in the Contract Drawings and Specifications. Sufficient lead times shall be provided by the contractor and the construction schedule will clearly indicate both lead times and delivery dates.

C. Relationship of Contract Documents: Drawings, Specifications and other Contract Documents in the Project Manual are intended to be complementary. What is required by one shall be as if required by all. What is shown or required, or may be reasonably inferred to be required, or which is usually and customarily provided for similar work, shall be included in the Work.

D. Discrepancies in Contract Documents: In the event of error, omission, ambiguity or conflict in Drawings or Specifications, Contractor shall bring the matter to attention of the Architect in a timely manner during the bidding period, for determination and direction by the Architect in accordance with provisions of the Contract General Conditions.

E. Construction Interfacing and Coordination: Layout, scheduling and sequencing of Work shall be solely the Contractor's responsibility.

1. Contractor shall verify, confirm and coordinate field measurements so that new construction correctly and accurately interfaces with conditions existing prior to construction.

2. Contractor shall bring together the various parts, components, systems and assemblies as required for the correct interfacing and integration of all elements of Work. Contractor shall coordinate Work to correctly and accurately connect abutting, adjoining, overlapping and related elements, including work under separate contracts by University, utility agencies and companies.

1.5 COORDINATION OF SUBCONTRACTS AND SEPARATE CONTRACTS

A. Superintendence of Work: Contractor shall appoint a field superintendent and a project manager, who shall directly and full time supervise and coordinate all Work of the Contract.

B. Subcontractors, Trades and Materials Suppliers: Contractor shall require all subcontractors, trades, crafts and suppliers to coordinate their portions of Work with the Contractor's field superintendent to prevent scheduling, sequencing, dimensional and other conflicts and omissions.

C. Coordination with Work Under Separate Contracts: Contractor shall coordinate and schedule Work under the Contract with work being performed for Project under separate contracts by University, serving utilities and public agencies. Contractor shall make direct contacts with parties responsible for work of the Project under separate contracts, in order to provide timely notifications and to facilitate information exchanges.

1.6 MECHANICAL AND ELECTRICAL COORDINATOR

DELETE THIS ARTICLE IF NOT APPLICABLE

A. Mechanical and Electrical Coordinator: Contractor shall employ and pay for services of a person, technically qualified and administratively experienced in field coordination for the type of mechanical and electrical Work required for this Project, for the duration of the Work.

1. Work out all "tight" conditions involving work of various sections in advance before installation. If necessary, and before work proceeds in these areas, prepare supplementary drawings for review showing all work in "tight" areas.

2. Provide supplementary drawings and additional work necessary to overcome "tight" conditions at no increase in contract price. Refer to Section 01 33 00, "Submittal Procedures."

3. Coordinated layout shop drawings shall be dimensionally accurate and detailed, giving complete dimensions of all locations, elevations, and clearances. Show exact locations of the following:
    a. Ductwork
    b. Piping, including fire protection systems.
    c. Valves and piping specialties, including all air vents and drains.
    d. Dampers
    e. Access doors
    f. Control and electrical panels
    g. Adjustable frequency controllers
    h. Motor control centers and transformers
    i. Disconnect switches
    j. Elevator equipment
    k. Electrical cable trays and main conduits
    l. Owner-furnished, Contractor-installed equipment
    m. Hand and Tool space for installation and maintenance work including component removal and replacement
    n. Access to components for installation and maintenance work (ladder, scissor lift access, etc.). Notify University of potential operational conflict for routine maintenance operations based on physical installed location of components.

4. Coordinated layout shop drawings shall show actual architectural and structural constraints and site conditions.

5. Coordination:
   a. Fully coordinate work between all trades with actual architectural, structural, and site conditions.
   b. Coordinate all adjustments required. Clearly identify by circling these adjustments on the coordinated layout shop drawings.
   c. If Contractor has specific questions regarding coordination of the installation with structural, architectural and site conditions and work between trades, submit same with appropriate shop drawings documenting areas in question with Contractor's proposed installation.

6. Submission and review of coordinated layout shop drawings:
   a. Prepare reproducible drawings.
   b. Submit to each trade for review of space allocated to all trades.
   c. Revise drawings to compensate for review by each trade.
   d. Review revisions with each trade.
   e. Submit to Architect for review.
   f. Review of coordinated layout shop drawings is only for verification that Contractor has performed coordination work as specified herein.
      (1) Review does not include verification of exact dimensions, clearances, arrangements and/or compliance with codes.

7. Final coordinated layout shop drawings shall show that all trades affected have made reviews and shall be signed by each trade at completion of coordination.
   a. General Contractor is to assure that each trade has coordinated work with other trades.
   b. Include stamp with labeled space for each trade to sign on each submittal indicating that layout shop drawing has been coordinated.
   c. No layout shop drawing will be reviewed without stamped and signed coordination assurance by General Contractor.

8. Coordinated layout shop drawings showing work of all trades are required. Individual trade layout shop drawings will not be accepted.

1.7 SUBMITTALS

A. Coordination Documents: Coordinate shop drawings, diagrams and other specified in various product Sections of the Contract Specifications. Submit coordination drawings and schedules as specified below, prior to submitting shop drawings, product data, and samples.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 – EXECUTION

3.1 COORDINATION REQUIRED

A. Coordinate Work specified in Division 13 - Special Construction, Division 21 – Fire Suppression, Division 22 – Plumbing, Division 23 – Heating, Ventilating, and Air Conditioning (HVAC), Division 25 – Integrated Automation, Division 26 – Electrical, Division 27 – Communications, Division 28 – Electronic Safety and Security, coordinate within each Division, between these Divisions and with Work specified in other Divisions.

B. Coordinate progress schedules, including dates for submittals and for delivery of products.

C. Conduct meetings with suppliers, installers and others concerned with the Work including University's Representative, to establish and maintain coordination of layout, sequencing and completion of various elements of Work.

D. Conduct meetings with installers and others concerned with the Work, to properly integrate various mechanical and electrical systems, to facilitate construction and to provide proper access and work space for maintenance, renovation and improvement of system components. Include participation by representatives of University, including maintenance personnel.

E. Assist in resolution of conflicts by providing technical advice, coordination drawings and three dimensional representations of integrated system components, including computer and physical models as necessary.

F. At construction progress meetings, report on progress of Work to be adjusted under coordination requirements and any necessary changes in sequencing and scheduling of Work.

G. Contractor shall record and transmit minutes of coordination meetings and reports to University's Representative, Architect, Architect's consultants (as applicable) and to meeting participants.

3.2 COORDINATION DOCUMENTS

A. Coordination Drawings and Models: Contractor shall prepare coordination drawings and three-dimensional models (BIM or similar), in computer form and in physical form as necessary, to organize layout and installation of mechanical and electrical products for efficient use of available space, for proper sequence of installation, for integration with building structure, for future maintenance and renovation, and to identify potential conflicts between systems and elements.

B. System Services: Contractor shall identify on coordination drawings and models all plumbing and electrical power and signal services required for each component of each system.

1. Contractor shall certify that characteristics of services and controls are correct for each component.

2. Certification shall be in written form and signed by Contractor and mechanical and electrical coordinator.

C. Responsibility and Services Matrix: Contractor shall prepare a schedule or matrix identifying elements of mechanical and electrical Work requiring coordination, as specified in all Divisions of the Contract Specifications.

1. Include identification of parties having responsibilities related to each element of Work and describe what that responsibility shall be.

2. Include required off-site and on-site tests and inspections for various elements of Work.

3. Include identification of administrative activities related to each element of mechanical and electrical Work, such as product data, shop drawings, coordination drawings, samples, mock-ups, test reports for each element of Work.

4. Include identification of elements of Work requiring temporary services.

D. Maintenance and Disposition of Coordination Documentation: Maintain coordination documents, including models, for duration of the Work, recording all changes. After review of original and revised documents and models by University's Representative and Architect, submit documents and models as part of Project record documents. See Section 01 78 39, Project Record Documents.

3.3 COORDINATION OF SUBMITTALS

A. Submittal Reviews by Mechanical and Electrical Coordinator: In addition to specified review actions by Contractor, specified in Section 01 33 00 - Submittals Procedures, all product data, shop drawings and samples shall be reviewed by the mechanical and electrical coordinator for proper coordination of various elements of Work, as described in the preceding Article titled "Coordination Documents."

1. Include Owner-furnished/Contractor-installed (OFCI) products.

2. Include products to be provided (furnished and installed) under separate contracts by University, to the extent that information is provided in the Contract Documents and supplemental instructions from University's Representative.

3. Review by Contractor shall be completed prior to submission of product data, shop drawings and samples to Architect for review.

4. Indicate review actions by Contractor by signed review stamp and other appropriate notations on submittals.

5. Coordinate with other review actions to be taken by Contractor, as specified in Section 01 33 00 - Submittals Procedures.

B. Field Conditions: Contractor shall verify field dimensions and clearances and relationship to available space and anchoring provisions. Report conflicts in writing to the Architect and the University's Representative.

C. Product Characteristics: Contractor shall:

1. Verify compatibility of equipment and other elements requiring plumbing, HVAC and electrical services and signals with services to be provided.

2. Verify motor voltages and control characteristics.

3. Coordinate controls, interlocks, wiring of pneumatic switches, and relays.

4. Coordinate wiring and control diagrams.

5. Review the effect of changes in one element of the Work of other elements of the Work. Identify conflicts and report conflicts in written and graphic form to the Architect and the University's Representative.

6. Verify information provided in maintenance and operating instructions and coordinate preparation of maintenance and operation data. See Section 01 78 23 - Operation and Maintenance Data.

3.4 COORDINATION OF SUBSTITUTIONS AND MODIFICATIONS

A. Review of Proposed Substitutions: See Section 01 63 00 - Product Substitution Procedures. Contractor shall review Contractor's proposals and requests for substitution prior to submission to Architect.

1. Contractor shall verify compliance with Contract Documents and shall certify compatibility with other elements of the Work, including proper integration with building structure, load limitations, operating and maintenance space and accessibility provisions, and suitability for available building services, including plumbing and electrical power and signal systems.

2. Contractor shall prepare and submit recommendation for action regarding proposals, including identification of related changes in other elements of the Work.

3.5 SYSTEM AND EQUIPMENT START-UP

A. Observations of System and Equipment Activation and Start-Up: Contractor shall observe activation and start-up of systems and equipment, including all Work specified in Divisions 2 through 49 with connections to utilities, building services and controls.

1. Contractor shall verify that utilities, building services and control systems are properly connected, complete and functional within criteria of manufacturer and criteria indicated in the Contract Documents.

2. Contractor shall verify that activated elements are properly anchored and that operating components operate properly according to the component's intended design.

3. Contractor shall verify that activated elements of the Work are in operable condition according to normal operating characteristics required by the manufacturer and the Contract Documents.

4. Should adjustments be necessary to activated elements, Contractor shall advise the Architect and University's Representative of necessary actions and shall observe that proper actions are performed to achieve required operating characteristics.

B. Observations of System and Equipment Demonstrations: Contractor shall observe performance demonstrations including equipment demonstrations to Architect and University's Representative. Record times and additional information required for operation and maintenance manuals.

C. Documentation of Observations of Activation, Start-Up, Adjustment and Demonstration: Contractor shall keep written record of activation, start-up, operational tests and inspections and necessary adjustments and re-tests and re-inspections.

1. Documentation shall include record of time and date of activation, start-up, operational tests and inspections and shall include measured results of tests and inspections.

2. Documentation shall be submitted to University's Representative and Architect.

3.6 INSPECTION AND ACCEPTANCE OF EQUIPMENT

A. Contract Completion Review:

1. Prior to Contract Completion review, Contractor shall verify that each component and system has been properly adjusted, cleaned, lubricated, inspected and tested, and is ready operation and use.

END OF SECTION 01 31 00

PRECONSTRUCTION MEETING AGENDA

Project Name____

Contract No.______________________________ **Project No.**________________________

Campus__________________________________ **Date**____________________________

Room___________________________________ **Time**____________________________

1. Introductions: Statement of Purpose (by Chairman) - Administration of Construction. Each person present will introduce himself (or herself) and briefly explain the role he or she will play in relation to the project, if any. Also print your name, organization and phone number on the sign-in sheet being circulated (copies will be distributed).

2. Explain/discuss:

A. Responsibility of each party involved in the project.

B. Inspection procedures.

C. Testing procedures labs. Soils:________Materials:__________

D. Progress schedule.

E. Submittals and approvals. Contractor to Architect, transmittals only to Inspector. Architect return approved submittals with a copy to Inspector.

F. Routing of correspondence. Contractor, Architect, Inspector, University, Construction Administrator. Each copies the others.
Changes, disputes, complaints, claims, etc.

G. Prevailing wage, CSU Labor Compliance Program (if applicable), EEO, Preliminary Notices, Insurance.

H. Payment procedure. Bid cost breakdown. (Forms 702.12, 702.17 and 702.21.

I. Change Order procedure: Change Proposal, Cost Request Bulletin, approved design and cost, Field Instruction, T&M, Change Order.

J. Field Instructions (Form 702.22).

K. Check and Final Inspections. (Punch List form 702.07).

L. Guarantee (Form 702.19), as-built plans, turn-over items, balance report.

M. Notice of Completion.

N. Retention payment.

3. **Review**:

A. Special coordination problems, utilities, traffic, parking, fence, keys.

B. Restrictions of contract operations, if any.

C. Ingress and egress to project to site.

D. Use of State property.

E. Demolition work.

F. Safety.

G. Utilities shut down.

4. Confirm:

A. Survey benchmarks and their location, and critical layout control.

B. Points of tie-in to existing utilities.

C. Existing condition of site and adjacent areas and structures (Site Survey Form 702.08).

D. Source of temporary utilities.

5. Determine:

A. Contractor's plan of operation.

B. Field office location.

C. Project Manager:____

Superintendent:____

D. Contractor's off-hour contact in case of emergency:

______________________________ Phone #______________________

E. Security arrangements contemplated by the Contractor.

F. Starting date of project:____

6. Resolve:

A. Questions of contract requirements.

B. Job meeting procedure with Architect, Inspector, and Contractor.

Time and date:____

C. Shop drawings and submittals requirements.

D. Special tests:____

E. Anticipated changes.

F. Color submittals. Who confirms?______________________________

7. University to outline campus requirements.

8. Anyone have anything to add while this group is assembled?

SECTION 01 31 19 - PROJECT MEETINGS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 REQUIREMENTS INCLUDED

A. Preconstruction meeting.

B. Construction progress meetings.

C. Pre-installation conferences.

1.3 RELATED REQUIREMENTS

A. Section 01 31 00 – Coordination: General requirements for coordinating the work, meetings required to properly coordinate the work are defined.

B. Section 01 32 26 - Construction Progress Reports: General requirements for construction progress reports, to be reviewed at construction progress meetings.

C. Section 01 33 00 - Submittals Procedures: Status of submittals to be reviewed at construction progress meetings.

D. Section 01 45 00 - Quality Control: General requirements for construction quality, to be reviewed at construction progress meetings

E. Section 01 77 00 - Contract Closeout Procedures: Contract closeout meeting.

1.4 PRECONSTRUCTION MEETING

A. Preconstruction Meeting: University's Representative will administer a preconstruction meeting immediately prior to Contractor mobilization onto the project site.

1. Representatives of the Trustees, the Contractor, Architect, and Architect's Consultants, Special Inspector, Inspector of Record, Commissioning Agent and other campus representatives, as appropriate, will attend.

2. Contractor and first tier subcontractors, as appropriate, shall attend.

B. Schedule: Schedule preconstruction meeting within fourteen (14) days of construction start date established in the Notice to Proceed.

C. Location: Preconstruction meeting will be held at a location as directed by the University's Representative.

D. Agenda: Preconstruction meeting shall cover the following topics as a minimum.

1. Special Project Procedures: Site access restrictions, if any, and requirements to avoid disruption of operations at adjoining facilities. University to present requirements for use of premises.

2. Designation of Key Personnel: Contractor shall designate key personnel and provide a name and address list that includes the following.
    a. Contractor: Project Manager and Superintendent.
    b. Major subcontractors: Principal/Project Manager and Superintendent.
    c. Major materials suppliers: Contact person.
    d. After hours or weekend contact

3. Subcontractors List: Distribute and discuss list of subcontractors and suppliers.

4. Coordination: Review requirements for Contractor's coordination of Work. Review sequence and schedule for work being performed for University under separate contracts. Discuss coordination of construction to minimize impacts on continuing Campus operations.

5. Project Communication Procedures: Review requirements and administrative requirements for written and oral communications.

6. Construction Schedule: Distribute and discuss initial construction schedule and critical work sequencing of major elements of Work, including coordination of Owner-Furnished/Contractor-Installed (OFCI) products and work under separate contracts by serving utility agencies and companies and University.

7. Campus and Site Security: Review requirements for Contractor to develop and implement site security.

8. Safety Program: Review requirements for Contractor to develop and implement safety program in compliance with Contract General Conditions.

9. Site Access by University's Representative and Architect: Review requirements and administrative procedures Contractor may wish to institute for identification and reporting purposes.

10. Permits and Fees: Review Contract requirements and review schedule and process for obtaining any required permits and paying fees. (Parking permits)

11. Project Layout: Review requirements for laying out of Work, including surveying requirements.

12. Construction Facilities: Designate storage and staging areas, construction office areas and parking areas and review site access requirements.

13. Temporary Utilities: Requirements for establishing and paying for temporary water, power, lighting and other utility services during construction, including metering and allowances. Refer to Section 01 51 00 - Temporary Utilities.

14. Construction Progress Schedules: Review requirements for preparation and updating of construction progress and submittals schedules.

15. Payment Procedures: Review requirements for preparation and submission of applications for progress payments and for final payment.

16. Change Procedures: Review requirements and administrative procedures for Change Orders, Field Instructions and Contractor's Requests for Interpretation (RFI).

17. Submittals Administration: Review administrative procedures for shop drawings, product data and samples submittals and review of preliminary Submittals Schedule.

18. Materials and Equipment: Review substitution or equal product requirements; review schedule for major equipment purchases and deliveries; review materials and equipment to be provided by University (OFCI products).

19. Testing and Inspection: Review tests and inspections to be performed by the following.
    a. Independent testing and inspection agency.
    b. Manufacturers and installers.
    c. Serving utilities and public agencies.
    d. Authorities having jurisdiction.
    e. Commissioning Agent

20. Operation and Maintenance Data: Format and content of operation and maintenance manuals. Refer to Section 01 78 23 - Operation and Maintenance Data.

21. Instruction of University's Personnel: Review requirements and scheduling of instruction of personnel specified in Section 01 79 00 - Demonstration and Training and in various Sections through all Divisions of the Specifications.

22. Starting and Adjusting Procedures: Review requirements of starting and adjusting operating components. Refer to Section 01 75 00 - Starting and Adjusting Procedures.

23. Project Record Documents: Review requirements and procedures for preparing, reviewing and submitting project record drawings and specifications in Section 01 78 39.

24. Construction Cleaning: Review requirements for progress and final cleaning specified in Section 01 74 00 - Cleaning Requirements.

25. Contract Closeout: Review requirements specified in Section 01 77 00 - Contract Closeout Procedures, including procedures for filing of Notice of Completion, final payment and submittals.

1.5 CONSTRUCTION PROGRESS MEETINGS

A. Construction Progress Meetings: Meetings will be held to review progress and quality of construction. The essence of the discussion of each meeting shall be entered into the written record (minutes) of the meeting by the Architect or the University Representative designee.

B. Schedule: Construction progress meetings shall be periodically scheduled throughout progress of the Work. Frequency shall be as determined necessary for progress of Work. Generally, it is intended that construction progress meetings be held at weekly intervals.

C. Administration: Architect shall make physical arrangements for meetings. Architect shall prepare agenda with copies for participants, preside at meetings, record minutes and distribute copies within two working days to University's Representative, Contractor, participants and those affected by decisions made at meetings. Each discussion item at construction progress meetings shall be numerically identified and carried through subsequent meeting minutes until resolved.

D. Attendance: Contractor's project manager and jobsite superintendent shall attend each meeting. Contractor's subcontractors and suppliers may attend as appropriate to subject under discussion. University's Representative will attend each meeting. Architect's consultants will also attend, as appropriate to agenda topics for each meeting and as provided in University-Architect Agreement.

E. Suggested Agenda for Each Construction Progress Meeting:

1. Meeting Minutes: Review and correct, if necessary, minutes of previous meeting.
   a. Unless published minutes are challenged in writing prior to the next regularly scheduled progress meeting, they will be accepted as properly stating the activities and decisions of the meeting.
   b. Persons challenging published minutes shall reproduce and distribute copies of the challenge to all indicated recipients of the particular set of minutes.
   c. Challenge to minutes shall be settled as priority portions of "old business" at the next regularly scheduled meeting.

2. Old Business: Active discussion topics carried over from previous meetings.

3. Progress of the Work: Since last meeting and proposed progress.
   a. Identify potential problems which might impede progress. This shall include upcoming University Holidays or University required alternate work schedules.
   b. Develop corrective measures and procedures, including but not necessarily limited to additional manloading to regain planned schedule.
   c. Review three-week "look ahead" construction schedule, including identification of conflicts and delays.

4. Ordering Status: Review status of long-lead time equipment and materials delivery affecting construction progress.

5. RFI Status: Review status of Requests for Interpretation (RFI) status.

6. Submittals Status: Review shop drawings, product data and samples submission and review status.

7. Contract Modifications: Pending Change Orders and Field Orders. Review status of proposed substitutions.

8. New Business: New topics of discussion affecting construction progress and quality.
   a. Inspections – Both outstanding issues to discuss and upcoming inspections to be scheduled.

3. Quality Control: Review maintenance of quality standards and identification of non-conforming Work, including proposed remedial measures to be taken by Contractor.

4. Project Record Documents: Status of project record drawings and specifications.

5. Environmental and Safety Issues.
   a. SWPPP controls and measures
   b. Air Pollution issues
   c. Noise Controls
   d. Site Safety
   e. Public Pedestrian and Traffic Control Measures

6. Other items affecting progress and quality of the Work.

F. Meeting Time and Location: As mutually agreed by the Architect, the Contractor, and the University's Representative at on-site location. Typically held weekly at the beginning or end of each week.

G. Special Meetings: As necessary, the Architect, the Contractor, or the University's Representative may convene special meetings to discuss specific construction issues in detail and to plan specific activities.

1.6 PRE-INSTALLATION CONFERENCES

A. Pre-Installation Conferences: When specified in individual product specification Sections, convene a pre-installation conference prior to commencing Work specified in individual product Sections.

1. Require attendance by representatives of firms whose activities directly affect or are affected by Work specified in the Section.

2. Review conditions of installation, preparation and installation procedures and coordination with related Work and work under separate contracts.

1.7 CONTRACT COMPLETION MEETING

A. Contract Closeout Meeting: As specified in Section 01 77 00 - Contract Closeout Procedures.
   a. It is recommended that this meeting be scheduled four (4) to six (6) weeks prior to the scheduled contract completion date. Refer to Section 01 77 00 for additional information on Punch List meetings, final completion submittals and final payments.

PART 2 - PRODUCTS

A. Contractor shall provide all necessary products as necessary to facilitate a productive meeting. This shall include any product samples, manufacture's literature, mockups, details, or any similar item.

PART 3 - EXECUTION

Not applicable to this Section.

END OF SECTION 01 31 19

California Polytechnic State University

Prolog Converge User List

Name	Trade	E-mail Address	Company	Primary Phone	Address	City	State	Zip

Attach additional pages as needed

SECTION 01 32 00 - ELECTRONIC PROJECT MANAGEMENT SYSTEM

1.00 GENERAL

1.01 DESCRIPTION

A. This Section is in addition to the Contract General Conditions.

B. The Contractor shall be required to use the University's Electronic Project Management (EPM) system, Prolog Website and/or Prolog Manager, by Meridian Project Systems for electronic construction management document control and communications between the University, Architect of Record, other project-related consultants, and Contractor. The system will be maintained and owned by the University but operated collaboratively by the Project Team.

C. The EPM system will contain information the following information available to the contractor and project team:

1. Submittal Information and Logs including weekly and monthly schedule updates
2. Requests for Information and Logs
3. Inspection Requests / Reports
4. Non-Compliance Inspection Reports
5. Project Photographs
6. Project Meeting Minutes
7. Electronic Drawings, Sketches, ASIs
8. Other Documentation as determined by the University's Representative.

D. All Request For Information (RFIs) and Inspection Requests, shall submitted by the Contractor to the University electronically, via Prolog Website.

E. The University will **NOT** accept faxed and/or hand written documentation of RFIs, RFI Sketches, and/or Inspection Requests.
1. The Contractor shall be solely responsible for data entry via the Electronic Project Management System Website for the generation of project related documentation.
2. The contractor shall be solely responsible for the scanning of sketches / drawings as necessary for the electronic submittal and attachment of necessary information related to RFIs.
3. Contractor shall supply field personnel with all necessary computer equipment necessary to enter documentation electronically.

F. Submittals may be submitted via hard copy per Section 01 33 00 Submittals with electronic documents for all paperwork and shall include photographs of physical product submittals.

1.02 CONTRACTOR'S RESPONSIBILITIES

A. The Contractor shall have sufficient computer(s) with capabilities to access the system at their on site and off site project offices.

B. At the pre-construction meeting, The Contractor shall submit to the University a comprehensive list of users utilizing the enclosed form. The University will use this information for no other purpose than to establish the necessary accounts for use by each individual.

C. The Contractor shall pay the University $500.00 for each user to be registered with the Electronic Project Management System. This payment will be due during the final closeout of the project at which time the University will access the final number of Contractor users assigned to the project.

Commented [CJM1]: This may reduce the collaboration of the system if contractors are having to pay for the service. If the project can pay for the service directly then remove this section.

D. Contractor shall complete the User List provided in Specification Section 01 32 00.1 for initial user set up and assignment.

E. These personnel shall have sufficient computer skills required to access the Internet, log on to the EPM system, and utilize the system. The Contractor may request the University to provide training and technical support to the Contractor's personnel for use of the EPM system. The cost of the training session will be borne by the attendees. The Contractor shall plan on an average of 4-hours training for each of the Contractor's personnel who will be using the system. Having the above capability in place onsite is a condition precedent to processing the Contractor's first payment request.

1.03 OFFICIAL RECORDS

A. The documentation and records maintained on the EPM system will be the "Official Records" for the project. This documentation shall be the records for the adjudication of any and all disputes. At the conclusion of the project all records can be made available via Adobe "pdf" at the request of the Contractor.

END OF SECTION 01 32 00

SECTION 01 32 10 - COLLABORATIVE CONSTRUCTION PLANNING PROCESS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Section Includes:
1. Definitions
2. Basic Requirements of Contractor's Scheduling System
3. Collaborative Schedule Overview/Background
4. LPS Facilitation
5. Collaborative Schedule Process
    a. Required Participants
    b. LPS Implementation Material and Tools
    c. Preconstruction Meeting
    d. Master Milestone Schedule
    e. Phase Pull Scheduling
    f. 6 Week Make Ready Planning
    g. Weekly Work Plan
    h. Workable Backlog
    i. Daily Work Planning Huddles
6. Deliverables
7. Responsibility for Completion

1.3 DEFINITIONS

A. Constraint – In the context of the Last Planner® System, an input, directive, resource or other requirement that will prevent a task or an assignment from starting, advancing or completing as planned.

B. Constraint Log – A list of constraints, each one with an identification of the individual or champion who promises to remove it by an agreed upon date.

C. Last Planner® System (LPS) – A system for project production planning and control aimed at creating a work flow for reliable execution.

D. Last Planner – The person who conducts the final planning of a task or activity and makes the work resource assignments for those in production.

E. Milestone Plan – A master plan schedule developed collaboratively by a project team that identifies major milestones in the project as well as each team members' milestones and their timing.

F. Pareto chart - Named after Vilfredo Pareto, this chart contains both bars and a line graph, where individual values are represented in descending order by bars, and the cumulative total is represented by the line.

G. Percent Planned Complete (PPC) – Metric used in the Last Planner® System to gauge plan reliability. Defined as the ratio of the number of actual activities completed in a given time period over the number of actual activities planned (typically weekly).

1.4 BASIC REQUIREMENTS OF CONTRACTOR'S SCHEDULING SYSTEM

A. This Section specifies CSU expectations, administrative and procedural requirements for planning and scheduling. The CSU requires that Lean Construction planning principles and techniques shall be utilized as described herein. This specification section requires the integrated and coordinated use of the Reliable Production plan based on the LPS and conventional CPM scheduling. The CSU requires a high level of use of the LPS and expects the Contractor to have experience at all levels of planning and scheduling using these systems.

B. The Schedule shall be prepared, updated and maintained using Primavera Project Planner, Microsoft Project, or other platform as approved by the Trustees.

1.5 COLLABORATIVE SCHEDULE OVERVIEW / BACKGROUND

A. The primary function of the LPS is the collaborative planning process that involves Last Planners, the persons executing the work, for planning in greater detail as the team gets closer to doing the work. The LPS is an opposite way of thinking when compared to conventional push scheduling principles, where the work that *should* be done is planned in weekly meetings emphasizing adherence to the master schedule milestones. In contrast to traditional CPM scheduling, the LPS incorporates pull planning principles where only the work that *can* and *will* be done is considered and promised by Last Planners themselves. At its core, LPS is a system view versus individual optimization, where the Last Planners' active engagement in this systemic process is a fundamental requirement. The LPS is a team sport.

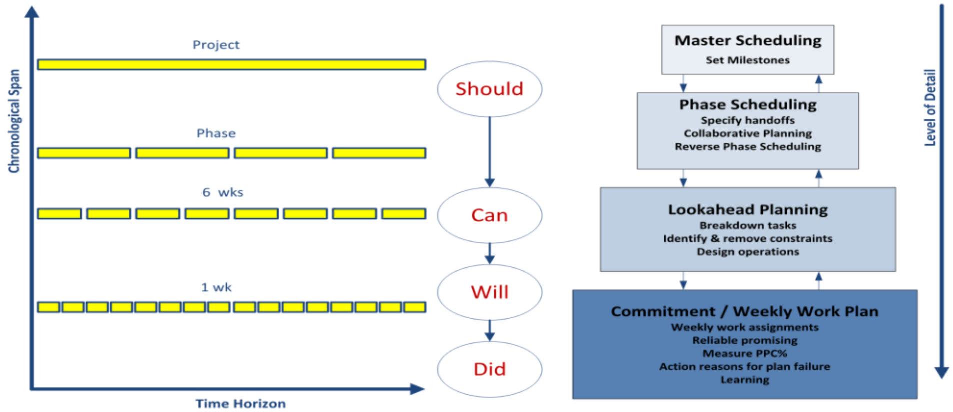
B. The purpose of the Collaborative Schedule process is to build a reliable project Master Schedule within a collaborative team environment. The primary goal is to 1) establish, solidify and maintain the Milestones within the Master Schedule, and 2) support the teams and work flow improvements necessary to produce safe, reliable and interruption-free project delivery, increasing the reliability of project production planning and improving project performance.

C. The Collaborative Schedule process is iterative and constantly measured with metrics. When executed successfully, the Weekly Work Plans can be easily formulated and Monthly Schedule Updates are naturally produced.

D. The Collaborative Schedule process makes detailed plans by those who execute and manage the work. The System promotes conversations between trade foremen and project management, at appropriate levels of detail, to resolve issues before they become critical. As an activity nears execution the team collaboratively acts to remove constraints and verify that the promises made are tied to milestones. The promises made must be firm commitments, timely and without ambiguity.

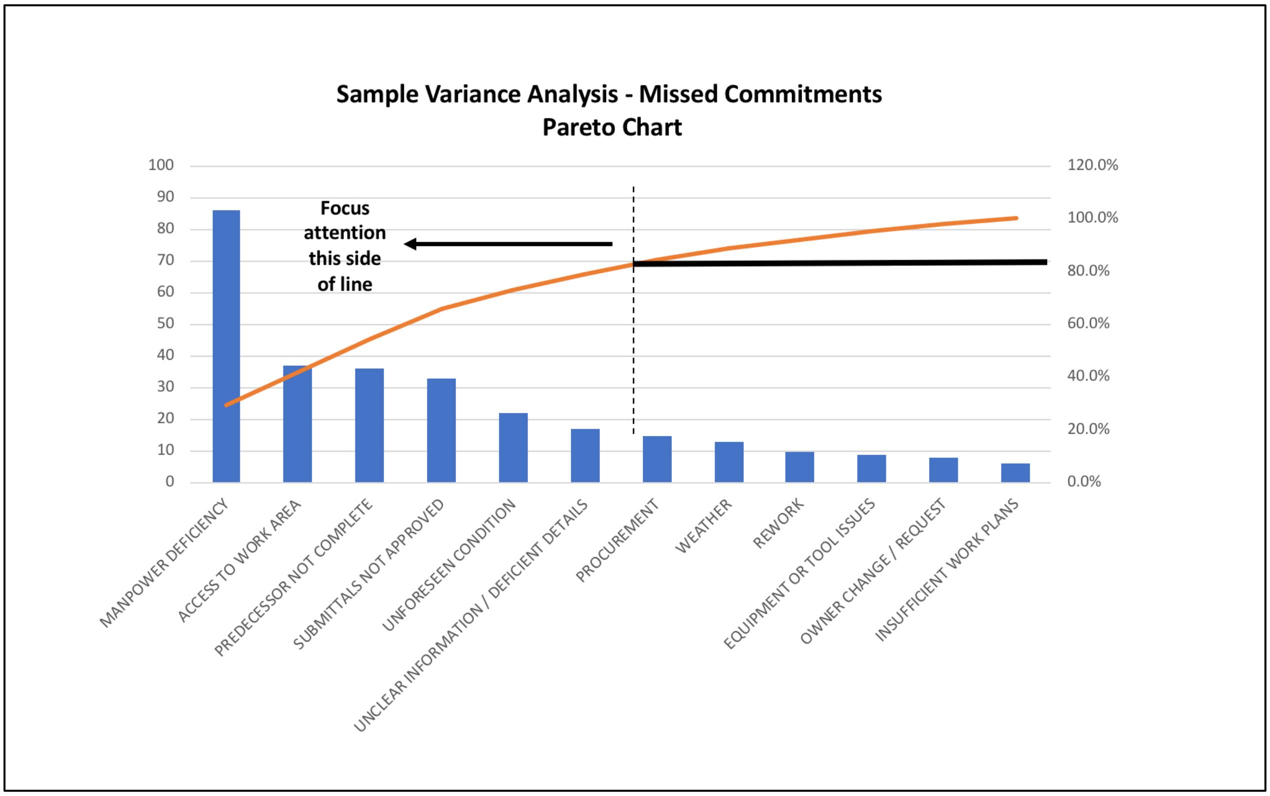
E. Use planning procedures described herein to create a Master Schedule, a Look-Ahead Schedule, and a commitment-based Weekly Work Plan schedule through front-end planning using LPS and Lean Construction Planning techniques.

F. LPS Planning Process Overview



The key elements of LPS are:

1. **Master Schedule Planning**: Setting milestones and strategy for the entire project including identification of long-lead items and major constraints. Incorporates critical path method (CPM) logic at a high level to determine overall project duration, what project work <u>*should*</u> be done. The milestone plan is used to develop the overall sequencing and flow of the work on the project. This will be a CPM-based schedule.
2. **Phase Pull Planning**: Strategically planning segments of work (typically 3 to 4 months in duration) in order to produce progressively detailed Weekly Work Plans. Collaborative reverse phase (pull) schedule planning by those who will be doing the work, at sufficient level of detail to specify handoffs, and identify and resolve operational conflicts. This effort will identify schedule activity durations and what project work <u>*can*</u> be done. Phase pull planning often results in modifications to the CPM logic used for initial project planning.
3. **6-Week Make-Ready Planning**: Look-ahead scheduling and constraint removal (roadblock removal process) in support of the progressively detailed planning process to assure that work is made ready for installation. This effort will identify what project work <u>*will*</u> be done. The plan is updated weekly constraints that threaten reliable workflow are identified and captured for the team's action to remove them.
4. **Weekly Work Planning**: Team collaboration to plan each day's work, conditions for handoff and acceptance, sequencing and synchronizing next week's work. Commitments are made to perform work in a certain manner and a certain sequence. What project work are we to <u>*do*</u> this week. Weekly work planning is the point of maximum progressive detailing to create reliable work plans. Plan reliability at this level is promoted by making only quality assignments and reliable promises so that the production unit will be shielded from upstream uncertainty.
5. **Daily Huddles**: Daily team check-ins, discussions based on the Weekly Work Plan. How are we doing? What do we need to maintain the plan in progress?
6. **Percent Plan Complete**: Number of activities completed divided by the total number of planned activities. At the end of each week, assignments are reviewed for completeness in order to measure the reliability of the planning.
7. **Reasons for Missed Commitments Variance Analysis**: Charted using Excel and/or Pareto charts (see example below) to identify trends, learning and understanding what needs to be fixed in order to improve next week's PPC.



8. **Learning:** By measuring percent of promises complete (PPC), tracking reasons for variance, diving deep into reasons for plan failures, and developing/implementing lessons learned to improve future plan reliability. Analyzing reasons for plan failures and acting on these reasons is the basis of learning.
9. **Reliable Promising**: Projects are essentially made up of an extensive set of reliable promises. LPS makes the planning processes and work flow highly reliable, and builds necessary trust within a collaborative team environment.

1.6 LAST PLANNER FACILITATION

A. LPS Facilitator

1. No later than ten (10) days after Contract NTP, the Contractor shall identify a facilitator to provide one day of training in the LPS and submit the facilitator's relevant qualifications for Trustees' review and approval. The facilitator may be in-house for the Contractor if the facilitator's qualifications meet the Trustees' approval. In general, it is expected that the facilitator has successfully led LPS on a number of projects previously. A listing of past experience and the project outcomes should be provided as part of the facilitator's qualifications. The facilitator is to be jointly chosen/approved by the contractor and Trustees. If an outside facilitator is selected, this shall be a shared service between contractor and Trustees with the cost split 50/50 with no contractor mark-up allowed. All subcontractors and general contractor personnel responsible for executing this specification shall attend.
2. The LPS facilitator shall attend Pull Planning sessions commensurate with the experience of the General Contractor, and as necessary to assure that all process and requirements detailed herein are satisfied.

3. Teams may have variable past experience with LPS. For teams that have never participated in a project delivered using LPS, six (6) facilitated sessions at the beginning of the project are typically required for project teams without LPS experience to develop working competency. This is not considered a minimum, but is a suggested training level. Teams more experienced with LPS may require a reduced level of facilitation.
4. If the cumulative Percent Planned Complete (PPC) drops below 70 percent, then a LPS facilitator needs to be brought back to assist the team in identifying why the plan is not reliable and assist in identifying and implementing appropriate countermeasures to improve reliability of the plan.

1.7 COLLABORATIVE SCHEDULE PROCESS

A. Required Participants
1. Since LPS is a collaborative process, all those that have a planning role on the project need to participate in the scheduling process at the appropriate time. It is expected that there will be different participants required at different times in the project timeline depending on their respective scope of work and the timing of when it will be planned and performed. The right people need to participate at the right time for the plan to be informed and reliable. These individuals will be expected to participate in all phases of LPS as described in this section.
2. Required Project Stakeholders (the list is not exhaustive, participation by others may be required):
a. General Contractor
b. CSU Project Manager
c. All subcontractors, and/or discipline-specific trades
d. Project Manager from each trade and/or subcontractor
e. General foremen and superintendents from each trade subcontractor
f. Key project engineers and/or construction coordinators
g. Vendor and/or suppliers with key materials, as necessary
h. Off-site fabricators, as necessary
i. Third-party support (testing, inspection, commissioning agents, LEED certification specialists, etc.)
j. Architects and engineers
k. University representatives including IT, campus police, purchasing and others as necessary. The University end-users shall be represented through separate user group meetings to identify items of concern, constraints, etc.
l. LPS Facilitator.

B. LPS Implementation Materials and Tools
1. These forms are included by reference:
a. Short-Term Production Plan
b. Constraint Log
2. Large Meeting Room – Job Site Trailer, large enough for 30 individuals, ideally from a single location, “The Big Room”.
3. Standard conference room white board (20’ horizontal length suggested).
4. Walls dedicated to visual system aids.
5. Standard 3-inch size “sticky notes”, dedicated color for each Trade Contractor and/or design discipline.

6. Weekly work plan boards for the 6-week make work ready plan that are freestanding and contain columns and rows for “sticky notes”. Boards should have 7 columns, one for each day of the week, and approximately 20 rows of 4 inch by 4 inch squares for standard 3-inch size “sticky notes”. The surface of the boards should allow good adhesion of the “sticky notes”. Boards should also be easily movable to accommodate rolling planning done weekly.
7. Microsoft Office Suite, specifically Microsoft Excel, for creation of Weekly Work Plans and other necessary LPS elements. Microsoft Project or Primavera scheduling software, for creation and documentation of milestone schedule, milestone relationships and 6-week Look-Ahead schedules.
8. Display well-maintained outputs for the group to use at Daily Huddles and Weekly Coordination Meetings in The Big Room.
9. The four primary Visual Outputs tools of LPS are:
    a. Weekly Work Plans (WWP) –boards described above.
    b. Percent Plan Complete (PPC) trend over time.
    c. Reasons for Variance Pareto, graph, or pie chart.
    d. Constraint Log.

C. Preparatory Meeting
1. Submit a Draft Master Milestone Schedule (High-Level Master Milestone) and an initial 6 Week look-ahead schedule at the Preparatory Meeting which will include mobilization activities, first collaborative pull planning session, etc.

D. Master Milestone Schedule
1. Prior to the first pull planning Milestone session, the General Contractor will prepare a high-level master milestone CPM schedule in advance of the session for the entire project to identify major project milestones and general sequence of how the project may be executed. These milestones should include required delivery dates for major long-lead equipment items like switch gear, transformers, chillers, etc.
2. As part of the Milestone Plan pull planning session, the major project milestones (e.g., CSU constraints and contract milestone requirements, foundation poured, topping out, weathered in, permanent power) developed through the CPM schedule will serve as the dates to work the Milestone Plan.
3. Milestones have zero duration and represent the completion or start of a particular activity or action.
4. Milestones used in the Milestone Plan should be completion milestones for the most part. Select start milestones for critical activities may also be appropriate to include in the plan.
5. Milestones for trade contractors should represent completion of major trade activities and for completion of trade work in a specific area of the project (e.g., floor, gridline or elevation).
6. Each trade should have multiple milestones and with sufficient detail to identify interim trade milestones at least every 6 weeks to help develop more reliable make work ready planning.
7. The team works backward from the final project milestone to pull towards the milestone plan.
8. The collaboratively developed milestone plan is used to validate or challenge the required CPM schedule, and collaboratively inform necessary changes to the CPM schedule.

9. Include milestones for each trade contractor, each phase, key submittal approvals, key release dates for long-lead equipment and material, shipment/arrival of key materials and/or equipment, key inspections, occupancy, commissioning, project completion, etc.
10. Any constraints that are identified that will prevent a task or an assignment from starting, advancing, or completing as planned need to be captured in a Constraint Log. The log should clearly identify the constraint, by what date it needs to be removed to not impact project production, and the member of the project team that has been assigned responsibility to lead the efforts to remove the constraint. The constraint log should be maintained and updated throughout the project and displayed visually in The Big Room so that all project team members can see it.
11. Master Milestone Schedule CPM Format
    a. Activities shall be coded in a logical manner to allow for sorting and grouping of like characteristics, including but not limited to such items as: phase, work shift, project area, activity type (e.g., submittal, agency review, and construction activity), trade, etc.
    b. Include activities and milestones as requested for work completed by University under separate contract, University-furnished materials, move-in, etc.
    c. The schedule duration shall be calculated for the Initial Construction Schedule, Contract Construction Schedule, and subsequent schedule updates.
    d. Contractor's Superintendent and Project Manager shall be integrally involved in production of the Initial Master Schedule and each subsequent update.
    e. Failure by Contractor to include any element of the work required for performance of the Contract shall not relieve Contractor of the obligation to complete the entire Work of the Contract in accordance with the Contract Completion Date.

E. Phase Pull Scheduling
1. Phase Pull Scheduling generates a detailed schedule magnifying the master schedule into more detailed project components strategically planning segments of work and activities in order to produce progressively detailed Weekly Work Plans.
2. The purpose of Phase Pull Scheduling is to produce a plan:
    - for completing a phase of work that everyone involved understands and supports, and
    - from which scheduled activities are drawn into the look-ahead process to be exploded into operational detail and made ready for assignment in weekly work plans.

    a. The project milestones shall be placed at the top of the visual phase plan which is developed at the wall.
    b. The level of detail in the Phase Schedule is determined by the requirement that the Phase Schedule specify the handoffs between subcontractors involved in doing the work.
    c. The phase plan will consist of activity tags completed for each trade by the Last Planner for that trade.
    d. Activities should be no longer than 10 days in duration. Any task longer than 10 days can be broken down into smaller discrete activities which allow for better planning.
    e. Identify the specific task to be completed with an action verb, identify what is required to release the work (predecessor or constraint), location, crew size, and duration.

f. Each discipline or trade is responsible for completing and placing its own tags on the plan.

g. The phase will pull, starting from the completion of the interim milestone associated with the end of the phase, and working backwards (right to left). After the pull backwards, then a forward pass (left to right) needs to be conducted to ensure that the plan demonstrates a logical building sequence and that the sum of the total activity durations is within the allowable time to meet the project milestone(s), and to identify opportunities to resolve conflict and improve production flow. Do not double count durations for concurrent tasks.

h. The completion of a “phase” should be sequenced so that the “phase” completion releases new work.

i. Participants: All team members involved in planning and execution of work during the Phase Schedule.

j. A Phase Pull schedule is produced for a typical duration of approximately three (3) months using an appropriate interim milestone as the completion point of the phase.

k. After the initial Phase Pull plan, subsequent Phase Pull plans should be developed every six (6) weeks to reflect the next three (3) months of project production. This will allow for a fully informed rolling 6-week Make-Work Ready Plan on the project.

l. The Phase Pull plan should be visually displayed in The Big Room for all to see and to inform subsequent 6-week Make-Work Ready Plans.

3. Phase Pull Schedule Format

a. Activities shall be organized in a logical manner to allow for grouping of like characteristics, including, but not limited to such items as: phase, work shift, project area, activity work stream. The use of swim lanes on the Phase Pull plan may also be included to designate different work areas, phases, or work streams.

b. An appropriate number of interim milestones will be used to help develop the Phase Pull schedule activities. Milestones from the Preparatory meeting shall be incorporated.

c. Identify work days and non-work days on the Phase Pull production schedule.

d. Contractor shall work in conjunction with each subcontractor and supplier to ensure that all relevant submittal, procurement, delivery and installation dates for the various trades are accurately represented in the Phase Pull schedule.

e. Include activities related to critical project submittals and approvals.

F. 6-Week Make Ready Planning

1. The 6-Week Make Ready / Look Ahead Plan is a visual plan of activities that need to be accomplished over the upcoming six weeks. Standard 3-inch “sticky notes” placed on 6 weekly activity boards are used to establish the rolling 6-week Make Ready Plan.

a. Once an activity has entered into the Look Ahead Plan, it is the team’s task to make that activity ready for execution by the scheduled time, remove constraints, and execute the work within the expected duration.

b. Look ahead week activities should be planned as a whole identifying operations to be planned jointly by multiple trades with respect to hand-offs and work areas.

c. The Last Planners create the make-work ready plan that consists of weekly work plans with daily tags for each crew on site identifying what and where they will be working for the next 6 weeks and the size of the respective crew.

d. The quality of the work assignments/activities needs to be in greater detail and accuracy for the upcoming two weeks of work.

e. After the initial 6-Week Make Ready Plan has been developed, the next 6th week of work is planned as part of the weekly schedule meeting in the Big Room to provide a rolling 6-week plan.

f. Any new constraints that are identified during the 6-Week Make Ready planning are identified need to be captured in the constraint log with an assigned champion to remove them and required completion date.

g. Participants: All Team members involved in planning and execution of work during the next 6 weeks.

G. Weekly Work Plan

1. Weekly Work Planning is tactical team collaboration to plan each day's work during the next week including defining work areas and zones, conditions for handoff and acceptance between trades and disciplines, and crew sizing.

a. Weekly Work Plan updates shall occur as planned by the project team. These may coincide with the Weekly Owners Meeting where it makes sense to do so.

b. The Weekly Work Plan needs to be highly reliable to produce effective work flow and production on the project.

c. Specify tasks planned to be done next week and on which days.

d. The five minimum requirements to control quality of input into the Weekly Work Plan are:

1) What is the Task?
2) What will be done? (e.g. install wire way sections 1, 2, 3)
3) Where it will be done? (e.g. Column A/1, above AC Box)
4) When it will be done? (e.g. Tuesday and Wednesday)
5) Who will do it? (e.g., company, crew size)

e. Identify make ready actions by assessing their feasibility prior to making assignments in the weekly work plan so as to shield production workers from uncertainty.

f. Synchronize tasks made ready relative to the promises of the team members.

g. The conditions for hand off and acceptance are clearly communicated within the team and all constraints removed.

h. Optimization of the team capabilities to plan, synchronize, execute, learn and improve.

i. At the end of each week, assignments are reviewed for completeness in order to measure the reliability of the planning system. Analyzing reasons for plan failures and acting on these reasons is the basis of learning.

j. Participants: All Team members involved in planning and execution of work during the next 6 weeks.

2. Weekly Work Plan Meeting Typical Agenda.

a. Review constraint log and note any overdue constraints and impact (5 minutes)

b. Review 6-week look-ahead plan (15 minutes)

c. Review the new week – Note activities that are starting up in week 6.

d. Review weeks 2-5 only by new exceptions that pop up. (Team should have been looking at weeks 2-5 for the last 5 weeks.)

e. Review last week's performance (5 min.)

f. Last week's PPC: The number of activities completed since the last weekly meeting divided by the total number of planned activities which were supposed to occur.
g. Current week's PPC
h. The Percent Plan Complete Statistic shall be kept on a Project Log showing each weeks Percent Plan Complete Statistic for each week of the project schedule until completion.
i. Trend chart
j. Variance chart, and reasons for variance: charted in Pareto or pie charts to see trends and facilitate learning, knowing what needs to be fixed in order to improve next week's PPC
k. Finalize next week's Weekly Work Plan (30 minutes)
l. Plus/Delta (2 minutes)

H. Workable Backlog

1. Capacity limitations of a production unit may prevent the Last Planner from assigning all work shown in the first week of the Look-ahead that satisfy the definition, soundness, and sequence criteria.
2. There may be more work made ready than a production unit can reasonably be expected to complete in any week.
3. Overloading a production unit is held against the performance of the Last Planner as assigned work that remains incomplete counts against the plan reliability measure.
4. Ready work that cannot be assigned is recorded as Workable Backlog on the Weekly Work Plan.
5. Should a production unit for any reason not be able to complete an assignment on their Weekly Work Plan, or should they complete assignments sooner than expected, the Workable Backlog will provide them with other work so they need not be idle or wind up doing out-of-sequence work
6. Items in workable backlog must meet the same quality criteria as do priority assignments for the week.

I. Daily Work Planning Huddles

1. Daily Huddles are meetings where team members quickly give the status of the previous shift's accomplishments and failures, plus the current shift's plan of work for that day.
2. Daily Huddle discussions must be directly connected to the team's Weekly Work Plan.
3. Transparency and reliable commitments are measured in the Daily Huddles for the Last Planners themselves to see and interact with directly.
4. This is the rallying point for "our plan," which has "my input" accurately reflected. This is the heart of LPS, it is of utmost importance for the team to establish and drive healthy Daily Huddle discipline.

1.8 DELIVERABLES

A. Schedule Deliverables:

1. Master Milestone Schedule / Baseline schedule – Due Prior to NTP
2. Initial Phase Pull Plan – Due 15 days after NTP
3. Updated 6-Week Make Work Ready Look Ahead Schedule – Due every week

4. Weekly Metric Report (Percent Plan Complete for week, variance analysis for week's missed commitments, Current Constraint Log)
   a. If weekly PPC is less than 70%, specify what specific efforts the Contractor will undertake to improve its weekly work plan reliability.
5. Monthly Master Schedule Updates – Due Every Month
   a. The updated Contract Construction Schedule shall accurately represent the as-built condition of all completed and in-progress work activities as of the schedule data date.
   b. The level of detail shall be sufficient to describe and forecast the scheduled completion dates for the phase milestones used in the Phase Pull Planning.
   c. Planned percent complete (PPC) for the month and cumulative to date for the project on a weekly basis displayed in a graphical format.
   d. If the average weekly PPC is less than 70%, specify what specific efforts the Contractor will undertake to improve its weekly work plan reliability.
   e. Variance analysis for missed commitments with bar chart, Pareto chart or pie chart that visually shows trends for the month and trends for the project-to-date. Discuss proposed countermeasures to address root causes of most frequent causes of variance that will be implemented during the next month. These may include actions required by the Trustees.
   f. The current status of the phase milestones as established in the Milestone Plan.
   g. The reason phase milestones may not have been accomplished and their delaying factors.
   h. What mitigation efforts the Contractor or Trustees will undertake to complete the phase milestones without adversely impacting the overall project milestones leading to successful completion of the project by the finish milestone.
   i. Any changes made to the sequencing, durations, working time, etc. made to accomplish the phase milestones.
   j. Current Constraint Log with all outstanding items that have the potential to prevent a task or assignment from starting, advancing or completing as planned. This should include a constraint removal need by date to avoid adversely impacting the schedule and the name of the assigned individual to champion the efforts to remove the constraint.
   k. Current and anticipated delays not resolved by approved change order, including:
      - Cause of the delay – Contractor or Trustees
      - Corrective action and schedule adjustments to correct the delay
      - Known or potential impact of the delay on other activities, milestones, and Project completion date
   l. Pending items and status thereof including but not limited to:
      - Pending change orders
      - Time extension requests
      - Other items
   m. Contract completion date status:
      - If ahead of Construction Schedule, the number of Days ahead
      - If behind Construction Schedule, the number of Days behind.

B. All schedule submittals including the updated progress schedules will be reviewed jointly by the Trustees, the Architect, and the Contractor. Review of the Contractor's schedules shall not constitute approval or acceptance of the Contractor's construction means, methods, or sequencing, or a positive determination by the Trustees and/or the Architect of the Contractor's ability to complete the Work in a timely manner.

1.9 RESPONSIBILITY FOR COMPLETION

A. Should any monthly or weekly update of the Contract Construction Schedule indicate that the Contract Completion Date has extended, Contractor shall work with the team and submit a written action plan to meet the Contract Completion Date. Contractor and the Trustees shall initiate corrective actions, as approved by the CSU Project Manager, at no additional cost, unless agreed otherwise for shared delays. These actions shall include, but not be limited to, one or more of the following:

1. Identify and remove constraints and barriers to the project production work flow.
2. Identify root causes for missed commitments and develop and implement countermeasures to address these.
3. In conjunction with the Last Planners, re-sequence activities in order to improve work flow production and subsequent completion of these activities.
4. In conjunction with the Last Planners, increase construction manpower in certain or all trades in order to bring the completion date into compliance with Contract requirements.
5. In conjunction with the Last Planners, increase the number of labor shifts, working hours per shift, or working days per week as required to bring the completion date into compliance with Contract requirements.
6. Arrange and pay for acceleration of fabrication schedules for long-lead material items.
7. Arrange and pay for alternate shipping or delivery methods in order to expedite material procurement.
8. Arrange and pay for acceleration of design / architectural responses, changes, and /or resolutions.

B. Comments provided by the CSU Project Manager concerning the Initial Construction Schedule, Contract Construction Schedule, or any schedule update shall not relieve Contractor from the responsibility for compliance with the entire requirements of the Contract Documents.

END OF SECTION 01 32 10

DRAFTING THE SUPPLEMENTARY GENERAL CONDITIONS.
If incorporating the previous Specification Section 01 32 10 – COLLABORATIVE CONSTRUCTION PLANNING PROCESS into your Project Documents, then use the following Supplementary General Conditions in lieu of the Contract General Conditions for the articles Contract Time and Schedule. In the chart below, the first column lists the Contract General Conditions for each Contract Type (delivery method), and in the columns to the right are the referenced articles/topics for which the section names and numbers are provided for each Contract Type. Using Contract Time for Design-Build delivery method in the SGC below, section b, Starting and Completion Date, there is a reference to a section named "Guarantee", so Campus would replace Article xx.xx with Article 39.06.

Also, it is required to select the appropriate contract entity whether Contractor, Construction Manager (CM), or Design-Builder and use consistently throughout. All areas highlighted in yellow need review and appropriate selection of the relevant contract section numbering or contract entity.

Contract Type	Contract Time Section No.	Schedule Section No.	Change Orders Section No.	Delay in Completion LDs Section No.	Guarantee Section No.
CM at Risk	4.16	4.17	6.01	7.02	8.11
Design-Bid-Build Major	4.15	4.16	6.01	7.02	8.06
Design-Bid-Build Minor	4.14	4.15	6.01	7.02	8.05
Design-Bid-Build Auxiliary	4.15	4.16	6.01	7.02	8.06
Collaborative Design-Build	36.15	36.16	38.01	39.02	40.06
Design-Build	35.15	35.16	37.01	38.02	39.06

Supplementary General Conditions to the Contract General Conditions

Definitions—Reference the Definitions found in the Contract General Conditions for the delivery method chosen, and, if not in the Contract General Conditions, check Division One Section 01 32 10, Collaborative Construction Planning Process, section 1.3—Definitions.

Article xx.xx, Contract Time, delete and replace with the following:

a. Time of the Essence.
All time limits specified in this Contract are the essence of the Contract.

b. Starting and Completion Date.
The Trustees shall designate in the Notice to Proceed the starting date of the Contract on which Contractor/CM/Design-Builder *(select appropriate party to be used throughout based on contract type)* shall immediately begin and thereafter diligently prosecute the Work to completion. Contractor/CM/Design-Builder agrees to complete the Work on the date specified for completion of Contractor/CM/Design-Builder's performance in the Contract unless such time is adjusted, in writing, by change order by the Trustees. Contractor/CM/Design-Builder may complete the Work before the completion date if it will not interfere with the Trustees or other contractors engaged in related or adjacent Work. The Work shall be regarded as completed on the acceptance date noted on the Trustees' Notice of Completion. This date shall be used as the date the guarantee period begins as defined in Article xx.xx, Guarantee.

c. Adjustment of Contract Time Due to Acts of God, etc.
Contractor/CM/Design-Builder shall not be assessed with liquidated damages, nor the cost of engineering and inspection, during any delay in the completion of the Project caused by acts of God, the public enemy, fire, flood, epidemic, quarantine restriction, strike, freight embargo, discovery of archaeological or paleontological artifacts, and unusual action of the elements; provided that Contractor/CM/Design-Builder shall notify the Trustees in writing of the causes of the delays within 24 hours from the beginning of any such delay. The Trustees shall determine the facts with regard to the delay and the reasonable period of time by which the date of completion should be extended by reason thereof, if any. The Trustees' findings thereon

shall be final and conclusive. There shall be no compensation to Contractor/CM/Design-Builder for costs associated with this kind of delay.

d. The term "unusual action of the elements" is limited to extraordinary, adverse weather conditions and conditions immediately resulting therefrom which cause a cessation in the progress of the Work which will delay the time of completion of the Contract. Contractor/CM/Design-Builder shall have no right to an adjustment in the time of completion due to weather conditions or industrial conditions which are normal for the locality of the site. The time for completion of the Contract has been calculated with consideration given to the average climatic range and usual industrial conditions prevailing in the locality of the site.

e. Adjustment of Contract Time Due to Acts of the Trustees.
If Contractor/CM/Design-Builder is delayed in completing the Contract by reason of any act or omission of the Trustees not provided by the Contract, or by reason of changes made pursuant to Article xx.xx, Change Orders, without reaching agreement as to any time adjustments, the time for completion of the Contract may be extended for a period commensurate with the delay. Contractor/CM/Design-Builder shall notify the Trustees in writing of the causes of the delay within seven (7) Days from the beginning of the delay.

f. Contractor/CM/Design-Builder's Duty to Fully Prosecute Work.
No extension of time will be granted for any of the causes for which extensions may be granted unless Contractor/CM/Design-Builder demonstrates to the satisfaction of the Trustees that Contractor/CM/Design-Builder has made every reasonable effort to fully prosecute the Work and complete the Work within the Contract Time. The causes of delay shall be subject to the same determinations as stated in Article xx.xx.c, Adjustment of Contract Time Due to Acts of God, etc., above. Contractor/CM/Design-Builder shall refer to Article xx.xx, Schedule.

g. Trustees' Adjustment of Contract Time.
Even though Contractor/CM/Design-Builder has no right to an extension of time for completion, the Trustees may extend the time at the request of Contractor/CM/Design-Builder, if they determine it to be in the best interest of the State. If the time is extended, the Trustees may, in lieu of assessing liquidated damages, charge Contractor/CM/Design-Builder, its successors, heirs, assigns, or sureties, and deduct from the final payment for the Work all or any part, as they may deem proper, the value of the lost use of the completed Project, and of the actual cost to the Trustees of engineering, inspection, superintendence, and other overhead expenses which are directly chargeable to the Contract, and which accrue during the period of such extension. Such costs will not exceed liquidated damages.

h. Adjustment of Contract Time Due to Reasons beyond Trustees' Control.
Should the Trustees be prevented or enjoined from proceeding with Work either before or after the start of construction by reason of any litigation or other reason beyond their control, Contractor/CM/Design-Builder shall not be entitled to make or assert any claim for damage by reason for said delay; but time for completion of the Work will be extended to such reasonable time as the Trustees may determine will compensate Contractor/CM/Design-Builder for time lost by such delay. Any such determinations will be set forth in writing.

i. Liquidated Damages.
Attention is directed to Article xx.xx, "Delay in Completion--Liquidated Damages."

Article xx.xx, Schedule, delete and replace with the following:

a. Time is of the essence of this Contract, including the time of beginning, the rate of progress, and the time of completion of the Work. The Work shall be prosecuted at such time, in such manner, and on such part or parts of the Project as may be required to complete the Project as contemplated in the Contract Documents and Contractor/CM/Design-Builder's Construction Schedule.

b. Contractor/CM/Design-Builder shall prepare and submit to the Trustees' Construction Administrator the Contractor/CM/Design-Builder's initial construction schedule prior to Notice to Proceed. The Contractor/CM/Design-Builder's initial Construction Schedule shall comprise a high-level milestone critical path method. Contractor/CM/Design-Builder's initial Construction Schedule shall show the dates on which each major part or division of the Work is expected to be started and completed. The initial Construction Schedule shall also show all major dates for submission and approval of submittals required by the Contract as well as required delivery dates for major pieces of long-lead material or equipment. Contractor/CM/Design-Builder shall also submit a separate listing of all submittals required under the Contract and noting the anticipated date that each submittal will be submitted. Contractor/CM/Design-Builder shall submit a projected monthly cash flow schedule with the initial Construction Schedule and shall revise the cash flow schedule with each Construction Schedule revision. The cash flow schedule is Contractor/CM/Design-Builder's estimate of the dollar value of Contract Work completed and billable each month of the Project. Contractor/CM/Design-Builder's initial Construction Schedule shall begin with the effective date of the Notice to Proceed and conclude with the date of acceptance.

c. Contractor/CM/Design-Builder shall identify the number of work days that reflects anticipated rain delay during the performance of the Contract. The duration shall reflect the average climatic range prevailing in the locality of the site. Weather data shall be based on information provided by the National Oceanic and Atmospheric Administration (NOAA).

d. Contractor/CM/Design-Builder may submit an initial Construction Schedule that shows the Work completed in less time than the specified Contract Time. However, the acceptance of such a Construction Schedule will not change the Contract Time. The Contract Time shall control in any determination of liquidated damages or extension of the Contract Time. Buffer, or schedule contingency, is the unused time within the Construction Schedule and the difference in time between the Project's early completion date and the required Contract completion date. This buffer is not for the exclusive use of either the Trustees or the Contractor/CM/Design-Builder, but is jointly owned by both and is a resource available to and shared by both parties as needed to meet Contract milestones and the Contract completion date.

e. Comments made by the Trustees on Contractor/CM/Design-Builder's initial Construction Schedule during review will not relieve Contractor/CM/Design-Builder from compliance with the requirements of the Contract Documents. The review is only for general conformance with the scheduling requirements of the Contract Documents. Upon the Trustees' request, Contractor/CM/Design-Builder shall participate in the review of Contractor/CM/Design-Builder's initial Construction Schedule submissions (including the original submittal, all update submittals, and any re-submittals). The Trustees may request the participation of subcontractors in these reviews, as determined necessary by the Trustees. All revisions shall be resubmitted within fifteen (15) Days after the Trustees' review.

f. The submittal of a fully revised and acceptable Contractor/CM/Design-Builder's initial Construction Schedule shall be a condition precedent to the processing of the second monthly payment application, unless the Trustees grant a time extension due to unusual circumstances.

g. Contractor/CM/Design-Builder shall submit the required monthly Master Schedule Updates to the Construction Administrator with a copy to the Project Manager/Construction Inspector five (5) Days prior to the submittal of Contractor/CM/Design-Builder's monthly payment request. The submittal of the monthly Progress Schedule that satisfies the requirements of this Article, accurately reflects the status of the Work, revises the cash flow schedule, and incorporates all changes into the Construction Schedule, shall be a condition precedent to the processing of the monthly payment application. Progress Schedules shall also be submitted at such other times as the Trustees may direct. If Contractor/CM/Design-Builder fails to comply or is late in compliance with this requirement, and the Trustees find it to be in their best interest to process the monthly payment, an amount not exceeding $10,000 shall be retained from each monthly progress payment until compliance is achieved.

h. Adjustment of Contract Times for Completion.
In addition to the provisions in the Contract General Conditions, the Contract Time for completion of the Work will be adjusted in accordance with these procedures.

Whenever Contractor/CM/Design-Builder submits a request for an adjustment of the Contract Time for completion for Trustees-requested changes, differing site conditions, weather impacts, or delays or alleged delays in removal of Trustees' constraints, Contractor/CM/Design-Builder shall also submit an evaluation of the Constraints Log identifying items that have impacted the schedule that are not within the responsibility of the Contractor/CM/Design-Builder. Contractor/CM/Design-Builder shall also submit information regarding Planned Percent Complete for the Project to date and for time periods of interest, with a detailed variance analysis for missed commitments in the weekly work planning, clearly indicating items beyond the responsibility of the Contractor/CM/Design-Builder that have adversely impacted the schedule. The Trustees will not grant time extensions unless substantiated by the analysis.

Contractor/CM/Design-Builder shall determine the impact based on the date or dates when the change or changes were issued, or the date or dates when the alleged delay or delays began.

If the Construction Administrator finds, after review of the analysis, that Contractor/CM/Design-Builder is entitled to any extension of time for completion, the Contract Time for completion will be adjusted accordingly by the Construction Administrator, and Contractor/CM/Design-Builder shall then revise the Milestone Plan accordingly.

No time extensions shall be granted nor indirect costs paid, unless Contractor/CM/Design-Builder can clearly demonstrate the delay on the basis of the Progress Schedule current as of the month the change is issued or the delay occurred, and which delay cannot be mitigated, offset, or eliminated through revising the intended sequence of Work or other means. Contractor/CM/Design-Builder shall include field instructions and change orders in the revised Construction Schedule. Failure to include field instructions or change orders shall waive rights to a Contract time extension or delay damages.

i. As a condition precedent to the release of retained funds, Contractor/CM/Design-Builder shall, after completion of the Work has been achieved, submit a final Contractor/CM/Design-Builder's As-Built Milestone Baseline Plan with Percent Planned Complete (PPC) graphically displayed for the duration of the project.

-End of Supplementary General Conditions-

Overview of the Last Planner® System

Collaborative Construction Progress Planning – Spec. 013200

David Umstot, PE

31 October 2018

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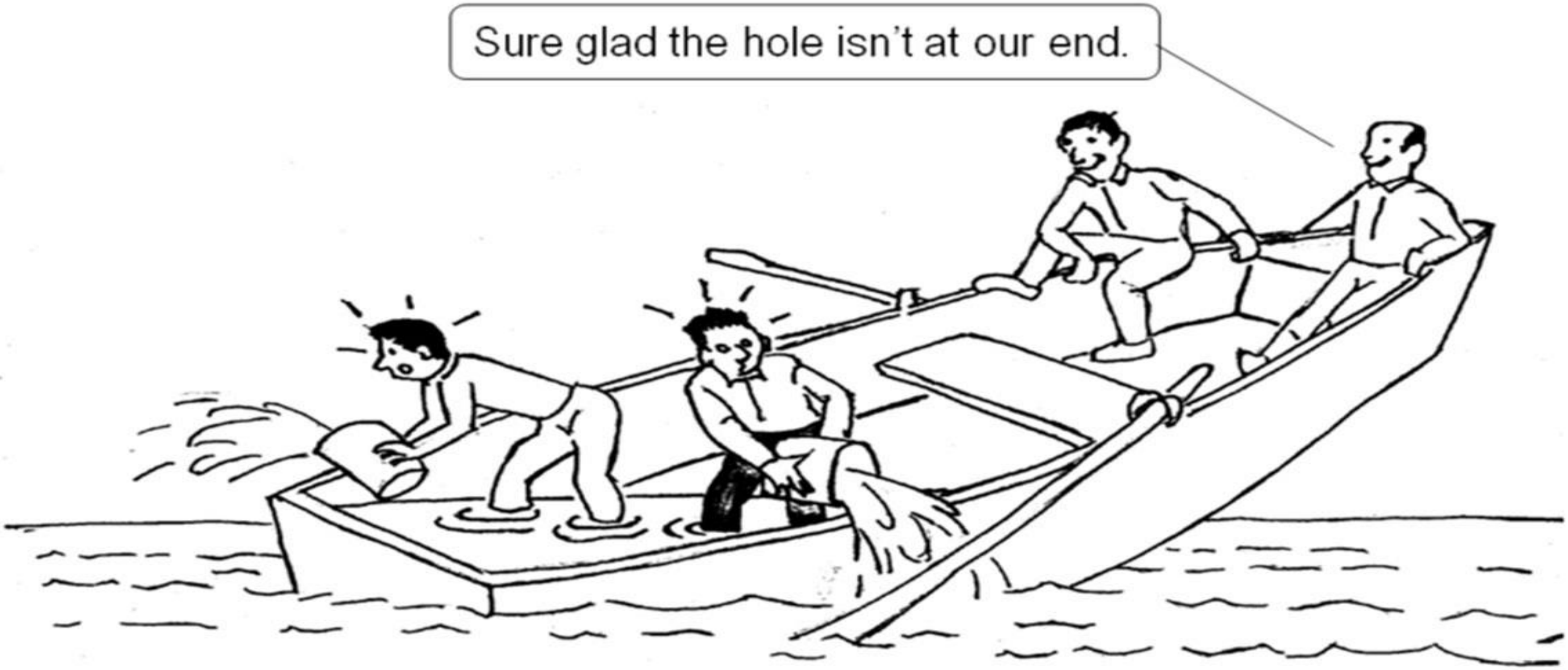
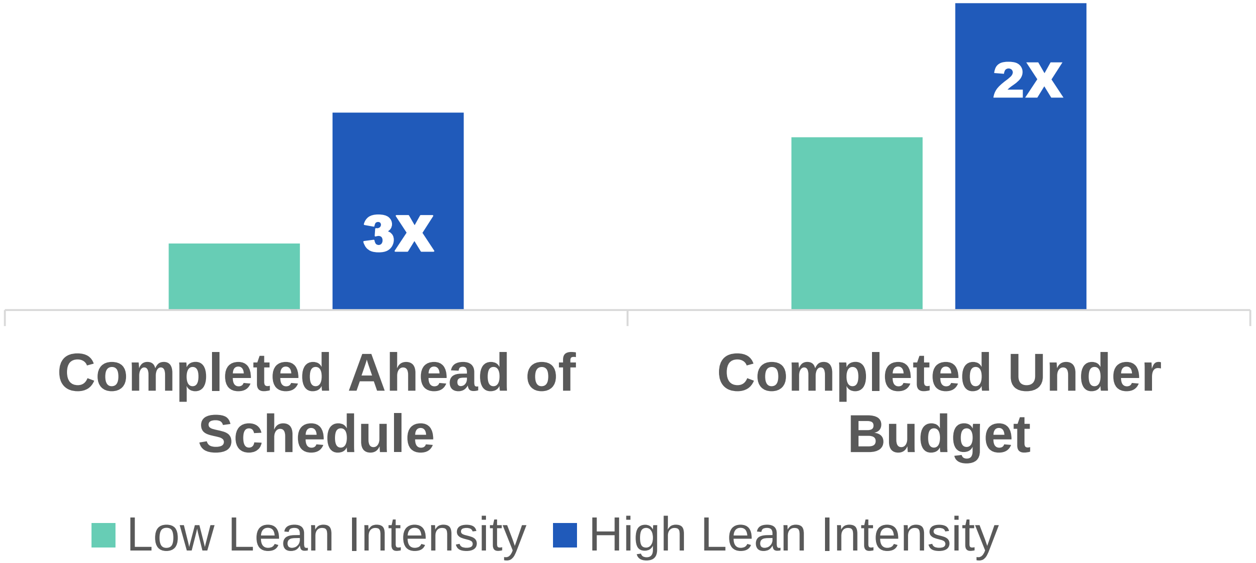
Sink or Swim Together?



Why use Last Planner®?

DODGE DATA & ANALYTICS



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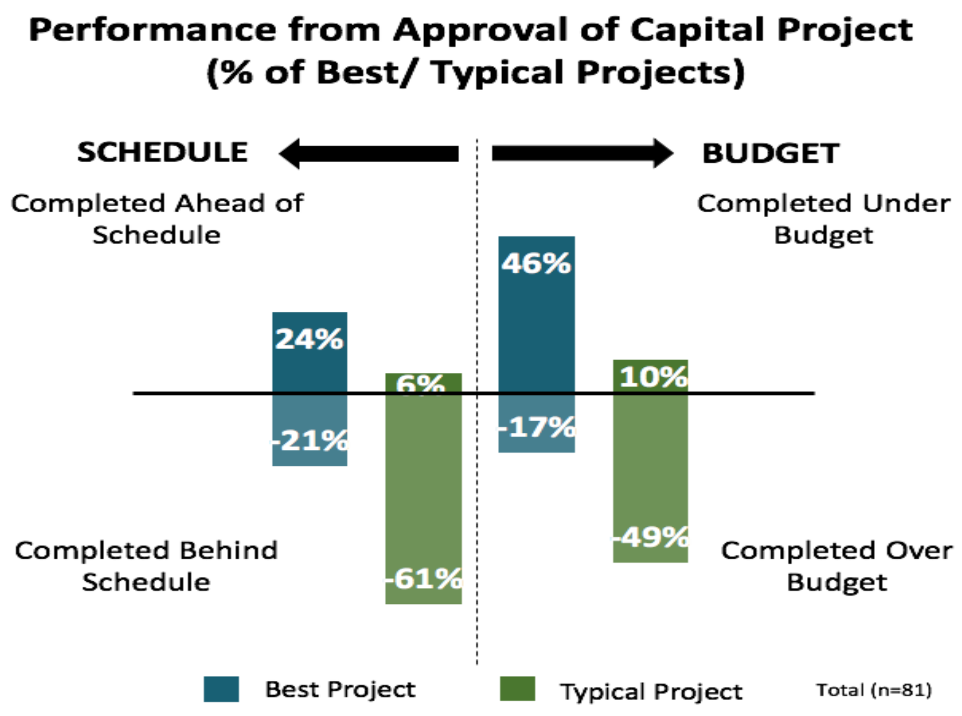

DODGE DATA & ANALYTICS

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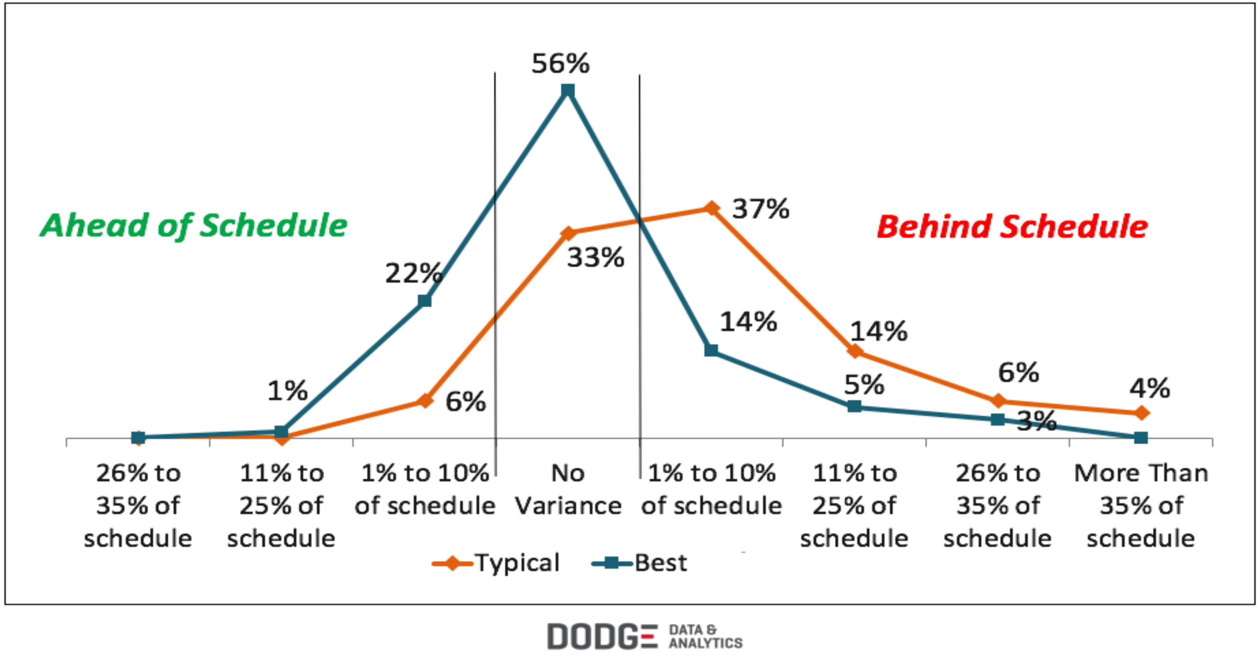
Schedule Performance

Variance of Final Schedule vs. Allocated Capital Schedule



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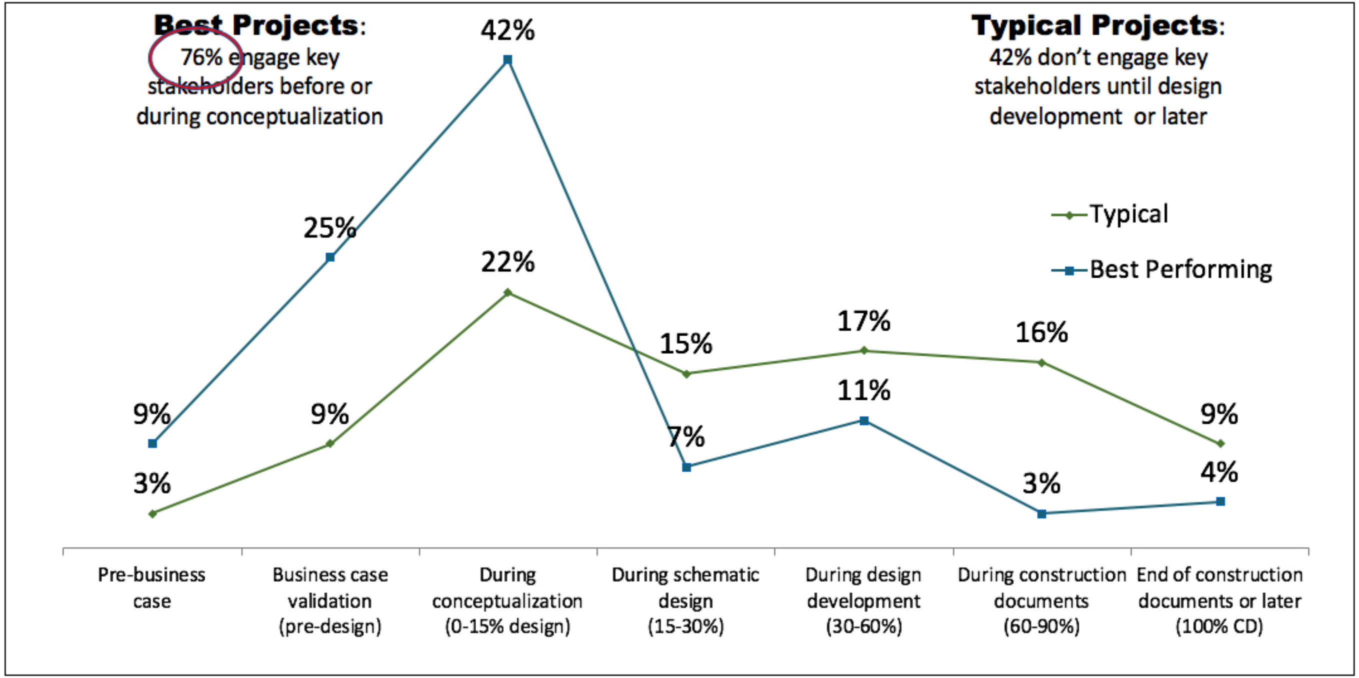
Timing of Key Stakeholder Engagement



DODGE DATA & ANALYTICS

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Workflow & Risk

- Workflow losses are real, lead to adversarial relations, and are difficult to quantify, so…
- Everyone protects themselves by adding contingency and/or holding back labor to keep utilization high.
- This further reduces workflow predictability and increases project risk.
- By their/our actions, we increase that risk and shift it along.

KEY CONCEPTS:

1. Traditional planning systems are unable to produce predictable workflow.
2. Workflow reliability directly affects system speed and cost.
3. All plans are forecasts, all forecasts are wrong, the longer the forecast the more wrong it is, the more detailed the more wrong it is.

Last Planner® System Benefits:

- Improves communication & reliability
- Fosters an enjoyable environment, trust & collaboration
- Promotes early stakeholder engagement
- Improves visibility of the project plan (transparency)
- Creates team buy in
- Rapid learning through metrics, revealing areas for improvement
- Improves planning in both design & construction phases

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Last Planner® System Overview

Hand-offs Matter

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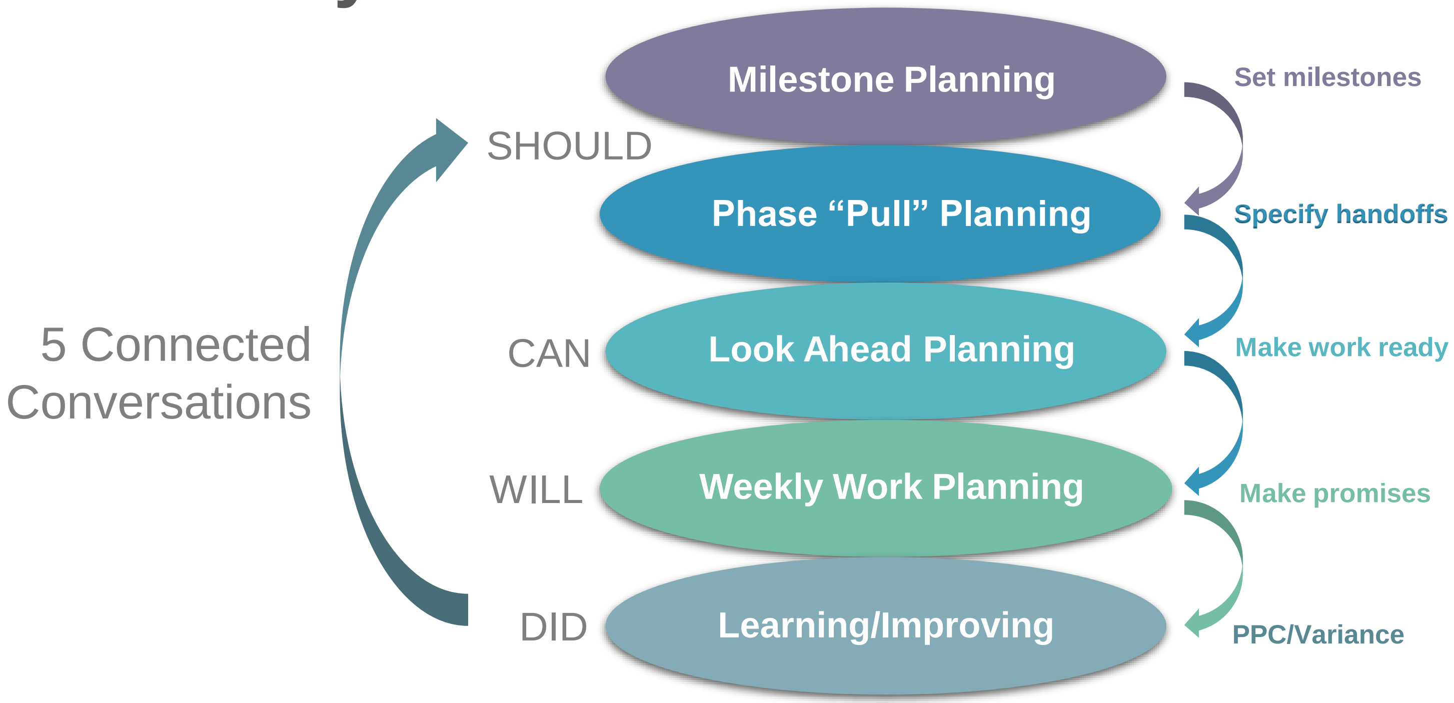
Last Planner® System



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Last Planner® System

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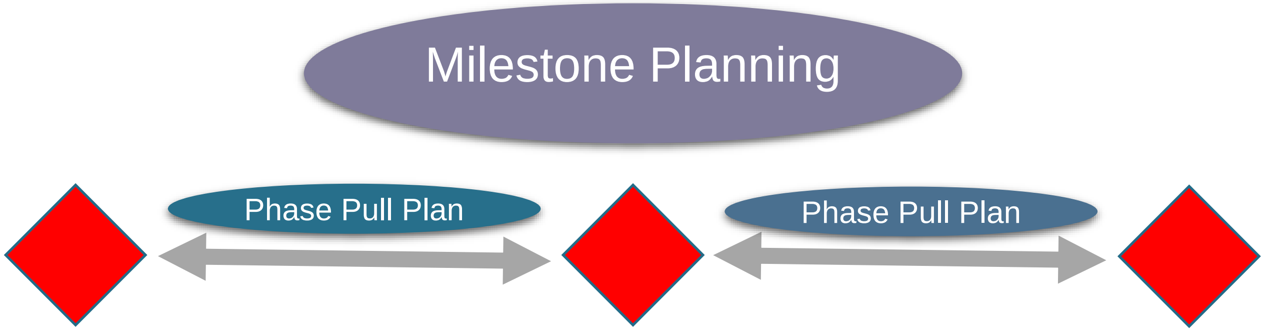
Who is the Last Planner?

- Person closest to work, with authority to make decisions creates schedule
- A Last Planner can make the reliable commitment to complete the work



- Define the overall road map and gain alignment
- Identify milestones important to client and stakeholders – especially immovable dates
- Informs the Phase Pull Planning

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Milestone Planning

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Phase Pull Planning

Phase Pull Planning

- Phase of the work (8-12 wks)
- Informed by the **Milestone Plan**
- Work out the structure and durations
- After – add dates and transfer to the **Look Ahead Plan**

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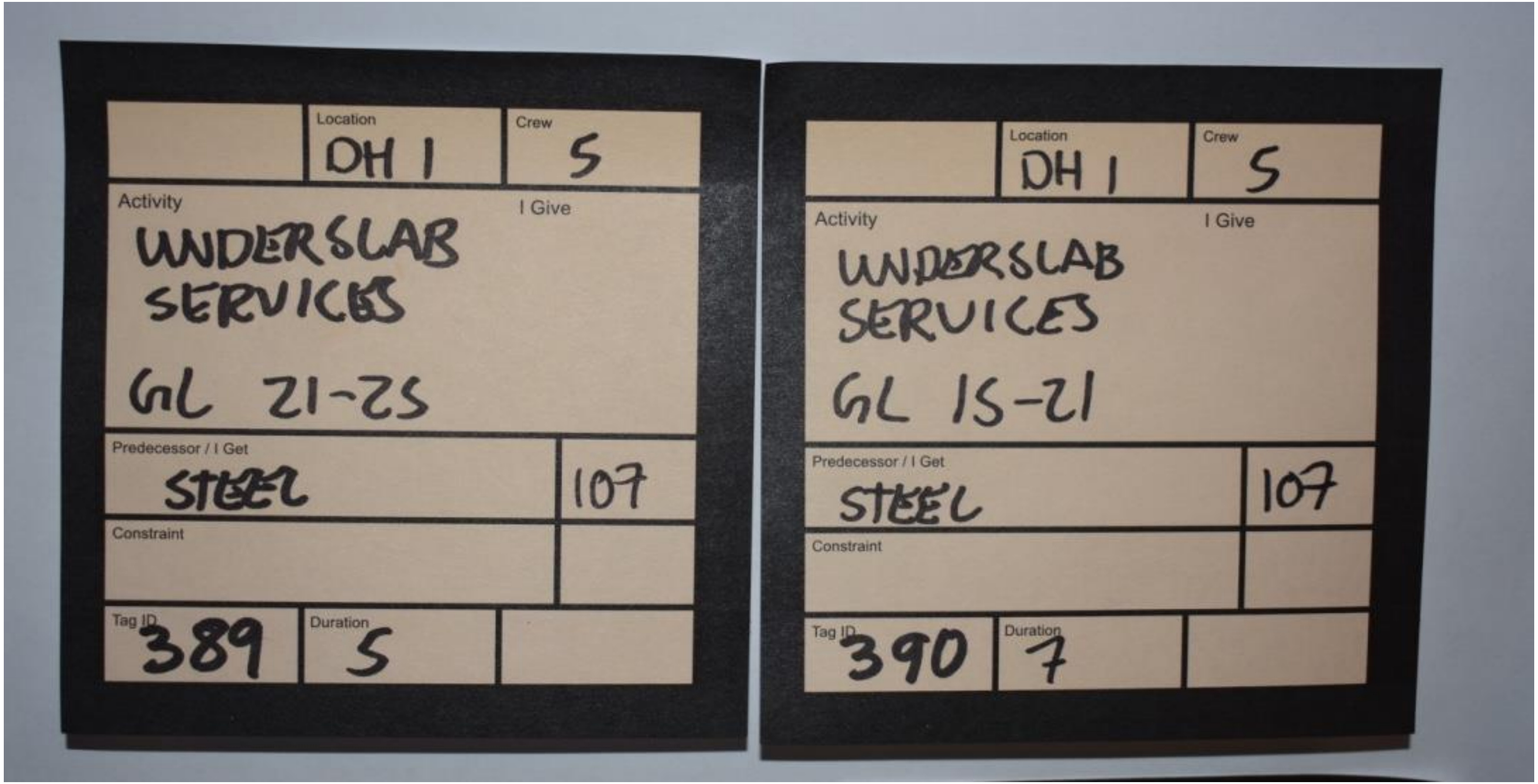

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Pull Example

Promise

Promise

Promise

Mike G	4 Crew	3days
Vaults Set Zone 1		
Vaults on Site		

Dave S	3Crew	3 days
Backfill Vaults Zone 1		
Vaults Set		

Mike G	4 Crew	5 days
Underground Piping Zone 1		
Vaults Backfilled		

Dave S	3 Crew	3days
Backfill UG Piping Zone 1		
UG Piping		

Request

Request

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Start at End

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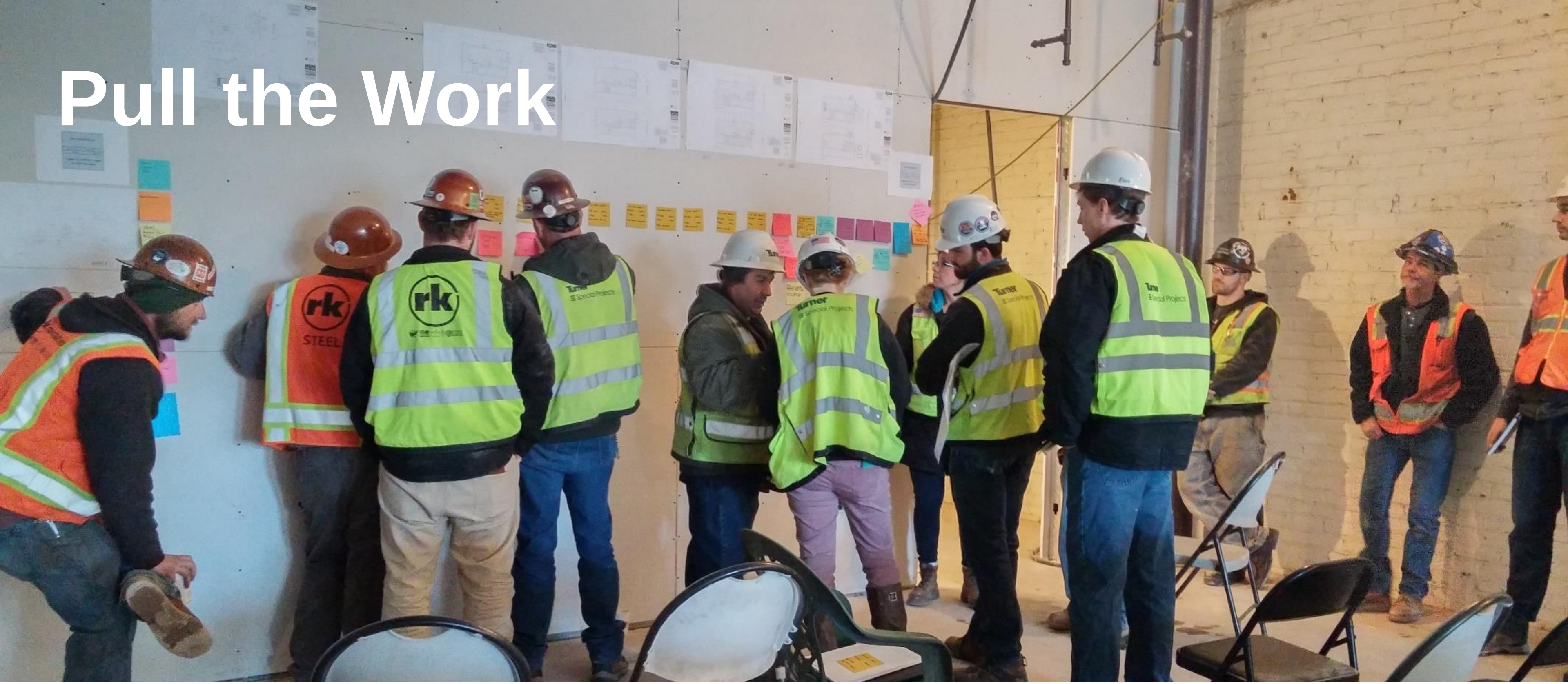
Pull the Work



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Arrive at the Start

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Look Ahead Planning

Constraint Log

Update Date:

Project:
Project No.:
Responsible Person:

Constraint Number	Activity Number	Constraint Description	RFI No.	Responsible Person	Responsible Company	Date Identified	Date Need Resolution	Date Resolution Promised	Actual Date Resolved

Constraint Description

Responsible Person & CO

Date Needed

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Look Ahead Planning

Boards in Weekly Meeting Space

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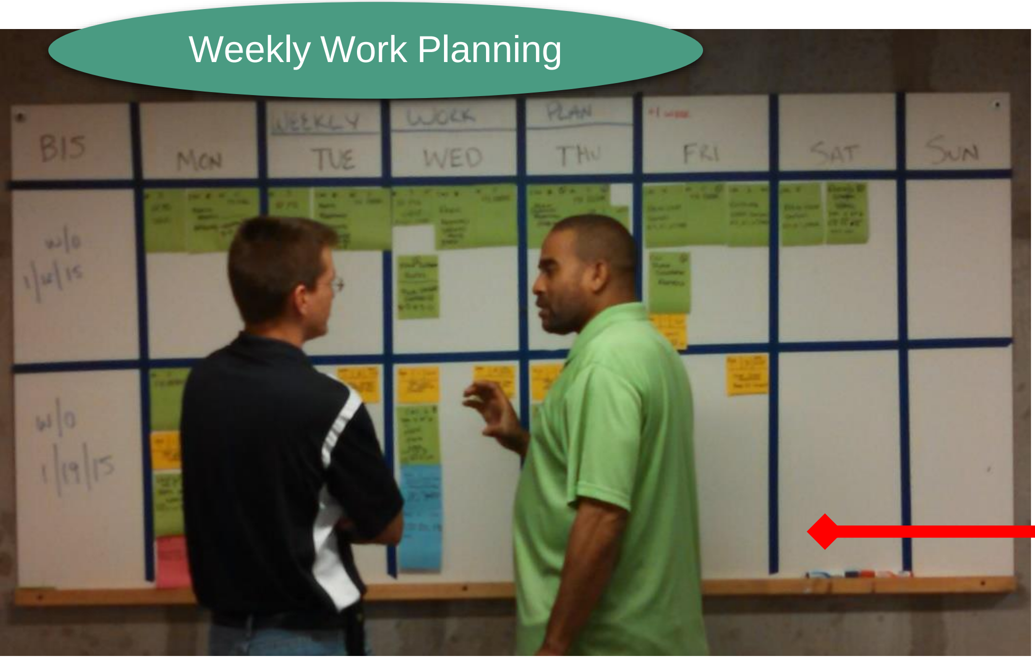
EAST
WEST

2/28 S
2/29 M
3/1 T
3/2 W
3/3 TH
3/4 F
3/5 S
Q1 Basement
Q1 Ground
Q1 Level 1
Q1 Level 2
Q1 Level 3
Q1 Level 4
Q1 Level 5
Q1 Roof
Q2 Ground
Q2 Level 1
Q2 Level 2
Q2 Level 3
Q2 Level 4
Q2 Level 5
Q2 Roof
Footings Area A
Footings Area B



Weekly Work Plan Informs the Daily Huddle

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Daily Huddle

1. What did I complete?
2. What will I complete?
3. What needs to be re-planned?

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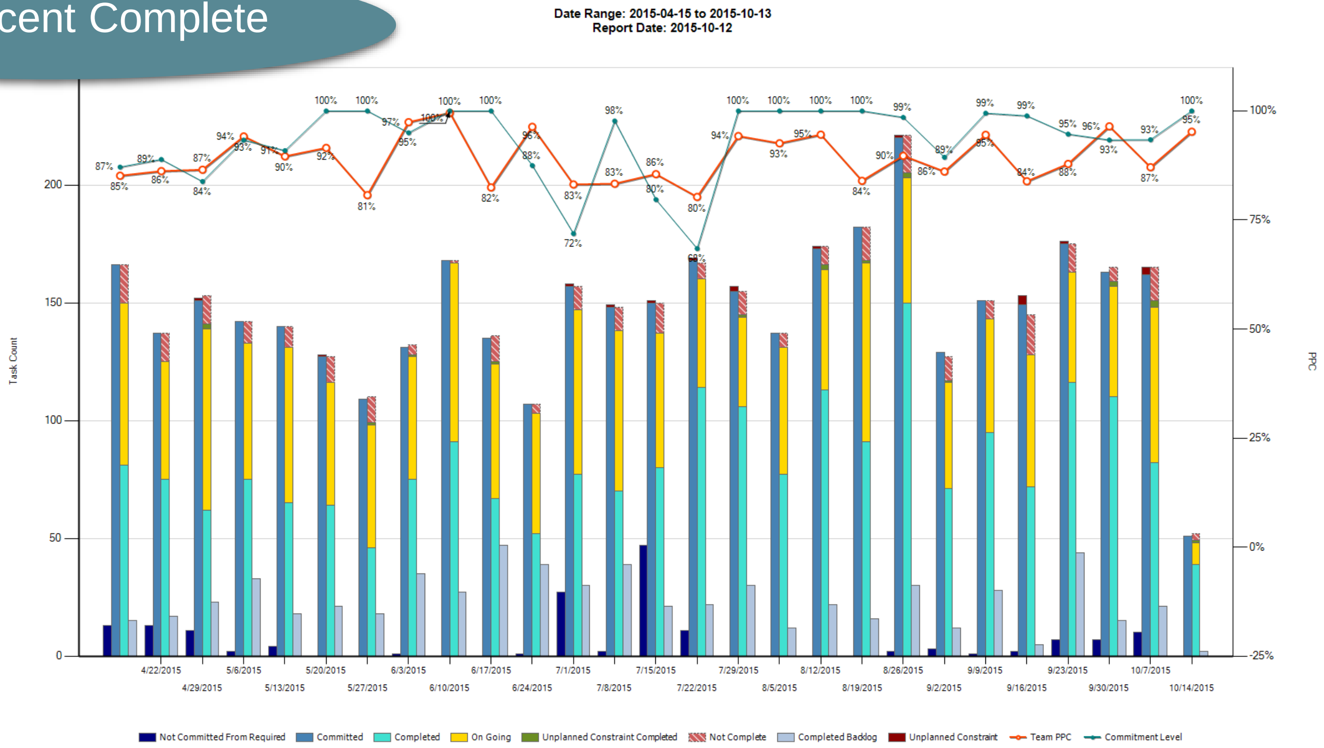
Plan Percent Complete



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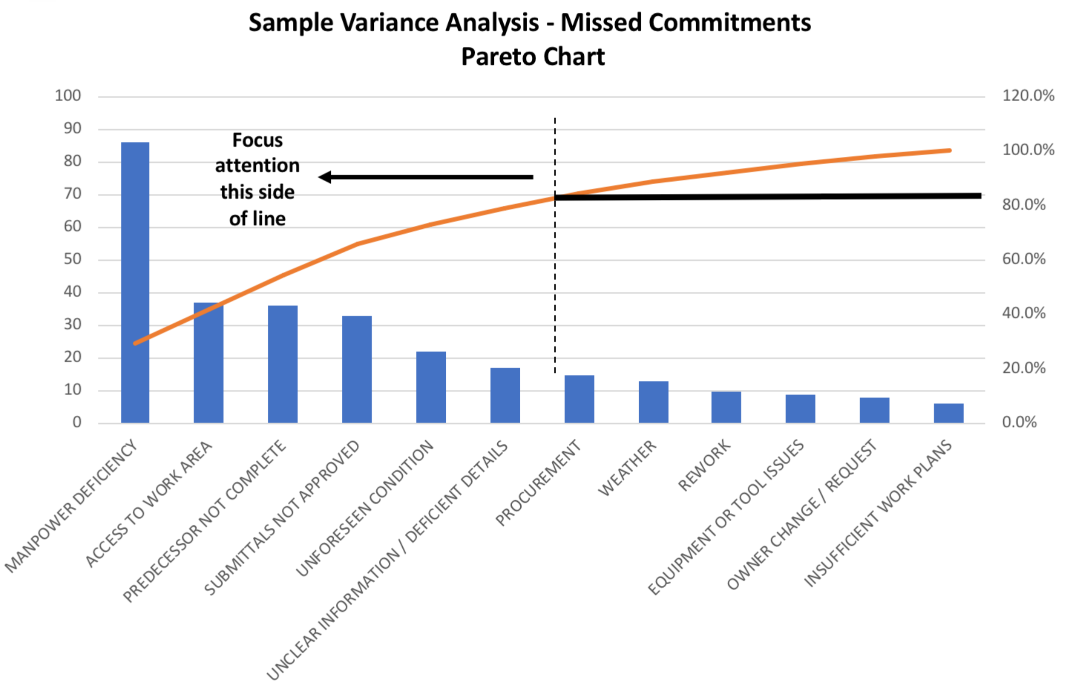
Reasons for Variance



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Future Training Opportunities

- CSU Dominguez Hills - Friday, November 16, 2018
- Cal Poly San Luis Obispo – Tuesday, December 11, 2018
- CSU East Bay – Wednesday, February 27, 2019
- CSU Sacramento – Friday, March 15, 2019

Questions?

David Umstot, PE, CEM

Umstot Project and Facilities Solutions, LLC

david.umstot@umstotsolutions.com

619.201.8483 (O)

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SECTION 01 32 26 - CONSTRUCTION PROGRESS REPORTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Construction progress reports.

1.3 RELATED SECTIONS

A. Section 01 31 19 - Project Meetings: Review of construction progress and submittals status at Project meetings.

B. Section 01 77 00 - Contract Closeout Procedures: Notice by Contractor of progress of the Work sufficient for Contract Completion review and Acceptance by University.

1.4 CONSTRUCTION PROGRESS REPORTS

A. Daily Log: Contractor shall maintain a written daily log at the job site with the following information as a minimum:

1. Date.

2. Weather conditions.

3. Subcontractors and trades performing Work under the Agreement on the Site, and number of workers each and number of hours worked by each worker.

4. Others on the Site performing work for University under separate contracts.

5. List of visitors to site, giving name, company or agency affiliation and telephone number.

6. Descriptions of situations and circumstances which could delay normal progress of Work or which could be basis of claim for change in Contract Time or Contract Sum.

7. Changes to Work and who authorized changes.

8. Contractor shall describe all damage, deterioration, soiling, staining or similar conditions in accordance with 01-45-00 QUALITY CONTROL. Contractor may note responsible party for the damage.

8. Comments, as Contractor determines are appropriate for Project record.

B. Submission of Logs: Submit one copy of daily logs to University's Representative and Architect at weekly intervals, for review at Construction Progress Meetings.

PART 2 - PRODUCTS

Not applicable to this Section.

PART 3 - EXECUTION

Not applicable to this Section.

END OF SECTION

SECTION 01 33 00 - SUBMITTAL PROCEDURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Administrative requirements

B. Construction Progress Schedule Submittal

C. Contractor's review of submittals.

D. Architect's review of submittals.

E. Product data submittals.

F. Shop drawing submittals.

G. Sample submittals.

H. Manufacturer’s Instructions

I. Reports of results of tests and inspections.

J. Operations and Maintenance Data submittals

K. Certificates

1.3 RELATED SECTIONS

A. Section 01 31 13 – Project Coordination

B. Section 01 31 26 – Electronic Communications Protocol

C. Section 01 45 00 - Quality Control: Test and inspection reports.

D. Section 01 77 00 - Closeout Procedures: Submittals for occupancy, Acceptance and Final Payment.

E. Section 01 78 23 - Operation and Maintenance Data: Requirements for preparation and submission.

1.4 DEFINITIONS

A. Shop Drawings, Product Data and Samples: Instruments prepared and submitted by Contractor, for Contractor's benefit, to communicate to Architect the Contractor's understanding of the design intent, for review and comment by Architect on the conformance of the submitted information to the general intent of the design. Shop drawings, product data and samples are not Contract Documents. Drawings, diagrams, schedules and illustrations, with related notes, are specially prepared for the Work of the Contract, to illustrate a portion of the Work. Approval of the shop drawings does not relieve the contractor of their duty to perform or install any component of the project.

B. Product Data: Standard published information ("catalog cuts") and specially prepared data for the Work of the Contract, including standard illustrations, schedules, brochures, diagrams, performance charts, instructions and other information to illustrate a portion of the Work.

C. Samples: Physical examples that demonstrate the materials, finishes, features, workmanship and other characteristics of a portion of the Work. Accepted samples shall serve as quality basis for evaluating the Work.

D. Other Submittals: Technical data, test reports, calculations, surveys, certifications, special warranties and guarantees, operation and maintenance data, extra stock and other submitted information and products shall also not be considered Contract Documents but shall be information from Contractor to Architect to illustrate a portion of the Work for confirmation of understanding of design intent.

1.5 ADMINISTRATIVE REQUIREMENTS

A. Administrative Requirements for Submittals: Submittals shall be made in accordance with requirements specified herein and in other Divisions of the Specifications. See also the Contract General Conditions for additional requirements; especially those regarding requests for alternatives or equals and for substitutions.

   1. All required submittals, with the exception of O&M manuals, close-out submittals, and mock-ups required to be installed concurrent with specific construction activities, shall be submitted within 90 calendar days after Notice to Proceed.

B. Contractor Coordination of Submittals: Coordinate preparation and processing of submittals with performance of construction activities. Transmit each submittal sufficiently in advance of performance of related construction activities to avoid delay.

   1. Coordinate each submittal with fabrication, purchasing, testing, delivery, other submittals and related activities that require sequential activity.

   2. Coordinate transmittal of different types of submittals for related elements of the Work so processing will not be delayed by the need to review submittals concurrently for coordination.
      a. The Architect will return without action submittals requiring coordination with other submittals until related submittals are coordinated.

C. Submittal Log: Prior to proceeding with affected work, Contractor shall prepare and submit a Submittal Log, which lists submittal items per the product specifications for review and approval by University's Representative and Architect. Contractor shall allow seven (7) calendar days for Trustees review. Submittal Log shall identify all specified submittals to be made and shall serve as checklist for submittals.

1. Maintain accurate submittal log for duration of Contract. Indicate current status of all submittals at all times. Submit log at progress meeting and as otherwise requested by University Representative or Architect.

2. Format shall be suitable for Project and shall be subject to acceptance by University's Representative and the Architect. Comply with directions by University's Representative and the Architect for scope and format of Submittals List.

3. Submittals list shall include the following submittal types and headings:
SD = Shop Drawings are required
PD = Product Data required
SA = Samples required
CO = Color samples required
SS = Site Sample installations are required
LM = List of Materials
RD = Record Drawings required
CE = Certificates are required
PR = Manufacturer's instructions or specifications required
OM = Operation and Maintenance manuals are required
EQ = Maintenance materials/equipment are required
WA = Warranties and/or guarantees are required
LR = Laboratory Reports are required
FT = Factory Test reports are required
ST = Site Test reports required
RP = Submittal to the Architect for record purposes only and not for review or approval
O = Other submittal requirements as specified in Section

2. Sample Table:

Section	SD	PD	SA	CO	SS	LM	RD	CE	PR	OM	EQ	WA	LR	FT	ST	RP	O
05 12 00	x					x											
09 25 00		x			x	x		x					x		x		
10 81 00		x	x														

D. Transmission of Submittals: Submittals shall be processed electronically (with exceptions such as product and material samples or otherwise designated or approved by the University Representative). Transmit all submittals from Contractor to Architect via Electronic Project Management (EPM) system, unless otherwise directed, using a transmittal form for each one. Submittals received from sources other than the Contractor will be returned without action. Include all information specified below for identification of submittal and for monitoring of review process.

1. Architect will provide example Letter of Transmittal, if requested.

2. Submittals shall be concurrently made available via EPM to University's Representative for review.

E. Timing of Submittals: Make submittals sufficiently in advance of construction activities to allow shipping, handling and review by the Architect and Architect's consultants. Allow sufficient review time so that installation will not be delayed as a result of the time required to process submittals, including time for resubmittals.

1. The Architect will make desired corrections and consolidate relevant Trustees comments within fifteen (21) calendar days and return the submittal to the Contractor via EPM system. Submittals, which require coordination with other submittals, may require more than fifteen (21) calendar days review time. Submittals that require selection of colors will be reviewed. Color selection may not be provided until all submittals requiring color selection have been received and reviewed, and color selections have been approved by the Trustees.

2. Make corrections required by the Architect and submit via EPM system for final review and distribution.

3. If an intermediate submittal is necessary, process the same as the initial submittal.

4. No extension of Contract Time will be authorized because of failure to transmit submittals to the Architect sufficiently in advance of the Work to permit processing.

F. Submittals Identification:

1. Provide a space on all submittals electronically approximately four-inches by five-inches on the label or beside the title block on Shop Drawings to record the Contractor's review and approval markings and the action taken. Include the following information on the label for processing and recording action taken:

   a. Project name and Trustees project number
   b. Submission date
   c. Name and address of Architect
   d. Name and address of Contractor
   e. Name and address of subcontractor
   f. Name and address of supplier
   g. Name of manufacturer
   h. Number and title of appropriate Specification Section
   i. Drawing number and detail references, as appropriate.

2. Identify each element on submittal by reference to Drawing sheet number, detail, schedule, room number, assembly or equipment number, Specifications article and paragraph, and other pertinent information to clearly correlate submittal with Contract Drawings. On the submittal transmittal form or separate sheet record deviations from Contract Document requirements, including minor variations and limitations. Include Contractor's certification that information submitted complies with requirements of the Contract Document. The Architect’s review of such submittals or shop drawings or product data shall not relieve the Contractor of responsibility for deviations from the drawings or specifications.

3. Identify each submittal by Specification Section number followed by a number indicating sequential submittal for that Section. Resubmittals shall use same number as original submittal, followed by a letter indicating sequential resubmittal. For example:

09 26 13-01-01	First submittal for Section 09 26 13 - Gypsum Veneer Plastering.
09 26 13-02-01	Second submittal for Section 09 26 13 - Gypsum Veneer Plastering.
09 26 13-02-02	Resubmittal of second submittal for Section 09 26 13 - Gypsum Veneer Plastering.
09 26 13-02-03	Second resubmittal of second submittal for Section 09 26 13 - Gypsum Veneer Plastering.

4. Place a permanent label or title block on each submittal electronically for identification. Indicate the name of the entity that prepared each submittal on the label or title block.

G. Grouping of Submittals: Unless otherwise specifically permitted by the Architect, make all submittals in groups containing all associated items. The Architect may reject partial submittals as incomplete or hold them until related submittals are made.

H. Unsolicited Submittals: Unsolicited submittals may be returned without being reviewed.

I. Record Submittals: When record submittals are specified, submit in accordance with the Electronic Project Management System requirements. Record submittals will not be reviewed but will be retained for historical and maintenance purposes.

J. Revisions: Revisions to original submittal list and schedule will only be accepted by University Representative and Architect when revisions are required by circumstances not reasonably anticipated by Contractor during preparation of original schedule. Submit revisions not later than 20 calendar days following the date that the need for revision became necessary.

1.6 CONSTRUCTION PROGRESS SCHEDULE SUBMITTAL

A. Submit as specified in the Contract General Conditions under Schedule and Section 01 32 26 for Construction Progress Documentation.

1.7 CONTRACTOR'S REVIEW OF SUBMITTALS

A. Contractor's Review of Submittals: Prior to submission to Architect for review, Contractor shall review each submittal for completeness and conformance to specified requirements. Contractor shall stamp each submittal with a review action stamp and sign each copy of submittal. Submittals without stamp and signature will not be reviewed and will be returned. Electronic signatures are acceptable but will need to be authenticated during the submittal process. Contractor's submittal action stamp shall certify the following actions by Contractor:

1. Field measurements have been determined and verified.

2. Conformance with requirements of Contract Drawings and Specifications is confirmed.

3. Catalog numbers and similar data are correct.

4. Work being performed by various subcontractors and trades is coordinated.

5. Field construction criteria have been verified, including confirmation that information submitted has been coordinated with the work being performed by others for University and actual site conditions.

6. All deviations from requirements of Drawings and Specifications have been identified and noted.

7. Contractor shall certify that submittals have been reviewed and approved:

Stamp Submittals utilizing the following language:

> "The undersigned certifies this submittal has been reviewed and approved with respect to the means, methods, techniques, sequences, and procedures of construction, and safety precautions and programs incidental thereto; and also warrants that this submittal complies with the Contract Documents and comprises no variation thereto.
>
> Signature:______________________________Date:_______________
> Name Printed:_____________________________Title______________
> Contractor Name:____

8. Submittals not certified by being stamped and signed by Contractor will be returned without action, as will submittals which, in University Representative's or Architect's opinion, have not been adequately reviewed and coordinated by Contractor.

B. Changes in Work: Changes in the Work shall not be authorized by submittal review actions. No review action, implicit or explicit, shall be interpreted to authorize changes in the Work. Changes shall only be authorized by separate written direction from the University Representative, in accordance with the Contract General Conditions.

C. Allow sufficient review time so that installation will not be delayed as result of time required to process submittals, including time for resubmittals.

1. Coordinate each submittal with fabrication, purchasing, testing, delivery, other submittals, and related activities that require sequential activity.
2. Coordinate transmittal of different types of submittals for related elements of Work so processing will not be delayed by need to review submittals concurrently for coordination.
    a. University Representative and Architect reserve right to withhold action on submittal requiring coordination with other submittals until related submittals are received.
3. Allow additional time if processing must be delayed to permit coordination with subsequent submittals.
4. If intermediate submittal is necessary, process same as initial submittal.
5. Allow same time for reprocessing each submittal as allowed for processing original submittal.
6. No extension of Contract Time will be authorized because of failure to transmit submittals to University Representative sufficiently in advance of Work to permit processing.

D. Package each submittal appropriately for transmittal and handling. Transmit each submittal from Contractor to University Representative using Submittal Transmittal form attached at the end of this section.

1. Submittals received from sources other than Contractor will be returned without action.
2. Number each submittal and resubmittal as indicated in approved Submittal Schedule.
3. Submittals forwarded without a completed Submittal Transmittal form will be returned without review.
4. Submittals shall be submitted electronically unless they are related to materials and products.

1.8 REVIEW OF SUBMITTALS BY UNIVERSITY'S REPRESENTATIVE AND ARCHITECT

A. Review of Submittals by University's Representative and Architect: Submittals shall be a communication aid between Contractor and Architect by which interpretation of Contract Documents requirements may be confirmed in advance of construction.

1. Reviews by University's Representative, Architect and Architect's consultants shall be only for general conformance with the design concept of the Project and general compliance with the information given in the Drawings and Specifications.

2. The Architect’s review shall not be construed as an “approval,” or to relieve the Contractor(s) and material suppliers of responsibility for errors or omissions in the submitted documents.

3. Acceptance of a specific item does not include acceptance of the assembly of which the item is a component.

4. Except for submittals for record, information or similar purposes, where action and return is required or requested, the Architect will review each submittal, mark to indicate action taken, and return promptly via EPM system.

B. Review Action: Architect will stamp each submittal with a uniform, self-explanatory action stamp.

1. Stamp will be appropriately marked as follows to indicate the action taken:

a. Action 1 (no exception taken): Means fabrication, manufacture, or construction may proceed providing submittal complies with Contract Documents.

b. Action 2 (make corrections noted; no resubmission required): Means fabrication, manufacture, or construction may proceed providing submittal complies with Architect's notations and Contract Documents. (Note: If Contractor cannot comply with notations, make revisions and resubmit.)

c. Action 3 (make corrections noted; submit corrected copy): Means fabrication, manufacture, or construction may proceed; however, submittal did not fully demonstrate full extent of all conditions, details and coordination with other surrounding work and therefore requires additional information and rework as noted. Resubmit shop drawings for final Action 1 or 2. Should Contractor proceed with fabrication, manufacturing or construction, it shall do so at its own risk.

d. Action 4 (rejected, revise and resubmit): Means submittal does not comply with design intent of Contract Documents. Do not use submittals stamped Action 3. Make revisions and resubmit.

e. Action 5 (rejected, submit specified item): Means submittal varies from specified item or system specified in Contract Documents and is not acceptable for use on the project. Do not use submittals stamped Action 4. Make revisions and resubmit.

f. Action 6 (resubmit with related assembly items): Means submittal of related assembly item(s) are required in conjunction with the submittal for proper review.

g. Action 7 (rejected; incorrect transmittal): Means the Submittal Transmittal form specified for use on the Project was not included, incomplete, or incorrectly completed.

h. Action 8 (No Action): Means documents have not been reviewed by Architect and submittal is returned to Contractor for several possible reasons: submittal not requested, submittal not complete, submittal not coordinated, or submittal bears no resemblance to design intent.

i. Action 9 (submitted to consultant for review): This code is for the use of the Architect to indicate routing to various A/E consultants. Any submittals marked Action 6 by Architect will be returned to Contractor without review.

j. Record Submittals: Specifications require certain information and calculations be submitted for record purposes only. Such submittals will not be acted upon, stamped or returned to Contractor.

2. Do not permit submittals marked "Rejected, Revise and Resubmit" to be used at the Project site, or elsewhere where Work is in progress.

3. Note: Any work performed prior to receiving a fully approved submittal shall be done at the Contractor's risk and shall be subject to being replaced if Contract requirements are not met.

C. Contract Requirements:

1. Review actions by Architect and Architect's consultants shall not relieve the Contractor from compliance with requirements of the Contract Drawings and Specifications.

a. Acceptance of submittals with deviations shall not relieve Contractor from responsibility for additional costs of changes required to accommodate such deviations.

b. Deviations included in submittals without prior acceptance will be considered an exception from review of submittals whether noted or not on returned copy.

2. No review action, implicit or explicit, shall be interpreted to authorize changes in the Work. Changes shall only be authorized by separate written Change Order or Field Instruction, in accordance with the Contract General Conditions.

3. When professional certification of performance criteria of materials, systems or equipment is required by Contract Documents, University Representative and Architect shall be entitled to rely upon accuracy and completeness of such calculations and certifications.

4. Notations by University Representative or Architect which increase contract cost or time of completion shall be brought to University Representative’s and Architect's attention before proceeding with Work.

D. Resubmittals:

1. Subject to same terms and conditions as original submittal.

2. University Representative and Architect will accept not more than one resubmittal.

a. Should additional resubmittals be required, Contractor shall reimburse Trustees for University Representative and Architect's accounts for time spent in processing additional resubmittals at rate of 2.5 times rate of Direct Personnel Expense (DPE). Direct Personnel Expense is defined as direct salaries of University Representative’s and Architect's personnel engaged on Project and portion of costs of mandatory, and customary contributions and benefits related thereto, including employment taxes and other statutory employee benefits, insurance, sick leave, holidays, vacations, pensions, and similar contributions and benefits.

1.9 PRODUCT DATA SUBMITTALS

A. Product Data: Catalog cuts, photographs, illustrations, standard details, standard schedules, performance charts, material characteristics, color and pattern charts, test data, roughing-in diagrams and templates, standard wiring diagrams and performance curves and listings by Code authorities and nationally-recognized testing and inspection services. Where product data must be specially prepared because standard manufacturer data is not suitable for use, submit according to requirements for shop drawings specified below.

B. Modifications to Standard Product Data: Modify manufacturer's standard catalog data to indicate precise conditions of the Project.

1. Provide space for review action stamps and, if required by authorities having jurisdiction, license seal of Engineer and/or design consultant, if applicable.

2. Mark each copy to show applicable choices and options. Where manufacturer’s product data includes information on several products, some of which are not required, mark copies to highlight applicable information.

3. Include the following information:

a. Manufacturer's literature with recommendations,
b. Compliance with recognized trade association standards,
c. Compliance with recognized testing agency standards,
d. Application of testing agency labels and seals,
e. Notation of dimensions verified by field measurement,
f. Notation of coordination requirements,
g. Environmental Product Declaration (EPD)'s information.

Environmental Product Declaration: Independently verified document created and verified in accordance with International Organization for Standardization (ISO) 14025 for Type III environmental declarations that identifies the global warming potential emissions of the facility- specific material or product through a product stage life cycle assessment.

The legislation was introduced as Assembly Bill (AB) 262. It targets the embedded carbon emissions of certain construction materials used in public works projects. AB 262 requires that these materials have a global warming potential that falls below a limit set by the Department of General Services.

The following materials or products are subject to the Buy Clean California Act, and shall have EPD's submitted for all products listed below:

Material or product	Material specifications: CSI Uniformat
Carbon steel rebar	Section 03 20 00, "Bar Reinforcement"
Structural steel	Section 05 12 00, "Structural Steel"
Flat glass	Section 08 80 00, "Glazing"
Mineral wool board insulation	Section 07 21 13.19 "Mineral Board Insulation"

4. Do not submit product data until compliance with requirements of the Contract Documents has been confirmed.

5. Proceed with installation only using reviewed copy of product data with appropriate action stamp as indicated in Section 1.8 B1 above. Do not permit use of unmarked copies of product data in connection with construction.

C. Copies: Submit electronic copies of catalog pages with applicable data highlighted and cross-referenced to Drawings and Specifications requirements. Paper copies may be reproduced however, will not be acceptable replacement of an electronig submittal unless specifically authorized by the University Representative. Distribution of approved submittals shall be electronic unless otherwise noted.

1.10 SHOP DRAWINGS SUBMITTALS

A. Shop Drawings: Drawings, diagrams, schedules and other graphic depictions to illustrate fabrication and installation of a portion of the Work. Shop Drawings shall include fabrication and installation drawings, setting diagrams, schedules, patterns, templates and similar drawings. Include the following information:

1. Identification of products and materials included

2. Compliance with referenced standards

3. Notation of coordination requirements

4. Dimensions

5. Notation of dimensions established by field measurement.

B. Coordination: Show all field dimensions and relationships to adjacent or critical features of Work.

C. Preparation of Shop Drawings: Prepare and submit electronically newly prepared information, drawn to accurate scale. Highlight, encircle, or otherwise indicate deviations from the Contract Documents. Do not reproduce Contract Documents or copy standard information as the basis of Shop Drawings. Standard information prepared without specific reference to the Project is not considered Shop Drawings.

1. Provide space for review action stamps and, if required by governing authorities having jurisdiction, license seal of Architect and Architect's design consultant, if applicable.

2. Prepare shop drawings submitted in electronic format that shall be printable on minimum sheet size of 17-inches by 22-inches, or smaller if a multiple of 8-1/2 inches by 11-inches. Maximum size shall be 30-inches by 42-inches.

3. Except as otherwise approved by the University Representative, submit all shop drawings electronically using the Contractor’s Electronic Project Management system.

4. Do not use Shop Drawings without an appropriate final review stamp indicating action taken in connection with construction.

D. Distribution of Reviewed Shop Drawings: Electronic distribution of reviewed shop drawings will be by Contractor and must be stamped by the Architect.

1.11 SAMPLE SUBMITTALS

A. Samples: Full-size, fully-fabricated samples cured and finished as specified and physically identical with the material or product proposed. Samples shall include partial sections of manufactured or fabricated components, cuts or containers of materials, color range sets, and swatches showing color, texture and pattern.

1. Mount, display, or package Samples in the manner specified to facilitate review of qualities indicated. Prepare Samples to include the following:

a. Project name and location

b. Manufacturer and supplier.

c. Name, finish, and composition of material.

d. Location where material is to be used.

e. Specification Section number.

f. Submittal number.

g. Contractor's review stamp.

h. Space for Architect's review stamp.

i. Compliance with recognized standards

j. Availability and delivery time.

2. Submit Samples for review of kind, color, pattern, and texture, for a final check of these characteristics with other elements, and for a comparison of these characteristics between the final submittal and the actual component as delivered and installed.

3. Submit actual samples. Photographic or printed reproductions will not be accepted.

4. Field samples specified in individual Sections are special types of samples. Field samples shall be full-size examples erected on site to illustrate finishes, coatings, or finish materials and to establish the standard by which the Work will be evaluated.

B. Preliminary or Selection Submittals: Where samples are for selection of color, pattern, texture or similar characteristics from a range of standard choices, submit full set of choices for the specified material or product.

1. Preliminary submittals will be reviewed and returned with the Architect's mark indicating selection and other action.

C. Quantity: Except for samples illustrating assembly details, workmanship, fabrication techniques, connections, operation and similar characteristics, submit three sets. One sample will be returned marked with the action taken.

1. Maintain sets of samples, as returned, at the Project site, for quality comparisons throughout the course of construction.

2. Unless otherwise noted, full-size and complete samples will be returned and may be incorporated into field mock-ups. Samples may be incorporated into the Work (completed construction) only with written approval of the Architect and the University Representative in advance of sample preparation.

3. Other samples shall be produced and mounted on cardstock in 8-1/2" by 11" format, three-hole punched and suitable for inclusion in product sample binders. Contractor shall provide

binders as directed.

4. Contractor shall prepare and distribute additional samples to subcontractors, manufacturers, fabricators, suppliers, installers, and others as necessary for performance of the Work.

5. Accepted samples will form standard of comparison for finished Work. Defects and deviations in excess of those in accepted samples, are unacceptable and are subject to rejection of completed Work.

D. Color Samples: Architect will review and select colors for Project only after all colors are received, so that colors may be properly coordinated.

E. Review of Field Samples: Review by Architect of field samples will be made for the following products if not otherwise required and if requested by Contractor.

THE FOLLOWING ARE EXAMPLES. EDIT TO SUIT ACTUAL PRODUCTS USED FOR PROJECT.

1. Casework.

2. Portland cement concrete paving: Trowel finish, imprinted texture, colors, abrasive blasting, exposed aggregate and acid washing.

3. Exterior plaster finish color and texture.

4. Gypsum board textures and finishes.

5. Gypsum plaster textures and finishes.

6. Field-applied paint colors and finishes: Draw-downs and brush-outs.

7. [_____].

1.12 MANUFACTURER'S INSTRUCTIONS

A. Manufacturer's Instructions: Submit manufacturer's instructions for preparation, mixing, assembly, handling, application and installation of products, as applicable and as specified in product sections of the Specifications.

1. Include applicable ICBO ES Evaluation Reports. Evaluation Reports shall be current and shall be annotated for applicable products.

2. Include applicable Safety Data Sheets (SDS), for Project record only.

3. Include written recommendations, as applicable, from manufacturer for Project conditions.

4. Identify conflicts between manufacturers' instructions and Contract Documents.

B. Copies: Electronic distribution will be required. If requested and agreed to by the University Representative, copies may be distributed as necessary.

C. Reviews by Architect and University's Representative: Manufacturer's instructions shall be for information and will not be reviewed by Architect or University's Representative. Contractor shall be responsible for proper coordination and installation of product or material specified.

1.13 REPORTS OF RESULTS OF INSPECTIONS AND TESTS

A. Reports of Results of Inspections and Tests: Submit technical data, test reports, calculations, surveys, and certifications based on field tests and inspections by independent inspection and testing agency and by authorities having jurisdiction.

1. Reports of results of inspections and tests shall not be considered Contract Documents.

2. Refer to Section 01 45 00 - Quality Control for additional requirements.

1.14 OPERATION AND MAINTENANCE DATA SUBMITTALS

A. Operation and Maintenance Data Submittals: Refer to requirements specified in Section 01 78 23 - Operation and Maintenance Data. Include operation and maintenance data submittals in Construction Progress Schedule. Refer to Contract General Conditions.

1.15 CERTIFICATES

A. When specified in individual specification Sections, submit manufacturers' certificates to Architect through Electronic Project Management system for review as specified.

B. Submit in form of letter or company standard forms, signed by officer of manufacturer.

C. Each certification shall include the following:

1. Project name and location.

2. Contractor's name and address.

3. Quantity and date or dates of shipment or delivery to which certificate applies.

4. Manufacturer's name.

D. Indicate material or product conforms to or exceeds specified requirements. Submit supporting reference data, affidavits, and certifications as appropriate.

E. Certificates may be recent or previous test results on material or product, but must be acceptable to University Representative and Architect.

PART 2 - PRODUCTS

Not applicable to this Section.

PART 3 - EXECUTION

Not applicable to this Section.

END OF SECTION

SECTION 01 34 00

REQUESTS FOR INTERPRETATION (RFI)

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Procedures for submitting requests for interpretation (RFI).

B. Limitations on use of RFI to obtain interpretation and clarification.

1.3 RELATED SECTIONS

A. Section 01 31 00 - Coordination: Requirements for organizing and coordinating the Work.

B. Section 01 33 00 - Submittals Procedures: Restriction on use of submittals for changes in materials, products, equipment and systems.

C. Section 01 63 00 - Product Substitution Procedures: Procedures for requesting substitutions of materials, products, equipment and systems.

1.4 DEFINITIONS

A. Request for Interpretation: A document submitted by the Contractor requesting clarification of a portion of the Contract Documents, hereinafter referred to as an RFI.

1.5 PREBID REQUESTS FOR INTERPRETATION (PREBID RFIS)

A. Bidders shall use the enclosed form titled, "Request for Interpretation of Contract Documents During Bid." to submit written requests for interpretation or corrections by e-mail or fax to the University:

Cal Poly State University
Attn: [Project Manager Name]
E-mail: [email]@calpoly.edu
Fax: (805) 756-####

B. To expedite the interpretation process, interpretations will be e-mailed to bidders as addenda.

C. Information shall be printed or typed including: Company name, address, phone, Email address, contact person, date, time of request, and question or clarification.

D. If bidders have several questions that will not fit on one form, submit additional pages, numbering each page.

E. Deadline for Requests for Interpretation during bid period: Requests for interpretation shall be received by the Trustees not later than ten (10) calendar days before the date bids will be opened.

F. The person submitting the request shall be responsible for its delivery.

1.6 CONTRACTOR'S REQUESTS FOR INTERPRETATION (RFIs) POST AWARD

A. Contractor's Requests for Interpretation (RFIs): Should Contractor be unable to determine from the Contract Documents the exact material, process, or system to be installed; or when the elements of construction are required to occupy the same space (interference); or when an item of Work is described differently at more than one place in the Contract Documents; the Contractor shall request that the Architect make an interpretation of the requirements of the Contract Documents to resolve such matters. Contractor shall comply with procedures specified herein to make Requests for Interpretation (RFIs).

B. Submission of RFIs: RFIs shall be prepared and submitted electronically utilizing the Electronic Project Management System. Refer to specification section 01 32 00. *[IF IS NOT USED – REVISE THIS SECTION]*

1. Forms shall be completely filled in, and if supplemental drawings or other information is prepared by hand, it shall be fully legible.

2. Each RFI shall be given a discrete, consecutive number. Revised RFI shall include the original number with the addition of a decimal and subsequent revision number. For instance, Revision #1 to RFI 029 should be noted as RFI 029.1.

3. Each page of the RFI and each attachments to the RFI shall bear the University's project name, project number, date, RFI number and a descriptive title.

4. Contractor shall attest to good faith effort to determine from the Contract Documents the information requested for interpretation. Frivolous RFI or simply passing on the RFI to the University without first vetting the RFI shall be subject to reimbursement from Contractor to University for fees charged by University, the Architect, Architect's consultants and other design professionals engaged by the University.

C. Subcontractor-Initiated and Supplier-Initiated RFIs: RFIs from subcontractors and material suppliers shall be submitted through, be reviewed by and be attached to an RFI prepared, signed and submitted by Contractor. RFIs submitted directly by subcontractors or material suppliers will be returned unanswered to the Contractor.

1. Contractor shall review all subcontractor- and supplier-initiated RFIs and take actions to resolve issues of coordination, sequencing and layout of the Work. Coordination of the work, sequence and layout are not the responsibility of the University or Architect.

2. RFIs submitted to request clarification of issues related to means, methods, techniques and sequences of construction or for establishing trade jurisdictions and scopes of subcontracts will be returned without interpretation. Such issues are solely the Contractor's responsibility.

3. Contractor shall be responsible for delays resulting from the necessity to resubmit an RFI due to insufficient or incorrect information presented in the RFI.

D. Requested Information: Contractor shall carefully study the Contract Documents, in particular, Article 5 of the Contract General Conditions, to ensure that information sufficient for interpretation of requirements of the Contract Documents is not included. RFIs that request interpretation of requirements clearly indicated in the Contract Documents will be returned without interpretation.

1. In all cases in which RFIs are issued to request clarification of issues related to means, methods, techniques and sequences of construction, for example, pipe and duct routing, clearances, specific locations of Work shown diagrammatically, apparent interferences and similar items, the Contractor shall furnish all information required for the Architect or University's Representative to analyze and/or understand the circumstances causing the RFI and prepare a clarification or direction as to how the Contractor shall proceed.

2. If information included with this type RFI by the Contractor is insufficient, the RFI will be returned unanswered.

E. Unacceptable Uses for RFIs: RFIs shall not be used to request the following:

1. Approval of submittals (use procedure specified in Section 01 33 00 - Submittals Procedures)

2. Approval of substitutions (refer to Section 01 63 00 - Product Substitution Procedures)

3. Changes that entail change in Contract Time and Contract Sum (comply with provisions of the Contract General Conditions, as discussed in detail during pre-construction meeting)

4. Different methods of performing Work than those indicated in the Contract Drawings and Specifications (comply with provisions of the Contract General Conditions).

F. Disputed Requirements: In the event the Contractor believes that a clarification by the University's Representative results in additional cost or time, Contractor shall comply with Article 5 of the Contract General Conditions.

G. RFI Log: The Electronic Project Management System will maintain a log of RFIs. Only the University may close any outstanding RFI.

H. Review Time: Architect will return RFIs to Contractor and University's Representative within seven calendar days of receipt. RFIs received after 12:00 noon shall be considered received on the next regular working day for the purpose of establishing the start of the seven-calendar day response period. Additional time shall be granted depending on the complexity of the request.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

Not Applicable to this Section.

END OF SECTION

CALIFORNIA POLYTECHNIC STATE UNIVERSITY

REQUEST FOR INTERPRETATION OF CONTRACT DOCUMENTS DURING BID

To: **California Polytechnic State University**
Attn: *[First Last Name] ([email]@calpoly.edu, Fax (805) 756-7566)*
Project: *[Project title and Project Number]*

Date:__________________________ **Time:**_________________________

Company:____

Contact Person:____

Address:____

____

Telephone:____ **Email:**________________________________

Plan Sheet:____ **Specification Section:**___________________

INTERPRETATION REQUESTED:

____

____

____

____

RESPONSE:

Date:__________________________ **Time:**_________________________

Company:____

Contact Person:____

INTERPRETATION:

____

____

____

____

SECTION 01 35 00 - SPECIAL PROCEEDURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

1. Environmental protection procedures
2. Smoke/odor control procedures
3. Noise control procedures
4. Dust and air pollution control procedures
5. Hazardous materials procedures
6. Welding and burning mitigation procedures
7. Erosion and sediment control procedures (Storm Water Pollution Protection Plan)
8. Disposal operations procedures
9. Cultural resources procedures
10. Alteration project procedures.

1.3 RELATED SECTIONS

A. Section 01 73 29 - Cutting and Patching: General requirements for procedures and limitations for cutting and patching the work.

1.4 SUBMITTALS

A. Refer to Section 01 33 00 – Submittal Procedures

B. Environmental Protection Plan – Submit within 30 days of commencement in Notice to Proceed.

C. State Water Pollution Prevention Plan (SWPPP): Submit Notice of Intent to the Regional Water Quality Control Board (RWQCB) with copies to Trustees Representative and Campus Environmental Health and Safety.

D. Submit notification in writing to the San Luis Obispo County Air Pollution Control District (SLOAPCD) with a copy to the Trustees Representative, 10 days prior to the start of Demolition.

E. Submit notification in writing to the San Luis Obispo County Air Pollution Control District (SLOAPCD) with a copy to the Trustee's Representative, 14 days prior to the start of road construction.

1.5 ENVIRONMENTAL PROTECTION PROCEDURES

A. Environmental Protection Procedures, General: Requirements specified in this Section are in addition to those of Article 4.03 of the Contract General Conditions.

1. During the progress of the work, keep the premises in a neat and clean condition and protect the environment from potentially polluting construction activities both on site and off site, throughout and upon completion of the construction project.

2. In coordination with the Campus, develop an Environmental Protection Plan in detail and submit to University's Representative for approval within 30 calendar days from the date of commencement specified in the Notice to Proceed. Distribute approved plan to all employees and to all subcontractors and their employees. Environmental Protection Plan shall include, but not be limited to, the following items:
   a. Copies of required permits
   b. Proposed sanitary landfill site
   c. Other proposed disposal sites
   d. Noise Control
   e. Dust Control
   f. Erosion and Sediment Control
   g. Copies of any agreements with public or private landowners regarding equipment, materials storage, borrow sites, fill sites, or disposal sites. Such agreements made by Contractor shall be invalid if their execution causes violation of local or regional grading or land use regulations.

B. Environmental Protection: Provide protection, operate temporary facilities and conduct construction in ways and by methods that comply with environmental regulations, and minimize the possibility that air, waterways and subsoil might be contaminated or polluted, or that other undesirable effects may result.

1. Avoid use of tools and equipment that produce harmful noise. Restrict use of noise making tools and equipment to hours that will minimize disruptions to the general public, staff and students near the site.

2. Comply with noise control requirements specified below.

C. Construction Operations: All construction operations shall comply with all applicable Federal, State and local Codes, ordinances, statutes and regulations pertaining to water, air, solid waste and noise pollution. It shall be Contractor's responsibility to identify and determine necessary measures to be taken to comply with such Codes, ordinances, statutes and regulations.

D. Definitions of Contaminants:

1. Sediment: Soil and other debris that have been eroded and transported by runoff water

2. Solid waste: Rubbish, debris, garbage and other discarded solid materials resulting from construction activities, including a variety of combustible and non-combustible wastes, such as ashes, waste materials that result from construction or maintenance and repair work, leaves and tree trimmings

3. Chemical waste: Includes petroleum products, bituminous materials, salts, acids, alkalis, herbicides, pesticides, disinfectants, organic chemicals and inorganic wastes. Some of the above may be classified as "hazardous"

4. Sanitary wastes:
   a. Sewage: Domestic sanitary sewage
   b. Garbage: Refuse and scraps resulting from preparation, cooking, dispensing and consumption of food.

E. Hazardous Materials: See also Section below titled "HAZARDOUS MATERIALS PROCEDURES."

1. Except as otherwise specified, in the event the Contractor encounters on the site material reasonably believed to be asbestos, polychlorinated biphenyl (PCB), lead containing/based paint or other hazardous materials which have not been rendered harmless, the Contractor shall immediately stop Work in the area affected and report the condition to the Trustees in writing.

2. Work in affected areas shall not thereafter be resumed except by written agreement of the Trustees and Contractor if in fact the material is asbestos, PCB, lead containing/based paint or other hazardous materials and has not been rendered harmless.

3. Work in affected areas shall be resumed in the absence of asbestos, PCB, lead containing/based paint or other hazardous materials, or when such materials have been rendered harmless.

F. Protection of Natural Resources: It is intended that the natural resources within the Project boundaries and outside the limits of permanent work performed under this Contract be preserved in their existing condition or be restored to an equivalent or improved condition upon completion of the work. Confine construction activities to areas defined by the public roads, easements, and work area limits shown on the drawings. Return construction areas to their pre-construction elevations except where surface elevations are otherwise noted to be changed. Maintain natural drainage patterns. Conduct construction activities such that ponding of stagnant water conducive to mosquito breeding habitat will not occur at any time.

1. Land resources protection: Do not remove, cut, deface, injure or destroy trees or shrubs outside the work area limits. Do not remove, deface, injure or destroy trees within the Project area without permission from University's Representative. Such improvements shall be removed and replaced, if required, by the Contractor at no change in Contract Time and Contract Sum.

2. Landscaping protection: Protect trees that are located near the limits of Project area which may possibly be defaced, bruised or injured or otherwise damaged by the Contractor's operations. No ropes, cables or guys shall be fastened to or be attached to any existing nearby trees or shrubs for anchorages. Refer to additional requirements specified in Section 01 56 00 - Temporary Barriers and Controls.
   a. Trimming: Refer to Section 01 56 69 - Tree and Plant Protection.
   b. Excavations around trees: Refer to Section 01 56 69 - Tree and Plant Protection.

c. Repair and restoration: Repair or replace trees or other landscape feature scarred or damaged by equipment or construction operations as specified below. Repair and restoration plan shall be reviewed and approved by University's Representative prior to its initiation.

3. Temporary construction:
   a. Remove all signs of temporary construction facilities such as haul roads, work areas, structures, foundations of temporary structures, stockpiles of excess or waste materials, or any other vestiges of construction as directed by the University's Representative.
   b. Level all temporary roads, parking areas and any other areas that have become compacted or shaped.
   c. Unpaved areas where vehicles have been operated shall receive suitable surface treatment or shall be periodically wetted down to prevent construction operations from producing dust damage and nuisance to persons and property, at no additional cost to the Trustees.
   d. Keep haul roads clear at all times of any object that creates an unsafe condition. Promptly remove any contaminants or construction materials dropped from construction vehicles. Do not drop mud and debris from construction equipment on public streets. Sweep clean turning areas and pavement entrances as necessary.

4. Water resources: Comply with all applicable Federal, State and local Codes, ordinances, statutes and regulations pertaining to discharge (directly or indirectly) of pollutants to underground and natural waters.
   a. Perform all Work under the Contract in a manner that any adverse environmental impacts are reduced to a level that is acceptable to University's Representative and authorities having jurisdiction.
   b. Refer to Division 02 - Site Construction, earthwork Sections, and Civil Drawings for specific requirements on control of Stormwater and disposal of water from dewatering activities.

5. Oily Substances: At all times, special measures shall be taken to prevent oily or other hazardous substances from entering the ground, drainage areas or local bodies of water in such quantities as to affect normal use, aesthetics or produce a measurable impact upon the areas. All soil or water that is contaminated with oily substances due to Contractor's operations shall be disposed of in accordance with applicable regulations, at no change in Contract Time and Contract Sum.

1.6 SMOKE/ODOR CONTROL PROCEDURES

G. Smoke/Odor Control: Protect primary fresh air intakes to existing buildings from exhaust from internal combustion engines, paint and solvent fumes and other noxious fumes and vapors.

1. Implement control methods such as snorkels from engine exhaust to within 50 feet from existing building air intakes. Provide carbon filters on air intakes as necessary, including periodic replacement of filters to ensure effectiveness.

2. All other activities generating fumes shall be limited to minimum distance of 50 feet from air intake grilles.

3. If fume-generating procedures must occur within 50 feet of an air intake, Contractor shall do the following:
   a. Notify University's Representative at least 14 calendar days in advance of such activities.

b. Perform Work when it least impacts the University (evenings, weekends or particularly windy days).
c. Provide carbon filter media, plastic barriers, or other control methods to ensure fresh air only enters into the building ventilation system.

1.7 NOISE CONTROL PROCEDURES

A. Noise Control Procedures, General: Requirements of this Section are in addition to those of Article 4.03 of the Contract General Conditions. Maximum noise levels within 1,000 feet of classrooms, laboratories, residences, businesses, adjacent buildings and other populated areas:

1. Noise levels for trenchers, pavers, graders and trucks: Not exceeding 90 dBA at 50 feet as measured under noisiest operating conditions.

2. Noise levels for all other equipment: Not exceeding 85 dBA at 50 feet.

B. Noise Control of Equipment:

1. Equip jackhammers with exhaust mufflers and steel muffling sleeves.

2. Use air compressors of a quiet type such as a "whisperized" compressor. Compressor hoods shall be closed while equipment is in operation.

3. Use electrically-powered rather than gasoline or diesel powered fork-lifts where feasible.

4. Provide portable noise barriers around jack hammering, with barriers constructed of 3/4 inch plywood lined with 1-inch thick ductliner type fiberglass on Work side.

C. Noise Control of Construction Operations:

1. Keep noisy equipment as far as possible from noise-sensitive site boundaries.

2. Machines shall not be left idling.

3. Use electric power in lieu of internal combustion engine power whenever possible.

4. Maintain equipment properly to reduce noise from excessive vibration, faulty mufflers, or other sources. All engines shall have properly functioning mufflers.

D. Scheduling of Noisy Operations: Schedule construction activities to minimize time of noisy operations and disruption to occupants of adjoining facilities. Notify University's Representative in advance of performing Work creating unusual noise and schedule such Work at times mutually agreeable.

E. Accessory Noise: Do not play radios, tape recorders, televisions, and other similar items at construction site.

1.8 DUST AND AIR POLLUTION CONTROL PROCEDURES

A. Dust and Air Pollution Control Procedures, General: Requirements of this Section are in addition to those of Article 4.03 of the Contract General Conditions. Employ measures to prevent or minimize creation of dust and air pollution. Contractor shall appoint a dust control monitor to oversee and implement all measures specified in this Article.

1. Unpaved areas shall be wetted down, to eliminate dust formation, a minimum of twice a day to reduce particulate matter. When wind velocity exceeds 15 mph, site shall be watered down more frequently.

2. Store all volatile liquids, including fuels or solvents in closed containers.

3. No on-site burning of debris, lumber and other scrap shall be permitted.

4. Properly maintain equipment to reduce gaseous pollutant emissions.

5. Exposed areas, new driveways and sidewalks shall be seeded, treated with soil binders or paved, as appropriate, as soon as possible.

6. Cover stockpiles of soil, sand and other loose materials if not currently being utilized and the the end of each work day.

7. Cover trucks hauling soil, debris, sand or other loose materials.

8. Sweep project area streets and walks at least once weekly or as needed to maintain a clean road or walkway. Refer to Section 01 74 00 - Cleaning Requirements.

1.9 HAZARDOUS MATERIALS PROCEDURES

A. Identified Hazardous Materials:

THE FOLLOWING IS AN EXAMPLE ONLY. DETERMINE IF HAZARDOUS MATERIAL STUDIES ARE APPLICABLE TO PROJECT AND, IF SO, IDENTIFY REPORT(S).

1. Limited hazardous materials investigations have been conducted for the University by {insert name of environmental consultant}, the results of which are in a document titled "[_*TITLE*_]" dated [_*DATE*_]. This report is furnished to Contractor as Information Available to Contractor. The report is included in the Project Manual as Appendix [_____].

2. Contractor shall perform hazardous materials abatement in compliance with requirements described in the document identified above. Costs and time associated with abatement of hazardous materials identified in this report shall be included in the Contract Sum and Contract Time.
   a. Comply with California Code of Regulations, Title 8, Sections 1529, 1532.1 and 5208.

3. Architect assumes no responsibility relating to existence of any hazardous materials, and Architect assumes no responsibility or liability for performance of Work described in the report identified above.

B. Unidentified Hazardous Materials:

1. Information regarding known asbestos containing material (ACM) is available from University's office of Environmental Health & Safety. Contact the University's Representative to request this information.

2. Except as otherwise specified, in the event that Contractor encounters on the project site material reasonably believed to be asbestos, polychlorinated biphenyl (PCB), lead containing/based paint or other hazardous materials which have not been rendered harmless, the Contractor shall immediately stop work in the area affected and report the condition to University's Representative.

3. Work in the affected area shall not be resumed except by written agreement between University and Contractor if in fact the material is asbestos, PCB, or other hazardous materials and has not been rendered harmless.

4. Work in the affected area shall be resumed in the absence of asbestos, PCB lead containing/based paint or other hazardous materials, or when such materials have been rendered harmless.

C. Notification and Disclosure: Refer to Contract General Conditions for Asbestos Notification and Disclosure requirements. Refer to [_*HAZARDOUS_MATERIALS_ABATEMENT_DOCUMENT*_] for information available to Contractor.

1. In the event that hazardous materials are discovered on site during performance of the Work, Contractor shall notify the University's Representative and request directions for abatement of hazardous materials.

2. University will ensure that the identified hazardous waste and/or hazardous materials are handled and disposed in the manner specified by the State of California Hazardous Substances Control Law (Health and Safety Code Division 20, Chapter 6.5).

1.10 WELDING AND BURNING MITIGATION PROCEDURES

A. Welding and Burning Mitigation Procedures: Eliminate welding and burning of steel as much as possible. Where unavoidable, perform welding and burning with all possible precaution to avoid fire hazard. Provide a fire watch for minimum of 30 minutes after burning stops. Provide adequate protection for all adjacent surfaces.

B. Hot work permit issued by the University is required for all activity where sparks, heat or flame are used which pose a threat to start a fire.

1.11 EROSION AND SEDIMENT CONTROL PROCEDURES

A. Erosion and Sediment Control Procedures: Refer to runoff control requirements specified in Section 01 57 00 - Temporary Controls. Obtain and comply with Storm Water Pollution Protection Plan (SWPPP) and project-specific requirements indicated on Civil Drawings and Specifications.

1.12 DISPOSAL OPERATIONS PROCEDURES

A. Solid Waste Management:

1. Supply solid waste transfer containers. Daily remove all debris such as spent air filters, oil cartridges, cans, bottles, combustibles and litter. Take care to prevent trash and papers from blowing off of the construction site. Encourage personnel to use refuse containers. Convey contents to a sanitary landfill.

2. Washing of concrete containers where wastewater may reach adjacent property, storm drains or natural water courses will not be permitted. Remove any excess concrete to the sanitary landfill.

B. Chemical Waste and Hazardous Materials Management: furnish containers for storage of spent chemicals used during construction operations. Dispose of chemicals and hazardous materials in accordance with applicable regulations.

C. Garbage: Store garbage in covered containers, pick up as required and dispose of properly.

D. Grading Spoil and Landscape Debris: Dispose of vegetation, weeds, rubble, and other materials removed by the clearing, stripping and grubbing operations off site at a suitable disposal site in accordance with applicable Federal, State and local Codes, ordinances, statutes and regulations

E. Excavated Materials:

1. Native soil complying with the requirements of applicable Division 2 - Site Construction earthwork Section, may be used for backfill, fill and embankments as allowed in applicable by that section.

2. Remove all material which is excavated in excess of that required for backfill. Dispose of unsuitable excavated material from the site and dispose of it legally.
   a. Excess suitable backfill material shall be hauled off site. No additional compensation will be paid to the Contractor for such off haul. Include all such costs in the Contract Sum.
   b. Unsuitable backfill material shall be disposed of off-site in accordance with applicable regulations, in a disposal site indicated in the Environmental Protection Plan.
   c. Remove rubbish and materials unsuitable for backfill immediately following excavation.
   d. Remove material in excess of that required for backfill immediately following backfill operations.

1.13 CULTURAL RESOURCES PROCEDURES

A. Cultural Resources Procedures: Requirements specified in this Section are in addition to those required by Article 4.03 of the Contract General Conditions.

1. Project does not pass through any known archaeological sites. However, it is conceivable that unrecorded archaeological sites could be discovered during construction.

2. In the event that artifacts, human remains, or other cultural resources are discovered during subsurface excavations at locations of the Work, the Contractor shall protect the discovered

items, cease work for a distance of 35 feet radius in the area, notify the Architect and University Representative and comply with applicable law.

3. Trustees may retain an Archaeologist to monitor and recover data and artifacts during period that work has ceased.

4. All items found which are considered to have archaeological significance are the property of the University.

1.14 ALTERATION PROJECT PROCEDURES

A. Coordinate the work of trades and schedule elements of alterations and renovation work by procedures and methods to expedite completion of the work.

B. In addition to demolition specifically shown, cut, move or remove items as necessary to provide access or to allow alterations and new work to proceed. Include such items as:

1. Repair or removal of hazardous or unsanitary conditions.

2. Removal of abandoned items and items serving no useful purpose, such as abandoned piping, conduit and wiring.

3. Removal of unsuitable or extraneous materials not marked for salvage, such as abandoned furnishings and equipment, and debris such as rotted wood, rusted metals and deteriorated concrete.

4. Cleaning of surfaces, and removal of surface finishes as needed to install new work and finishes.

C. Patch, repair and refinish existing items to remain, to the specified condition for each material, with a smooth and clean transition to adjacent new items of construction.

D. Assign the work of moving, removal, cutting and patching, to trades qualified to perform the work in a manner to minimize the possibility of damage to each type of work, and provide means of returning surfaces to appearance of new work.

E. Perform cutting and removal work with minimal disruption and in a manner to avoid damage to adjacent work.

F. Cut finish surfaces such as masonry, tile, plaster or metals, using methods that terminate surfaces in a straight line at a natural point of division.

G. Perform cutting and patching as specified in Section 01 73 29 - Cutting and Patching.

H. Protect existing finishes, equipment, and adjacent construction that is scheduled to remain, from damage.

1. Protect existing and new work from weather and extremes of temperature.

2. Maintain existing interior work above 60 degrees F or at a temperature recommended by any

product manufacture.

3. Provide weather protection, waterproofing, heat and humidity control as needed to prevent damage to remaining work and to new work.

1.15 RULES FOR PERFORMING ELECTRICAL WORK ON CAMPUS

A. To ensure the safety of personnel working on electrical systems, it is critical that safe work practices are followed and that outages are well planned and carefully scheduled and coordinated. The following rules apply to contractors performing electrical work on the University campus:

1. Electrical work shall not be performed on energized systems.

2. Panel covers and dead fronts shall not be removed while a panel is energized.

3. Circuit breakers shall not be cycled to switch existing loads on and off while a panel is in service.

4. All work up to the point of connection at an existing panel shall be completed without penetrating the enclosure. Once this new work is complete up to the existing panel, an outage shall be scheduled to facilitate termination at the panel.

5. Contractors shall schedule electrical outages with the responsible University Representative. Work shall be planned and scheduled so as to minimize the number of outages required.

6. The University Electricians will de-energize systems at the point of connection on the scheduled outage day and time. Lock Out/Tag Out will be performed by both the University Electricians and the qualified contractor performing the work.

7. During an outage, the individual working on the system shall be responsible for confirming that an "***electrically safe work condition***" has been established per NFPA 70E procedures using appropriate PPE.

8. Once the work at the point of connection is complete an inspection shall be scheduled with the campus building inspector and/or University Electricians.

9. Upon acceptance of the inspection, the locks shall be removed and the system shall be re-energized by the University Electricians.

10. Upon re-energization, the contractor shall test all new installations to confirm they are functioning properly prior to completing work and leaving the work location.

PART 2 - PRODUCTS

2.1 PRODUCTS FOR PATCHING, EXTENDING AND MATCHING

A. Provide same products or types of construction as that in existing structure, as needed to patch, extend or match existing.

B. Generally the Contract Documents will not define products or standards of workmanship present in existing construction; determine products by inspection and necessary testing, and determine quality of workmanship by using existing as a sample for comparison.

C. The presence of a product, finish, or type of construction requires that patching, extending or matching shall be performed as necessary to make work complete and consistent with identical standards of quality.

PART 3 - EXECUTION

3.1 CUTTING AND PATCHING

A. Perform cutting and patching as specified in Section 01 73 29 - Cutting and Patching.

END OF SECTION 01 35 00

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SECTION 01 35 05
SAFETY AND HEALTH PROCEDURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Procedures for health and safety protection and requirements for reporting accidents.

1.3 RELATED SECTIONS

A. Section 01 35 01 - Hazardous Material Procedures: Protection from asbestos containing materials (ACM), polychlorinated biphenyl (PCB), lead containing paint or other hazardous materials.

B. Section 01 56 00 - Temporary Barriers and Enclosures: Protective barriers.

1.4 SUBMITTALS

A. Accident Reporting: A copy of each accident report, which the Contractor or subcontractors submit to their insurance carriers, shall be forwarded to the Architect and to the Trustees' Representative as soon as possible, but in no event later than seven (7) calendar days after the day the accident occurred.

B. Other Submittals: If agreed to in writing at the preconstruction safety meeting, other submittals shall be required. One such submittal that may be included is a plan of action for handling hazardous materials to contain the following:

1. Number, type, and experience of employees to be used for the Work

2. Description of how safety and health regulations and standards shall be met

3. Type of protective equipment and work procedures to be used

4. Emergency procedures for accidental spills or exposures.

PART 2 - PRODUCTS

2.1 GENERAL

A. Special facilities, devices, equipment, clothing, and similar items used by the Contractor in the execution of the Work shall comply with all applicable regulations. The contractor is responsible for insuring a safe work environment for its employees, its subcontractors as well as University representatives, for the full duration of the work.

PART 3 - EXECUTION

3.1 STOP WORK ORDERS

A. Stop Work Orders:

1. When the Contractor or its subcontractors are notified by the University's Representative of an incident of noncompliance with the provisions of the Contract, and the action(s) to be taken, the Contractor shall immediately, or within 48 hours after receipt of a notice of violation, correct the unsafe or unhealthy condition to prevent harm.

2. If the Contractor fails to comply promptly, all or any part of the work performed may be stopped by with a "Stop Work Order." When, in the opinion of the University's Representative, satisfactory corrective action has been taken to correct the unsafe and unhealthy condition, a start order will be given immediately.

3. The Contractor shall not be allowed any extension of time or compensation for damages by reason of or in connection with such work stoppage.

3.2 PROTECTION

A. Protection: Contractor shall take all necessary precautions to prevent injury to the public, building occupants, or damage to property of others.

1. For the purposes of the Contract, the public or building occupants shall include all persons not employed by the Contractor or a subcontractor working under the Contractor's direction.

2. Work shall not be performed in any area occupied by the public or Owner's employees unless specifically permitted by the Contract or the Owner and unless adequate steps are taken for the protection of the public and the Owner's employees.

3. Whenever practicable, the work area shall be fenced, barricaded, or otherwise blocked off from the public or building occupants to prevent unauthorized entry into the work area.

B. Alternate Precautions: When the nature of the Work prevents isolation of the work area, and the public or building occupants may be in or pass through, under or over the work area, alternate precautions such as the posting of signs, the use of signal persons, the erection of barricades or similar protection around particularly hazardous operations shall be used as appropriate.

C. Public Thoroughfare: When Work is to be performed over a public thoroughfare such as a sidewalk, lobby, or corridor, the thoroughfare shall be closed, if possible, or other precautions taken such as the installation of screens or barricades. When the exposure to heavy falling objects exists, as during the erection of building walls or during demolition, special protection of the type detailed in 29 CFR 1910/1926 shall be provided.

D. Hazardous Conditions: Storing, positioning or use of equipment, tools, materials, scraps, and trash in a manner likely to present a hazard to the public or building occupants by its accidental shifting, ignition, or other hazardous qualities is prohibited.

END OF SECTION

SECTION 01 35 53

SECURITY

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Contractor Security requirements.

1.3 SECURITY (Also refer to Contract General Conditions)

A. Protect the Work from theft, vandalism and unauthorized entry. Contractor shall have sole responsibility for job site security.

B. Maintain security throughout construction until the University’s occupancy or acceptance.

C. Provide keying different from permanent keying of locks and include organized, locked and supervised storage for receiving and dispensing items of finish hardware throughout the construction.

D. Provide the Project Inspector with keys necessary to gain access to locked areas of the Work. The Project Inspector will be responsible for such keys and will return them to the Contractor upon acceptance of the project or area as complete.

1.4 ENTRY CONTROL

A. Restrict entrance of persons and vehicles into project site.

B. Allow building entrance only to authorized persons with proper identification.

1.5 PERMANENT KEYS

A. Immediately upon receipt of permanent keys for whatever purpose (finish hardware, mechanical equipment, casework, dispensers, lockers, switches, equipment items, etc.), tag or otherwise clearly identify keys according to one approved system and turn them over to the University prior to any opportunity of access to keys by parties other than the University.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION

SECTION 01 41 00 - REGULATORY REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 AUTHORITY AND PRECEDENCE OF CODES, ORDINANCES AND STANDARDS

A. Authority: All codes, ordinances and standards referenced in the Drawings and Specifications shall have the full force and effect as though printed in their entirety in the Specifications.

B. Precedence:

1. Where specified requirements differ from the requirements of applicable codes, ordinances and standards, the more stringent requirements shall take precedence.

2. Where the Drawings or Specifications require or describe products or execution of better quality, higher standard or greater size than required by applicable codes, ordinances and standards, the Drawings and Specifications shall take precedence so long as such increase is legal.

3. Where no requirements are identified in the Drawings or Specifications, comply with all requirements of applicable codes, ordinances and standards of authorities having jurisdiction.

1.3 APPLICABLE CODES, LAWS AND ORDINANCES

A. Applicable Codes, Laws and Ordinances: Refer also to Section 01 11 00 - Summary of the Work regarding permits and licenses.

1. Performance of the Work shall meet or exceed the minimum requirements of California Code of Regulations (CCR), Title 24, including but not limited to the following:

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CONFIRM CURRENT, APPLICABLE EDITION OF TITLE 24 COMPONENTS.

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a. 2016 Building Standards Administrative Code, Part 1, Title 24 C.C.R.
b. 2016 California Building Code, Vol. 1 & 2 (CBC), Part 2, Title 24 C.C.R. (2012 International Building Code, as amended by California)
c. 2016 California Electrical Code (CEC), part 3, title 24 C.C.R. (2011 National Electrical Code, as amended by California)
d. 2016 California Mechanical Code (CMC), Part 4, Title 24 C.C.R. (2012 Uniform Mechanical Code, as amended by California)
e. 2016 California Plumbing Code (CPC), Part 5, Title 24 C.C.R. (2012 Uniform Plumbing Code, as amended by California)
f. 2016 California Energy Code, Part 6, Title 24 C.C.R.

g. 2016 California Fire Code (CFC), Part 9, Title 24 C.C.R. (2012 International Fire Code, as amended by California)
h. 2016 California Referenced Standards, Part 12, Title 24 C.C.R. (partial list, see CBC chapter 35 & CFC Chapter 45)
i. NFPA 13 Sprinkler Systems (CA amended)
j. NFPA 14 Standpipe & Hose (CA Amended)
k. NFPA 17 Dry Chemical Systems, 2017 Edition
l. NFPA 17a Wet Chemical Systems, 2017 Edition
m. NFPA 20 Pumps for Fire Protection, 2017 Edition
n. NFPA 24 Fire Service Mains and The Appurtenances, 2017 Edition
o. NFPA 72 National Fire Alarm Code (California Amended), 2017 Edition
p. NFPA 253 Critical Radiant Flux of Floor Covering Systems, 2011 Edition
q. NFPA 2001 Clean Agent Fire Extinguishing Systems, 2012 Edition
r. Reference Code Section for NFPA Standards - CBC (SFM) Chapter 35

2. Performance of the Work shall also comply with applicable requirements of California Code of Regulations (CCR) as follows:

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CONFIRM CURRENT, APPLICABLE EDITION OF TITLE 19 AND TITLE 22.

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a. Title 19 - Public Safety, 2017 edition.
b. Title 22 - Social Security, 2017 edition.

3. References on the Drawings or in the Specifications to "code", "Code" or "building code" similar terms, not otherwise identified, shall mean the codes specified above, together with all additions, amendments, changes, and interpretations adopted by code authorities of the jurisdiction having authority over the Project.

4. The applicable edition of all codes shall be that adopted at the time of issuance of permits by the authority having jurisdiction and shall include all modifications and additions adopted by that authority. The applicable date of laws and ordinances shall be that of the date of performance of the Work.

B. Other Applicable Laws, Ordinances and Regulations:

1. Work shall be accomplished in conformance with all applicable laws, ordinances, rules and regulations of Federal, State, County, City and special district agencies and jurisdictions having authority over the Project.

2. Performance of the Work shall be accomplished in conformance with all rules and regulations of public utilities, utility districts and other agencies serving the facility.

3. Where such laws, ordinances, rules and regulations require more care or greater time to accomplish Work, or require better quality, higher standards or greater size of products, Work shall be accomplished in conformance to such requirements with no change to the Contract Time and Contract Sum, except where changes in laws, ordinances, rules and regulations occur subsequent to the execution date of the Agreement.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

Not Applicable to this Section.

END OF SECTION

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SECTION 01 42 00 - REFERENCE STANDARDS AND ABBREVIATIONS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Use of references in Drawings and Specifications, including requirements for copies of reference standards at Project site.

B. Definitions of terms used in Specifications and Drawings, including abbreviations, acronyms, names and terms which may be used in Specifications.

1.3 RELATED SECTIONS

A. Section 01 41 00 - Regulatory Requirements: Identification of applicable building Code and other codes, ordinances and regulations applicable to performance of the Work.

1.4 USE OF REFERENCES

A. References: The Drawings and Specifications contain references to various standards, standard specifications, codes, practices and requirements for products, execution, tests and inspections. These reference standards are published and issued by the agencies, associations, organizations and societies listed in this Section or identified in individual product specification Sections.

1. Wherever term "Agency" occurs in Standard Specifications, it shall be understood to mean the term used for University for purposes of the Contract.

2. Wherever term "Engineer" occurs in Standard Specifications, it shall be understood to mean Architect or other responsible design professional for purposes of the Contract.

3. Where reference is made to Standard Details, such reference shall be to the Standard Details accompanying the Standard Specifications.

B. Relationship to Drawings and Specifications: Such references are incorporated into and made a part of the Drawings and Specifications to the extent applicable.

C. Referenced Grades Classes and Types: Where an alternative or optional grade, class or type of product or execution is included in a reference but is not identified on the Drawings or in the Specifications, provide the highest, best and greatest of the alternatives or options for the intended use and prevailing conditions.

D. Copies of Reference Standards:

1. Reference standards are not furnished with the Drawings and Specifications because it is presumed that the Contractor, subcontractors, manufacturers, suppliers, trades and crafts are familiar with these generally-recognized standards of the construction industry.

2. Copies of reference standards may be obtained from publishing sources.

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REVIEW LIST AGAINST ACTUAL TYPE AND SIZE OF PROJECT AND SELECT ACCORDINGLY. COMPLETE LIST IS COSTLY AND MAY ONLY BE WARRANTED ON LARGE, COMPLICATED PROJECTS.
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E. Jobsite Copies:

1. Contractor shall obtain and maintain at the Project site copies of reference standards identified on the Drawings and in the Specifications in order to properly execute the Work.

2. At a minimum, the following shall be readily available at the site, as applicable to the Work:
   a. State Building Codes: As referenced in Section 01 41 00 - Regulatory Requirements.
   b. Safety Codes: Occupational Safety and Health Act (OSHA) regulations and State of California, California Administrative Code, California Code of Regulations (CCR), Title 8 - Industrial Relations, Chapter 4, Subchapter 7, General Industry Safety Orders (Cal-OSHA), to extent applicable to the Work.
   c. General Standards:
      1) CCR Title 24, Part 2, Volume 3: Current edition California Building Code (CBC) Material, Testing and Installation Standards.
      2) CCR Title 24, Part 12: California Referenced Standards Code, current edition.
      3) Underwriters Laboratories, Inc. (UL) Building Products Listing.
      4) Factory Mutual Research Organization (FM) Approval Guide.
      5) American Society for Testing and Materials (ASTM) Standards in Building Codes.
      6) American National Standards Institute (ANSI) standards.
   d. Fire and Life Safety Standards: All referenced standards pertaining to fire rated construction and exiting.
   e. Common Materials Standards: American Concrete Institute (ACI), American Institute of Steel Construction (AISC), American Welding Society (AWS), Gypsum Association (GA), National Fire Protection Association (NFPA), Tile Council of America (TCA) and Woodwork Institute of California (WIC) standards to the extent referenced within the Contract Specifications.
   f. Research Reports: ICC Evaluation Service, Inc. (ICC-ES), formerly ICBO Evaluation Service, Inc. (ICBO ES) Research Reports and National Evaluation Service, Inc. Reports (NER), for products not in conformance to prescribed requirements stated in California Building Code (CBC).
   g. Product Listings: Approval documentation, indicating approval of authorities having jurisdiction for use of product within the applicable jurisdiction.

F. Edition Date of References:

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CONSULT WITH UNIVERSITY'S REPRESENTATIVE AND DETERMINE WHICH OPTION IN PARAGRAPH BELOW APPLIES.

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1. When an edition or effective date of a reference is not given, it shall be understood to be the current edition or latest revision published as of the date of the Agreement, Contract Drawings and Contract Specifications.

2. All amendments, changes, errata and supplements as of the effective date shall be included.

G. ASTM and ANSI References: Specifications and Standards of the American Society for Testing and Materials (ASTM) and the American National Standards Institute (ANSI) are identified in the Drawings and Specifications by abbreviation and number only and may not be further identified by title, date, revision or amendment. It is presumed that the Contractor is familiar with and has access to these nationally- and industry-recognized specifications and standards.

1.5 DEFINITIONS OF TERMS

A. Basic Contract Definitions: Words and terms governing the Work are defined in the Contract General and Supplementary Conditions, as referenced in the Agreement.

B. Words and Terms Used on Drawings and in Specifications: Additional words and terms may be used in the Drawings and Specifications and are defined as follows:

1. "Applicable:" As appropriate for the particular condition, circumstance or situation.

2. "Approve(d):" Approval action shall be limited to the duties and responsibilities of the party giving approval, as stated in the Conditions of the Contract. Approvals shall be valid only if obtained in writing and shall not apply to matters regarding the means, methods, techniques, sequences and procedures of construction. Approval shall not relieve the Contractor from responsibility to fulfill Contract requirements.

3. "And/or:" If used, shall mean that either or both of the items so joined are required.

4. "Directed:" Limited to duties and responsibilities of the University's Representative or Architect as stated in the Contract General Conditions, meaning "as instructed by the University's Representative or Architect, in writing, regarding matters other than the means, methods, techniques, sequences and procedures of construction. Terms such as "directed", "requested", "authorized", "selected", "approved", "required", and "permitted" mean "directed by the University's Representative or Architect", "requested by the University's Representative or Architect", and similar phrases. No implied meaning shall be interpreted to extend the responsibility of the University's Representative, Architect or other responsible design professional into the Contractor's supervision of construction.

5. "Equal" or "Equivalent:" As determined by Architect or other responsible design professional as being equivalent, considering such attributes as durability, finish, function, suitability, quality,

utility, performance and aesthetic features.

6. "Furnish:" Means "supply and deliver, to the Project site, ready for unloading, unpacking, assembly, installation, and similar operations."

7. "Indicated:" The term indicated refers to graphic representations, notes, or schedules on the Drawings, or other Paragraphs or Schedules in the Specifications, and similar requirements in the Contract Documents. Terms such as "shown", "noted", "scheduled", and "specified" are used to help the reader locate the reference. There is no limitation on location.

8. "Install:" Describes operations at the Project site including the actual unloading, unpacking, assembly, erection, placing, anchoring, applying, working to dimension, finishing, curing, protecting, cleaning and similar operations.

9. "Installer:"
   a. "Installer" refers to the Contractor or an entity engaged by the Contractor, such as an employee, subcontractor, or sub-subcontractor for performance of a particular construction activity, including installation, erection, application and similar operations. Installers are required to be experienced in the operations they are engaged to perform.
   b. "Experienced Installer:" The term "experienced," when used with "installer" means having a minimum of 5 previous Projects similar in size to this Project, knowing the precautions necessary to perform the Work, and being familiar with requirements of authorities having jurisdiction over the Work.

10. "Jobsite:" Same as site, Area of Work, or other similar term referencing the physical property where the work is to be carried out upon.

11. "Necessary:" With due considerations of the conditions of the Project and as determined in the professional judgment of the Architect or other responsible design professional as being necessary for performance of the Work in conformance with the requirements of the Contract Documents, but excluding matters regarding the means, methods, techniques, sequences and procedures of construction.

12. "Noted:" Same as "Indicated."

13. "Per:" Same as "in accordance with," "according to" or "in compliance with."

14. "Products:" Material, system or equipment.

15. "Project Site:" Same as "Site." See definition of “Jobsite” above.

16. "Proper:" As determined by the Architect or other responsible design professional as being proper for the Work, excluding matters regarding the means, methods, techniques, sequences and procedures of construction, which are solely the Contractor's responsibility to determine.

17. "Provide:" Means "furnish and install, complete and ready for the intended use."

18. "Regulation:" Includes laws, ordinances, statutes and lawful orders issued by authorities having jurisdiction, as well as and rules, conventions and agreements within the construction industry

that control performance of the Work.

19. "Required:" Necessary for performance of the Work in conformance with the requirements of the Contract Documents, excluding matters regarding the means, methods, techniques, sequences and procedures of construction, such as:
    a. Regulatory requirements of authorities having jurisdiction.
    b. Requirements of referenced standards.
    c. Requirements generally recognized as accepted construction practices of the locale.
    d. Notes, schedules and graphic representations on the Drawings.
    e. Requirements specified or referenced in the Specifications.
    f. Duties and responsibilities stated in the Bidding and Contract Requirements.

20. "Scheduled:" Same as "Indicated."

21. "Selected:" As selected by the University's Representative, Architect or other responsible design professional from the full selection of the manufacturer's products, unless specifically limited in the Contract Documents to a particular quality, color, texture or price range.

22. "Shown:" Same as "Indicated."

23. "Site:" Same as "Site of the Work" or "Project Site;" the area or areas or spaces occupied by the Project and including adjacent areas and other related areas occupied or used by the Contractor for construction activities, either exclusively or with others performing other construction on the Project. The extent of the Project Site is shown on the Drawings, and may or may not be identical with the description of the land upon which the Project is to be built.

24. "Supply:" See "Furnish."

25. "Testing Laboratory" or "Testing Laboratories:" An independent entity engaged to perform specific inspections or tests, at the Project Site or elsewhere, and to report on, and, if required, to interpret, results of those inspections or tests. Refer to Section 01458 - Testing Laboratory Services.

26. "Testing and Inspection Agency:" Same as "Testing Laboratory."

1.6 ABBREVIATIONS, ACRONYMS, NAMES AND TERMS, GENERAL

A. Abbreviations, Acronyms, Names and Terms: Where acronyms, abbreviations, names and terms are used in the Drawings, Specifications or other Contract Documents, they shall mean the recognized name of the trade association, standards generating organization, authority having jurisdiction or other entity applicable.

B. Abbreviations, General: The following are commonly-used abbreviations which may be found on the Drawings or in the Specifications. Refer to the Drawings for additional abbreviations or acronyms. This is a partial list. If there is any discrepancy or confusion, notify the University in writing by RFI:

AC or ac	Alternating current or air conditioning (depending upon context)
AMP or amp	Ampere
C	Celsius

CFM or cfm	Cubic feet per minute
CM or cm	Centimeter
CY or cy	Cubic yard
DC or dc	Direct current
DEG or deg	Degrees
F	Fahrenheit
FPM or fpm	Feet per minute
FPS or fps	Feet per second
FT or ft	Foot or feet
Gal or gal	Gallons
GPM or gpm	Gallons per minute
IN or in	Inch or inches
Kip or kip	Thousand pounds
KSI or ksi	Thousand pounds per square inch
KSF or ksf	Thousand pounds per square foot
KV or kv	Kilovolt
KVA or kva	Kilovolt amperes
KW or kw	Kilowatt
KWH or kwh	Kilowatt hour
LBF or lbf	Pounds force
LF or lf	Lineal foot
M or m	Meter
MPH or mph	Miles per hour
MM or mm	Millimeter
PCF or pcf	Pounds per cubic foot
PSF or psf	Pounds per square foot
PSI or psi	Pounds per square inch
PSY or psy	Per square yard
SF or sf	Square foot
SY or sy	Square yard
V or v	Volts

C. Abbreviations and Acronyms for Industry Organizations: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities indicated in Gale Research's "Encyclopedia of Associations" or in Columbia Books' "National Trade & Professional Associations of the U.S."

D. Undefined Abbreviations, Acronyms, Names and Terms: Words and terms not otherwise specifically defined in this Section, in the Instructions to Bidders, in the Contract General Conditions, on the Drawings or elsewhere in the Specifications, shall be as customarily defined by trade or industry practice, by reference standard and by specialty dictionaries such as the following:

1. Dictionary of Architecture and Construction, Third Edition (Cyril M. Harris, McGraw-Hill Book Company, 2000).

2. The American Institute of Architects (AIA) Document M101, "Glossary of Construction Industry Terms."

3. Encyclopedia of Associations, published by Gale Research Co., commonly available in public libraries.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

Not Applicable to this Section.

END OF SECTION

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SECTION 01 45 00 - QUALITY CONTROL

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Regulatory requirements for testing and inspection.

B. Contractor's quality control.

C. Quality of the Work.

D. Inspections and tests by authorities having jurisdiction.

E. Inspections and tests by serving utilities.

F. Inspections and tests by manufacturer's representatives.

1.3 RELATED SECTIONS

A. Section 01 31 00 - Coordination: Coordination of Work under Contract.

B. Section 01 41 00 - Regulatory Requirements: Compliance with applicable codes, ordinances and standards.

C. Section 01 45 29 - Testing Laboratory Services: Selection of independent testing and inspection laboratory; tests and inspections conducted by testing laboratory.

D. Section 01 61 00 - Basic Product Requirements: Product options, substitutions, transportation and handling requirements, storage and protection requirements, and system completeness requirements.

1.4 REGULATORY REQUIREMENTS FOR TESTING AND INSPECTION

A. Building Code Requirements: Comply with requirements for testing and inspections in the California Building Code (CBC), as interpreted by authorities having jurisdiction. Additional requirements for testing and inspection, as adopted by authorities having jurisdiction, shall be included in the Contract Sum and Contract Time.

B. Requirements of Fire Regulations: Comply with testing and inspection requirements of the Fire Marshal having jurisdiction. All tests and inspections shall be included in Contract Sum and Contract Time.

1.5 CONTRACTOR'S QUALITY CONTROL

A. Contractor's Quality Control: Contractor is solely responsible for the quality of Work. Contractor shall ensure that products, services, workmanship and site conditions comply with requirements of the Drawings and Specifications by coordinating, supervising, testing and inspecting the Work and by utilizing only suitably qualified personnel.

B. Quality Requirements: Work shall be accomplished in accordance with quality requirements of the Drawings and Specifications, including, by reference, all Codes, laws, rules, regulations and standards. When no quality basis is prescribed, the quality shall be in accordance with the best accepted practices of the construction industry for the locale of the Project, for projects of this type.

C. Quality Control Personnel: Contractor shall employ and assign knowledgeable and skilled personnel as necessary to perform quality control functions to ensure that the Work is provided as required.

D. Coordination of Field Quality Control: The General Contractor is solely responsible to coordinate and schedule field quality assurance activities of University's testing and inspection agency and inspectors from authorities having jurisdiction. The University will not coordinate or schedule quality assurance activities on behalf of the Contractor.

1.6 QUALITY OF THE WORK

A. Quality of Products: Unless otherwise indicated or specified, all products shall be new, free of defects and fit for the intended use.

B. Quality of Installation: All Work shall be produced plumb, level, square and true, or true to indicated angle, and with proper alignment and relationship between the various elements.

C. Protection of Existing and Completed Work: Take all measures necessary to preserve and protect existing and completed Work free from all damage, deterioration, soiling and staining, until Acceptance by the University. This includes but is not limited to rain and wind conditions, subcontractor damage of installed or existing products, or similar conditions. All damage shall be deemed non-conforming until such time the work is corrected to the satisfaction of the contract documents.

D. Standards and Code Compliance and Manufacturer's Instructions and Recommendations: Unless more stringent requirements are indicated or specified, comply with manufacturer's instructions and recommendations, reference standards and building code research report requirements in preparing, fabricating, erecting, installing, applying, connecting and finishing Work.

E. Deviations from Standards and Code Compliance and Manufacturer's Instructions and Recommendations: Document and explain all deviations from reference standards and building code research report requirements and manufacturer's product installation instructions and recommendations, including acknowledgement by the manufacturer that such deviations are acceptable and appropriate for the Project.

F. Verification of Quality: Work shall be subject to verification of quality by University or Architect in accordance with provisions of the Contract General Conditions.

1. Contractor shall cooperate by making Work available for inspections and observations by University's Representative, Architect and their consultants.

2. Such verification may include mill, plant, shop, or field inspection, as required.

3. Provide access to all parts of the Work, including plants where materials or equipment are manufactured or fabricated.

4. Provide all information and assistance as necessary, including that from subcontractors, fabricators, materials suppliers and manufacturers, for verification of quality by University's Representative or Architect.

5. Contract modifications, if any, resulting from such verification activities shall be governed by applicable provisions in the Contract General Conditions.

G. Observations by Architect and Architect's Consultants: Periodic and occasional observations of Work in progress will be made by Architect and Architect's consultants as deemed necessary to review progress of Work and general conformance with the plans and specifications.

H. Limitations on Inspection, Test and Observations: Employment of an independent testing and inspection agency and observations by Architect and Architect's consultants shall not relieve Contractor of the obligation to perform Work in full conformance to all requirements of Contract Documents and applicable Building Code and other regulatory requirements.

I. Rejection of Work: The University reserves the right to reject any and all Work not in conformance to the requirements of the Contract Documents.

J. Correction of Non-Conforming Work: Non-conforming Work shall be modified, replaced, repaired or redone by the Contractor at no change in Contract Sum or Contract Time.

K. Acceptance of Non-Conforming Work: Acceptance of non-conforming Work, without specific written acknowledgement and approval of the University's Representative, shall not relieve the Contractor of the obligation to correct such Work.

L. Contract Adjustment for Non-conforming Work: Should University's Representative determine that it is not feasible or not in University's interest to require non-conforming Work to be repaired or replaced, an equitable reduction in Contract Sum shall be made by agreement between University's Representative and Contractor. If an equitable amount cannot be agreed upon, a Field Instruction will be issued and the amount in dispute resolved in accordance with applicable provisions of the Contract General Conditions.

M. Non-Responsibility for Non-Conforming Work: Architect and Architect's consultants disclaim any and all responsibility for Work produced that is not in conformance with the Contract Drawings and Contract Specifications.

1.7 INSPECTIONS AND TESTS BY AUTHORITIES HAVING JURISDICTION

A. Inspections and Tests by Authorities Having Jurisdiction: Contractor shall cause all tests and inspections required by authorities having jurisdiction to be made for Work under this Contract.

1. Except as specifically noted, scheduling, coordinating and conducting such inspections and tests shall be solely the Contractor's responsibility.

2. All time required for inspections and tests by authorities having jurisdiction shall be included in the Contract Time.

3. Costs for inspections and tests by authorities having jurisdiction will be paid by University. Any re-test or additional cost incurred due to initial failure or a lack of preparedness shall be charged to the Contractor and a deductive change order processed.

1.8 INSPECTIONS AND TESTS BY SERVING UTILITIES

A. Inspections and Tests by Serving Utilities: Contractor shall cause all tests and inspections required by serving utilities to be made for Work under the Contract.

1. Except as specifically noted, scheduling, coordinating and conducting such inspections and tests shall be solely the Contractor's responsibility. All time required for inspections and tests by serving utilities shall be included in the Contract Time.

2. Except as specifically noted, all costs for inspections and tests by serving utilities shall be included in the Contract Sum. All costs for retest due to the original test failure will be paid for by the Contractor.

1.9 INSPECTIONS AND TESTS BY MANUFACTURER'S REPRESENTATIVES

A. Inspections and Tests by Manufacturer's Representatives: Contractor shall cause all specified tests and inspections to be conducted by materials or systems manufacturers. Additionally, all tests and inspections required by materials or systems manufacturers as conditions of warranty or certification of Work shall be made, the cost of which shall be included in the Contract Sum.

1. Scheduling, coordinating and conducting such inspections and tests shall be solely the Contractor's responsibility. All time required for inspections and tests by manufacturer's representatives shall be included in the Contract Time.

2. All costs for inspections and tests by manufacturer's representatives shall be included in the Contract Sum.

1.10 INSPECTIONS BY INDEPENDENT TESTING AND INSPECTION AGENCY

A. Inspections by independent Testing Laboratory: Refer to Section 01 45 29 - Testing Laboratory Services.

PART 2 - PRODUCTS

Not applicable to this Section.

PART 3 - EXECUTION

Not applicable to this Section.

END OF SECTION

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SECTION 01 45 05 - MOCK UPS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Full scale mock-ups for visual qualities.

1.3 RELATED SECTIONS

****
THE FOLLOWING ARE EXAMPLES ONLY. LIST SECTIONS APPLICABLE TO PROJECT, REQUIRING MOCK-UPS.
****

A. Section 07 10 00 – Dampproofing and Waterproofing

B. Section 08 80 00 - Glazing

C. Section 09 21 00 – Plaster and Gypsum Board Assemblies

D. Section 09 30 00 - Tiling

E. Section 09 60 00 - Flooring

F. Section 09 68 00 - Carpeting

G. Section 09 90 00 – Painting and Coating

H. Section 10 14 00 – Signage

I. Section 10 26 00 - Wall and Door Protection

J. Section 12 30 00 – Casework

K. Section 12 36 00 - Countertops

L. Section 22 40 00 - Plumbing Fixtures

M. Section 26 50 00 – Lighting

N. Section 32 05 23 - Portland Cement Concrete Paving (for review of color and finish)

O. Section 32 13 73 – Concrete Paving Joint Sealants

P. Section [_____] - [_Exterior Cladding for water test____].

Q. Section [_____] - [_____].

R. Section [_____] - [_____].

1.4 DEFINITIONS

A. Mock-Ups: Full-size, physical example assemblies to illustrate finishes and materials.

1. Mock-ups are used to verify selections made under Sample submittals, to demonstrate aesthetic effects and, where indicated, qualities of materials and execution, and to review construction, coordination, testing, or operation; they are not Samples.

2. Mock-ups establish the standard by which the Work will be judged.

1.5 SUBMITTALS

A. Product Data and Shop Drawings: For each product or system that will be incorporated in the mock-ups, submit required submittals as specified in applicable product Section of the Specifications.

1.6 QUALITY ASSURANCE

A. Mock-Ups: Before installing portions of the Work requiring mock-ups, build mock-ups for each form of construction and finish required to comply with the following requirements, using materials indicated for the completed Work:

1. Build mock-ups in location and of size indicated or, if not indicated, as directed by University's Representative.

2. Notify University's Representative and Architect minimum of seven days in advance of dates and times when mock-ups will be constructed.

3. Demonstrate the proposed range of aesthetic effects and workmanship.

4. Obtain review and acceptance of mock-ups by Architect and University's Representative before starting Work, including fabrication and installation construction.

5. Maintain mock-ups during construction in an undisturbed condition as a standard for judging the completed Work.

6. Demolish and remove mock-ups when directed, unless otherwise indicated.

PART 2 - PRODUCTS

2.1 MOCK-UPS FOR VISUAL QUALITIES

A. Mock-Ups for Visual Qualities: Before installing portions of the Work requiring a mock-up, build the mock-ups with each form of construction and finish required to comply with the following requirements, using materials indicated for the completed Work:

1. Construct field mock-ups as indicated on the Drawings, indicating assemblies and interfaces of materials.

2. Construct mock-ups at location where directed by University's Representative.

3. Demonstrate the proposed range of visual effects, qualities and workmanship.

4. Provide structural substrate for mock-ups as suitable. Mock-ups shall be free standing and self-supporting.

5. Maintain mock-ups during construction in an undisturbed condition as a standard for judging completed Work.

6. Demolish and legally dispose of mock-ups when directed, unless otherwise indicated.

PART 3 - EXECUTION

3.1 CONSTRUCTION OF MOCK-UPS FOR VISUAL QUALITIES

A. Mock-Ups for Visual Qualities, General: Construct mock-ups as noted on the Drawings and specified in individual product Sections of the Specifications, including but not limited to the the following:

****
THE FOLLOWING ARE EXAMPLES. EDIT TO SUIT PROJECT REQUIREMENTS.
****

1. Casework:

   a. Typical base cabinet, plastic laminate countertop and wall cabinet, including ceiling and wall trim.
   b. [_*DESCRIPTION*_].

2. Ceramic tile: Toilet room, floor and wall tile including:

   a. Intersections of floor-wall and wall-ceiling.
   b. Transition from wall tile finish to gypsum board finish (including wallcovering, where applicable).
   c. Inside corner of tile covered wall.
   d. Shower curb.

3. [_*DESCRIPTION*_].

    a. [_*DESCRIPTION*_].
    b. [_*DESCRIPTION*_].
    c. [_*DESCRIPTION*_].

B. [_*DESCRIPTION*_] Room Mock-Ups:

****
THE FOLLOWING IS EXAMPLE TEXT ONLY. EDIT TO SUIT PROJECT REQUIREMENTS. TEXT SHOULD DESCRIBE THE SPECIFIC LEVEL OF FINISH REQUIRED OF THE PROJECT AND TO THE EXTENT THE MOCK UP SHOULD BE CONSTRUCTED.
****

1. Construct mock-up where indicated on the Drawings or, if not indicated, where designated by University's Representative.

2. Construct wall and ceiling framing for gypsum board finish, gypsum board finish, paint, door frames and doors (with hardware), floor fill at Corridor door, floor coverings and base, wallcoverings, dummy lighting fixtures, dummy electrical and signal outlets, dummy plumbing fixtures, casework and trim.

3. Remove, reconstruct and refinish products as necessary to achieve fit, finish and tolerances acceptable to University's Representative and Architect.

****
ADD ADDITIONAL PARAGRAPHS AS NECESSARY TO DESCRIBE MOCK-UPS.
****

END OF SECTION

SECTION 01 45 23 - TESTING AND INSPECTING SERVICES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Administrative and procedural requirements for quality assurance services.

1. Quality assurance services include inspections and tests and related actions including reports, performed by independent agencies, and governing authorities. They do not include Contract enforcement activities performed by the Trustees or Architect.

2. Inspection and testing services are required to verify compliance with requirements specified or indicated. These services do not relieve the Contractor of responsibility for compliance with Contract Document requirements.

1.3 RELATED SECTIONS

A. Section 01 45 00 - Quality Control: General requirements for inspections and tests.

B. Individual Product Specifications Sections: Specific requirements for inspections and tests.

1.4 RESPONSIBILITIES

A. Testing Agency: For the purpose of this specification “Testing Agency” shall include any persons designated by the University to ensure the work is conforms to the contract documents. This definition shall include the Inspector of Record and Special Inspector. Trustees will engage and pay for the services of an independent agency to perform inspections and tests specified as the Trustees' responsibility for both on-site and off-site testing and inspecting. Refer to paragraph E for travel restrictions.

1. Where the Trustees have engaged a testing agency or other entity for testing and inspection of a part of the Work, and the Contractor is also required to engage an entity for the same or related element, the Contractor shall not employ the entity engaged by the Trustees, unless otherwise agreed in writing with the Trustees.

B. Retesting: The Contractor is responsible for the cost of retesting where results of required inspections, tests or similar services prove unsatisfactory and do not indicate compliance with Contract Document requirements, regardless of whether the original test was the Contractor's responsibility.

1. Cost of retesting construction revised or replaced by the Contractor is the Contractor's

responsibility, where required tests were performed on original construction.

C. Associated Services: The Contractor shall cooperate with agencies performing required inspections, tests and similar services and provide reasonable auxiliary services as requested.

D. Coordination: The Contractor, Project Manager/Inspector, and each agency engaged to perform inspections, testing and similar services shall coordinate the sequence of activities to accommodate required services with a minimum of delay. In addition the Contractor shall coordinate activities to avoid the necessity of removing and replacing construction to accommodate inspections and tests.

1. The Contractor is responsible for communicating to the Project Manager/Inspector the scheduling times for inspections, tests, taking samples and similar activities. The contractor shall formally request all inspections.

E. Payment for Testing Agency Services:

1. Unless otherwise specified, Trustees will pay for tests and inspections performed by Testing Agency, as specified in individual product Sections of the Specifications. Overtime costs due to scheduling for the convenience of the Contractor or to make up for Work behind schedule shall be deducted by Change Order from Contract Sum.

2. When tests and inspections are required on an overtime basis, initial payment will be made by the Trustees. All costs for overtime testing and inspections shall be deducted by Change Order from Contract Sum.
   a. The deducted cost will be for the full amount and shall not include any mark-up factor.

3. Unless otherwise specified, Contractor shall be back-charged for mileage and travel time for inspection services requiring more than 100 miles from Project site to test products purchased or fabricated by Contractor.

   a. Testing Agency shall forward all billings and records of such costs to University's Representative for approval.
   b. Such costs, if determined by University's Representative to be attributable to the Contractor under this provision, shall be deducted by Change Order from Contract Sum.

4. Contractor shall pay all costs for repeated observations, re-inspection or retesting by Testing Agency due to non-conforming Work. Costs shall be deducted by Change Order from Contract Sum.
   a. The deducted cost will be for the full amount and shall not include any mark-up factor.

5. Additional Tests, Inspections and Related Services: Contractor shall be charged costs for additional tests, inspections and related services, due to the following. Such costs shall be deducted by Change Order from Contract Sum.

   a. Work is not ready to inspect when inspectors arrive.
   b. Failure to properly schedule or notify testing and inspection agency or authorities having jurisdiction.
   c. Changes in sources, lots or suppliers of products after original tests or inspections.
   d. Changes in means methods, techniques, sequences and procedures of construction that

necessitate additional testing, inspection and related services.

e. Changes in mix designs for concrete and mortar after review and acceptance of submitted mix design.
f. Multiple off-site fabrication sites.
g. Fabrication and installation errors.
h. Inefficient, sporadic, or poorly organized manufacturing that causes additional testing costs to be incurred.

F. Segregation in Billing of Overtime Services: Billings for overtime services shall have straight time and overtime costs segregated and shall have substantiation by detailed explanations justifying necessity of services on overtime basis.

G. Obligation to Perform Work According to Contract Documents: Employment of Testing Agency shall in no way relieve Contractor of obligation to perform Work in accordance with requirements of Contract Documents and applicable Codes.

H. Limits on Testing Agency's Authority:

1. Testing Agency may not release, revoke, alter, or enlarge on requirements of Contract Documents.

2. Testing Agency may not approve or accept any portion of the Work. Inspection includes only a confirmation that the installation meets the details in the plans and specifications.

3. Testing Agency may not assume any duties of Contractor and will not assist or direct work in any way nor will testing agency coordinate work of other contractors.

4. Testing Agency shall have the authority to stop work when it in direct conflict with contract documents, code or when work poses a risk to the safety of personnel.

I. Contractor shall make the Work in all stages of progress available for University personal and continuous observation by the Testing Agency.

1. Testing Agency shall have free access to any and all parts of the Work at all times.

2. Contractor shall provide the Testing Agency with reasonable facilities for Testing Agency to obtain such information as Testing Agency determines is necessary for Testing Agency to be kept fully informed of the progress and manner of performance of the Work and character of products, according to Testing Agency's duties and responsibilities.

3. Observation and inspection of the Work by Testing Agency shall not relieve Contractor from any obligation to fulfill the requirements of the Contract.

J. Retesting: When materials tested fail to meet requirements herein specified, they shall be promptly corrected or removed and replaced at the expense of the Contractor and retested in a manner required by University's Representative. Costs involved in retesting shall be deducted by Change Order from Contract Sum.

1.5 TESTS AND INSPECTIONS

A. Tests and Inspections, General: All construction work shall be subject to inspection by the Trustees and the Architect and all such construction or work shall remain accessible and exposed for inspection purposes until approved by the Trustees.

1. The Trustees will provide project personnel, including inspectors, to be available at the project site.

2. Approval as a result of an inspection shall not be construed to be an approval of a violation of the provisions of the building code or of other ordinances of the jurisdiction, including plans and specifications. Inspections presuming to give authority to violate or cancel the provisions of code, or of plans and specifications shall not be valid.

3. It shall be the duty of the contractor to cause the work to remain accessible and exposed for inspection purposes. Neither the Inspector nor the Trustees or Architect shall be liable for expense entailed in the removal or replacement of any material required to allow inspection.

B. Inspection Requests: It shall be the duty of the Contractor doing the work to notify the Inspector that such work is ready for inspection. The Trustees require that such work is ready for inspection. The Trustees require that every request for inspection be filed at least two business days before such inspection is desired. Such requests shall be in writing.

C. Approval Required: Work shall not be done beyond the point indicated in each successive inspection without first obtaining the approval of the Inspector. The Inspector, upon notification, shall make the requested inspections and shall either indicate in writing that portion of the construction is satisfactory as completed, or shall notify the Contractor that same fails to comply with plans and specifications. Any portions of Work that do not comply shall be corrected by the Contractor, and such portion shall not be covered or concealed until authorized by the Inspector.

1. There shall be a final inspection and approval of all buildings and structures when completed and ready for occupancy and use.

D. Inspection Coordination: Contractor shall provide, on a weekly basis, an anticipated Inspection Requirements Schedule, coordinated with the three-week look ahead schedule, showing the anticipated inspection needs for the following three weeks to facilitate appropriate campus coordination and interface as well as mobilization of required inspection staffing.

E. Required Inspections: Reinforcing steel, structural framework, or interior wall and/or ceiling support framing of any part of any building or structure shall not be covered or concealed without first obtaining the approval of the Inspector.

1. Listed below are the minimum inspection requirements:

Sect #	**Section Name**	**Test**	**Inspection**	**Paid By**
01 45 0 0	Quality Control		• Final Inspection	University

Sect #	Section Name	Test	Inspection	Paid By
03 30 0 0	Cast-In-Place Concrete	• Molded Cylinder Test • Concrete Consistency	• Welding Continuous Inspection • Structural Concrete • Special Inspection • Epoxy Anchor Concrete • Embeds	University
05 12 0 0	Structural Steel Framing	• Bolt Connection Test • Welding • Lamellar Tearing • Prior Testing of Base Material	• Bolt torque • Welding	University
05 30 0 0	Metal Decking	• Mill Analysis & Test Reports	• Welding • Crimping • Closures	University
05 41 0 0	Structural Metal Stud Framing		• Field Welds • Material & Equipment • Screws and spans	University
07 11 1 3	Bituminous Dampproofing	• Site Flood Test	• Periodic Inspection	University
07 53 2 3	EPDM Roofing	• Defect Testing	• Periodic Inspection • Manufactures Rep	University
07 84 0 0	Firestopping		• General Inspection	University
07 95 1 3	Expansion Joint Cover Assemblies	• Exterior Joint Covers • Water Test	• Manufacturer's Field Inspection • Mill Certification	University
08 51 1 3	Aluminum Windows	• Air Infiltration • Water Infiltration	• Fenestration • Solar Gain	University
09 29 0 0	Gypsum Board		• Rough Gypsum Board Inspection • Screws	University
14 21 0 0	Electric Traction Elevators		• Inspections & Permits	Contractor
14 24 0 0	Hydraulic Elevators		• Inspections & Permits	Contractor

Sect #	Section Name	Test	Inspection	Paid By
22 05 0 0	Common Work Results for Plumbing		• Rough Plumbing Inspection	University
			• General Pipe Testing • Pressure testing on supply • DWV stack test • Gas line pressure testing •	Contractor
23 05 0 0	Common Work Results for HVAC		• Rough Mechanical Inspection	University
23 08 0 0	Commissioning of HVAC	• Commissioning		Contractor & University
26 05 00	Common Work Results for Electrical		• Rough Electrical Inspection	University
		• Electrical Testing	•	Contractor
26 08 0 0	Commissioning of Electrical Systems	• Commissioning		Contractor & University
27 05 0 0	Common Work Results for Communications		• Rough Telecom Inspection	University
27 30 0 0	Voice Communications			Contractor
31 23 0 0	Excavation and Fill	• Soils Density Tests • Field Density Tests		University
31 23 1 3	Subgrade Preparation	• Backfilling & Compaction		University
32 11 2 3	Aggregate Base Courses	• Compaction		University
32 12 1 6	Asphalt Paving	• Compaction	• Slope and flow	University
32 13 1 3	Concrete Paving	• Slump Test/ Compression Test	• Slope and flow	University
32 16 1 3	Concrete Curbs and Gutters	• Slump Test/ Compression Test	• Slope and flow	University

Sect #	Section Name	Test	Inspection	Paid By
32 84 0 0	Planting Irrigation	• Hydrostatic Test • Operational Testing		University
32 93 0 0	Planting		• Preliminary Inspection • Field Inspection	University
33 11 0 0	Water Utility Distribution Piping	• Hydrostatic Tests • Pressure Tests • Leakage Tests		University
33 41 0 0	Storm Utility Drainage Piping	• Hydrostatic Test on Watertight Joints		University

(ARCHITECT TO REVISE/ ADD TO LIST AS REQUIRED BY PROJECT TECHNICAL DETAILS)

a. Footings
b. Underground utilities
c. Rebar
d. Fire sprinklers.
e. Ceiling above t-bar
f. Welding
g. Roof/metal deck
h. Roofing
i. Insulation
j. Rated wall penetrations
k. Rated doors and access panels
l. High voltage cable installation
m. High pot high voltage cables

2. The Contractor shall be responsible for reviewing all of the Contract Documents for any additional inspection requirements.

1.6 SUBMITTALS

A. Reports: Trustees' independent testing agency shall submit a certified written report of each inspection, test or similar service, to the Architect, the Trustees, the Contractor, and the Project Manager/ Inspector.

B. Report Data: Written reports of each inspection, test or similar service shall include, but not be limited to:

Date of issue
Project title and number
Name, address and telephone number of testing agency
Dates and locations of samples and tests or inspections
Names of individuals making the inspection or test
Designation of the Work and test method

Identification of product and Specification Section
Complete inspection or test data
Test results and an interpretation of test results
Ambient conditions at the time of sample-taking and testing
Comments or professional opinion as to whether inspected or tested
Work complies with Contract Document requirements
Name and signature of laboratory inspector
Recommendations on retesting.

1.7 SCHEDULES FOR TESTING

A. Testing and Inspection Schedule: After discussion with University's Representative and Testing Agency in advance of performance of testing and inspection services, Contractor shall determine dates and times necessary for Testing Agency to schedule performance of required tests and inspections and determine due dates for issuance of reports.

1. Integrate Testing and Inspection Schedule with the Weekly Project Meetings specified in Section 01 31 19 – Project Meetings

2. Determine and indicate in Testing and Inspection Schedule necessary time for preparation and submission of reports of tests and inspections.

B. Revising Testing and Inspection Schedule: When changes of the construction schedule are necessary during construction, coordinate all such changes of schedule with the testing Agency as required.

C. Adherence to Testing and Inspection Schedule: When the Testing Agency is ready to test according to the determined schedule but is prevented from testing or taking specimens due to incompleteness of the work, all extra costs for testing attributed to the delay may be back-charged to the Contractor and shall not be borne by the University.

1.8 CONTRACTOR'S RESPONSIBILITIES

A. Contractor's Responsibilities for Inspections and Tests:

1. Notify Project Inspector and Testing Agency formally with an Inspection Request form a minimum of two (2) business days in advance of expected time for operations requiring inspection and testing services.
   a. State Fire Marshal Inspections - Notify Project Inspector a minimum of three (3) business days in advance of expected State Fire Marshal Inspections.
   b. Architect/Engineer Review – Notify Project Inspector, University Project Manager and Archiect/Engineer a min of five (5) business days in advance of all quality assurance reviews.

2. Deliver to Testing Agency or designated location, adequate samples of materials proposed to be used which require advance testing, together with proposed mix designs or other approved submittal documentation.

3. Cooperate with University's Representative, Testing Agency, Project Inspector, Architect, Architect's consultants and other responsible design professionals. Provide access to Work areas and off-site fabrication and assembly locations, including during weekends and after normal

work hours.

4. Provide incidental labor and facilities to provide safe access to Work to be inspected and tested, to obtain and handle samples at the Work site or at source of products to be tested, and to store and cure test samples.

5. Provide at least 14 days in advance of first inspection or test of each type, a schedule of tests or inspections indicating types of tests or inspections and their scheduled dates.

1.9 INSPECTIONS AND TESTS BY OTHERS

A. Inspections by Others: Refer to Section 01 45 00 - Quality Control for requirements regarding observations and inspections by University's Representative, Architect and Project Inspector.

B. Tests by Others: Refer to Section 01 45 00 - Quality Control and individual product Specifications Sections for requirements regarding tests and inspections by product manufacturers and others, including serving utilities.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

3.1 REPAIR AND PROTECTION

A. Repair and Protection: Upon completion of inspection, testing, sample-taking and similar services, repair damaged construction and restore substrates and finishes to eliminate all deficiencies, including deficiencies in visual qualities of exposed finishes. Comply with Contract Document requirements for "Cutting and Patching."

1. Protect construction exposed by or for quality control and quality assurance service activities, and protect repaired construction.

2. Repair and protection of all installed and stored materials is the Contractor's responsibility, regardless of the assignment of responsibility for inspection, testing or similar services.

END OF SECTION

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CSU The California State University
OFFICE OF THE CHANCELLOR

Statement of Required Project Special Inspections

Project: ....
Location: ....
Owner: ....
Architect ....

This Statement of Special Inspections is submitted in fulfillment of the requirements of CBC Sections 1704 and 1705 and summarizes the special inspections and tests required for this project. Additional tests and inspections may be called for at the discretion of the (deputy) building official.

This statement includes:

- Attachment A: List of the Testing Agencies and Inspectors retained to conduct the tests and inspections.
- Attachment B: Schedule of Special Inspections and tests applicable to this project:

☐ Special Inspections per Sections 1704 and 1705
☐ Special inspections for Seismic Resistance
☐ Special inspections for Wind Resistance

Special inspections and testing shall be performed in accordance with the approved plans and specifications, this statement, and CBC sections 1704, 1705, 1707, and 1708. No less than CBC minimum requirements shall be observed.

Interim reports shall be coordinated by the project Inspector of Record (IOR) and submitted to the (Deputy) Building Official and Architect in accordance with CBC Section 1704.1.2.

A final report of special inspections and confirmation of resolution of discrepancies noted in the inspections shall be submitted by the IOR to the (Deputy) Building Official and Architect. The (Deputy) Building Official shall review and approve the final report as a prerequisite to the issuance of a Certificate of Occupancy.

Responsibility for payment for inspections and testing is defined in project agreements. Typically, it is trustee policy to pay for all initial and reasonable back check inspections.

This Statement has been developed with the understanding that the (Deputy) Building Official will personally or by delegation:

- Review and approve the qualifications of the Special Inspectors who will perform the inspections.
- Monitor special inspection activities on the job site to confirm that the Special Inspectors are qualified and are performing their duties as called for in this Statement of Special Inspection.
- Review submitted inspection reports, confirming resolution of discrepancies as the work progresses.
- Perform all inspections as required by the CBC building code and additionally as identified herein.

Architect:

____
Signature *Date*

Authorization by Trustees

____
Signature *Date*

(Deputy) Building Official Acceptance

____
Signature *Date*

Attachment A - Testing Agencies, and Inspectors

Testing agencies and special inspectors retained to conduct tests and inspection on this project.

Responsibility	Firm	Address, Telephone, e-mail
1. Special Inspection (except for geotechnical)	*To Be Determined*	
2. Material Testing	*To Be Determined*	
3. Geotechnical Inspections	*To Be Determined*	
4. Other	*To Be Determined*	
5. Other	*To Be Determined*	

☐ Additional pages attached (if checked)

Attachment B - Special Inspections for Seismic & Wind Resistance

Seismic Requirements (Section 1705.3.1)

Summary description of seismic force resisting system and designated seismic systems requiring special inspections as per Section 1705.3.:

[Sample: Steel moment frame on pier foundation]

Wind Requirements (Section 1705.4.1)

Summary description of main wind force resisting system and designated wind resisting components requiring special inspections in accordance with Section 1705.4.2.:

1. Roof cladding and roof framing connections.
2. Wall connections to roof and floor diaphragms and framing.
3. Roof and floor diaphragm systems, including collectors, drag struts and boundary elements.
4. Braced frame, moment frame and shear walls
5. Braced frame, moment frame and shear walls connections to foundations

[Sample: All elements identified in CBC 1705.4.2. above. No additional items.]

Attachment B - Schedule of Special Inspections *(cont.)*

Verification and Inspection. (1)	**C.**	**P**	**Notes**
1704.2.1:- Inspect fabricators fabrication and quality control procedures.	---	---	
Table 1704.3 - Steel			
1. Material verification of high-strength bolts, nuts, and washers.			
a. Identification markings to conform to ASTM standards specified in the approved construction documents.		X	
b. Manufacturer's certificate of compliance required.		X	
2. Inspection of high-strength bolting:			
a. Bearing type connections.		X	
b. Slip-critical connections	X	X	
3. Material verification of structural steel:			
a. Identification markings to conform to ASTM standards specified in the approved construction documents.	---	---	
b. Manufacturer's mill test reports	---	---	
4. Material verification of weld filler materials:			
a. Identification markings to conform to AWS designation listed in the WPS.	---	---	
b. Manufacturer's certificate of compliance required.	---	---	
5. Inspection of welding: a. Structural steel			
1) Complete and partial penetration groove welds.	X		
2) Multipass fillet welds	X		
3) Single-pass fillet welds>5/16"	X		
4) Single-pass fillet welds<=5/16"		X	
5) Floor and roof deck welds.		X	
b. Reinforcing steel			
1) Verification of weldability of reinforcing steel other than ASTM A 706.		X	
2) Reinforcing steel-resisting flexural and axial forces in intermediate and special moment frames, and boundary elements of special reinforced concrete shear walls, and shear reinforcement.	X		
3) Shear reinforcement.	X		
4) Other reinforcing steel		X	
6. Inspection of steel frame joint details for compliance with approved construction documents: a. Details such as bracing and stiffening. b. Member locations. c. Application of joint details at each connection.		X	

Verification and Inspection. (1)	**C.**	**P**	**Notes**
1704.3- Welded studs when used for structural diaphragms.		X	
1704.3- Welding of cold-formed sheet steel framing members.		X	
1704.3- Welding of stairs and railing systems		X	
Table 1704.4 - Concrete			
1. Inspection of reinforcing steel, including prestressing tendons and placement		X	
2. Inspection of reinforcing steel welding in accordance with Table 1704.3 Item 5b	---	---	
3. Inspect bolts to be installed in concrete prior to and during placement of concrete where allowable loads have been increased.	X		
4. Verifying use of required design mix.		X	
5. At time fresh concrete is sampled to fabricate specimens for strength tests, perform slump and air content tests and determine the temperature of the concrete.	X		
6. Inspection of concrete and shotcrete placement for proper application techniques.	X		
7. Inspection for maintenance of specified curing temperature and techniques.		X	
8. Inspection of prestressed concrete			
a. Application of prestressing forces	X		
b. Grouting of bonded prestressing tendons in the seismic-force-resisting system	X		
9. Erection of precast concrete members.		X	
10. Verification of in-situ concrete strength, prior to stressing of tendons in postensioned concrete and prior to removal of shores and forms from beams and structural slabs.		X	
11. Inspect formwork for shape, location, and dimensions of the concrete member being formed.		X	
Table 1704.5.1 - Level 1 Masonry Inspections.			
1. At the start of masonry construction verify the following to ensure compliance:			
a. Proportions of site-prepared mortar.		X	
b. Construction of mortar joints.		X	
c. Location of reinforcement, connectors, prestressing tendons, and anchorages.		X	
d. Prestressing technique.		X	
e. Grade and size of prestressing tendons and anchorages.		X	
2. Verify:			
a. Size and location of structural elements.		X	
b. Type, size, and location of anchors, including other details of anchorage of masonry to structural members, frames or other construction.		X	
c. Specified size, grade, and type of reinforcement.		X	

Verification and Inspection. (1)	**C.**	**P**	**Notes**
d. Welding of reinforcing bars.	X		
e. Protection of masonry during cold weather (temperature below 40 degrees F) or hot weather (temperature above 90 degrees F)		X	
f. Application and measurement of prestressing force.		X	
3. Prior to grouting verify the following to verify compliance.			
a. Grout space is clean.		X	
b. Placement of reinforcement and connectors and prestressing tendons and anchorages.		X	
c. Proportions of site-prepared grout and prestressing grout for bonded tendons.		X	
d. Construction of mortar joints.		X	
4. Verify grout placement to ensure compliance with code and construction document provisions.	X		
a. Observe grouting of prestressing bonded tendons.	X		
5. Observe preparation of required grout specimens, mortar specimens, and/or prisms.	X		
6. Verify compliance with required inspection provisions of the construction documents and the approved submittals.		X	
Table 1704.5.3 - Level 2 Masonry Inspections			
1. From the beginning of masonry construction the following shall be verified to ensure compliance:			
a. Proportions of site-prepared mortar, grout, and prestressing grout for bonded tendons.		X	
b. Placement of masonry units and construction of mortar joints.		X	
c. Placement of reinforcement, connectors and prestressing tendons and anchorages.		X	
d. Grout space prior to grouting.	X		
e. Placement of grout.	X		
f. Placement of prestressing grout.	X		
2. Verify:			
a. Size and location of structural elements.		X	
b. Type, size, and location of anchors, including other details of anchorage of masonry to structural members, frames and other construction.	X		
c. Specified size, grade, and type of reinforcement.		X	
d. Welding of reinforcing bars.	X		

Verification and Inspection. (1)	**C.**	**P**	**Notes**
e. Protection of masonry during cold weather (temperature below 40 degrees F) or hot weather (temperature above 90 degrees F)		X	
f. Application and measurement of prestressing force.	X		
3. Preparation of any required grout specimens, mortar specimens, and/or prisms shall be observed.	X		
4. Compliance with required provisions of construction documents and the approved submittals shall be verified.		X	
1704.6: -Inspect pre fabricated wood structural elements and assemblies in accordance with Section 1704.2	---	---	
1704.6:- Inspect site built assemblies.	---	---	
1704.6.1: Inspect high-load diaphragms :			
1. Verify grade and thickness of sheathing.	---	---	
2. Verify nominal size of framing members at adjoining panel edges.	---	---	
3. Verify: • Nail or staple diameter and length, • Number of fastener lines, • Spacing between fasteners in each line and at edge margins.	---	---	
Table 1704.7 - Inspection of Soils			
1. Verify materials below footings are adequate to achieve the desired bearing capacity.		X	
2. Verify excavations are extended to proper depth and have reached proper material.		X	
3. Perform classification and testing of controlled fill materials.		X	
4. Verify use of proper materials, densities and lift thicknesses during placement and compaction of controlled fill.	X		
5. Prior to placement of controlled fill, observe subgrade and verify that site has been prepared properly.		X	
Table 1704.8 - Pile Foundations			
1. Verify pile materials, sizes and lengths comply with the requirements.	X		
2. Determine capacities of test piles and conduct additional load tests, as required.	X		
3. Observe driving operations and maintain complete and accurate records for each pile.	X		

Verification and Inspection. (1)	**C.**	**P**	**Notes**
4. Verify locations of piles and their plumbness. a. Confirm type and size of hammer. b. Record number of blows per foot of penetration. c. Determine required penetrations to achieve design capacity. d. Record tip and but elevations and record any pile damage.	X		
5. For steel piles, perform additional inspections in accordance with Section 1704.3.	---	---	
7. For specialty piles, perform additional inspections as determined by the registered design professional in responsible charge.	---	---	
8. For augered uncased piles and caisson piles, perform inspections in accordance with Section 1704.9.	---	---	
Table 1704.9 - Pier Foundations			
1. Observe drilling operations and maintain complete and accurate records for each pier.	X		
2. Verify locations of piers and their plumbness. Confirm: • Pier diameters, • Bell diameters (if applicable), • Lengths, embedment into bedrock (if applicable), • Adequate end strata bearing capacity.	X		
1704.10- Sprayed fire-resistant materials			
1. Inspect surface for accordance with the approved fire-resistance design and the approved manufacturer's written instructions.	---	---	
2. Verify minimum ambient temperature before and after application.	---	---	
3. Verify ventilation of area during and after application.		X	
4. Measure average thickness per ASTM E605 and Section 1704.10.3.	---	---	
5. Verify density of material for conformance with the approved fire-resistant design and ASTM E605.	---	---	
6. Test cohesive/adhesive bond strength per Section 1704.10.5.	---	---	
1704.11- Mastic and intumecent fire-resistant coating	---	---	
1704.12- Exterior insulation and finish systems (EIFS):-	---	---	
1704.13- Alternate materials and systems.	---	---	
1704.14- Smoke Control System	---	---	
1705.3: Seismic Resistance.			
1705.3 [4.3]:- Suspended ceiling systems and their anchorage.	---	---	

Verification and Inspection. (1)	**C.**	**P**	**Notes**
1705.4 Wind Resistance			
1705.4.2 1. Roof cladding and roof framing connections.	---	---	
2. Wall connections to roof and floor diaphragms and framing.	---	---	
3. Roof and floor diaphragm systems, including collectors, drag struts and boundary elements	---	---	
4. Vertical wind force-resisting systems, including braced frames, moment frames, and shear walls.	---	---	
5. Wind force-resisting system connections to the foundation.	---	---	
6. Fabrication and installation of systems or components required to meet the impact resistance requirements of Section 1609.1.2	---	---	
Special Inspections for Seismic Resistance.			
1707.2:- Special inspection for welding in accordance with AISC 341.	X		
1707.3:- Structural Wood.			
1. Inspect field gluing operations of elements of the seismic-force-resisting system.	X		
2. Inspect nailing, bolting, anchoring, and other fastening of components within the seismic-force-resisting system, including: • wood shear walls, • wood diaphragms, • drag struts, braces, • shear panels, • hold-downs.		X	
1707.4:- Cold-formed steel framing:			
1. Welding of elements of the seismic-force-resisting system.		X	
2. Inspection of screw attachments, bolting, anchoring, and other fastening of components within the seismic-force-resisting system including struts, braces, and hold-downs.		X	
1707.5: Pier Foundations			
1. Placement of reinforcing		X	
2. Placement of concrete	X		
1707.6:- Anchorage of storage racks and access floors 8 feet or greater in height.		X	
1707.7: Architectural Components			
1. Inspect erection and fastening of exterior cladding weighing more than 5 psf.		X	
2. Inspect erection and fastening of interior and exterior non-bearing walls weighing more than 15 psf.		X	
3. Inspect erection and fastening of interior and exterior veneer weighing more than 5 psf.		X	
1707.8: Mechanical and electrical components.			
1. Inspect anchorage of electrical equipment for emergency or stand-by power systems.		X	

Verification and Inspection. (1)	**C.**	**P**	**Notes**
2. Inspect anchorage of non-emergency electrical equipment		X	
3. Inspect installation of piping systems and associated mechanical units carrying flammable, combustible, or highly toxic contents.		X	
4. Inspect installation of HVAC ductwork that contains hazardous materials.		X	
5. Inspect installation of vibration isolation systems where required by Section 1707.8.		X	
1707.9: Verify that the equipment label and anchorage or mounting conforms to the certificate of compliance when mechanical and electrical equipment must be seismically qualified.	---	---	
1707.10- Seismic isolation system:- Inspection of isolation system per ASCE 7 – Section 17.2.4.8		X	
1708.1: Masonry Testing for Seismic Resistance			
1708.1.1- Verify certificates of compliance prior to construction.	---	---	
1708.1.2- Verification of f'_m and f'_{AAC} prior to construction.	---	---	
1708.1.2- Verification of f'_m and f'_{AAC} every 5000 square feet during construction.		X	
1708.1.4- Verification of proportions of materials in mortar and grout as delivered to the site.	---	---	
1708.3- Obtain mill certificates for reinforcing steel, verify compliance with approved construction documents, and verify steel supplied corresponds to certificate.	---	---	
1708.4- Structural Steel: - Invoke the QAP Quality Assurance requirements in AISC 341.	---	---	
1708.5- Obtain certificate that equipment has been tested per Section1708.5.	---	---	
1708.6- Obtain system tests as required by ASCE 7 Section 17.8	---	---	

Notation:
Column headers:

- C — Indicates continuous inspection is required.
- P — Indicates periodic inspections are required. The Notes and or contract documents should clarify.

Box entries:

- X — Is placed in the appropriate column to denote either "C" continuous or "P" periodic inspections.
- --- — Denotes an activity that is either a one-time activity or one whose frequency is defined in some other manner.

<u>Notes:</u>

(1) Additional detail regarding inspections and tests are provided in the project specifications and construction documents

The California State University
OFFICE OF THE CHANCELLOR

Attachment B - Schedule of Special Inspection *(cont.)*

Other Inspections

Other inspections not listed previously or additional notes
[none]

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SECTION 01 51 00 - TEMPORARY UTILITIES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Temporary utilities and services, including:

1. Heating and cooling during construction
2. Ventilation during construction
3. Temporary water service
4. Temporary sanitary facilities
5. Temporary power and lighting
6. Construction telephone service.
7. Contractor metering and payment of Electrical and Water service usage

B. Removal of temporary utilities.

1.3 RELATED SECTIONS

A. Section 01 11 00 - Summary of the Work: Contractor's use of site and premises.

B. Section 01 33 00 – Submittal Procedures

1.4 SUBMITTALS

A. Temporary Utilities: Submit reports of tests, inspections, applicable meter readings and similar procedures performed on temporary utilities.

1.5 TEMPORARY UTILITIES AND SERVICES

****

CHECK FOR PROJECT SPECIFIC REQUIREMENT

****

A. Temporary Utilities and Services, General: All utilities and other services necessary for proper performance of the Work shall be provided by Contractor, unless specifically noted otherwise. Refer to Contract General Conditions, Article 4.11. Temporary utilities and services shall conform to all applicable requirements of authorities having jurisdiction and serving utility companies and agencies, including the following:

1. Requirements of authorities having jurisdiction, including:
   a. Cal OSHA
   b. California Building Code (CBC) requirements
   c. Health and safety regulations
   d. Utility agency and company regulations
   e. Police, Fire Department and Rescue Squad rules
   f. Environmental protection regulations

2. Standards:
   a. NFPA Document 241 - Building Construction and Demolition Activities.
   b. ANSI A10 Series - Safety Requirements for Construction and Demolition.
   c. NECA Electrical Design Library - Temporary Electrical Facilities.
   d. Electrical Service: Comply with NEMA, NECA and UL standards and regulations for temporary electric service. Install service in compliance with California Electrical Code (CEC).
   e. Piped or Plumbed systems – Install service in accordance with California Plumbing Code (CPC)

B. Inspections: Arrange for authorities having jurisdiction to inspect and test each temporary utility before use. Obtain required certifications and permits.

C. Temporary Connections and Fees: Contractor shall arrange for services and pay all fees and service charges for temporary power, water, sewer, gas and other utility services necessary for the Work.

1. Contractor shall apply for and obtain permits for temporary utilities, including permits for temporary generators, from authorities having jurisdiction.

2. All costs for temporary connections, including fees charged by serving utilities, shall be included in Contract Sum. Rates to be included in the contract sum shall be as follows:
   a. Electrical power consumption will be billed monthly to the contractor at $0.13 per khw (kilowatt hour)
   b. Water consumption will be billed monthly to the Contractor at $6.50 per hcf (hundred cubic feet)

D. Permanent Connections and Fees: Contractor shall arrange for utility agencies and companies to make permanent connections. University will arrange for permanent utility account and pay permanent connection fees. After Contract Completion review and determination that Work is acceptable, University will pay utility service charges for services delivered through permanent connections, for normal quantities.

E. Use of Temporary Utilities: Enforce strict discipline in use of temporary utilities to conserve on consumption. Limit use of temporary utilities to essential and intended uses to minimize waste and abuse.

1.6 PROJECT CONDITIONS

A. Conditions of Use: Keep temporary services and facilities clean and neat in appearance. Operate in a safe and efficient manner. Take necessary fire prevention measures. Do not overload facilities, or permit them to interfere with progress. Do not allow hazardous, dangerous, or unsanitary conditions, or public nuisances to develop or persist on the site.

1.7 HEATING AND COOLING

A. Temporary Heating and Cooling: Provide and pay for temporary heating and cooling devices, fuel and related service charges to provide ambient temperatures as required to maintain conditions necessary for proper performance of construction activities.

B. Use of Permanent Heating and Cooling Systems: Permanent heating and cooling equipment may be used after completion, testing and inspection of systems and approval of code authorities having jurisdiction.

1. Prior to operation of permanent heating equipment for temporary heating purposes, verify that installation is approved for operation, equipment is lubricated and filters are in place.

2. Contractor shall provide and pay for operation, maintenance and regular replacement of filters and worn or consumed parts.

3. Immediately prior to Contract Completion review, change disposable filters and clean permanent filters of equipment used during construction.

4. Contractor use of the equipment does not start the warrantee period. Refer to Specification section 01 78 33 for additional information.

C. Temperature Criteria: Maintain interior ambient temperature of minimum 50 degrees F and maximum 80 degrees F, unless otherwise specified or approved by University's Representative.

1.8 VENTILATION DURING CONSTRUCTION

A. Ventilation During Construction: Provide and pay for temporary ventilation devices, energy and related service charges.

B. Use of Permanent Ventilation Systems: The Contractor may use permanent ventilation equipment after completion, testing and inspection of systems and approval by University's Representative and authorities having jurisdiction.

1. Prior to operation of permanent ventilation equipment for ventilation purposes during construction, Contractor shall verify that equipment is lubricated and filters are in place.

2. Contractor shall provide and pay for maintenance and regular replacement of filters and worn or consumed parts of permanent ventilation system using for ventilation during construction.

3. Immediately prior to Contract Completion review, Contractor shall change disposable filters and clean permanent filters of equipment used during construction.

4. Contractor use of the equipment does not start the warrantee period. Refer to Specification

section 01 78 33 for additional information.

C. Ventilation Criteria: Ventilate enclosed areas to assist cure of materials, to dissipate humidity and to prevent accumulation of dust, fumes, vapors and gases, as necessary for proper performance of the Work.

1.9 TEMPORARY WATER SERVICE

A. Temporary Water Service: Contractor shall locate, with the assistance of the University, and connect to existing water source for temporary construction water service. Contractor shall comply with the following:

1. Locate and connect to existing water source for temporary construction water service, as acceptable to University's Representative.

2. Extend branch piping with outlets located, so that water is available by use of hoses.

3. Temporary water service piping, valves, fittings and meters shall comply with requirements of the serving water utility and California Plumbing Code (CPC).

4. All costs to establish temporary construction water system shall be included in the Contract Sum, of if so specified, costs shall be paid from Allowance specified in Section 01 21 00 - Allowance Procedures.

B. Use of Permanent Water System: Permanent water system may be used for construction water after completion, sterilization, testing and inspection of system and approval by University's Representative and authorities having jurisdiction.

1. Contractor use of the permanent water system does not start the warrantee period. Refer to Specification section 01 78 33 for additional information.

1.10 TEMPORARY SANITARY FACILITIES

A. Temporary Sanitary Facilities: Provide and maintain adequate temporary sanitary facilities and enclosures for use by construction personnel.

1. Number of temporary toilets shall be suitable for number of workers.

2. Provide wash-up sink with soap, towels and waste disposal.

B. Use of Permanent Sanitary Facilities: Do not use permanent sanitary facilities unless approved by University's Representative. Immediately prior to Contract Completion review, thoroughly clean and sanitize permanent sanitary facilities used during construction.

1.11 TEMPORARY POWER AND LIGHTING

A. Temporary Power and Lighting, General: Comply with NECA Electrical Design Library - Temporary Electrical Facilities.

B. Temporary Power: Provide electric service as required for construction operations, with branch wiring and distribution boxes located to provide electrical service for performance of the Work.

1. Provide temporary electric feeder connected to University operated electric utility service at location determined by the Contractor and as approved by the University electric utility.

2. Temporary power conduit, raceways, fittings, conductors, panels, connections, disconnects, overcurrent protection, outlets and meters shall comply with requirements of the serving electric utility, California Electrical Code (CEC) and requirements of authorities having jurisdiction.

3. Contractor shall pay all costs to establish temporary electric service, or if so specified, costs of temporary power shall be paid from Allowance specified in Section 01 21 00 - Allowance Procedures.

4. As necessary in order to maintain construction progress, Contractor shall provide and pay all costs associated with generators used for temporary power.

C. Temporary Lighting: Provide temporary lighting as necessary for proper performance of construction activities and for inspection of the Work.

1. Provide branch wiring from power source to distribution boxes with lighting conductors, pigtails, and lamps as required.

2. Maintain lighting and provide routine repairs.

D. Protection: Provide weatherproof enclosures for power and lighting components as necessary. Provide overcurrent and ground-fault circuit protection, branch wiring and distribution boxes located to allow convenient and safe service about site of the Work. Provide flexible power cords as required.

E. Use of Permanent Power and Lighting Systems: Permanent power and lighting systems may be used after completion, testing and inspection of systems and approval by University's Representative and authorities having jurisdiction.

1. Contractor shall maintain lighting and make routine repairs and replacements as necessary.

2. Contractor shall pay for electricity consumed after permanent power system is operational and approved by authorities having jurisdiction.

F. Service Disruptions: When necessary for energizing and de-energizing temporary electric power systems, minimize disruption of service to those served by public mains. Schedule transfers at times convenient to University and to occupants.

G. Relamping: For permanent lighting used during construction, relamp all fixtures immediately prior to Contract Completion (punch list) review.

1.12 CONSTRUCTION TELEPHONE/DATA SERVICE

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VERIFY ALL COSTS WITH CAL POLY ITS

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A. Request and pay for telephone/data and fax facilities available for the duration of contract where the Contractor and its superintendent may be contacted.

B. Connect to and use University's phone and internet system.

1. Request and pay for phone/data installation through the Trustees Representative. Approximate costs are as follows:

a.	Tele/data lines to each trailer	$1,000 each
b.	Phone or Data connection	$ 85 each
c.	Telephone instrument	$ 350 each

2. Pay for phone sets, connection, data racks and servers and use costs.
3. Contractor will be billed directly for actual Telecommunications charges.

C. Option: Use of cellular telephone, with Trustees Representative approval.

1. Include voice message services. Contractor shall provide for cost of all services.

PART 2 - PRODUCTS

2.1 MATERIALS AND EQUIPMENT

A. Materials: Contractor shall provide new materials. If acceptable to the Architect, undamaged previously used materials in serviceable condition may be used. Provide materials that are suitable for the use intended. Their use and methods of installation shall not create unsafe conditions or violate requirements of applicable codes and standards.

B. Equipment: Contractor shall provide new equipment; or, if acceptable in writing by the Trustees, Contractor may provide undamaged, previously used equipment in serviceable condition. Provide equipment that is suitable for use intended.

PART 3 - EXECUTION

3.1 TEMPORARY UTILITIES INSTALLATION

A. Temporary Utilities Installation, General: Contractor shall engage the appropriate local utility company or personnel to install temporary service or connect to existing service.

1. Use Charges: Cost or use charges for temporary facilities are the Contractor's responsibility.

2. Allowance for Utilities Charges: When Contract includes an allowance for metering of utility services, whether through temporary or permanent facilities, unused amount shall be returned to the Trustees by deductive change order.

B. Water Service: Contractor may take water from the University's systems in such quantities and at such times as they are available. If this is done, Contractor shall provide all temporary materials necessary to extending the utility to where they will be used. Contractor shall install a meter and reimburse the University for any water used.

C. Temporary Electric Power Service: Contractor may take electricity from the University's system if available. If this is done, Contractor shall provide all equipment, including connections, and other materials necessary for extending the utility lines to where they will be used. Contractor shall coordinate the installation with the University's Representative. Contractor shall install a meter and reimburse the University for any power used. Where sub-metering is not possible or practical, a flat fee may be established and paid to the University.

1. When not available from the University, the Contractor must arrange and pay for electric service through the local utility or furnish his own portable power.

2. All permanent power used by the Contractor prior to Occupancy by the Trustees shall be metered and paid for by the Contractor.

D. Temporary Telephones: Contractor shall have telephone facility available at its business office for the duration of contract where the Contractor and its superintendent may be contacted. A pay phone for use of subcontractors is recommended.

E. Temporary Fire Protection: Until fire protection needs are supplied by permanent facilities, Contractor shall install and maintain temporary fire protection facilities of the types needed to protect against reasonably predictable and controllable fire losses. Contractor shall comply with NFPA 10 "Standard for Portable Fire Extinguishers," and NFPA 241 "Standard for Safeguarding Construction, Alterations and Demolition Operations." Contractor shall:

1. Locate fire extinguishers where convenient and effective for their intended purpose, but not less than one extinguisher on each floor at or near each usable stairwell.

2. Store combustible materials in containers in fire-safe locations.

3. Maintain unobstructed access to fire extinguishers, fire hydrants, temporary fire protection facilities, stairways and other access routes for fighting fires. Prohibit smoking in hazardous fire exposure areas.

4. Provide supervision of welding operations, combustion type temporary heating units, and similar sources of fire ignition.

F. Maintenance of Temporary Utilities and Services: Contractor shall maintain temporary utilities and services in good operating condition until removal. Contractor shall protect from utilities and services from environmental and physical damage.

3.2 TERMINATION AND REMOVAL OF TEMPORARY UTILITIES AND SERVICES

A. Termination and Removal of Temporary Utilities and Services: Unless the Trustees require that it be maintained longer, Contractor shall remove each temporary facility when the need has ended, or when replaced by authorized use of a permanent facility, or no later than Completion. Contractor shall complete or, if necessary, restore permanent construction that may have been delayed because of interference with the temporary facility. At Completion, Contractor shall clean and renovate permanent facilities that have been used during the construction period.

B. Removal of Temporary Underground Utilities and Restoration: Remove temporary underground utility installations to a minimum depth of two-feet below utility services. Contractor shall:

1. Backfill, compact and regrade site as necessary to restore areas or to prepare for indicated paving and landscaping.

2. Restore paving damaged by temporary utilities. Refer to requirements specified in Section 01732 - Cutting and Patching Requirements.

C. Cleaning and Repairs: Contractor shall clean exposed surfaces and repair damage caused by installation and use of temporary utilities and services. Where determined by University's Representative that repair of damage is unsatisfactory, Work, Contractor shall replace construction with matching finishes. Refer to requirements specified in Section 01 73 29 - Cutting and Patching Requirements.

END OF SECTION

SECTION 01 52 00 - CONSTRUCTION FACILITIES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Field offices and sheds.

B. Removal of construction facilities.

1.3 RELATED SECTIONS

A. Section 01 11 00 - Summary of the Work: Contractor's use of site and premises.

B. Section 01 51 00 - Temporary Utilities: Water, power and telephone services to construction facilities.

C. Section 01 52 05 - Construction Staging Areas: Locations for field offices and sheds.

D. Section 01 74 00 - Cleaning Requirements: Cleaning during construction and final cleaning.

1.4 MAINTENANCE OF CONSTRUCTION FACILITIES CONTROLS

A. Maintenance: Contractor shall maintain construction facilities in proper and safe condition throughout progress of the Work.

B. Replacement: In the event of loss or damage, Contractor shall promptly restore temporary construction facilities by repair or replacement at no change in the Contract Sum or Contract Time.

1.5 CONTRACTOR'S FIELD OFFICES AND SHEDS

A. Contractor's Field Office: Contractor shall provide a mobile field office of weather-tight construction, with lighting, power, ventilation, heating and cooling to house Contractor. Unless otherwise indicated on the Drawings, Contractor shall locate field office at in staging area described in Section 01 52 05 - Construction Staging Areas. Contractor shall comply with University's requirements transmitted through University's Representative.

1. Contractor shall provide temporary utilities to serve Contractor's field office. Refer to Section 01 51 00 - Temporary Utilities. Contractor shall connect to the University Telephone and Data service.

2. Contractor's Field Office shall present neat, business-like appearance at all times, internally and

externally.

3. Contractor shall ensure that neither Contractor's Field Office nor other jobsite facilities are used for living quarters.

B. Storage Sheds for Tools, Materials, and Equipment: Contractor shall provide weather-tight sheds, all with the following:
   1. Heat and ventilation appropriate for storage of products requiring controlled conditions,
   2. Adequate space for organized storage and access, and
   3. Lighting for inspection of stored materials.

C. Layout of Field Offices and Sheds: Within seven (7) calendar days of the Notice to Proceed, Contractor shall submit to University's Representative a proposed layout for field offices, sheds and storage areas. University's Representative will review and respond within seven (7) calendar days with comments and directions. Contractor shall comply with directions of University's Representative.

****
THE FOLLOWING ARE SAMPLE SPECIFICATIONS; EDIT TO SUIT PROJECT REQUIREMENTS.
****

1.6 UNIVERSITY'S CONSTRUCTION MANAGEMENT FIELD OFFICE

A. General: Contractor shall provide a field office for use by University’s Inspector of Record and/or Construction Management team for the duration of the Contract, equipped and furnished as specified below. [*_DEPENDING ON THE PROJECT SIZE, A SINGLE CONSTRUCTION TRAILER MAY BE UTILIZED AND SHARED BETWEEN THE CONTRACTOR AND IOR, CM…_]*

   1. Contractor shall pay for all temporary water and power services, in accordance with Section 01 51 00 - Temporary Utilities.

   2. Contractor shall provide and pay for twice weekly cleaning services, including trash removal and restocking of toilet facility consumables. Contractor shall provide and pay for emptying sewage holding tank and related services on an as-needed basis, but not less frequently than each week.

   3. Contractor’s initial progress payment for Work under the Contract will not be approved until University's Field Office is fully equipped and functional.

   4. Unless otherwise directed in writing by University's Representative, University's Field Office, including furnishings and equipment provided by Contractor, shall remain operational until execution or recording of Notice of Completion.

   5. With 14 days of written direction by University's Representative or within 14 days of execution or recording of Notice of Completion, whichever is earliest, Contractor shall take possession and remove University's Field Office from the campus.

   6. University's representatives shall have the right to use University's Construction Management

Field Office, including furnishings and equipment, for the purpose of construction contract administration, testing and inspection for Work under this and any other contract, or other University business, at no change in Contract Sum and Contract Time.

B. Construction: Contractor shall provide the following:

1. Field office of pre-fabricated, weather-tight construction, approximately 12 feet wide by 60 feet long, with lockable entrances, operable windows and serviceable finishes. Set field office on foundations suitable for normal office loadings, with tie-downs to resist wind and seismic forces. Provide field office of non-combustible construction where located within 30 feet of building lines. Comply with NFPA 241. Field office shall be capable of maintaining 68 to 78 degrees Fahrenheit interior year round.

2. Field office with two exit doors, with cylinder locks and latch guards.

3. Within field office, provide the following rooms:
   a. Two private offices, approximately 120 square feet each.
   b. Conference room of minimum 400 square feet.
4. Private toilet facilities, complete with water closet, lavatory with hot and cold running water, medicine cabinet with mirror and dispensers for toilet paper and paper towels.

5. Each private office and conference room with operable windows, at least one on each side equipped with blinds, insect screens.

6. All plumbing, HVAC, power, lighting systems and telecommunications wiring and outlets as necessary for complete and habitable use.

7. Properly configured, NEMA-polarized electrical outlets which prevent insertion of 110- to 120-volt plugs into higher-voltage outlets. Equip outlets with ground-fault circuit interrupters (GFCI), having reset button and pilot light in accordance with all applicable building codes.

8. Ceiling-mounted fluorescent lighting fixtures, capable of providing uniform lighting of minimum 50 lumens at level 30-inches above floor.

9. Provide heating and air conditioning unit mounted on end of trailer; of sufficient function, capacity and ductwork for equal distribution of air conditioning to all rooms. Roof-mounted units are not acceptable. Unit must be capable of maintaining 68 to 78 degrees F interior to year-round.

C. Furnishings: Contractor shall provide the following furnishings.

1. Door mats: One per entrance, heavy-duty cocoa mat suitable for heavy use and removal of dirt and mud.

2. Coat rack: Wall mounted, with shelf and hanging rod with twelve hangers.

3. Folding tables: Four each 36-inches by 72-inches and two each 30-inches by 72-inches, heavy duty, with wood grain plastic laminate top.

4. Folding chairs: Twelve each, heavy duty, with padded seats.

5. Desks, per office: One each, 36-inches by 72-inches, double pedestal, painted steel with resilient writing surface top.

6. Desk chairs, per desk: One each, ergonomic design, heavy duty, wheeled pedestals, with adjustable back angle, seat angle and arm height.

7. File cabinets: Four 4-drawer, legal-size vertical file cabinets, with lockable drawers.

8. Bookcases: Four each, 84-inches high by 36-inches wide by 13-inches deep, with five adjustable shelves.

9. Plan racks: Two each, factory-manufactured mobile stand by PlanHold or equal, with 24 removable drawing clamps each.

10. Plan tables: Field-fabricated by Contractor, with top constructed from 35-inch by 84-inch solid core, 1-3/8 inch thick with tempered hardboard faces, and wood or steel support structure, located where directed by University's Representative.

11. Markerboards: Four each, 36-inches wide by 48-inches high, with white markerboard suitable for oil- or water-base markers.

12. Tackboards: Four each, 36-inches wide by 48-inches high, with wood fiberboard core and burlap grain vinyl facing, color as selected by University's Representative.

D. Equipment: Contractor shall provide the following equipment. University shall be permitted to remove any equipment from field office and use elsewhere. All equipment shall be new and no substitutions or deviations from specified descriptions will be acceptable. Equipment will be returned by University prior to Contract close-out. At Contract close-out, University shall have option to purchase equipment at depreciated, fair-market value negotiated with Contractor.

1. Fire extinguisher: Portable, UL-listed and labeled, complying with NFPA 10 and NFPA 241 for classification, extinguishing agent and size as necessary for location and class of fire exposure, minimum UL Rating 4A-60BC (nominal 10 pound capacity).

2. Drinking water: Containerized, hot and chilled water tap-dispenser with paper cup dispenser, with bottled water units and paper cup supply as necessary. Contractor shall provide weekly restocking of water and paper cups.

3. Refrigerator: Minimum 3.2 cubic feet capacity, compact refrigerator with internal freezer compartment, white color.

4. Microwave oven: Countertop design, white color.

5. Coffee maker: One each, 12-cup capacity.

6. Color printer/FAX/copier: One each, to be located in private offices, as manufactured by Hewlett-Packard, H-P OfficeJet Model G85 or current equivalent model, 3-year manufacturer's "Next Day Exchange" warranty, with black and tri-color ink cartridges. Contractor shall provide all consumables, including inkjet-suitable paper, for duration of Contract. Printer/fax/copier shall connect to personal computer and service printer for computer as well as fax machine and copier.

7. Telecommunications:
   a. Provide three (3) telecommunication lines minimum for each office, connected to campus network system.
   b. Provide mounting backboard in a secure location for the Owner installation of a network distribution rack to campus system. Contractor shall make all final connections and label all jacks and cables according to the California State University Telecommunications Infrastructure Planning Standards (CSU TIPS). Standards may be found at the following web site, http://www.calstate.edu/cpdc/ae/gsf/TIP_Guidelines/, or may be obtained by written request.

E. Miscellaneous: Contractor shall provide the following. University shall be permitted to remove any miscellaneous products from field office and for use elsewhere. All miscellaneous products shall be new and will be returned by University prior to Contract close-out. At Contract close-out, University shall have option to purchase miscellaneous products at depreciated, fair-market value negotiated with Contractor.

1. Flashlights: Two each, MagLite tubular aluminum flashlights, for three D-size batteries. Include replacement batteries.

2. Hardhats: Five each, Class B hardhats, Fibre-Metal or equal.

3. First Aid Supplies: Comply with industrial safety regulations.

PART 2 - PRODUCTS

Not applicable to this Section.

PART 3 - EXECUTION

3.1 INSTALLATION OF CONSTRUCTION FACILITIES

A. Layout of Field Offices and Sheds: Within seven (7) calendar days of the Notice to Proceed, Contractor shall submit to University's Representative a proposed layout for field offices, sheds and storage areas. University's Representative will review and respond within five working days with comments and directions. Contractor shall comply with directions of University's Representative.

1. Coordinate with requirements specified in Section 01 52 05 - Construction Staging Areas.

2. Coordinate installation of construction fencing as specified in Section 01 56 00 - Temporary Barriers and Enclosures.

B. Installation of University's Field Office: Provide field office ready for use within 20 working days of commencement date stated in Notice to Proceed or Notice of Award, whichever is earliest.

3.2 REMOVAL OF CONSTRUCTION FACILITIES

A. Removal of Construction Facilities: Unless otherwise mutually agreed by University's Representative and Contractor, remove temporary materials, equipment, services, and construction prior to Contract Completion review.

1. Coordinate removal with requirements specified in Section 01 51 00 - Temporary Utilities, Section 01 52 00 - Construction Facilities, Section 01 55 00 - Vehicular Access and Parking and Section 01 56 00 - Temporary Barriers and Enclosures.

2. Completely remove in-ground construction facilities. Backfill, compact and regrade site as necessary to restore areas or to prepare for indicated paving and landscaping.

B. Cleaning and Repairs: Clean and repair damage caused by installation or use of temporary construction facilities on public and private rights-of-way.

END OF SECTION

SECTION 01 52 05 - CONSTRUCTION STAGING AREAS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Contractor Staging Area requirements.

1.3 RELATED SECTIONS

A. Section 01 11 00 - Summary of the Work: Contractor's use of site and premises.

B. Section 01 52 00 - Construction Facilities: Field offices and sheds.

C. Section 01 54 01 - Security

D. Section 01 55 00 - Vehicular Access and Parking: Construction parking.

E. Section 01 56 00 - Temporary Barriers and Enclosures: Temporary construction barriers, enclosures and passageways.

F. Section 01 57 00 - Temporary Controls: Storm water pollution prevention measures; video record of existing conditions to be used to determine restoration Work.

G. Section 01 58 00 - Project Identification and Signage: Directional and informational signage.

H. Section 01 74 00 - Cleaning Requirements: Periodic cleaning and cleaning for Substantial Completion review.

1.4 SUBMITTALS

A. Shop Drawings: Prior to site mobilization, Contractor shall prepare and submit for review by University's Representative a site plan indicating detailed layout of Contractor Staging Area, including:

1. Temporary utilities
2. Temporary fencing and gates
3. Temporary offices and sheds
4. Construction aids
5. Vehicular access ways, haul routes and on-site parking
6. Temporary barriers and enclosures
7. Storm water pollution prevention measures

PART 2 - PRODUCTS

Not applicable to this Section.

PART 3 - EXECUTION

3.1 CONTRACTOR STAGING AREA REQUIREMENTS

A. Contractor Staging Areas: Refer to reference drawings included in the set of Contract Drawings for location of Contractor Staging Areas.

1. Contractor shall use only site areas designated specifically by University as Contractor Staging Area for the Project.
2. Contractor Staging Area for the Project shall be clearly indicated by use of signage, delineators or other means acceptable to clearly identify the area**.** Contractor shall remove equipment placed or located outside of areas designated for Contractor Staging Area to within Contractor Staging Area at no change in Contract Time and Contract Sum.
3. Contractor shall keep access to Contractor Staging Areas and other construction access ways and thoroughfares clear at all times. Contractor shall provide traffic and parking control signage acceptable to University's Representative.
4. Contractor shall not impede access to/from any designated fire truck or emergency vehicle access lane at any time unless specifically granted by the University.

B. Cleanliness: Contractor shall keep Contractor Staging Area clear of trash and debris and in neat order. Contractor shall be responsible for cleanliness and order of assigned Contractor Staging Areas, as acceptable to University's Representative. Contractor shall clean and organize the area at no additional cost.

3.2 REMOVAL OF CONSTRUCTION FACILITIES AND TEMPORARY CONTROLS

A. Removal of Construction Facilities and Temporary Controls: Unless otherwise mutually agreed by University's Representative and Contractor, Contractor shall remove temporary materials, equipment, services, and construction prior to Contract Completion review. Contractor shall coordinate removal with requirements specified in Section 01 51 00 - Temporary Utilities, Section 01 52 00 - Construction Facilities, Section 01 55 00 - Vehicular Access and Parking and Section 01 56 00 - Temporary Barriers and Enclosures.

B. Cleaning and Repairs: Contractor shall clean and repair damage caused by installation or use of temporary facilities on public and private rights-of-way to the level of finish that was existing prior to the installation of temporary facilities. If there is no record of the prior condition, the finish shall be considered as new.

C. Removal of Temporary Utilities and Restoration: Contractor shall remove temporary underground utility installations. Backfill, compact and regrade site as necessary to restore areas or to prepare for indicated paving and landscaping.

END OF SECTION

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SECTION 01 54 00 - CONSTRUCTION AIDS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

****
EDIT LIST OF CONSTRUCTION AIDS BELOW TO SUIT PROJECT REQUIREMENTS.
****

A. Construction aids, including but not limited to:

1. Temporary lifts and hoists
2. Debris chutes
3. Temporary stairs
4. Scaffolding

1.3 RELATED SECTIONS

A. Section 01 11 00 - Summary of the Work: Contractor's use of site and premises

B. Section 01 56 00 - Temporary Barriers and Enclosures: Temporary construction barriers, enclosures and passageways

****
EDIT THE FOLLOWING TO SUIT PROJECT REQUIREMENTS.
****

C. [Section 14 21 00 - Electric Traction Elevators:] [Section 14 24 00 - Hydraulic Elevators:] Use of building elevators for construction activities.

1.4 CODES AND REGULATIONS

A. Safety Regulations: Contractor shall comply with requirements of all applicable Federal, State and local safety rules and regulations. Contractor shall be solely responsible for jobsite safety.

1.5 TEMPORARY LIFTS AND HOISTS

A. Temporary Lifts and Hoists: Contractor shall provide facilities for hoisting materials and personnel. Mobile lifts and truck cranes and similar devices used for hoisting materials are considered "tools and equipment" and not temporary facilities.

B. Temporary Elevator Usage: [Refer to [Section 14 21 00 - Electric Traction Elevators] [Section 14 24 00 - Hydraulic Elevators] for use of building elevator[s] during construction.

1. Contractor shall provide protective coverings, barriers, devices, signs, or other procedures to protect elevator car and entrance doors and frame.

2. Contractor shall clean and restore elevator cars used during construction.

3. If, despite such protection, elevators become damaged, Contractor shall engage (and Contract Sum shall include) elevator Installer to restore damaged work so no evidence remains of correction Work.

4. Contractor shall return items that cannot be refinished in field to the shop, make required repairs and refinish entire unit, or provide new units as required.

1.6 DEBRIS CHUTES

A. Debris Chutes: Contractor shall provide chutes as necessary for debris removal. Contractor shall:

1. Construct debris chutes of substantial materials. Use cylindrical, laminated fiber forms (Sonotube□ or equal) to minimize noise of debris removal.

2. Provide controls at debris chutes to minimize spread of dust and debris.

3. Limit use of debris chutes to times to minimize disruption of activities in adjacent spaces.

1.7 TEMPORARY STAIRS AND SCAFFOLDING

A. Temporary Stairs: Until permanent stairs are available, Contractor shall provide temporary stairs where ladders are not adequate. Contractor shall cover finished, permanent stairs with protective covering of plywood or similar material so finishes will be undamaged at time of Contract Completion review.

B. Permanent Stair Usage: Use of permanent stairs will be permitted, as long as Contractor cleans and maintains stairs in a condition acceptable to University's Representative.

1. Contractor shall provide protective coverings, barriers, devices, signs, or other procedures to protect stairs and to maintain means of egress.

2. If, despite such protection, stairs become damaged, Contractor shall restore damaged areas as acceptable to University's Representative.

3. Contractor shall coordinate usage of existing stairs at occupied facilities with University's Representative.

C. Scaffolding: Contractor shall provide scaffolding as necessary for access and proper performance of the Work. Design and installation of scaffolding shall be solely Contractor's responsibility.

PART 2 - PRODUCTS

Not applicable to this Section.

PART 3 - EXECUTION

3.1 MAINTENANCE OF CONSTRUCTION AIDS

A. Maintenance: Contractor shall use all means necessary to maintain construction aids in proper and safe condition throughout progress of the Work.

B. Replacement: In the event of loss or damage, Contractor shall promptly restore construction aids by repair or replacement at no change in the Contract Sum or Contract Time.

3.2 REMOVAL OF CONSTRUCTION AIDS

A. Removal of Construction Aids: Unless otherwise mutually agreed by University's Representative and Contractor, Contractor shall remove construction aids prior to Contract Completion review. Contractor shall coordinate removal with requirements specified in Section 01 51 00 - Temporary Utilities, Section 01 52 00 - Construction Facilities, Section 01 55 00 - Vehicular Access and Parking and Section 01 56 00 - Temporary Barriers and Enclosures.

B. Cleaning and Repairs: Contractor shall clean and repair damage caused by installation or use of construction aids.

END OF SECTION

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SECTION 01 55 00 - VEHICULAR ACCESS AND PARKING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Requirements for vehicular access to Work areas

B. Requirements for construction parking

1.3 RELATED SECTIONS

A. Section 01 11 00 - Summary of the Work: Contractor's use of site and premises.

B. Section 01 52 00 - Construction Facilities: Coordination of access to field offices and sheds.

C. Section 01 52 05 - Construction Staging Areas: Layout of construction staging area, including locations for vehicular access and construction parking.

D. Section 01 56 00 - Temporary Barriers and Enclosures: Requirements for temporary construction barriers, enclosures and passageways, applicable to construction parking areas.

E. Section 01 58 00 - Project Identification and Signage: Directional and informational signage.

F. Section 01 57 00 - Temporary Controls: Storm water pollution prevention measures; video record of existing conditions to be used to determine restoration Work.

G. Section 01 74 00 - Cleaning Requirements: Cleaning during construction and final cleaning.

1.4 PROTECTION OF EXISTING CONDITIONS

A. Protection of Adjacent Facilities: Contractor shall restrict Work to limits indicated on the Drawings and as specified in Section 01 11 00 - Summary of the Work. Contractor shall protect existing, adjacent facilities from damage, including soiling and debris accumulation.

1.5 SITE ACCESS

A. Site Access: Use of designated existing on-site streets and driveways for construction traffic is permitted. Contractor shall review access routes with University Representative and comply with University Representative's directions.

1. Contractor shall ensure that tracked vehicles shall not use paved areas.

2. Contractor shall provide unimpeded access for emergency vehicles. Contractor shall maintain 20-foot (6 m) width driveways with turning space between and around combustible materials.

3. Contractor shall provide and maintain access to fire hydrants free of obstructions.

4. Contractor shall clean and restore paving and other site features after construction use.

B. Traffic Control:

1. Contractor shall comply with all on-campus traffic regulations, including speed limits. Contractor shall pay all parking and traffic fines.

2. Blockage of site roadways and access to site parking lots and parking structures shall be only with approval of University's Representative. Contractor shall comply with University's restrictions on blocking roadways and parking areas.

3. Contractor shall employ a minimum of two (2) trained and equipped flag persons to regulate traffic when construction operations or traffic encroach on vehicular and pedestrian traffic lanes.

4. Contractor shall provide signage, cones and other suitable devices to direct traffic. Contractor shall use flares and lights during hours of low visibility to delineate traffic lanes and to guide traffic.

5. Large vehicles shall have University public safety escort. Contractor shall provide minimum 48 hours written notice through University Representative.

6. Contractor shall submit a detailed traffic management plan to the University for review and approval fourteen (14) days minimum before any required full road closure and seven (7) days' notice prior to any single lane closures. Traffic plan shall clearly indicate vehicle, bicycle and pedestrian paths.

1.6 TRAFFIC SIGNS AND SIGNALS

A. Traffic Signs and Signals: Contractor shall provide temporary signs and signals as required by authorities having jurisdiction and in compliance with University's requirements transmitted through University Representative. Contractor shall relocate signs and signals as necessary during construction.

1.7 CONSTRUCTION PARKING

A. Construction Parking:

1. Contractor shall not park on public roadways unless approved by campus police and fire authorities.

2. Contractor shall maintain clear access ways and parking for emergency vehicles, as required by campus police and fire authorities.

3. Contractor shall provide on-site parking for construction purposes.

4. Contractor shall obtain and pay for parking permits for on-campus parking, including use permits for campus parking lots and for parking on the construction site itself. This requirements does not apply to construction equipment (fork lifts, excavators, backhoes cranes, etcetera)

   a. General Parking: Any vehicle parked on campus not actively used to carry tools, equipment, and supplies must display a valid general permit.

      1. Fee: Current rate for General Daily, Weekly or Quarterly. For more information on general parking rules, regulations and rates visit

https://afd.calpoly.edu/parking/parkingoncampus/permits/general.php or contact the University Representative.

2. Replacement fee: Equivalent to current General rate.
3. Sold/Issued through University Police, Bldg. #036.

b. Construction Area – Designated parking within the construction site or undesignated parking near project buildings or work area (sidewalks, greenbelts, dirt area).

1. Fee: $10.00 per permit flat rate (cost subject to change).
2. Duration: 6 months.
3. Replacement rate: $10.00 (cost subject to change).
4. Rate for projects less than (4) days charged Daily General permit rate.
5. Limited number of permits available for parking during project hours; number to be determined by University Representative and provided to University Police.
6. Limited to work trucks with tools, equipment, and supplies.
7. Issued through University Representative.

c. Depending on lot availability, Contractor may rent lay down area for field office and/or staging. Contractor(s) requiring lay down area(s) will make this request through the University Representative. If approved Contractor will enter into a rental agreement with University Police/Parking Services. Rates are based on proximity to the campus core and academic term, but will not exceed the current Residential permit rate per space/space equivalent.

PART 2 - PRODUCTS

Not applicable to this Section.

PART 3 - EXECUTION

3.1 MAINTENANCE OF PARKING AND ACCESS ROADS

A. Maintenance: Contractor shall maintain traffic and parking areas in a sound condition. Contractor shall repair breaks, potholes, low areas, standing water and other deficiencies, to maintain paving and drainage in original or specified condition.

B. Cleaning of Roadways and Parking Areas: Contractor shall keep public and private rights-of-way and parking areas clear of construction-caused soiling, dust and debris, especially debris hazardous to vehicle tires. Contractor shall perform cleaning as frequently as necessary. Contractor shall coordinate with requirements specified in Section 01 57 00 - Temporary Controls and Section 01 74 00 - Cleaning Requirements.

END OF SECTION

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SECTION 01 56 00 - TEMPORARY BARRIERS AND ENCLOSURES

PART 1 GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Temporary construction barriers, enclosures and passageways.

1. Dust and debris barriers.
2. Security barriers
3. Temporary chain link fencing.
4. Covered passageways.
5. Protection of completed Work.

B. Removal of construction facilities and temporary controls.

1.3 RELATED SECTIONS

A. Section 01 11 00 - Summary of the Work: Contractor's use of site and premises

B. Section 01 51 00 – Temporary Utilities: Temporary sanitary facilities, power and lighting

C. Section 01 52 00 – Construction Facilities: Installation of Construction Facilities

D. Section 01 52 05 - Construction Staging Areas: Submittals, staging and removal

E. Section 01 54 00 – Construction Aids: Temporary lifts, hoists, stairs, scaffolding

F. Section 01 54 01 - Security

G. Section 01 55 00 - Vehicular Access and Parking: Construction parking restrictions

H. Section 01 56 39 - Tree and Plant Protection: Requirements for barriers and covers at existing trees, shrubs and ground covers

I. Section 01 57 00 - Temporary Controls: General requirements for protection of existing conditions and run-off control

J. Section 01 58 00 - Project Identification and Signage: Directional and informational signage.

1.4 CODES AND REGULATIONS

A. California Building Code (CBC): Comply with California Building Code (CBC) Chapter 33, Section 3303, Protection of Pedestrians During Construction or Demolition.

B. Fire Regulations: Comply with requirements of fire authorities having jurisdiction, including California Fire Code (CFC) Article 87 during performance of the Work.

C. Safety Regulations: Comply with requirements of all applicable Federal, State and local safety rules and regulations. Contractor shall be solely responsible for jobsite safety.

D. Barricades and Barriers: As required by governing authorities having jurisdiction, provide substantial barriers, guardrails and enclosures around Work areas and adjacent to embankments and excavations for protection of workers and the public.

1. In and around existing facilities, walkways or throufares where the public must exit, use or otherwise occupy, provide and submit to the University an impairment plan. This plan shall define the pathways to be protected and maintained and shall consider egress occupant load. Contractor shall coordinate with the University for specific conditions.

1.5 PROTECTION OF EXISTING CONDITIONS

A. Protection of Adjacent Facilities: Contractor shall restrict Work to limits indicated on the Drawings and as specified in Section 01 11 00 - Summary of the Work: Protect existing, adjacent facilities from damage, including soiling and debris accumulation.

B. Protection of Existing Furniture, Fixtures and Equipment: As applicable, provide temporary enclosures, barriers and covers to protect existing furniture, fixtures and equipment remaining in Project area during construction.

1.6 MAINTENANCE OF CONSTRUCTION FACILITIES AND TEMPORARY CONTROLS

A. Maintenance: Use all means necessary to maintain temporary barriers and enclosures in proper and safe condition throughout progress of the Work, including all non-working times such as nights, holidays and weekends.

B. Replacement: In the event of loss or damage, promptly restore temporary barriers and enclosures by repair or replacement at no change in the Contract Sum or Contract Time.

1.7 TEMPORARY BARRIERS, ENCLOSURES AND PASSAGEWAYS

A. Temporary Barriers, General: Provide temporary fencing, barriers and guardrails as necessary to provide for public safety, to prevent unauthorized entry to construction areas and to protect existing facilities and adjacent properties from damage from construction operations.

1. Refer to temporary fencing and phasing plan in the Drawings. Comply with requirements indicated.
2. Note requirements for continued occupancy and use of existing buildings and site areas during construction.
3. Comply with applicable requirements of California Building Code (CBC) and authorities having jurisdiction, including industrial safety regulations. Review requirements with University's Representative.
4. Maintain unobstructed access to fire extinguishers, fire hydrants, temporary fire-protection facilities, stairways, and other access routes for firefighting.
5. Paint temporary barriers and enclosures with appropriate colors, graphics, and warning signs to inform personnel and public of possible hazard.

6. Where appropriate and necessary, provide pedestrian lighting (1cd minimum) warning lighting, including flashing red or amber lights.

B. Temporary Chain-Link Fencing: Provide temporary portable chain-link fencing with windscreen. See Section 01 52 05 – Construction Staging Area for requirements for layout of fencing.

1. Portable Chain-Link Fencing: Minimum 2-inches (50-mm) 11-gauge, galvanized steel, chain-link fabric fencing; minimum 6-feet (2.4 m) high with galvanized steel pipe posts; minimum 2-3/8-inches- (60-mm-) OD line posts and 2-7/8-inches- (73-mm-) OD corner and pull posts, with 1-5/8-inches- (42-mm-) OD top and bottom rails.

a. Provide concrete or galvanized steel bases for supporting posts.

b. Provide protective barriers at bases to prevent tripping by pedestrians or wind. Where required, install ground mounted posts or other means to prevent the fence from tipping.

2. Windscreen on Chain-Link Fencing: For screening of construction activities from view, equivalent to the following:

a. Specified manufacturer: None identified. Equivalent products of other manufacturers will be considered in accordance with the "or equal" provision specified in Section 01 61 00 - Basic Product Requirements.

b. Acceptable manufacturers: None identified. Equivalent products of other manufacturers will be considered in accordance with the "or equal" provision specified in Section 01 61 00 - Basic Product Requirements.

c. Windscreen fabric: Closed mesh weave of 30 warp by 16 fills per square inch.

1) Fiber: 5.6 ounce per square yard polypropylene fiber.

2) Shade factor: 78 percent.

3) Tensile strength: 360 pounds for warp and 190 pounds for fill, when tested according to ASTM D1682, grab method.

4) Tear strength: 110 pounds for warp and 70 pounds for fill, when tested according to ASTM D2263, trapezoidal method.

5) Color: Green

d. Fabric Fabrication:

1) Reinforce hems and seams with 2-3/4 inch black polypropylene folded binding tape, with tensile strength of 300 pounds.

2) Provide center reinforcing tape in addition to reinforced perimeter hems and panel seams.

3) Sew hems and seams with UV light resistant polyester thread.

4) Provide 9/32-inch brass grommets spaced at 12-inches on center in perimeter hems and center reinforcing tape.

e. Secure windscreen to fence at all grommets.

f. Locate windscreen on prevailing windward side of fence. Where prevailing wind direction cannot be determined, install fabric on the outside of fence.

g. **Do not cut the fabric to relieve wind pressure.** Install ground mounted posts or other means to prevent tipping.

C. Tarpaulins: Fire-resistive labeled with flame-spread rating of 15 or less.

D. Covered Passageways: Erect a structurally adequate, protective, covered walkways for passage of persons along adjacent passageways.

1. Coordinate installation details with University's requirements for continuing operations in adjoining facilities.
2. Review design and details with University's Representative.
3. Comply with applicable regulations of authorities having jurisdiction.
4. Construct covered walkways using scaffold or shoring framing.
5. Provide wood-plank overhead decking, protective plywood enclosure walls, handrails, barricades, warning signs, lights, safe and well-drained walkways, and similar provisions for protection and safe passage.
6. Extend back wall beyond the structure to complete enclosure fence.
7. Paint and maintain in a manner as directed by University's Representative.

E. Temporary Wood Fencing: Erect a structurally adequate, protective wood fencing in compliance with California Building Code (CBC) Chapter 33, Section 3303.7 - Pedestrian Protection. Wood fencing shall be provided as required by Table 33-A.

1. Materials: As required by CBC Section 3303.7.
2. Finishes: As acceptable to University's Representative. Fence where exposed to public view shall receive minimum of one coat wood primer and one coat semi-gloss paint, color(s) as directed by University's Representative.

F. Temporary Closures: Provide temporary closures for protection of construction, in progress and completed, from exposure, foul weather, other construction operations, and similar activities. Provide temporary weather-tight enclosure for building exterior.

1. Where heating or cooling is needed and permanent enclosure is not complete, provide insulated temporary enclosures. Coordinate closures with ventilating and material drying or curing requirements to avoid dangerous conditions and effects such as mold.
2. Vertical openings: Close openings of 25 sq. ft. (2.3 sq. m) or less with plywood or similar materials.
3. Horizontal openings: Close openings in floor or roof decks and horizontal surfaces with load-bearing, wood-framed construction.
4. Install tarpaulins securely using wood framing and other suitable materials.
5. Where temporary wood or plywood enclosure exceeds 100 sq. ft. (9.2 sq. m) in area, use fire-retardant-treated material for framing and main sheathing.

G. Temporary Partitions: Erect and maintain temporary partitions and temporary closures to limit dust and dirt migration, including migration into existing facilities, to separate areas from fumes and noise and to maintain fire-rated separations.

INCLUDE FOLLOWING THREE SUBPARAGRAPHS FOR ALTERATION PROJECTS, AND IF CONDITIONS AT NEW CONSTRUCTION INCLUDE INTERFACE WITH AN EXISTING FACILITY.

1. Dust barriers: Construct dustproof, floor-to-ceiling partitions of not less than nominal 4-inch (100-mm) studs, 2 layers of 3-mil (0.07-mm) polyethylene sheets, inside and outside temporary enclosure.
   a. Overlap and tape full length of joints.
   b. Include 5/8-inch thick gypsum board at temporary partitions serving as noise barrier.
   c. Insulate partitions to minimize noise transmission to adjacent occupied areas.
   d. Seal joints and perimeter of temporary partitions.
2. Dust barrier passages: Where passage through dust barrier is necessary, provide gasketed doors or heavy plastic sheets that effectively prevent air passage.
   a. Construct a vestibule and airlock at each entrance to temporary enclosure with not less than 48 inches (1219 mm) between doors.
   b. Maintain water-dampened foot mats in vestibule where passage leads to existing occupied spaces.
   c. Equip doors with security locks.
3. Fire-rated temporary partitions: Maintain fire-rated separations, including corridor walls and occupancy separations, by construction of stud partitions with gypsum board faces.
   a. Construction details shall comply with recognized time-rated fire-resistive construction according to the latest edition of the California Building Code (CBC). Typically, 1-hour rated partitions shall be 2x4 wood studs at 16-inches on center or 3-1/2 inch metal studs at 16-inches on center, with 5/8-inch thick Type X gypsum board at both faces, with joints filled, taped and topped.
   b. Seal partition perimeters with acceptable fire stopping and smoke seal materials.
   c. Construct fire-rated temporary partitions whenever existing time-rate fire-resistive construction is removed for 12 hours or more.

H. HVAC Protection: Provide dust barriers at HVAC return grilles and air inlets to prevent spread of dust and clogging of filters. Notify University Representative of all HVAC register barriers. If existing HVAC system is required to maintain proper temperature, refer to specification section 01 55 00, Temporary Utilities.

I. Temporary Floor Protection: Protect existing floors from soiling and damage.

1. Cover floor with 2 layers of 3-mil (0.07-mm) polyethylene sheets, extending sheets 18 inches (460 mm) up walls.
2. Cover polyethylene sheets with 3/4-inch (19-mm) fire-retardant plywood.
3. Provide floor mats to clean dust from shoes.
4. Other methods may be acceptable by written request and approval of the University Representative.

J. Landscape Barriers: Provide barriers around trees and plants designated to remain. Coordinate with requirements specified in Section 01 56 39 - Tree and Plant Protection.

1. Locate barriers as directed outside of drip lines of trees and plants.
2. Protect entire area under trees against vehicular traffic, stored materials, dumping, chemically injurious materials, and puddling or continuous running water.
3. Contractor shall pay all costs to restore trees and plants within barriers that are damaged by construction activities. Restoration shall include replacement with plant materials of equal quality and size. Costs shall include all fines, if any, levied by authorities having jurisdiction.

K. Barricades, Warning Signs and Lights, General: Comply with standards and code requirements for erection of structurally adequate barricades. Paint or utilize barricades with appropriate colors, graphics and warning signs to inform personnel and the public when protecting them against a hazard. Where appropriate and needed provide lighting, including flashing red or amber lights.

L. Guard Rails: Provide guard rails along tops of embankments and excavations. Along public walkways and areas accessible by the public, adjoining excavations, provide guardrails in addition to fencing.

1. Guardrails shall be substantially and durably constructed of lumber, firmly anchored by posts embedded in concrete, and complying with Code requirements for temporary barriers.
2. Guardrails shall comply with dimensional requirements and accommodate loads as prescribed by California Building Code (CBC) for permanent guardrails.

M. Security Closures and Lockup: Provide substantial temporary closures of openings in exterior surfaces and interior areas as appropriate to prevent unauthorized entrance, vandalism, theft and similar violations of security. Provide doors with self-closing hardware and locks.

1. Storage: Where materials and equipment must be stored, and are of value or attractive for theft, provide a secure lockup. Enforce discipline in connection with the installation and release of material to minimize the opportunity for theft and vandalism.

N. Weather Closures: Provide temporary weather-tight closures at exterior openings to prevent intrusion of water, to create acceptable working conditions, to protect completed Work and to maintain temporary heating, cooling and ventilation. Provide access doors with self-closing hardware and locks.

O. Temporary Access, Passage and Exit Ways: Construct temporary stairs, ramps, and covered walkways, with related doors, gates, closures, guardrails, handrails, lighting and protective devices, to maintain access and exit ways to existing facilities to remain operational.

1. Design and location of temporary construction shall be by Contractor, subject to review by University's Representative and authorities having jurisdiction.
2. Provide temporary lighting, illuminated interior exit signage, non-illuminated directional and instructional signage, and temporary security alarms for temporary exits and exit passageways.
3. Temporary measures shall suit and connect to existing building systems, and shall be approved by University's Representative and authorities having jurisdiction.

1.8 PROTECTION OF INSTALLED WORK

A. Protection of Installed Work, General: Provide temporary protection for installed products. Control traffic in immediate area to minimize damage.

B. Protective Coverings: Provide protective coverings at walls, projections, jambs, sills, and soffits of openings as necessary to prevent damage from construction activities, such as coatings applications, and as necessary to prevent other than normal atmospheric soiling.

C. Traffic Protection:

1. Protect finished floors, stairs and other surfaces from traffic, soiling, wear and marring.
2. Provide temporary covers of plywood, reinforced kraft paper or temporary rugs and mats, as necessary. Temporary covers shall not slip or tear under normal use.
3. Prohibit traffic and storage on waterproofed and roofed surfaces and on landscaped areas.
4. Protect newly fine graded, seeded and planted areas with barriers and flags to designate such areas as closed to pedestrian and vehicular traffic.

1.9 REMOVAL OF TEMPORARY BARRIERS AND ENCLOSURES

A. Removal of Temporary Barriers and Enclosures: Unless otherwise mutually agreed by University's Representative and Contractor, remove temporary materials, equipment, services, and construction prior to Contract Completion review. Coordinate removal with requirements specified in Section 01 51 00 - Temporary Utilities, Section 01 52 00 - Construction Facilities and Section 01 55 00 - Vehicular Access and Parking.

B. Cleaning and Repairs: Clean and repair damage, soiling and marring caused by installation or use of temporary barriers and enclosures.

PART 2 PRODUCTS (not applicable to this section)

PART 3 PART 1 – EXECUTION (not applicable to this section

END OF SECTION

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PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Requirements for protection of existing landscape plant materials, including trees, shrubs and ground covers. Contractor shall preserve, protect, and prune as necessary existing trees and shrubs, and other vegetation indicated to remain.

1.3 RELATED SECTIONS

A. Section 01 56 00 – Temporary Barriers and Enclosures: Barricades and barriers used to protect landscaping.

B. Section 01 57 00 – Temporary Controls: Runoff.

C. Division 2 - Site Construction: Landscaping specifications related to trees, shrubs and ground covers, as applicable.

1.4 WORK DESCRIPTION

A. Protection: All trees and plant materials to remain on site shall be protected from construction activities. Preserve, protect, and prune as necessary existing trees and shrubs and other vegetation indicated to remain.

B. Maintenance: Until Contract closeout, Contractor shall irrigate, fertilize, prune and clean as necessary to maintain all existing trees, shrubs and ground covers in healthy condition, within and adjacent to Project area.

1.5 DEFINITIONS

A. "Injury" is defined, without limitation, as any bruising, scarring, tearing, or breaking of roots, branches, or trunk.

B. "Dripline" is defined as the outermost limits of the tree or shrub canopy.

C. “Certified Arborist" is a consulting arborist certified by the International Society of Arboriculture, current certification in effect.

1.6 QUALITY ASSURANCE

A. General Responsibility: The Contractor shall be directly responsible for protection and welfare of existing trees within the Contract Limits which are noted to remain. This responsibility shall continue throughout the full construction period until the entire project is completed and accepted by the University's Representative and through completion of the guarantee period.

B. Qualifications of workmen: Trimming shall be performed only by a licensed arborist. Provide at least one person approved by the University's Representative who shall be present at all times during tree protection and trimming operations, who shall be thoroughly familiar with the type of work involved, and who shall direct all protection and trimming work.

C. Reference Standards: Published specifications, standards, tests, or recommended methods of trade, industry, or governmental organizations apply to work of this section.
   1. International Society of Arboriculture (ISA) "Guide for Plant Appraisal" prepared by the Council of Tree and Landscape Appraisers (CTLA).
   2. "Cabling, Bracing and Guying Standards for Shade Trees", as published by the National Arborist Association (NAA), 174 TR 101, Bedford, New Hampshire.
   3. ANSI A-300, Standards for Tree care Operations – Tree, Shrub, and Other Woody Plant Maintenance:
      Part 1 – Pruning, 2008.
      Part 2 – Fertilization, 2004.
      Part 3 – Supplemental Support Systems, 2006.

D. Sequencing Schedule: Coordinate and cooperate with other trades to enable the work to proceed as rapidly and efficiently as possible.

1.6 PRE: CONSTRUCTION MEETING:

A. Prior to construction of any nature on the site, Contractor shall arrange a site meeting with the University's Representative. The purpose of the meeting shall be to establish the conditions of all existing plant material to be preserved upon receipt of the site by the Contractor. Failure to call for said meeting implies acceptance by the Contractor of plant material to be preserved in its existing condition.

   1. Contractor shall document condition of all existing plant material to be preserved by means of individual color photos for each tree or plant to be preserved.
      a. Photos shall be labeled with plant botanical and common names, location on site, and date of photograph.
      b. Photos shall clearly indicate condition of tree or plant to be retained.

   2. Submit photographs to University for approval.

1.7 SUBMITTALS

A. Submittal procedures are specified in Division 1. Provide a minimum of four copies unless otherwise directed.

B. Provide the following submittals for review and approval:
   1. Pruning materials.
   2. Fencing materials.
   3. Maintenance plan.
   4. Photo documentation of existing plant material.

1.8 GUARANTEE

A. Contractor shall guarantee that all plants covered by the provisions of this Section will be healthy and in flourishing condition of active growth 1 year from the date of Final Completion.

B. During the warranty period the Contractor shall be liable for damages to all trees covered by the provisions of this Section, and shall pay compensation to the University as specified herein.

C. Contractor will not be held responsible for damages due to vandalism or acts of nature during the guarantee period. Immediately report such conditions to the University's Representative.

PART 2 - PRODUCTS

2.1 TREE PROTECTION FENCING

A. Tree Protection Fence:
1. Fencing Materials - 11 gauge galvanized 6' high chain-link fence with galvanized steel posts at 10' on center maximum**.**
2. Provide padlocked gate access to inside for care of tree.
3. Fence shall be sturdy and capable of acting as a barrier against objects, vehicles, etc., and designed so as to allow for relocations as required. It shall be continuously maintained and repaired as necessary.

2.2 PRUNING MATERIALS

A. Pruning materials shall be in accordance with current horticultural practices.

PART 3 – EXECUTION

3.1 PLANT MATERIAL PROTECTION

A. Protect existing plant material specified on the drawings during course of construction and maintenance period in a healthful state, free from damage or harm as the result of any work performed.
1. Protection shall be given to the roots, trunk, and foliage.
2. Plants shall not be allowed to deteriorate and shall be maintained in a healthy and vigorous condition.

B. Protect trees and plant materials to remain from trades working on site.
1. Insure that subcontractors are aware of and held responsible for damage to existing trees and plant material.
2. Insure that protective measures are carried out continually throughout the construction period.

C. Contractor is responsible for replacing damaged plant life with approved equivalent.

D. Water: Provide ample water supply of potable quality and sufficient quantity for all operations required under this Section.

E. Trees and plants, subject to the provisions of this Section, which have been injured shall be repaired immediately under the direction of an approved arborist. Repair shall include removal of rough edges and sprung bark and severely injured branches as directed by the University's Representative.

F. The Contractor shall maintain healthy living conditions for existing plant materials to be preserved.
   1. Such measures shall include monthly washing of leaves for the removal of dust, regular irrigation, root feeding, etc.

G. Install tree protection fencing as specified elsewhere in this section.

H. Drainage:
   1. During construction, the existing site surface drainage patterns shall not be altered within the area of the drip line of existing plant materials.
   2. Contractor shall not alter the existing water table within the area of the drip line of existing plant materials.

I. Do not permit any of the following within drip line of any existing tree or shrub to be preserved:
   1. Storage or parking of automobiles or other vehicles.
   2. Stockpiling of building materials or refuse of excavated materials.
   3. Skinning or bruising of bark.
   4. Use of trees as support posts, power poles, or signposts; anchorage for ropes, guy wires, or power lines; or other similar functions.
   5. Dumping of poisonous materials on or around trees and roots. Such material includes but is not limited to paint, petroleum products, contaminated water, or other deleterious materials.
   6. Cutting of tree roots by utility trenching, foundation digging, placement of curbs and trenches, and other miscellaneous excavation without prior written approval by the University's Representative.
   7. Damage to trunk, limbs, or foliage caused by maneuvering vehicles or stacking material or equipment too close to the tree.
   8. Compaction of the root area by movement of trucks or grading machines; storage of equipment, gravel, earth fill, or construction supplies; etc.
   9. Excessive water or heat from equipment, utility line construction, or burning of trash under or near shrubs or trees.
   10. Damage to root system from flooding, erosion, and excessive wetting and drying resulting from dewatering and other operations.

3.2 TREE PROTECTION FENCING

A. Install tree protection fencing around trees to be preserved with a caliper of 4" or larger, unless otherwise noted on plans.
   1. Install fencing at the dripline unless otherwise approved by the University's Representative.
   2. Fence can be erected around groups of adjacent trees where possible.
   3. Otherwise, fence to be erected around individual trees.

B. No construction, demolition, or work of any nature will be allowed within the fenced area without prior written approval by the University’s Representative.
   1. Approval by the University’s Representative for work within the fenced area shall not release the Contractor from any of the provisions specified herein for the protection of existing trees and plants to be preserved.
   2. During the course of construction of approved work within the fence area, no roots shall be cut without prior written approval by the University's Representative.

C. During the course of construction, relocation of the fence may be required to facilitate construction. The Contractor shall do so when authorized and as directed by the University’s Representative, at no additional expense to the University.

D. The fence shall be removed only at the end of construction, as approved by the University's Representative.

3.3 TEMPORARY IRRIGATION:

A. Irrigate existing trees and other plants to remain, as necessary to maintain their health before, during and after Work under the Contract, as directed by the University’s Representative.
   1. Maintain an irrigation schedule and document. Submit schedule to University's Representative for review and acceptance.
   2. Provide temporary piping, valves, hoses, emitters and spray heads as necessary until Contract closeout.

B. Soil Preparation: If soil within drip line of trees is compacted, then prior to watering or fertilizing trees, area within the drip lines shall be tilled to break up the top two inches of existing soil.

3.4 EXCAVATION AROUND TREES AND PLANTS:

A. Excavation within the drip lines of trees and plants to be preserved shall be done only where absolutely necessary, under the direction of a Certified Arborist and with prior approval from the University's Representative.

B. Where trenching for utilities is required within driplines, tunneling under and around roots shall be by hand digging. Main lateral roots and taproots shall not be cut. Smaller roots that interfere with installation of new work may be cut with prior approval from certified Arborist.

C. Where excavation of new construction is required within drip line of trees, hand excavation shall be employed to minimize damage to root system. Roots shall be relocated in backfill areas wherever possible. If large, main lateral roots are encountered, they shall be exposed beyond excavation limits as required to bend, and relocate without breaking. If encountered immediately adjacent to location of new construction and relocation is not practical, roots shall be cut approximately 6 inches back from new construction under the direction of a certified Arborist.

D. Exposed roots shall not be allowed to dry out before permanent backfill is placed. Temporary earth cover shall be provided, or roots shall be packed with wet peat moss or four layers of wet, untreated burlap and temporarily supported and protected from damage until permanently relocated and covered with backfill. The cover over the roots shall be wetted to the point of runoff daily.

E. Branching structure shall be thinned in accordance with National Arborists Association "Pruning Standards and Principles" to balance loss of root system caused by damage or cutting of root system. Thinning shall not exceed 30 percent of existing branching structure.

3.5 TREE TRIMMING

A. If tree trimming is necessary to protect the health of the tree, as determined by the University Representative, or if required to clear for construction, it shall be done only under the direction of a Certified Arborist engaged at the Contractor's expense.

B. In company with the University's Representative, University and the Arborist, ascertain the limbs and roots which are to be trimmed. Clearly mark them to designate the approved point of cutting.

C. Dead and damaged trees that are determined by the Certified Arborist to be incapable of restoration to normal growth pattern shall be removed.

D. Cut evenly, using proper tools and skilled workmen, to achieve neat severance with the least possible damage to the tree.

E. In the case of root cuts, apply wet burlap or other protection, approved as noted herein, to prevent drying out, and maintain in a wet condition as long as necessary for temporary protection.

3.6 REPAIR COMPENSATION

A. Damage:
  1. Damage to existing tree crowns or roots over 1" in diameter shall be immediately reported to the University's Representative in writing.
  2. A Certified Arborist shall direct all repairs to trees damaged by construction operations. Repairs shall be made promptly after damage occurs to prevent progressive deterioration of damaged trees. Repairs shall be at the Contractor's expense.

B. Irreparable Damage: Any tree to be protected which is irreparably damaged owing to the Contractor's negligence or failure to provide adequate protection shall be compensated for in accordance with the following schedule of values using the "tree caliper" method (greatest trunk diameter, measured 18 inches above the ground):
  1. For trees with diameters up to and including 6 inches, compensation shall be the actual cost of replacement with item similar in species, size, and shape, including:
    a. Actual cost of item boxed out of the ground.
    b. Transportation or delivery of boxed item to site.
    c. Planting and staking.
    d. Maintenance, including watering, fertilizing, pruning, pest control, and other care to bring replacement to same general condition of original item.
  2. For trunks up to:
     7" .... $ 1,200

8" 1,700
9" 2,200
10" 2,600
11" 3,100
12" 3,600
13" 4,100
14" 4,600
15" 5,000
16" 5,500
17" 6,000
18" and over, add for each caliper inch ... $600

C. Damaged tree limbs or trees which have died as a result of injury during construction shall remain the property of the University and shall remain or be removed by the Contractor as directed by the University's Representative.

3.7 MAINTENANCE

A. Contractor shall be responsible to perform periodic inspections of existing trees and shrubs to be preserved and submit written proposals to the University's Representative for additional maintenance work as may be required to ensure the health and general well-being of the trees.
1. Contractor shall retain, at the direction of the University's Representative additional specialists as may be required to perform this work.
2. The maintenance and quality of the plant materials shall be subject to monthly checks by the University. The dates of these checks shall be outlined in the University's Representative's field notification relating to the establishment of the plant maintenance period. Additional checks shall be scheduled as determined by the University's Representative.

B. Plant material must be maintained throughout the duration of the construction period in a healthful manner. Plant material identified which requires special pruning, insect control, fertilization or other remedial health action will be treated during this period. Methods and rates of pesticide and fertilizer application will be reviewed by the University's Representative prior to approval.

C. Tree Basins and Mulching:
1. Construct tree basins at base of existing trees to be preserved so that applied water will remain on top of the root-ball. Size of basin shall be as recommended by the arborist.
2. Tree basins shall be mulched with 1-inch neutralized fir bark.

D. Watering:
1. Plant materials shall be watered on a regular basis, at a rate consistent with their particular requirements. Verification of the proposed watering schedule shall be reviewed by the University's Representative prior to commencement of the maintenance.
2. The maintenance of the plant materials shall comply with standard horticultural practice for the correct watering, fertilizing, pruning and spraying of the specimen boxed trees.

3. Irrigation: During construction the existing trees to be preserved shall, at the direction of the University's Representative, be given deep watering (be irrigated). Quantities and lengths of time are variable and shall depend upon seasonal rainfall.

E. Washing:
   1. Contractor shall perform periodic washing of leaves to remove dust.

F. Contractor shall keep plant material free from weeds and debris at all times.

3.8 CLEANUP

A. Upon completion of the work, restore ground to required elevations and remove excess material debris and equipment from the site to the satisfaction of the University's Representative.

END OF SECTION

SECTION 01 57 00 - TEMPORARY CONTROLS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Protection of existing conditions, including video record of existing conditions.

B. Life safety and fire protection.

C. Security.

D. Runoff control.

E. Protection of installed Work.

1.3 RELATED SECTIONS

A. Section 01 11 00 - Summary of the Work: Contractor's use of site and premises.

B. Section 01 52 00 - Construction Facilities: Field offices and sheds.

C. Section 01 54 00 - Construction Aids: Temporary lifts and hoists; temporary stairs and scaffolding.

D. Section 01 55 00 - Vehicular Access and Parking: Vehicle access and parking control at Work areas.

E. Section 01 56 00 - Temporary Barriers and Enclosures: Requirements for dust and debris barriers.

1.4 CODES AND REGULATIONS

A. Fire Regulations: Comply with requirements of fire authorities having jurisdiction, including California Fire Code (CFC) Article 87 during performance of the Work.

B. Safety Regulations: Contractor shall be solely responsible for jobsite safety. Minimum requirements shall include the following.
   1. Comply with requirements of all applicable Federal, State and local safety rules and regulations, including but not limited to all OSHA regulations.
   2. [_*INCLUDE THIS LINE IF PROJECT WILL BE ENROLLED IN O.C.I.P. – IF OVER $10 MILLION, OCIP IS REQUIRED_]* Comply with requirements in the University’s Owner Controlled Insurance Program “Safety Manual," provided under separate cover by University's Representative.

C. Barricades and Barriers: As required by authorities having jurisdiction, provide substantial barriers, guardrails and enclosures around Work areas and adjacent to embankments and excavations for protection of workers and the public. See Section 01 56 00 - Temporary Barriers and Controls for additional requirements.

1.5 PROTECTION OF EXISTING CONDITIONS

A. Protection of Adjacent Facilities: Contractor shall restrict Work to limits indicated on the Drawings and as specified in Section 01 11 00 - Summary of the Work. Protect existing, adjacent facilities from damage, including soiling and debris accumulation.

B. Video Record of Existing Conditions: Contractor shall produce video record of all existing conditions within and adjacent to Project area.

1. Record by video or digital photography in a format that is easily transferred to the University. Video shall have sound to record comments to identify locations and describe conditions.

2. University's Representative will accompany Contractor during recording of existing conditions but will not direct recording process.

3. Video shall record state of existing features, including but not limited to:
   a. Paving.
   b. Landscaping.
   c. Building surfaces.
   d. Utilities.
   e. Lighting standards, fencing, signage and other site appurtenances.
   f. [_DESCRIPTION_].

4. Contractor shall retain one copy and deliver one copy of video record to University's Representative within seven calendar days after the video record was produced.

5. Video record shall be used to verify restoration of existing conditions after completion of construction activities.

6. Existing feature not recorded shall be restored as directed by University's Representative, including reconstruction and refinishing as determined necessary by University's Representative.

1.6 FIRE PROTECTION

A. Fire Protection Responsibility: Protection of Project from fire shall be solely Contractor's responsibility.

B. Fire Protection Provisions, General: Maintain, at a minimum, the Work in conditions to minimize fire hazards and provide adequate fire protection devices, such as suitable fire extinguishers, blankets, warning signs and storage containers.

1. Store combustible materials in containers in fire-safe locations.

2. Maintain unobstructed access to fire extinguishers, fire hydrants, temporary fire protection facilities, stairways and other access routes for fighting fires. Prohibit smoking in hazardous fire exposure areas.

3. Provide supervision of welding operations, combustion type temporary heating units, and similar sources of fire ignition.

C. Special Fire Protection Provisions: During hazardous construction activities, maintain adequate fire protection devices immediately available for use at the location of such activities.

D. Fire Protection Equipment: Until fire protection is provided by permanent fire protection systems and equipment, install and maintain temporary fire protection equipment as necessary to protect against ignition and spread of fires. Comply with NFPA 10 "Standard for Portable Extinguishers" and NFPA 241 "Standard for Safeguarding Construction, Alteration and Demolition Operations."

E. Temporary Fire Sprinkler Provisions: Where existing fire sprinkler system is affected by demolition and
re-construction activities, provide either temporary fire protection measures acceptable to authorities having jurisdiction or modify existing system as necessary to maintain fire protection. Include extensions and additions to standpipe system, for Fire Department connections. Comply with California Fire Code (CFC) Article 87 during all phases of the Work.

F. Fire Extinguishers for Protection During Construction: Comply with NFPA 10 and 241 for classification, extinguishing agent and size required by location and class of fire exposure.

1. Provide hand carried, portable UL-rated, Class "A" fire extinguishers for temporary offices and similar spaces.

2. In other locations, provide hand-carried, portable, UL-rated, Class "ABC" dry chemical extinguishers, or a combination of extinguishers of NFPA recommended classes for the exposures.

G. Installation of Fire Extinguishers for Protection During Construction: Locate fire extinguishers in field offices, storage sheds, tool houses, other temporary buildings and throughout the Work site. Comply with directions of Fire Marshal having jurisdiction.

1. In the area under construction, provide at least one fire extinguisher for each 5,000 square feet of building floor area.

2. Locate fire extinguishers no greater than 75 feet travel distance apart.

1.7 SECURITY

A. Security Responsibility: Security of the Project area shall be solely the Contractor's responsibility until completion of the Work. Reference Contract General Conditions Article 4.08-c, Protection of Facilities.

B. Security Provisions, General: Provide security program and facilities to protect Work from unauthorized entry, vandalism, and theft. It is the contractor’s responsibility to protect Work while Work is being conducted and during times of non-work. This includes all days, nights, weekends, and holidays from the issuance of the Notice to proceed to the final building occupancy by the University.

C. Guard Service: At Contractor's discretion, employ guards to protect the site after working hours.

1.8 RUNOFF CONTROL

A. Erosion and Sedimentation Control: Erosion and sedimentation control provisions shall meet or exceed minimum requirements of authorities having jurisdiction. When provisions are indicated on Drawings, they are minimum requirements. If no sedimentation control system is shown, then Contractor shall design and provide system to prevent siltation of adjacent property as required by authorities having jurisdiction.

   1. Implement erosion and sedimentation control provisions prior to commencing site clearing, grading, backfilling and compacting or other construction activities which will expose soil to erosion and potential for sediment-laden runoff.

   2. Ensure that sediment-laden water does not enter drainage systems.

   3. Maintain erosion and sedimentation control provisions until Contract Completion review is completed for landscaping, or sooner if approved by authorities having jurisdiction.

   4. Implementation, maintenance, replacement and additions to erosion and sedimentation control provisions shall solely be the responsibility of the Contractor. As construction progresses and seasonal conditions dictate, more erosion and sedimentation controls may be required. If so, Contractor shall provide additional provisions over and above minimum requirements as necessary.

B. Drainage: Grade site and other Work areas to drain.

   1. Provide temporary drainage ditches and diversion measures as necessary to protect construction.

   2. Provide erosion control measures as necessary and as required by authorities having jurisdiction. Comply with local water quality control requirements, as applicable.

C. De-Watering: Maintain excavations free of water. Provide and operate pumping equipment as necessary.

   1. Removal of contaminated water from excavations, dewatering of contaminated groundwater and discharging of contaminated soils via surface erosion is prohibited.

   2. Dewatering of non-contaminated groundwater shall be performed only after Contractor obtains a National Pollutant Discharge Elimination System Permit from the State or Regional Water Quality Control Board having authority. Costs of such permit shall be included in the Contract Sum.

D. Runoff Control: Storm water runoff and other waters may be encountered at various times during construction. Contractor, by signing the Agreement, acknowledges that risks arising from storm water runoff and other waters have been investigated and considered, and Contract Sum and Contract Time include all costs associated with runoff control.

   1. It shall be responsibility of Contractor to protect Work from detrimental effects of all waters encountered.

   2. It shall be responsibility of Contractor to protect Work from detrimental effects of runoff.

3. Should damage to the Work due to surface or other water occur prior to acceptance of the Work by University's Representative, Contractor shall repair or replace Work at no change in Contract Time or Contract Sum.

E. National Pollutant Discharge Elimination System: Contractor shall comply with requirements of environmental protection and storm drainage authorities having jurisdiction.

1. Project Area and other areas affected by Work under the Contract shall be maintained in such condition that anticipated storm runoff does not carry wastes and other pollutants off the site.

2. Discharges of material other than storm water will be allowed only when necessary for performance of the Work and where such discharge does not cause the following:
   a. Cause or contribute to a violation of applicable water quality standard.
   b. Cause or threaten to cause pollution, contamination or nuisance, as determined by authorities having jurisdiction. Potential pollutants include but are not limited to:
      1) Solid or liquid chemical spills.
      2) Wastes from paints, stains, sealants, adhesives, limes, pesticides, herbicides, wood preservatives and solvents.
      3) Asbestos fibers, paint flakes or fragments of plaster and drywall.
      4) Fuels, lubricants, hydraulic fluids, coolants, battery electrolytes.
      5) Vehicle or equipment, degreasing, steam cleaning and wash water.
      6) Concrete, mortar and plaster mix and cleaning water.
      7) Detergents and floatable wastes.
      8) Superchlorinated potable water line flushings.
   c. Contain hazardous substances in a quantity reportable under Federal Regulations 40 CFR Parts 117 and 302.

3. During performance of the Work, disposal of such materials shall occur at a temporary on-site location, physically separated from potential storm water runoff, with ultimate disposal in compliance with all applicable local, regional, State and Federal requirements.

4. Contractor shall obtain and comply with Storm Water Pollution Prevention Plan (SWPPP). Contractor shall be responsible for payment of the permit and all fines for non-compliance with the SWPPP, at no change in Contract Sum.

F. Pavement clearing and cleaning: Keep site access ways, parking areas and building access and exit facilities clear of mud.

1. Remove mud, soil and debris and dispose in a manner which will not be injurious to persons, property, plant materials and site.

2. Comply with runoff control requirements stated above and as required by authorities having jurisdiction.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

Not Applicable to this Section.

END OF SECTION

SECTION 01 58 00 - PROJECT IDENTIFICATION SIGNAGE

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. On-site Project identification and temporary informational signs provided by University and maintained by Contractor during Contract.

1.3 RELATED SECTIONS

A. Section 01 52 00 – Construction Facilities: Coordination of signage with field offices and sheds.

B. Section 01 54 01 – Security: Personnel identification badges.

C. Section 01 55 00 – Vehicular Access and Parking: Coordination of signage with construction parking

D. Section 01 56 00 – Temporary Barriers and Enclosures: Temporary wood barriers and enclosures with directional signage.

1.4 SUBMITTALS

A. Shop Drawings: In compliance with directions from University's Representative, Contractor shall prepare and submit site plan locating temporary project identification and informational signs furnished by University.

PART 2 - PRODUCTS

2.1 SIGN MATERIALS

A. Sign Structure and Framing: Contractor shall provide new materials, wood or metal, structurally adequate to support sign panel and suitable for specified finish.

B. Sign Surfaces: Sign shall be designed and provided by the University. Sign shall be installed by the contractor.

1. For Contractor information: Sign surfaces shall be minimum 3/4-inch thick, exterior grade, softwood plywood with medium or high-density phenolic sheet overlay, standard large sizes to eliminate joints.
2. Contractor shall provide backing or additional support as required to span across framing members and provide even, smooth surface without waves or buckles.

C. Rough Hardware: Rough hardware shall be hot-dip galvanized steel.

D. Paint, Sign Structure: Paint used for Sign Structure shall be exterior quality, primer and flat finish paint, adequate to resist weathering and fading for scheduled construction period.

2.2 PROJECT IDENTIFICATION SIGN

A. Project Identification Sign: As directed, Contractor shall install one University furnished painted Project Identification Sign of the size and construction indicated on graphic to be provided by Architect.
   1. Graphic design, text, style of lettering, and colors of sign shall be as directed by University Marketing Department.
   2. Sign shall identify project name, project number, University's name, with University Project Manager's Name and contact information, Architect's name and Contractor's name.
   3. Sign shall include corporate logos of parities identified on sign.

C. Project Address Signs: Contractor shall provide Project name and street address signs, minimum of four feet wide, to identify Project to facilitate deliveries.
   1. Graphic design and colors of sign shall match Project Identification Sign.
   2. Text on sign shall be as directed.

D. Sign Painting: Sign Panels shall be shop painted and field installed by Contractor.

2.3 PROJECT INFORMATIONAL SIGNS

A. Restrictions: Contractor shall not display signs other than Project Identification Sign specified above and Project Informational Signs specified below without written approval of University's Representative.

B. Project Informational Signs: Informational signs, necessary for conduct of construction activities or required by governmental authorities having jurisdiction, may be displayed when in conformance to sign construction and graphic requirements specified in this Section.
   1. University's Representative may review such signs. If so, review will be for sign construction, and graphic designs only.
   2. Adequacy of signage for safety and conformance to requirements of authorities having jurisdiction and trade practices shall be solely Contractor's responsibility.

C. Sign Painting: Contractor shall ensure that informational signage shall be produced by professional sign painters and be of size and lettering style consistent with use. Colors shall be as required by authorities having jurisdiction and, if not otherwise required, of colors consistent with Project graphics.
   1. Sign Face Finish: Sign face finish shall be gloss enamel.
   2. Structure Finish: Sign structure finish shall be paint exposed surfaces of supports and framing members one coat of primer and one coat of exterior paint, flat finish.

PART 3 – EXECUTION

3.1 PROJECT IDENTIFICATION SIGN INSTALLATION

A. Project Identification Sign Construction: Contractor shall construct sign support structure and install panels in durable manner, to resist high winds.

B. Project Identification Sign Installation: Contractor shall erect Project Identification Sign on site at a lighted location of high public visibility, adjacent to the main entrance to the site, as approved by University's Representative.
   1. Contractor shall install sign at height for optimum visibility, on ground-mounted poles or attached to portable structure on skids.
   2. Portable structure shall resist overturning force of wind.

C. Street Address Signs: Contractor shall locate and install signs at each access point from public streets.

D. Field Painting: Contractor shall paint all surfaces and edges of sign face and support structure for finished appearance.

3.2 PROJECT INFORMATIONAL SIGN INSTALLATION

A. Project Informational Signs Construction: Contractor shall construct sign support structure and install panels in durable manner, to resist high winds.

B. Project Informational Sign Installation:
   1. Contractor shall locate signs as necessary for construction activities and as required by authorities having jurisdiction.
   2. Contractor shall install informational signs for optimum visibility, on ground-mounted posts or temporarily attached to surfaces of structures.
   3. Attachment methods shall leave no permanent disfiguration or discoloration on completed work.

C. Field Painting: Contractor shall paint all surfaces and edges of sign face and support structure for finished appearance.

3.3 SIGNS MAINTENANCE

A. Signs Maintenance: Contractor shall maintain signs and supports in a neat, clean condition. Contractor shall repair all damage and weathering to structure, framing and signage.

B. Sign Relocation: Contractor shall relocate signs as required by progress of the work.

3.4 REMOVAL

A. Project Identification Sign Removal: Contractor shall remove Project Identification Sign when directed. Contractor shall coordinate removal with requirements specified in Section 01 51 00 – Temporary Utilities, Section 01 52 00 – Construction Facilities, Section 01 55 00 – Vehicular Assess and

Parking and Section 01 56 00 – Temporary Barriers and Enclosures.

B. Project Informational Signs Removal: Contractor shall remove all informational signs, framing, supports and foundations prior to Contract Completion review. Contractor shall coordinate removal with requirements specified in Section 01 51 00 – Temporary Utilities, Section 01 52 00 – Construction Facilities, Section 01 55 00 – Vehicular Access and Parking and Section 01 56 00 – Temporary Barriers and Enclosures.

END OF SECTION

SECTION 01 61 00 - BASIC PRODUCT REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. General requirements for products used for the Work, including:
1. General characteristics of products
2. Product options
3. System completeness
4. Transportation and handling requirements
5. Storage and protection of products
6. Installation of products.

1.3 RELATED SECTIONS

A. Section 01 33 00 - Submittals Procedures: Requirements applicable to submittals for "or equal" and substitute products.

B. Section 01 41 00 - Regulatory Requirements: Codes and standards applicable to product specifications; minimum requirements.

C. Section 01 42 00 - Reference Standards and Abbreviations: References to various standards, standard specifications, codes, practices and other requirements.

D. Section 01 64 00 - Owner-Furnished Products: Requirements for installing products furnished by University.

E. Section 01 65 00 - Product Delivery Requirements: General requirements for delivery of products to Project site.

F. Section 01 66 00 - Product Storage and Handling Requirements: General requirements for storage and handling of products.

1.4 GENERAL PRODUCT REQUIREMENTS

A. Products, General: "Products" include items purchased for incorporation in the Work, whether purchased for the Project or taken from previously purchased stock, and include materials, equipment, assemblies, fabrications and systems.
1. Named Products: Items identified by manufacturer's product name, including make or model designations indicated in the manufacturer's published product data.

2. Materials: Products that are shaped, cut, worked, mixed, finished, refined or otherwise fabricated, processed or installed to form a part of the Work.

3. Equipment: A product with operating parts that are motorized or manually operated and require connections such as wiring or piping.

B. Specific Product Requirements: Refer to requirements of Section 01 45 00 - Quality Control and individual product Specifications Sections in Divisions 2 through 49 for specific requirements for products.

C. Minimum Requirements: Specified requirements for products are minimum requirements. Refer to general requirements for quality of the Work specified in Section 01 45 00 - Quality Control and elsewhere herein.

D. Product Selection: Provide products that fully comply with the Contract Documents, are undamaged and unused at installation. Comply with additional requirements specified herein in Article titled "PRODUCT OPTIONS".

E. Standard Products: Where specific products are not specified, provide standard products of types and kinds that are suitable for the intended purposes and that are usually and customarily used on similar projects under similar conditions. Products shall be as selected by Contractor and subject to review and acceptance by the Architect.

F. Product Completeness: Provide products complete with all accessories, trim, finish, safety guards and other devices and details needed for a complete installation and for the intended use and effect. Comply with additional requirements specified herein in Article titled "SYSTEM COMPLETENESS".

G. Code Compliance: All products, other than commodity products prescribed by Code, shall have a current ICBO Evaluation Service (ICBO ES) Research Report or National Evaluation, Inc. Report (NER). Refer to additional requirements specified in Section 01 41 00 - Regulatory Requirements.

H. Interchangeability: To the fullest extent possible, provide products of the same kind from a single source. Products required to be supplied in quantity shall be the same product and interchangeable throughout the Work. When options are specified for the selection of any of two or more products, the product selected shall be compatible with products previously selected.

I. Product Nameplates and Instructions:

1. Except for required Code-compliance labels and operating and safety instructions, locate nameplates on inconspicuous, accessible surfaces. Do not attach manufacturer's identifying nameplates or trademarks on surfaces exposed to view in occupied spaces or to the exterior.

2. Provide a permanent nameplate on each item of service-connected or power-operated equipment. Nameplates shall contain identifying information and essential operating data such as the following example:
   Designation of product as identified on the Plans and Specifications
   Name of manufacturer
   Name of product
   Model and serial number
   Capacity

Operating and Power Characteristics
Labels of Tested Compliance with Codes and Standards

3. For each item of service-connected or power-operated equipment, provide operating and safety instructions, permanently affixed and of durable construction, with legible machine lettering. Comply with all applicable requirements of authorities having jurisdiction and listing agencies.

4. Permanent nameplate shall be constructed from metal with lettering punched or indented into the material. "Permanent" marker or other inks shall not be used.

J. Plumbing Product Requirements: Comply with requirements specified in Division 22 - Plumbing

K. Mechanical Product Requirements: Comply with requirements specified in Division 23 - Mechanical.

L. Electrical Product Requirements: Comply with requirements specified in Division 26 - Electrical.

1.5 PRODUCT OPTIONS

A. Product Options: Refer to Contract General Conditions, Article 5.04. Provisions of Public Contract Code Section 03400 shall apply, as supplemented by the following general requirements.

B. Products Specified by Description: Where Specifications describe a product, listing characteristics required, with or without use of a brand name, provide a product that has the specified attributes and otherwise complies with specified requirements.

C. Products Specified by Performance Requirements: Where Specifications require compliance with performance requirements, provide product(s) that comply and are recommended by the manufacturer for the intended application. Verification of manufacturer's recommendations may be by product literature or by certification of performance from manufacturer.

D. Products Specified by Reference to Standards: Where Specifications require compliance with a standard, provided product shall fully comply with the standard specified. Refer to general requirements specified in Section 01 42 00 - Reference Standards and Abbreviations regarding compliance with referenced standards, standard specifications, codes, practices and requirements for products.

E. Products Specified by Identification of Manufacturer and Product Name or Number:
   1. Sole, source, no other product shall be accepted: Provide the specified product(s) of the specified manufacturer. No substitutions shall be allowed.

   2. "Acceptable Manufacturers": Product(s) of the named manufacturers, if equivalent to the specified product(s) of the specified manufacturer, will be acceptable in accordance with the requirements specified herein in the Article titled "'OR EQUAL' PRODUCTS."

   3. Unnamed manufacturers: Products of unnamed manufacturers will be acceptable only as follows:
      a. Unless specifically stated that equals will not be accepted or considered, the phrase "or equal" shall be assumed to be included in the description of specified product(s). Equivalent products of unnamed manufacturers will be accepted in accordance with the "or equal"

provision specified herein, below.

b. If provided, products of unnamed manufacturers shall be subject to the requirements specified herein in the Article titled "'OR EQUAL' PRODUCTS."

4. Quality basis: Specified product(s) of the specified manufacturer shall serve as the basis by which products by named acceptable manufacturers and products of unnamed manufacturers will be evaluated. Where characteristics of the specified product are described, where performance characteristics are identified or where reference is made to industry standards, such characteristics are specified to facilitate evaluation of products by identifying the most significant attributes of the specified product(s).

F. Products Specified by Combination of Methods: Where products are specified by a combination of attributes, including manufacturer's name, product brand name, product catalog or identification number, industry reference standard, or description of product characteristics, provide products conforming to all specified attributes.

G. "Or Equal" Provision: Where the phrase "or equal" or the phrase "or approved equal" is included, product(s) of unnamed manufacturer(s) may be provided as specified above in subparagraph titled "Unnamed manufacturers."

1. The requirements specified herein in the Article titled "'OR EQUAL' PRODUCTS" shall apply to products provided under the "or equal" provision.

2. Use of product(s) under the "or equal" provision shall not result in any delay in completion of the Work, including completion of portions of the Work for use by University or for work under separate contract by University.

3. Use of product(s) under the "or equal" provision shall not result in any costs to University, including design fees and permit and plan check fees.

4. Use of product(s) under the "or equal" provision shall not require substantial change in the intent of the design, in the opinion of the Architect. The intent of the design shall include functional performance and aesthetic qualities.

5. The determination of equivalence will be made by the Architect, and such determination shall be final.

H. Visual Matching: Where Specifications require matching a sample, the decision by the Architect on whether a proposed product matches shall be final. Where no product visually matches, but the product complies with other requirements, comply with provisions for substitutions for selection of a matching product in another category.

I. Selection of Products: Where requirements include the phrase "as selected from manufacturer's standard colors, patterns and textures", or a similar phrase, selections of products will be made by indicated party or, if not indicated, by the Architect. The Architect will select color, pattern and texture from the product line of submitted manufacturer, if all other specified provisions are met.

1.6 "OR EQUAL" PRODUCTS

A. "Or Equal" Products: Products are specified typically by indicating a specified manufacturer and specific products of that manufacturer, with acceptable manufacturers identified with reference to this "or equal" provision. If Contractor proposes to provide products other than the specified products of the specified manufacturer, provisions of any relevant Supplementary General Conditions, Contract General Conditions Article 5.04, and Public Contract Code section 3400 shall apply. Contractor shall submit if and when directed by Architect, complete product data, including drawings and descriptions of products, fabrication details and installation procedures. Include samples where applicable or requested.

1. Submit a minimum of four copies. Form and other administrative requirements shall be as directed by the Architect.

2. Include appropriate product data for the specified product(s) of the specified manufacturer, suitable for use in comparison of characteristics of products.
   a. Include a written, point-by-point comparison of characteristics of the proposed equal product with those of the specified product.
   b. If the proposed equal is accepted, Contractor shall include a detailed description in written or graphic form as appropriate, indicating all necessary changes or modifications to other elements of the Work and to construction to be performed by the University and others under separate contract with University.

3. "Or Equal" product submissions shall include a statement indicating the equal's effect on the Construction Schedule. Contractor shall indicate the effect of the proposed products on overall Contract Time and, as applicable, on completion of portions of the Work for use by University or for work under separate contract by University.

4. "Or Equal" product submissions shall include signed certification that the Contractor has reviewed the proposed products and has determined that the products are equivalent or superior in every respect to product requirements indicated or specified in the Contract Documents, and that the proposed products are suited for and can perform the purpose or application of the specified product indicated or specified in the Contract Documents.

5. "Or Equal" product submissions shall include a signed waiver by the Contractor for change in the Contract Time or Contract Sum because of the following:
   a. "Or equal" product failed to perform adequately.
   b. "Or equal" product required changes in on other elements of the Work.
   c. "Or equal" product caused problems in interfacing with other elements of the Work.

6. If, in the opinion of the Architect, the "or equal" product request is incomplete or has insufficient data to enable a full and thorough review of the proposed products, the proposed products may be summarily refused and determined to be unacceptable.

B. Product Substitutions: For products not governed by the "or equal" provision, comply with substitution provisions of the Contract General Conditions (Article 5.04-d, Substitutions) and requirements specified in Section 01630 - Product Substitution Procedures.

1.7 SYSTEM COMPLETENESS

A. System Completeness
   1. The Contract Drawings and Specifications are not intended to be comprehensive directions on

how to produce the Work. Rather, the Drawings and Specifications are instruments of service prepared to describe the design intent for the completed Work.

2. It is intended that all equipment, systems and assemblies be complete and fully functional even though not fully described. Provide all products and operations necessary to achieve the design intent described in the Contract Documents.
3. Refer to related general requirements specified in Section 01 41 00 - Regulatory Requirements regarding compliance with minimum requirements of applicable codes, ordinances and standards.

B. Omissions and Misdescriptions: Contractor shall report to Architect immediately when elements essential to proper execution of the Work are discovered to be missing or misdescribed in the Drawings and Specifications or if the design intent is unclear.

1. Should an essential element be discovered as missing or misdescribed prior to receipt of Bids, an Addendum will be issued so that all costs may be accounted for in the Contract Sum.
2. Should an obvious omission or misdescription of a necessary element be discovered and reported after execution of the Agreement, Contractor shall provide the element as though fully and correctly described, and a no-cost Change Order shall be executed.
3. Refer to related general requirements specified in Section 01 31 00 - Coordination regarding construction interfacing and coordination.

1.8 TRANSPORTATION, DELIVERY AND HANDLING

A. Transportation, Delivery and Handling, General: Contractor shall comply with manufacturer's instructions and recommendations for transportation, delivery and handling, in addition to the following.

B. Transportation: Contractor shall transport products by methods to avoid product damage.

C. Delivery:

1. Contractor shall schedule delivery to minimize long-term storage and prevent overcrowding construction spaces. Contractor shall coordinate with installation to ensure minimum holding time for items that are flammable, hazardous, easily damaged, or sensitive to deterioration, theft and other losses.
2. Contractor shall deliver products in undamaged condition in manufacturer's original sealed container or packaging system, complete with labels and instructions for handling, storing, unpacking, protecting and installing.
3. The Contractor shall not have products of any type that are intended to be received by the Contractor delivered to the University’s Mail and Package receiving Warehouse.

D. Handling:

1. Contractor shall provide equipment and personnel to handle products by methods to prevent soiling, marring or other damage.
2. Contractor shall promptly inspect products on delivery to ensure that products comply with Contract Documents, quantities are correct, and to ensure that products are undamaged and properly protected.

1.9 STORAGE AND PROTECTION

A. Storage and Protection, General: Contractor shall store and protect products in accordance with manufacturer's instructions, with seals and labels intact and legible.

1. Contractor shall periodically inspect to ensure products are undamaged, and are maintained under required conditions.
2. Contractor shall remove and replace products damaged by improper storage or protection with new products at no change in Contract Sum or Contract Time.
3. Contractor shall store sensitive products in weather tight enclosures or by other means recommended by the manufacturer.

B. Inspection Provisions: Contractor shall arrange storage to provide access for inspection and measurement of quantity or counting of units.

C. Structural Considerations: Contractor shall store heavy materials away from the structure in a manner that will not endanger supporting construction.

D. Weather-Resistant Storage:
1. Contractor shall store moisture-sensitive products above ground, under cover in a weather tight enclosure or covered with an impervious sheet covering. Contractor shall provide adequate ventilation to avoid condensation.

2. Contractor shall maintain storage within temperature and humidity ranges required by manufacturer's instructions.

3. For exterior storage of fabricated products, Contractor shall place products on raised blocks, pallets or other supports, above ground and in a manner to not create ponding or misdirection of runoff. Contractor shall place on sloped supports above ground.

4. Contractor shall store loose granular materials on solid surfaces in a well-drained area. Contractor shall prevent mixing with foreign matter.

E. Protection of Completed Work:
1. Contractor shall provide barriers, substantial coverings and notices to protect installed Work from traffic and subsequent construction operations.

2. Contractor shall remove protective measures when no longer required and prior to Contract Completion review of the Work.

3. Contractor shall comply with additional requirements specified in Section 01560 - Temporary Barriers and Enclosures.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

3.1 INSTALLATION OF PRODUCTS

A. Installation of Products:
1. Contractor shall comply with manufacturer's instructions and recommendations for installation of products, except where more stringent requirements are specified and necessary due to Project conditions or are required by authorities having jurisdiction.

2. Contractor shall anchor each product securely in place, accurately located and aligned with other Work.
3. Contractor shall clean exposed surfaces and provide protection to ensure freedom from damage and deterioration at time of Contract Completion review. Contractor shall refer to additional requirements specified in Section 01 74 00 - Cleaning Requirements and Section 01 56 00 - Temporary Barriers and Enclosures.

END OF SECTION

SECTION 01 63 00 - PRODUCT SUBSTITUTION PROCEEDURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. General requirements applicable to substitutions of materials, products, equipment and systems.

1.3 SUBSTITUTION OF MATERIALS AND EQUIPMENT

A. Substitutions, General: Catalog numbers and specific brands or trade names are used in materials, products, equipment and systems required by the Specifications to establish the standards of quality, utility and appearance required. Alternative products which are of equal quality and of required characteristics for the purpose intended may be proposed for use provided the Contractor complies with provisions of Contract General Conditions, Article 5.04., subject to the following provisions.

1. See Section 01 61 00 - Basic Product Requirements for requirements regarding product options.

2. Substitutions during the course of Work will only be authorized by properly executed Change Order or Field Instruction. Substitutions during the bidding and/or negotiation phases shall be added to the identified specification by Addendum.

3. **Note: the Trustees have no obligation to entertain substitutions.**

B. Substitution Provisions:

1. Documentation: Substitutions will not be considered if they are indicated or implied on shop drawing, product data or sample submittals. All requests for substitution shall be by separate written request from Contractor. See paragraph below for documentation required in the submission of request for substitution.

2. Cost and Time Considerations: Substitutions will not be considered unless a net reduction in Contract Sum or Contract Time results to University's benefit, including redesign costs, life cycle costs, plan check and permit fees, changes in related Work and overall performance of building systems.

3. Design Revision: Substitutions will not be considered if acceptance will require substantial revision of the Contract Documents or will substantially change the intent of the design, in the opinion of the Architect. The intent of the design shall include functional performance and aesthetic qualities.

4. Data: It shall be the responsibility of the Contractor to provide adequate data demonstrating

the merits of the proposed substitution, including cost data and information regarding changes in related Work.

5. Determination by Architect: Architect and University's Representative will determine the acceptability of proposed substitutions, and University's Representative will notify Contractor in writing of acceptance or rejection. The determination by the Architect regarding functional performance and aesthetic quality shall be final.

6. Non-Acceptance: If a proposed substitution is not accepted, Contractor shall immediately provide the specified product.

7. Substitution Limitation: Only one request for substitution will be considered for each product.

C. Request for Substitution Procedures: Comply with provisions of Contract General Conditions, Article 5.04 and the following.

1. Contractor shall prepare a request for substitution and submit the request to Architect through University's Representative for review and recommendation for acceptance. Acceptance and approval of substitutions shall be by University's Representative.
   a. Submit a minimum of five hard copies or submit electronically to the University's Representative.
   b. Present the request for substitution using form provided below.
   c. Comply with other administrative requirements shall be as directed by University's Representative.

2. Substitution requests shall include complete product data, including drawings and descriptions of products, fabrication details and installation procedures. Include samples where applicable or requested.

3. Substitution requests shall include appropriate product data for the specified product(s) of the specified manufacturer, suitable for use in comparison of characteristics of products.
   a. Include a written, point-by-point comparison of characteristics of the proposed substitute product with those of the specified product.
   b. Include a detailed description, in written or graphic form as appropriate, indicating all changes or modifications needed to other elements of the Work and to construction to be performed by the University and by others under separate contracts with University that will be necessary if the proposed substitution is accepted.

4. Substitution requests shall include a statement indicating the substitution's effect on the Construction Schedule. Indicate the effect of the proposed substitution on overall Contract Time and, as applicable, on completion of portions of the Work for use by University or for work under separate contracts by University.

5. Except as otherwise specified, substitution requests shall include detailed cost data, including a proposal for the net change, if any, in the Contract Sum.

6. Substitution requests shall include signed certification that the Contractor has reviewed the proposed substitution and has determined that the substitution, in combination with the cost or time savings, represents an equivalent or superior condition in every respect to product

requirements and value indicated or specified in the Contract Documents, and that the substitution is suited for and can perform the purpose or application of the specified product indicated or specified in the Contract Documents.

7. Substitution requests shall include a signed waiver by the Contractor for change in the Contract Time or Contract Sum because of the following:
    a. Substitution failed to perform adequately.
    b. Substitution required changes in on other elements of the Work.
    c. Substitution caused problems in interfacing with other elements of the Work.
    d. Substitution was determined to be unacceptable by authorities having jurisdiction.

8. If, in the opinion of the Architect, the substitution request is incomplete or has insufficient data to enable a full and thorough review of the intended substitution, the substitution may be summarily refused and determined to be unacceptable.

D. Contract Document Revisions:

1. Should a Contractor-proposed substitution or alternative sequence or method of construction require revision of the Contract Drawings or Specifications, including revisions for the purposes of determining feasibility, scope or cost, or revisions for the purpose of obtaining review and approval by authorities having jurisdiction, Architect or other consultant of University who is the responsible design professional will make revisions as approved in writing in advance by University's Representative.

2. Contractor shall pay for services of Architect, other responsible design professionals and University for researching and reporting on proposed substitutions or alternative sequence and method of construction when such activities are considered additional services to the design services contracts of Architect or other responsible design professional with University.

3. Contractor shall pay for costs of services by Architect, other responsible design professionals and University. These costs may include travel, reproduction, long distance telephone and shipping costs reimbursable at cost plus usual and customary mark-up for handling and billing.

4. Contractor shall pay such fees whether or not the proposed substitution or alternative sequence or method of construction is ultimately accepted by University and a Change Order is executed.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

Not Applicable to this Section.

SUBSTITUTION REQUEST FORM

SUBSTITUTION REQUEST NUMBER: ____

TO: ____

PROJECT: ____

SPECIFIED ITEM: ____

________________ __________ ___________ ______________________________________
Section Page Paragraph Description

The undersigned requests consideration of the following:

Proposed Substitution (Manufacturer, Model # or Name, Color, Etc.): ___________________________

____

History: ___New Product, ___Available 2-5 Years, ___Available 6-10 Years, ___Available 10+ Years

Provide UL, ITS, WHI, (or other) listing / rating of proposed substitution: ______________________

Attached data shall include, but not be limited to, product, specification, drawings, performance and test data adequate for evaluation of the request for the proposed substitution product and the specified product, with applicable portions of the proposed substitution and the specified product data clearly identified in a point-by-point direct comparison chart. Incomplete form and attachments will result in rejection of substitution request.

Requestor shall address the following items on this Substitution Request Form. Use a separate attached sheet attached as needed:

1. Reason for not providing specified item:
 ____
 ____
2. Will proposed substitution affect dimensions indicated on Drawings? ____(Yes) ____(No)
 If yes, how? ____
3. Will proposed substitution affect Electrical, Mechanical, Structural, Architectural, etc.? ____(Yes) ____(No) If yes, explain:
 ____
 ____
4. Is proposed substitution larger or smaller than specified product? ____(Yes) ____(No)
 If yes, state size of substitute product: ____

5. Does proposed substitution weight less/more than specified product? ____(Yes) ____(No)
   If yes, state weight of substitute product:

   ____

6. Will proposed substitution affect other trades and/or parts of the work? ____(Yes) ____(No)

   If yes, explain all effects:

   ____

   ____

7. Comparison between proposed substitution and specified product (Similarities / Differences)?

   ____

   ____

8. If Substitution Request is accepted, Owner will receive a credit of $____________. The Contract Sum will be adjusted accordingly.

9. Will proposed substitution affect the Contract Time? ____(Yes) ____(No)

   If yes, ____(Add) ____(Deduct) ________ calendar days.

____

INITIAL	UNDERSIGNED CERTIFIES:
______	Proposed substitution has been fully investigated and determined to be equal or superior in all respects to specified product.
______	Proposed substitution has same or better warranty as specified product.
______	Proposed substitution has same or better maintenance service and availability of replacement parts as specified product.
______	Proposed substitution will not affect or delay the Construction Schedule.
______	Claims for additional costs related to accepted substitution, which may subsequently become apparent, are hereby waived.
______	Proposed substitution will not affect dimensions and functional clearances.
______	Coordination, installation, and changes in the Work as necessary for installation of accepted substitution will be complete in all respects, at no additional cost to Owner.
______	Contractor will pay for all costs associated with changes to the project's design, including, but not limited to, architectural or engineering design fees, detailing, Agency approvals and construction costs caused by the requested substitution.
______	The function, appearance and quality of the proposed substitution is equivalent or superior to the specified item.

The undersigned certifies that the above is accurate and correct.

Signature: ____

Company: ____

Address: ____

____

Date: ____

Telephone: ____

Attachments: ________________Product Data ___Samples ___Tests ___Reports ___Other (Describe)

____

Architect's Review and Action:

______ Substitution Accepted – Make submittals in accordance with Specification Division 01 33 00.

______ Substitution Accepted as Noted - Make submittals in accordance with Specification Division 01 33 00.

______ Substitution Rejected – Provide specified product.

______ Substitution Request Received Too Late – Provide specified product.

By: ____ Date:________________________

Remarks: ____

____

END OF SECTION

SECTION 01 64 00 - OWNER FURNISHED PRODUCTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Requirements for installing Owner-furnished products, including providing miscellaneous items and accessories for a complete, functioning installation.

1.3 RELATED SECTIONS

A. Section 01 58 00 - Project Identification and Signage: Owner-furnished, Contractor-installed (OFCI) temporary signage.

1.4 PRODUCT HANDLING

A. Protection: Contractor shall use means necessary to protect the materials of this Section before, during, and after installation and to protect completed Work, including products installed by others.

B. Replacements: In the event of damage, Contractor shall immediately repair all damaged and defective Work to satisfaction of University's Representative, at no change in Contract Time and Contract Sum.

PART 2 - PRODUCTS

2.1 OWNER-FURNISHED/CONTRACTOR-INSTALLED (OFCI) PRODUCTS

A. Products Identified with Contractor Responsibility for Installation:

1. Contractor shall verify mounting and utility requirements for accepted products.

2. Contractor shall provide mounting and utility rough-ins for OFCI products.
    a. Rough-in locations, sizes, capacities and similar type shall be as indicated and required by product manufacturers.
    b. If the University substitutes items similar to those scheduled there shall be no change in rough-in cost, unless substitution occurs after rough-in has been completed or rough-in involves other mounting requirements, utilities of different capacity than those required by item originally specified.

3. For items Designated to Be Owner- or Vendor-Furnished: University or its vendor will furnish manufacturer's literature or information, shop drawings, or appropriate information for preparing required shop drawings.

B. Installation Instructions: Approved manufacturer's printed descriptions, specifications and recommendations shall govern the Work, unless specifically indicated otherwise.

C. Electrical Components: Contractor shall comply with requirements specified in Division 26 - Electrical, including California Electrical Code (CEC).

D. Plumbing and HVAC Components: Contractor shall comply with requirements specified in Division 22 – Plumbing and Division 23 - HVAC, including California Plumbing Code (CPC) and California Mechanical Code (CMC).

2.2 OWNER-FURNISHED/CONTRACTOR-INSTALLED PRODUCT REQUIREMENTS

A. Products Furnished by University and Installed by Contractor:

1. Contractor shall coordinate delivery of OFCI products. University will furnish products to coincide with construction schedule.

2. University will:
    a. Furnish standard integral components of products.
    b. Deliver products to site. Contractor shall assist University in offloading products.

3. The Contractor shall:
    a. Receive products at site and give written receipt for product at time of delivery, noting visible defects and omissions; if such declaration is not given, the Contractor shall assume responsibility for such defects and omissions.
    b. Store products until ready for installation and protect from loss and damage.
    c. Uncrate, assemble and set products in place.
    d. Install products in accordance with manufacturer's recommendations, instructions and shop drawings under supervision of manufacturer's representative where specified, supplying labor and material required and making mechanical, plumbing and electrical connections necessary to operate equipment.
    e. Where so specified, installation shall be only by installer approved by manufacturer. If known, approved installer is identified on the Drawings or in the Specifications.
    f. Provide and install backing for all products weighing 20 pounds or more.
    g. Treat all Owner or Vendor supplied products with the same care as all Contractor furnished items.

B. Products Furnished and Installed by University:

1. Contractor prepare; vendor install:
    a. General: Contractor shall coordinate deliveries of vendor-supplied products. Vendor will furnish products to coincide with the construction schedule.
    b. Vendor will:
        1) Furnish standard integral components of products.
        2) Deliver products to site.
        3) Make connections to roughed-in utilities.
    c. Contractor shall:
        1) Receive products at site and give written notice of receipt of each product at time of delivery, noting visible defects.

2) Provide rough-in of utility products in accordance with manufacturer's recommendations, instructions and shop drawings under supervision of the manufacturer's representative where specified.
3) Provide and install backing for all products weighing 20 pounds or more.

PART 3 - EXECUTION

3.1 SURFACE CONDITIONS

A. Inspection:

1. Prior to commencing Work, Contractor shall verify that Work specified in other Sections has been properly completed and installed as specified to allow for installation of all materials and methods required of this Section.

2. Contractor shall verify that new and existing products and conditions are satisfactory for installation or relocation of OFCI products. If unsatisfactory conditions exist, do not commence the installation until such conditions have been corrected.

B. Discrepancies:

1. In the event of discrepancy, Contractor shall immediately notify the University's Representative.

2. Contractor shall not proceed with installation in areas of discrepancy until all such discrepancies have been resolved.

3.2 INSTALLATION

A. Contractor shall relocate and reinstall existing products in accordance with Contract Documents and reviewed shop drawings, original manufacturer's instructions and recommendations if applicable and as directed.

B. Contractor shall install Owner-furnished products in accordance with reviewed shop drawings and manufacturer's printed instructions, as applicable.

3.3 ADJUSTING AND CLEANING

A. Contractor shall adjust products as necessary and as directed by University's Representative.

B. Contractor shall clean all new and relocated OFCI products.

C. Contractor shall protect OFCI products from damage until Contract Completion.

END OF SECTION

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SECTION 01 65 00 - PRODUCT DELIVERY REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Protect products scheduled for use in the work by means including, but not necessarily limited to, those described in this Section.

1.3 RELATED SECTIONS

A. Section 01 61 00 - Basic Product Requirements: Qualitative requirements for products.

B. Section 01 66 00 - Product Storage and Handling Requirements: Requirements for protection of products after delivery.

1.4 QUALITY ASSURANCE

A. Contractor's Quality Assurance: Contractor shall include within the Contractor's quality assurance program procedures as necessary to ensure protection of products upon delivery. Contractor shall be solely responsible for all products upon delivery to Work site and in off-site storage.

1. Contractor shall schedule delivery to minimize long-term storage at Project site and to prevent overcrowding of construction spaces.

2. Contractor shall coordinate delivery with installation time to ensure minimum holding time for items that are flammable, hazardous, easily damaged, or sensitive to deterioration, theft, and other losses.

3. Contractor shall inspect products on delivery to ensure compliance with the Contract Documents and to ensure that products are undamaged and properly protected.

B. Manufacturer's Requirements: Contractor shall determine and comply with manufacturer's instructions and recommendations for product handling.

C. Packaging: Contractor shall deliver products to Work site in manufacturer's original containers, with labels intact and legible.

1. Products delivered to Work site shall be in undamaged condition, in manufacturer's original sealed container or other packaging system, complete with labels and instructions for handling, storing, unpacking, protecting, and installing.

2. Contractor shall maintain packaged materials with seals unbroken and labels intact until time of use.

3. Products will be subject to rejection if they do not bear required identification or are unsuitably packaged.

D. Delivery:

1. Contractor shall address and deliver products to Project site. Do not deliver products to University campus shipping and delivery department. Address deliveries to Contractor and Project name. Do not address products "care of" University.

2. University will not be responsible for mis-addressed and mis-delivered products, including claims for damage and delay.

E. Damaged Products: In event of damage, Contractor shall promptly make replacements and repairs to packaging and contents, as acceptable to University's Representative, at no change in Contract Sum and Contract Time.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

Not Applicable to this Section.

END OF SECTION

SECTION 01 66 00 - PRODUCT STORAGE AND HANDLING REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Storage and protection requirements to ensure that products intended for use in the Work will not be damaged and will not deteriorate from time of delivery to time of incorporation into the Work.

1.3 RELATED SECTIONS

A. Section 01 52 00 - Construction Facilities: Requirements for storage sheds.

B. Section 01 52 05 - Construction Staging Areas: Locations for vehicular access and staging of products during Work.

C. Section 01 56 00 - Temporary Barriers and Enclosures: Requirements for temporary construction barriers, enclosures and passageways, applicable to protection of construction.

D. Section 01 61 00 - Basic Product Requirements: Qualitative requirements for products.

E. Section 01 65 00 - Product Delivery Requirements: Requirements for packaging and delivery of products.

1.4 QUALITY ASSURANCE

A. Contractor's Quality Assurance: Contractor shall include within the Contractor's quality assurance program procedures as necessary to ensure protection of products after delivery to Work site. Contractor shall be solely responsible for all products stored on site and in off-site storage.

1. Contractor shall protect stored products from damage.

2. Contractor shall store products to allow for inspection and measurement of quantity or counting of units.

3. Contractor shall store materials in a manner that will not endanger Project structure.

4. Contractor shall store products that are subject to damage by the elements, under cover in a weather tight enclosure above ground, with ventilation adequate to prevent condensation.

B. Manufacturer's Handling Requirements: Contractor shall determine and comply with product manufacturer's written instructions for handling products.

C. Manufacturer's Storage Requirements: Contractor shall determine and comply with product manufacturer's written instructions for temperature, humidity, ventilation, and weather-protection requirements for storage.

D. Storage: Contractor shall provide secure locations and enclosures at Project site for storage of materials and equipment. Contractor shall coordinate location with Contractor storage and staging areas. Refer to Section 01 52 00 - Construction Facilities and Section 01 52 05 - Construction Staging Areas.

1. Contractor shall maintain packaged materials with seals unbroken and labels intact until time of use.

2. Products will be subject to rejection if they do bear required identification or are unsuitably packaged.

E. Damaged Products: In event of damage, Contractor shall promptly make replacements and repairs to packaging and contents, as acceptable to University's Representative, at no change in Contract Sum and Contract Time.

F. Contractor shall not have products or materials that are intended for the Contractor delivered to the University’s Receiving and Distribution Warehouse. Contractor shall make arrangements to have products and/or materials delivered directly to the work site. Products and/or materials delivered to the University’s Receiving and Distribution Warehouse will not be accepted.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

Not Applicable to this Section.

END OF SECTION

SECTION 01 72 00

PREPARATION REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Requirements for preparation prior to installing, applying and placing products to determine acceptable conditions for the Work.

B. Layout of the Work and other engineering services necessary to accomplish the Work.

1.3 RELATED SECTIONS

A. Section 01 31 00 - Coordination: Requirements for proper sequencing and interfacing of the Work.

B. Section 01 31 19 - Project Meetings: General requirements for pre-installation conferences.

C. Section 01 73 29 - Cutting and Patching: Work performed to provide access for performing the Work.

D. Section 01 77 00 - Contract Closeout Procedures: Project record documents, including layout data.

E. Section 01 78 29 - Survey and Layout Data: Requirements for survey and layout data submittals.

****
FOLLOWING ARE EXAMPLES ONLY. LIST RELATED SECTIONS APPLICABLE TO PROJECT.
****

A. Section 02 41 13 - Selective Site Demolition: Removal of existing construction in preparation of performance of specified Work.

B. Section 02 42 00 – Removal and Salvage of Construction Materials: Removal of products in preparation for the Work.

C. Individual Division 2 through 48 Product Specification Sections: Specific requirements for preparation prior to performance of the Work.

1.4 LAYOUT OF WORK

A. Surveyor: Contractor shall select and pay for services of a land surveyor, registered in the State of California, for proper performance of the Work.

1. Services of surveyor shall be suitable for layout and verification of location of buildings and site elements.
2. For the Project record, Contractor shall submit the name, address and telephone number of land surveyor before starting survey Work.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

3.1 PREPARATION

A. Manufacturer's Requirements: Contractor shall determine product manufacturer's requirements and recommendations prior to commencing Work.

B. Preparations: Contractor shall perform preparation actions according to manufacturer's instructions and recommendations and according to specified procedures.

1. Contractor shall perform surface preparation as necessary to create suitable substrates for application, installation and placement of products.

2. Contractor shall notify University's Representative in writing of unsuitable conditions preventing proper performance of the Work.

C. Existing Utility Information: Contractor shall furnish information to serving utility that is necessary to adjust, move, or relocate existing utility structures, utility poles, lines, services, or other utility appurtenances located in or affected by construction. Contractor shall coordinate with University's Representative and with authorities having jurisdiction.

D. Existing Utility Interruptions: Contractor shall not interrupt utilities serving facilities occupied by University or others unless permitted under the following conditions and then only after arranging to provide temporary utility services according to requirements indicated:

1. Contractor shall notify University's Representative not less than two working days in advance of proposed utility interruptions.

2. Contractor shall not proceed with utility interruptions without written permission from University's Representative.

E. Field Measurements: Contractor shall take field measurements as required to fit the Work properly. Contractor shall recheck measurements before installing each product. Where portions of the Work are indicated to fit to other construction, Contractor shall verify dimensions of other construction by field measurements before fabrication. Contractor shall coordinate fabrication schedule with construction progress to avoid delaying the Work.

F. Space Requirements: Contractor shall verify space requirements and dimensions of items shown diagrammatically on Drawings.

G. Review of Contract Documents and Field Conditions: Immediately upon discovery of the need for clarification of the Contract Documents, Contractor shall submit a Request for Interpretation (RFI) to Architect. Contractor shall include a detailed description of problem encountered, together with recommendations for changing the Contract Documents. Contractor shall submit requests in accordance with requirements specified in Section 01340 - Requests for Interpretation (RFI), using form as directed by University's Representative.

H. Verification of Construction Layout: Before proceeding to lay out the Work, Contractor shall verify layout information shown on Drawings, in relation to the property survey and existing benchmarks, and locate survey reference points. If discrepancies are discovered, Contractor shall promptly notify University's Representative, Architect and Project Inspector.

3.2 FIELD ENGINEERING

A. Examination: Contractor shall verify locations of survey control and reference points prior to starting Work. If discrepancies are discovered, Contractor shall promptly notify University's Representative, Architect and Project Inspector.

B. Survey Control and Reference Points: Contractor shall locate and protect survey control and reference points. Control datum for survey shall be as indicated on Civil Drawings. Notwithstanding the data on Civil Drawings, Contractor shall use NAD 83 State Plane Coordinate System for survey control and reference points.

1. Business and Professions Code section 8771 provides for the preservation of Survey Monuments in construction projects. This legislation mandates that, prior to construction, monuments shall be referenced in the field and "Corner Records" shall be prepared for filing in the Office of the County Surveyor. Contractor shall ensure that these shall be performed prior to Contract Completion of the Work.

2. Contractor shall comply with requirements of authorities having jurisdiction for survey monument preservation on capital improvement projects where monument points are present.

3. Contractor shall be responsible for preparing and submitting proper documentation to the Office of the County Surveyor in compliance with Business and Professions Code section 8771.

4. Contract Completion and release of retainage shall be contingent upon obtaining documentation from Contractor's project surveyor or engineer that monuments have been set or restored and that Corner Records have been filed with and to the satisfaction of the County Surveyor.

5. All costs and actions necessary for compliance with Business and Professions Code section 8771 shall be included in the Contract Sum and Contract Time.

3.3 SURVEYING AND FIELD ENGINEERING SERVICES

A. Surveying and Field Engineering Services: Contractor shall provide surveying and field engineering services as necessary for performance of the Work. Refer to Section 01 78 29 - Survey and Layout Data.

1. Contractor shall be responsible for the accuracy and adequacy of surveying and field engineering services.

2. Contractor shall utilize recognized engineering practices.

3. Contractor shall check the location, level and plumb, of every major element as the Work progresses.

4. Contractor shall preserve construction survey stakes and marks for the duration of their usefulness.

5. If construction survey stakes are lost or disturbed, and require replacement, Contractor shall perform replacement at no change in Contract Sum and Contract Time.

6. Contractor shall excavate all holes necessary for line and grade stakes.

B. Surveying for Layout and Control of the Work: Contractor shall establish elevations, lines and levels for all Work under the Contract. Contractor shall locate and lay out by instrumentation and similar appropriate means:

1. Site improvements, including pavements, curbs, headers, sewers, storm drains, structures, and paving. Note on Project Record Drawings utility locations, slopes and invert elevations.

2. Stakes for cutting, filling, grading and topsoil placement, to establish finished grade or flow line indicated on Contract Drawings.

   a. Contractor shall preserve construction survey stakes and marks for the duration of their usefulness.
   b. If construction survey stakes are lost or disturbed, and require replacement, Contractor shall perform replacement at no change in Contract Sum and Contract Time.
   c. Contractor shall excavate all holes necessary for line and grade stakes.

3. Grid or axis for structures, building foundation, column locations and ground floor elevations.

4. Contractor shall establish benchmarks and control points to set lines and levels at each story of construction and elsewhere as needed to locate each element of Project.

5. Contractor shall establish dimensions within tolerances indicated. Contractor shall not scale Drawings to obtain required dimensions.

6. Contractor shall inform installers of lines and levels to which they must comply.

7. When deviations from required lines and levels exceed allowable tolerances, Contractor shall notify University's Representative, Architect and Project Inspector.

8. Contractor shall close site surveys with an error of closure equal to or less than the standard established by authorities having jurisdiction.

C. Monuments: Contractor shall establish a minimum of two permanent monuments on site, referenced to established control points. Contractor shall record locations, with horizontal and vertical data, on Project Record Drawings.

1. In accordance with Business and Professions Code section 8772, any monument set by a licensed land surveyor or registered civil engineer to mark or reference a point on a property or land line shall be permanently and visibly marked or tagged with the certificate number of the surveyor or civil engineer setting it, each number preceded by the letters "L.S." or "R.C.E." respectively, as the case may be, or, if the monument is set by a public agency, it shall be marked with the name of the agency and the political subdivision it serves.

2. Nothing in this Section shall prevent the inclusion of other information on the tag which will assist in the tracing or location of survey records which relate to the tagged monument.

3. Contractor shall ensure that centerline ties filed with the County Surveyor will be checked for compliance with this law.

D. Site Grading Verification: Upon completion of grading, Contractor shall survey graded areas and establish that elevations are correct and within acceptable tolerances for paving and finish grading.

E. Verification of Work: Contractor shall periodically verify layout and completed conditions of the Work by same means.

END OF SECTION

SECTION 01 73 00

EXECUTION REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. General requirements for installing, applying and placing products.

B. General requirements for correction of defective Work.

1.3 RELATED SECTIONS

A. Section 01 31 19 - Project Meetings: Pre-installation and coordination conferences where procedures for installing, applying and placing products are reviewed prior to performance of the Work.

B. Individual Division 2 through 49 Product Specification Sections: Specific requirements for installing, applying and placing products.

1.4 EXECUTION

A. Manufacturer's Requirements: Contractor shall determine product manufacturer's requirements and recommendations prior to commencing Work.

B. Execution: Contractor shall perform installation, application and placement actions according to manufacturer's instructions and recommendations and according to specified procedures.
   1. Contractor shall perform surface preparation as necessary to create suitable substrates for application, installation and placement of products.
   2. Contractor shall notify University's Representative in writing of unsuitable conditions preventing proper performance of the Work.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

3.1 INSTALLATION, APPLICATION AND PLACEMENT OF PRODUCTS

A. Manufacturer's Instructions: Contractor shall comply with manufacturer's written instructions and recommendations for installing, applying, placing and finishing products.

B. Installation, Application and Placement, General: Contractor shall locate the Work and components of the Work accurately, in correct alignment, orientation and elevation, as indicated.
   1. Contractor shall make vertical work plumb and make horizontal work level.
   2. Where space is limited, Contractor shall install components to maximize space available for maintenance and ease of removal for replacement.
   3. Contractor shall conceal pipes, ducts, and wiring in finished areas, unless otherwise indicated.
   4. Contractor shall maintain minimum headroom clearance of 80” in all spaces, unless otherwise directed.

5. Contractor shall install products at the time and under conditions that will ensure the best possible results. Contractor shall maintain conditions required for product performance until acceptance of the Work.
6. Contractor shall conduct construction operations so no part of the Work is subjected to damaging operations or loading in excess of that expected during normal conditions of occupancy.

C. Tools and Equipment: Contractor shall not use tools or equipment that produce harmful noise levels. Refer to Section 01 14 00 – Work Restrictions

D. Anchors and Fasteners: Contractor shall provide anchors and fasteners as required to anchor each component securely in place, accurately located and aligned with other portions of the Work.
1. Mounting Heights: Where mounting heights are not indicated, Contractor shall mount components at heights directed by Architect.
2. Contractor shall allow for building movement, including thermal expansion and contraction.

E. Joints: Contractor shall make joints of uniform width. Where joint locations in exposed work are not indicated, Contractor shall arrange joints for the best visual effect. Contractor shall fit exposed connections together to form hairline joints.

F. Hazardous Materials: Contractor shall use products, cleaners, and installation materials that are not considered hazardous.

G. Cleaning: Contractor shall comply with requirements specified in Section 01 74 00 - Cleaning Requirements. See individual product Specifications Sections for specific cleaning procedures to be performed.

H. Protection: Contractor shall provide barriers, covers and other protective devices as recommended by manufacturer and complying with general requirements specified in Section 01 56 00 - Temporary Barriers and Enclosures.
1. Contractor shall comply with manufacturer's written instructions for temperature and relative humidity.
2. See individual product Specifications Sections for specific protective measures to be provided.

I. Limiting Exposures: Contractor shall supervise construction operations to assure that no part of the construction, completed or in progress, is subject to harmful, dangerous, damaging, or otherwise deleterious exposure during the construction period.

3.2 OWNER-INSTALLED PRODUCTS

A. Site Access: Contractor shall provide access to Project site for University's construction forces and those performing work for University under separate contracts. Contractor shall coordinate with requirements specified in Section 01 55 00 - Vehicular Access and Parking.

B. Coordination: Contractor shall coordinate construction and operations of the Work with work performed by University by separate contract or with University's construction forces.
1. Construction schedule: Contractor shall inform University's Representative of Contractor's preferred construction schedule for University-installed work. Contractor shall adjust construction schedule based on a mutually agreeable timetable. Contractor shall notify University's Representative if changes to schedule are required due to differences in actual construction progress.
2. Pre-installation and coordination conferences: Contractor shall include University's construction forces at pre-installation and coordination conferences covering portions of the Work that are to receive University-installed work. If portions of the Work depend on University-installed products, Contractor shall attend pre-installation conferences conducted by University's construction forces.

3.3 CORRECTION OF THE WORK

A. Correction of the Work, General: Contractor shall repair or remove and replace defective construction. Contractor shall restore damaged substrates and finishes to match original and new surrounding construction.

1. Contractor shall comply with requirements in Section 01 73 29 - Cutting and Patching Procedures.
2. Repairing shall include replacing defective parts, refinishing damaged surfaces, touching up with matching materials, and properly adjusting operating equipment.
3. Contractor shall remove and replace damaged surfaces that are exposed to view if surfaces cannot be repaired without visible evidence of repair.
4. Contractor shall repair components that do not operate properly. Remove and replace operating components that cannot be repaired.
5. Contractor shall remove and replace chipped, scratched, and broken glass.

B. Restoration of Existing Conditions: Contractor shall restore permanent facilities used during construction to their original condition or to match new construction.

END OF SECTION

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SECTION 01 73 29

CUTTING AND PATCHING REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Requirements and procedural requirements for cutting and patching, including:
   1. Cutting and patching not required to be performed as part of the Work specified in other Sections.
   2. Cutting and patching existing construction altered or disturbed to accommodate new construction.
   3. Cutting and patching existing construction damaged or defaced during new construction as required to restore to existing or better condition at the time of award of Contract.
   4. Cutting and patching required to:
      a. Install or correct non-coordinated Work.
      b. Remove and replace defective and non-conforming Work.
      c. Remove samples of installed Work for testing.

B. Refer to other Sections and drawings for specific requirements of the extent and limitations applicable to cutting and patching, demolishing, or altering existing construction of individual parts of the Work.
   1. Requirements of this Section also apply to plumbing mechanical and electrical installations. (Refer to Division-22, Division 23 and Division-26 Sections for other requirements and limitations applicable to cutting and patching plumbing, mechanical and electrical installations).

1.3 RELATED SECTIONS

A. Section 01 11 00 - Summary of the Work: Work by University's construction forces or by others under separate contract with University.

B. Section 01 56 00 - Temporary Barriers and Enclosures: Dust-control barriers at cutting and patching locations.

C. Section 01 74 00 - Cleaning Requirements: Cleaning after cutting and patching Work.

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FOLLOWING IS EXAMPLE ONLY. LIST RELATED SECTIONS APPLICABLE TO PROJECT.
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D. Section 02 41 13 - Selective Demolition: Cutting and removal of existing construction.

E. Individual Division 2 through 48 Product Specification Sections:
   1. Cutting and patching incidental to Work specified in the Section.
   2. Coordination with Work specified in other Sections for openings required to accommodate Work specified in those other Sections.

1.4 SUBMITTALS

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EDIT REFERENCE BELOW TO SUIT PROJECT. VERIFY APPLICABLE SECTION NUMBER AND TITLE FOR DEMOLITION WORK.
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A. Written Requests for Cutting and Alteration: Coordinate with requirements specified in [*Section 02 41 13 - Selective Demolition*] [*Section* [*_No._*] - [_____]].
   1. Contractor shall submit written request in advance of cutting or alteration which affects:
      a. Structural integrity of any element of new or existing construction.
      b. Integrity of weather-exposed or moisture-resistant elements.
      c. Efficiency, maintenance, or safety of operational elements.

d. Visual qualities of elements exposed to view in the completed construction.
e. Work by University's construction forces or by others under separate contract with University.
f. Existing construction not otherwise indicated to be revised by Work under the Contract.

2. Contractor shall include in requests for cutting and alteration:
   a. Identification of Project.
   b. Location and description of affected Work. Include shop drawings as necessary to identify locations and communicate descriptions.
   c. Explanation of necessity for cutting and patching.
   d. Description of proposed Work and products to be used.
   e. Alternatives to cutting and patching.
   f. Effect on existing construction.
   g. Effect on work by University's construction forces or by separate contractors performing work for University.

3. Contractor shall include written evidence that those performing work under separate contract for University have been notified and acknowledge that cutting and patching work will be occurring. Contractor shall include written permission for intended cutting and patching, included scheduled times.

4. Contractor shall indicate date and time cutting and patching Work will be performed, including duration.

5. Contractor shall describe the extent of cutting and patching required and how it is to be performed.

6. Contractor shall describe anticipated results in terms of changes to existing construction; include changes to structural elements and operating components as well as changes in the building's appearance and other significant visual elements.

7. Contractor shall list products to be used and firms or entities that will perform work.

8. Contractor shall list utilities that will be disturbed or affected, including those that will be relocated and those that will be temporarily out-of-service. Contractor shall indicate how long service will be disrupted.

9. Where cutting and patching involves addition of reinforcement to structural elements, Contractor shall submit details to show how reinforcement is integrated with the original structure.

10. Approval by the Architect to proceed with cutting and patching does not waive the Architect's right to later require complete removal and replacement of a part of the Work found to be unsatisfactory.

11. Contractor shall minimize effects on University operations and on concurrent operations construction by other contractors.

1.5 QUALITY ASSURANCE

A. Requirements for Structural Work: Contractor shall not cut and patch structural elements in a manner that would reduce their load-carrying capacity or load-deflection ratio.

1. Contractor shall obtain approval from the Architect of the cutting and patching proposal before cutting and patching the following structural elements:
   Bearing and retaining walls
   Structural concrete
   Structural steel
   Lintels
   Timber and primary wood framing
   Structural decking
   Stair systems
      Miscellaneous structural metals
      Equipment supports
      Piping, ductwork, vessels and equipment

B. Operational and Safety Limitations: Contractor shall not cut and patch operating elements or safety-related components in a manner that would result in reducing their capacity to perform as intended, or result in increased maintenance, or decreased operational life or safety.
  1. Contractor shall obtain approval of the cutting and patching proposal before cutting and patching the following operating elements or safety-related systems:
     Primary operational systems and equipment
     Air or smoke barriers
     Water, moisture, or vapor barriers
     Membranes and flashings
     Fire protection systems
     Noise and vibration control elements and systems
     Control systems
     Communication systems
     Electrical wiring systems

C. Visual Requirements: Contractor shall not cut and patch construction exposed on the exterior or in occupied spaces, in a manner that would, in the Architect's opinion, reduce the building's aesthetic qualities, or result in visual evidence of cutting and patching. Contractor shall remove and replace work cut and patched in a visually unsatisfactory manner.

D. If possible Contractor shall retain the original installer or fabricator throughout construction phases to cut and patch the following categories of exposed work, or if it is not possible to engage the original installer or fabricator, Contractor shall engage another recognized experienced and specialized firm:
     Concrete finishes
     Masonry
     Stucco and ornamental plaster
     Acoustical ceilings
     Painting
     Wall covering
     HVAC enclosures, cabinets or covers

PART 2 - PRODUCTS

2.1 PATCHING MATERIALS

A. Patching Materials, General: As required for original installation and to match surrounding construction.
  1. Contractor shall provide same products or types of construction as that in existing structure, as needed to patch, extend or match existing.
  2. Generally the Contract Documents will not define products or standards of workmanship present in existing construction; Contractor shall determine products by inspection and necessary testing, and determine quality of workmanship by using existing as a sample for comparison.
  3. The presence of a product, finish, or type of construction requires that patching, extending or matching shall be performed as necessary to make work complete and consistent with identical standards of quality.

B. Patching at Paving: At Portland cement concrete (PCC) paving, Contractor shall use concrete mix with maximum 3/8-inch aggregate and minimum 3000 psi 28-day compressive strength. Contractor shall provide dowels to existing paving with min. 6” penetration into existing surface and reinforce new paving with minimum No. 3 reinforcing steel bars at 16-inches on center each way placed in the vertical center of the slab. Welded wire fabric reinforcement will not be acceptable.
  1. All PCC paving shall be cut and patched from score line to score line and shall match as closely as possible in color and texture of the adjacent finish.

C. Patching of Lawns and Grasses: Contractor shall restore areas trenched, disturbed or damaged. Contractor shall provide sod or seeded planting mix, to match existing lawn or grass area. Contractor shall properly barricade the area until such time as the planting mix establishes. Refer to sections 01 59 39 – Tree and Plant Protection

D. Patching of Building Finish Materials: Contractor shall match existing products and finishes. Contractor shall confirm colors, patterns and textures with Architect. Contractor shall custom cut new materials to fit and to match joint patterns with existing materials.
   1. Ceramic tile and acoustical panels: Contractor shall custom cut new materials to size to match existing construction.

E. Product Substitutions: For each proposed change in materials, Contractor shall submit request for substitution under provisions of Section 01 61 00 - Basic Product Requirements.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examination, General: Before cutting existing surfaces, Contractor shall examine surfaces to be cut and patched and conditions under which cutting and patching is to be performed. Contractor shall take corrective action before proceeding, if unsafe or unsatisfactory conditions are encountered. Contractor shall inspect existing conditions prior to commencing Work, including elements subject to damage or movement during cutting and patching.
   1. Before proceeding, Contractor shall meet at the site with parties involved in cutting and patching, including asbestos abatement, mechanical and electrical trades. Contractor shall review areas of potential interference and conflict. Contractor shall coordinate procedures and resolve potential conflicts before proceeding.
   2. Beginning of cutting or patching shall be interpreted to mean that existing conditions were found by Contractor to be acceptable.
   3. After uncovering existing Work, Contractor shall inspect conditions affecting proper accomplishment of Work.

3.2 PREPARATION

A. Temporary Supports: Contractor shall provide supports to ensure structural integrity of the Work. Contractor shall provide devices and methods to protect other portions of Project from damage.

B. Protection: Contractor shall protect existing construction during cutting and patching to prevent damage. Contractor shall provide protection from adverse weather conditions for portions of the Project that might be exposed during cutting and patching operations.

C. Contractor shall avoid interference with use of adjoining areas or interruption of free passage to adjoining areas.

D. Contractor shall take all precautions necessary to avoid cutting existing pipe, conduit or ductwork serving the building, but scheduled to be removed or relocated until provisions have been made to bypass them.

E. Weather Protection: Contractor shall provide protection from elements for areas which may be exposed by uncovering Work. Contractor shall maintain excavations free of water.

3.3 CUTTING AND PATCHING

A. Cutting and Patching, General: Contractor shall execute cutting, fitting, and patching, excavation and fill, as necessary to complete the Work. Contractor shall employ skilled workers to perform cutting and patching. Contractor shall proceed with cutting and patching at the earliest feasible time and complete without delay. Contractor shall:
   1. Coordinate installation or application of products for integrated Work. Avoid having to cut and patch new substrates and finishes.
   2. Uncover completed Work as necessary to install or apply products out of sequence.
   3. Cut, remove and replace defective and non-conforming Work.
   4. Cut and patch as necessary to provide openings in the Work for penetration of plumbing, fire protection,

HVAC and electrical Work.

5. Where partitions are removed, patch floors, walls, and ceilings with finish materials to match existing.
   a. Where removal of partitions results in adjacent spaces becoming one, re-work floors and ceilings to provide smooth and clean planes without breaks, steps, or bulkheads.
   b. Where extreme change of plane of one inch or more occurs, request instructions from Architect as to method of making transition.
6. Trim and refinish existing doors as necessary to clear new floor finishes.
7. By-pass utility services such as pipe or conduit, before cutting, where services are shown or required to be removed, relocated or abandoned. Cut-off pipe or conduit in walls or partitions to be removed. Cap, valve or plug and seal the remaining portion of pipe or conduit to prevent entrance of moisture or other foreign matter after by-passing and cutting. Update as-built set with photographs or notations for the actual conditions.

B. Cutting: Contractor shall:
1. Cut existing construction using methods least likely to damage elements to be retained or adjoining construction. Where possible review proposed procedures with the original installer; comply with the original installer's recommendations. Provide appropriate surfaces to receive final finishing. It is recommended to photograph the existing condition prior to cutting. This photo record shall serve as the pre-cut condition for comparison to the final patched outcome.
2. Execute cutting and patching of weather-exposed, moisture-resistant elements and surfaces exposed to view by methods to preserve weather, moisture and visual integrity.
3. Cut rigid materials using carbide tip saw blades, diamond grit abrasive saw blades, diamond core drills and hole saws, and similar cutters for smooth edges. Do not overcut corners.
   a. Core drill holes through concrete and masonry.
   b. Pneumatic tools will not be allowed without prior approval.
4. Provide fire and smoke seals at new penetrations to maintain fire rating at all penetrations.
5. Confirm and comply with all Asbestos and lead containing/based paint remedial procedures listed in Section 01 35 01 – Hazardous Material Procedures prior to any disturbance of any existing material.

C. Patching: Contractor shall patch with durable seams that are as invisible as possible. Contractor shall comply with specified tolerances. Contractor shall restore substrates and finishes with products to match existing construction and as specified in product Sections of the Specifications for new construction. Contractor shall:
1. Where feasible, inspect and test patched areas to demonstrate integrity of the installation.
2. Restore exposed finishes of patched areas and extend finish restoration into retained adjoining construction in a manner that will eliminate evidence of patching and refinishing.
3. Where removal of walls or partitions extends one finished area into another, patch and repair floor and wall surfaces in the new space to provide an even surface of uniform color and appearance. Remove existing floor and wall coverings and replace with new materials, if necessary to achieve uniform color and appearance.
   a. Where patching occurs in a smooth painted surface, extend final paint coat over entire unbroken containing the patch, after the patched area has received primer and second coat.
4. Patch, repair or re-hang existing ceilings as necessary to provide an even plane surface of uniform appearance.
5. Finish surfaces flush and textured to match surrounding finishes.
6. Fit work neat and tight allowing for expansion and contraction.
7. Butt new finished to existing exposed structure, pipes, ducts, conduit, and other penetrations through surfaces.

D. Finishing: Contractor shall refinish surfaces to match adjacent and similar finishes as used for the Project.
1. For continuous surfaces, Contractor shall refinish to nearest intersection or natural break.
2. For an assembly, Contractor shall refinish entire unit.

E. Penetrations at Fire-Rated Construction: At penetrations of fire rated walls, partitions, ceiling, or floor construction, Contractor shall completely seal voids with firestopping and smoke seal material in compliance with an applicable UL-listed assembly, to full thickness of the penetrated element. Refer to [Section 07 84 00 - Firestopping] [*Section* [*_No._*] - [*_TITLE_*]].

F. Restoration and Finishing: Contractor shall finish surfaces to match adjacent and similar finishes as used for the Project.
   1. Contractor shall restore Work with new products as specified in individual product Specifications Sections in Divisions 07 and 09.
   2. Contractor shall patch and replace any portion of an existing finished surface which is found to be damaged, lifted, discolored, or shows other imperfections, with matching material. Contractor shall**:**
      a. Provide adequate support of substrate prior to patching the finish.
      b. Refinish patched portions of painted or coated surfaces in a manner to produce uniform color and texture over the entire surface.
      c. When existing surface finish cannot be matched, refinish entire surface to nearest intersections.

G. Transition from Existing to New Construction:
   1. When new work abuts or finishes flush with existing work, Contractor shall make a smooth and clean transition. Contractor shall patched work shall match existing adjacent work in texture and appearance so that the patch or transition is invisible at a distance of five feet.
   2. When finished surfaces are cut in such a way that a smooth and clean transition with the new work is not possible, Contractor shall notify Architect. Contractor shall terminate existing surface in a neat manner along a straight line at a natural line of division, and provide trim appropriate to finished surface, or as otherwise directed by Architect.

H. Plaster Installation: Contractor shall comply with manufacturer's instructions and install thickness and coats as indicated.

3.4 CLEANING

A. Cleaning: Contractor shall thoroughly clean areas and spaces where cutting and patching is performed or used as access. Contractor shall remove completely paint, mortar, oils, putty and items of similar nature. Contractor shall thoroughly clean piping, conduit and similar features before painting or other finishing is applied. Contractor shall restore damaged pipe covering to its original condition.

END OF SECTION

SECTION 01 74 00 - CLEANING REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Cleaning during construction.

B. Cleaning for Contract Completion review and final acceptance of the Work.

1.3 RELATED SECTIONS

A. Additional Requirements: Cleaning for specific products or elements of Work are described in individual product Specification Sections in Divisions 2 through 49. Contractor shall comply also with University's Contractor Safety Handbook.

1.4 SUBMITTALS

A. Product List: Contractor shall submit complete list of all cleaning agents and materials for University's Representative's review and approval.

B. Cleaning Procedures: Contractor shall submit description of cleaning processes, agents and materials to be used for final cleaning of the Work. Processes and degree of cleanliness shall be as directed by University's Representative. All cleaning processes, agents and materials shall be subject to University's Representative's review and approval.

1.5 QUALITY ASSURANCE

A. Cleaning and Disposal Requirements, General: Contractor shall conduct cleaning and disposal operations in compliance with all applicable codes, ordinances and regulations, including environmental protection laws, rules and practices.

B. Cleaning Workers: Contractor shall employ experienced workers or professional cleaners for final cleaning. Contractor shall clean each surface or unit to the condition expected in a normal, commercial building cleaning and maintenance program. Contractor shall comply with manufacturer's instructions.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Cleaning Agents and Materials: Contractor shall use only those cleaning agents and materials which

will not create hazards to health or property and which will not damage or degrade surfaces. Contractor shall:
1. Use only those cleaning agents, materials and methods recommended by manufacturer of the material to be cleaned.
2. Use cleaning materials only on surfaces recommended by cleaning agent manufacturer.

PART 3 - EXECUTION

3.1 CLEANING DURING CONSTRUCTION

A. Garbage Control: Contractor shall control accumulation of debris, waste materials and rubbish. Contractor shall leave the site each day clean and neat. Contractor shall dispose of debris, waste and rubbish off-site in a legal manner.

B. Cleaning, General: Contractor shall clean sidewalks, driveways and streets frequently to maintain public thoroughfares free of dust, debris and other contaminants. This shall be at no cost to the University.

C. Cleaning of Existing Facilities: Contractor shall clean surfaces in existing buildings where alteration and renovation Work is being performed or where other construction activities have caused soiling and accumulation of dust and debris. Contractor shall:
1. Clean dust and soiling from floor surfaces.
2. Clean dust from horizontal and vertical surfaces, including lighting fixtures.
3. Replace HVAC filters.

D. Parking Area Cleaning: Contractor shall keep parking areas clear of construction debris, especially debris hazardous to vehicle tires.

E. Thoroughfare Clearing and Cleaning: Contractor shall keep site accessways, parking areas and building access and exit facilities clear of mud, soiling and debris. Contractor shall:
1. Remove mud, soil and debris and dispose in a manner which will not be injurious to persons, property, plant materials and site.
2. Comply with runoff control requirements stated above and as required by governing authorities having jurisdiction.

F. Cleaning Frequency: At a minimum, Contractor shall clean Work areas and site daily. Contractor shall leave the site each day clean and neat.

G. Failure to Clean: At any point during the course of Work, should cleaning by Contractor not be sufficient or acceptable to University's Representative, especially regarding paths of travel, University may engage cleaning service to perform cleaning and deduct costs for such cleaning from sums owed to Contractor.

3.2 CONTRACT COMPLETION REVIEW CLEANING, GENERAL

A. Contract Completion Review Cleaning, General: Contractor shall execute a thorough cleaning prior to Contract Completion review by University's Representative and Architect. Contractor shall complete final cleaning before submitting final Application for Payment. Contractor shall:

1. Conduct cleaning in compliance with regulations of authorities having jurisdiction and industrial safety standards for cleaning.
2. Employ professional building cleaners to thoroughly clean building.
3. Complete cleaning operations specified below before requesting inspection for Certification of Completion. Contractor shall:
    a. Clean exposed exterior and interior hard-surfaced finishes to a dust-free condition, free of stains, films and similar foreign substances. Restore reflective surfaces to their original reflective condition. Leave concrete floors broom clean. Vacuum carpeted surfaces.
    b. Wipe surfaces of mechanical and electrical equipment. Remove excess lubrication and other substances. Clean plumbing fixtures to a sanitary condition. Clean light fixtures and lamps.
    c. Clean the site, including landscape development areas, of rubbish, litter and foreign substances. Sweep paved areas broom clean; remove stains, spills and other foreign deposits.

B. Waste Disposal, Contractor shall:
1. Remove waste materials from the site and conduct disposal in a lawful manner.
2. Do not burn waste materials.
3. Do not bury debris or excess materials on the University property.
4. Do not discharge volatile, harmful or hazardous materials into drainage systems.
5. Where extra materials of value remaining after completion of associated work have become the University's property, arrange for disposition of these materials as directed.

3.3 INTERIOR CLEANING

A. Interior Cleaning, Contractor shall:
1. Clean each surface or unit to the condition expected in a normal, commercial building cleaning and maintenance program.
2. Remove labels that are not permanent labels.
3. Remove grease, mastic, adhesives, dust, dirt, stains, fingerprints, labels, and other foreign materials from all visible interior and exterior surfaces.
4. Remove dust from all horizontal surfaces not exposed to view, including light fixtures, ledges and plumbing fixtures.
5. Clean all horizontal surfaces to dust-free condition, including tops of door and window frames, tops of doors and interiors of cabinets and casework.
6. Remove waste and surplus materials, rubbish and temporary construction facilities, utilities and controls.

B. Accessories and Fixtures Cleaning: Contractor shall clean building accessories, including toilet partitions, fire extinguisher cabinets, lockers and toilet accessories, all plumbing fixtures and all lighting fixture lenses and trim.

C. Glass and Mirror Cleaning: Contractor shall clean and polish all glass and mirrors as specified in all Division 08 specifications. Contractor shall remove glazing compound and other substances that are noticeable vision-obscuring materials. Contractor shall replace chipped or broken glass and other damaged transparent materials.

D. Metalwork: Contractor shall clean and buff all metalwork, to be free of soiling and fingerprints. Mirror finished metalwork shall be buffed to high luster.

E. Floor Cleaning: Contractor shall clean floors to dust-free condition, free of stains, films and similar foreign substances.

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EDIT PRODUCTS AND FINISHES BELOW TO SUIT PROJECT CONDITIONS.

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1. Exposed concrete floors: Contractor shall thoroughly sweep and wet mop floors in enclosed spaces. Contractor shall mop concrete floors and, at concrete floors in occupied spaces, apply floor finish as specified for resilient flooring. At unoccupied spaces, Contractor shall leave concrete floors broom clean.

2. Ceramic tile flooring: Contractor shall thoroughly sweep and mop tile flooring. Contractor shall comply with specific requirements in tile and installation materials manufacturers for cleaning materials.

3. Resilient flooring: Contractor shall thoroughly sweep all resilient flooring. Contractor shall damp wash and wax (as appropriate) all resilient flooring. Contractor shall comply with specific requirements in applicable resilient flooring Sections, and notes of the Drawings.

4. Carpet cleaning: Contractor shall comply with accepted industry practices for cleaning commercial carpet, subject to review and acceptance by University's Representative. Contractor shall vacuum, spot clean and generally clean carpet using commercial carpet cleaning solution, scrubbers and solution extraction-type vacuuming equipment.

F. Ventilation System Cleaning: Contractor shall replace filters and clean heating and ventilating equipment used for temporary heating, cooling and ventilation.

3.4 EXTERIOR CLEANING

A. Building Exterior Cleaning: Contractor shall clean exterior of adjacent facilities where construction activities have caused soiling and accumulation of dust and debris. Contractor shall:
1. Remove labels that are not permanent labels.
2. Wash down exterior surfaces to remove dust.
3. Clean exterior surfaces of mud and other soiling.
4. Clean exterior side of windows, storefronts and curtainwalls, including window framing.

B. Glass and Mirror Cleaning: Contractor shall clean and polish all glass and mirrors as specified in all Division 08 specifications. Contractor shall remove glazing compound and other substances that are noticeable vision-obscuring materials. Contractor shall replace chipped or broken glass and other damaged transparent materials.

C. Site Cleaning: Contractor shall broom clean exterior paved surfaces. Contractor shall rake clean other surfaces of the grounds. Contractor shall:
1. Wash down and scrub where necessary all paving soiled as a result of construction activities. Thoroughly remove mortar droppings, paint splatters, stains and adhered soil.
2. Remove from the site all construction waste, unused materials, excess soil and other debris resulting from the Work. Legally dispose of waste.

3.5 PEST CONTROL

A. Pest Control: Contractor shall engage an experienced, licensed exterminator to inspect and rid the project area of insects, rodents and other pests. Pests shall not be allowed to roost, nest or otherwise inhabit the Work at any point during construction. All animal/insect debris shall be promptly cleaned and disposed of in accordance with all applicable regulations to prevent surfaces becoming stained, or compromised in any manner.
   1. Exterminator shall prepare and submit report of inspection and extermination.
   2. Extermination materials shall comply with applicable pest control regulations and not leave toxic residue harmful to humans.

3.6 CLEANING INSPECTION

A. Cleaning Inspection: Prior to Final Payment or acceptance by University for partial occupancy or beneficial use of the premises, Contractor and University's Representative shall jointly conduct an inspection of interior and exterior surfaces to verify that entire Work is acceptably clean. Punchlist shall be utilized for recording any deficiencies.

B. Inadequate Cleaning: Should final cleaning be inadequate, as determined by University's Representative, and Contractor fails to correct conditions, University may engage cleaning service under separate contract and deduct cost from Contract Sum.

END OF SECTION

01 74 19 – Construction Waste Management – Instruction to Project Manager

For each facility renovation project, the Project Manager will coordinate with the contractor or personnel to discuss the scope of the renovation.

- The scope of the renovation must be determined and the materials to be used and discarded during the renovation must be identified. Packaging will be a consideration in the materials that will be discarded.
- The approximate volume of each type of waste will be broken out. Separate categories may include cardboard, wood products and cabinetry, drywall, tile, etc.
- From this material flow, the five largest waste categories will be determined.
    - The Project Manager will coordinate proper waste disposal and landfill diversion for these waste categories. This will involve contacting the appropriate vendors, scheduling haul dates, and ensuring properly sized storage areas for the construction waste.
    - If necessary, a separate secured storage area will be secured for hazardous waste, such as paint.
    - Once the waste disposal has been coordinated, the renovation manager will write waste disposal instructions for each waste category and will distribute to the appropriate vendors.
- For regular maintenance activities, the facility manager will ensure that the proper materials are recycled or composted.

For all work, including renovation and maintenance, document solid waste disposal and diversion. Include the quantity by weight of waste generated; waste diverted through sale, reuse, or recycling; and waste disposed by landfill or incineration. Identify landfills, recycling centers, waste processors, and other organizations that process or receive the solid waste.

SECTION 01 74 19 - CONSTRUCTION WASTE MANAGEMENT

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section includes requirements and procedures for ensuring optimal diversion of construction and demolition (C&D) waste materials generated by the Work from landfill disposal within the limits of the Construction Schedule and Contract Sum.

1. California State law (Public Resources Code sections 40000 *et seq.*) requires the California State University to develop source reduction, re-use, recycling, and composting programs to divert 75% of all solid waste from landfill disposal by 2020. Construction waste materials generated by the Work are targeted to achieve and maintain these diversion rates.
2. The Work of this Contract requires that a minimum of 75% by weight of the construction and demolition materials generated in the Work is diverted from landfill disposal through a combination of re-use and recycling activities.
3. For LEED® projects, requirements for submittal of LEED documentation in compliance with the Materials and Resources category, Construction and Demolition Waste Management credit.
4. Requirements for submittal of Contractor's Construction Waste and Recycling Plan prior to the commencement of the Work.
5. Contractor's quantitative reports for construction waste materials as a condition of approval of the third progress payment.

1.3 DEFINITIONS

A. Class III Landfill: A landfill that accepts non-hazardous resources such as household, commercial, and industrial waste, resulting from construction, remodeling, repair, and demolition operations. A Class III landfill must have a solid waste facilities permit from CalRecycle and is regulated by the Enforcement Agency (EA).

B. Construction and Demolition Debris: Building materials and solid waste resulting from construction, remodeling, repair, cleanup, or demolition operations that are not hazardous as defined in California Code of Regulations, Title 22, and Section 66261.3 *et seq*. This term includes, but is not limited to, asphalt concrete, Portland cement concrete, brick, lumber, gypsum wallboard, cardboard and other associated packaging, roofing material, ceramic tile, carpeting, plastic pipe, and steel. The debris may be commingled with rock, soil, tree stumps, and other vegetative matter resulting from land clearing and landscaping for construction or land development projects.

C. C&D Recycling Center. A facility that receives only C&D material that has been separated for reuse prior to receipt, in which the residual (disposed) amount of waste in the material is less than 10% of the amount separated for reuse by weight.

D. Disposal. Final deposition of construction and demolition or inert debris into land, including stockpiling onto land of construction and demolition debris that has not been sorted for further processing or resale, if such stockpiling is for a period of time greater than 30 days; and construction and demolition debris that has been sorted for further processing or resale, if such stockpiling is for a period of time greater than one year, or stockpiling onto land of inert debris that is for a period of time greater than one year.

E. Enforcement Agency. Enforcement agency as defined [i.e. in Public Resources Code 40130].

F. Inert Disposal Facility or Inert Waste Landfill: A disposal facility that accepts only inert waste such as soil and rock, fully cured asphalt paving, uncontaminated concrete (including fiberglass or steel reinforcing rods embedded in the concrete), brick, glass, and ceramics, for land disposal.

G. Mixed Debris: Loads that include commingled recyclable and non-recyclable materials generated at the construction site.

H. Mixed Debris Recycling Facility: A processing facility that accepts loads of commingled construction and demolition debris for the purpose of recovering re-usable and recyclable materials and disposing the non-recyclable residual materials.

I. Recycling: The process of sorting, cleansing, treating and reconstituting materials for the purpose of using the altered form in the manufacture of a new product. Recycling does not include burning, incinerating or thermally destroying solid waste.

J. Reuse. The use, in the same or similar form as it was produced, of a material which might otherwise be discarded.

K. Separated for Reuse. Materials, including commingled recyclables, that have been separated or kept separate from the solid waste stream for the purpose of additional sorting or processing those materials for reuse or recycling in order to return them to the economic mainstream in the form of raw material for new, reused, or reconstituted products which meet the quality standards necessary to be used in the marketplace, and includes materials that have been “source separated.”

L. Solid Waste: All putrescible and non-putrescible solid, semisolid, and liquid wastes, including garbage, trash, refuse, paper, rubbish, ashes, industrial wastes, demolition and construction wastes, abandoned vehicles and parts thereof, discarded home and industrial appliances, dewatered, treated, or chemically fixed sewage sludge which is not hazardous waste, manure, vegetable or animal solid and semisolid wastes, and other discarded solid and semisolid wastes. "Solid waste" does not include hazardous waste, radioactive waste, or medical waste as defined or regulated by State law.

M. Source-Separated: Materials, including commingled recyclables, that have been separated or kept separate from the solid waste stream at the point of generation for the purpose of additional sorting or processing of those materials for reuse or recycling in order to return them to the economic mainstream in the form of raw materials for new, reused, or reconstituted products which meet the quality standards necessary to be used in the marketplace.

N. Waste Hauler: A company that possesses a valid permit from the local waste management authority to collect and transport solid wastes from individuals or businesses for the purpose of recycling or disposal in the locality.

1.4 SUBMITTALS

A. Contractor’s Construction Waste and Recycling Plan

1. Review Contract Documents and estimate the types and quantities of materials under the Work that are anticipated to be feasible for on-site processing, source separation for re-use or recycling. Indicate the procedures that will be implemented in this program to effect jobsite source separation, such as, identifying a convenient location where dumpsters would be located, putting signage to identify materials to be placed in dumpsters, etc.
2. Prior to commencing the Work, submit Contractor’s Construction Waste and Recycling Plan. Submit in format provided (**Section 01 74 19A**). The Plan must include, but is not limited to the following:
    a. Contractor’s name and project identification information;
    b. Procedures to be used;
    c. Materials to be re-used and recycled;
    d. Estimated quantities of materials;
    e. Names and locations of re-use and recycling facilities/sites;
    f. Tonnage calculations that demonstrate that Contractor will re-use and recycle a minimum 65% by weight of the construction waste materials generated in the Work.
3. Contractor’s Construction Waste and Recycling Plan must be approved by the Construction Administrator prior to the start of Work.
4. Contractor’s Construction Waste and Recycling Plan will not otherwise relieve the Contractor of responsibility for adequate and continuing control of pollutants and other environmental protection measures.

B. Contractor’s Reuse, Recycling, and Disposal Report

1. Submit Contractor’s Reuse, Recycling, and Disposal Report on the form provided (**Section 01 74 19B**) with each application for progress payment. Failure to submit the form and its supporting documentation will render the application for progress payment incomplete and delay progress payments. If applicable, include manifests, weight tickets, receipts, and invoices specifically identifying the Project for re-used and recycled materials:
    a. Reuse of building materials or salvage items on site (i.e. crushed base or red clay brick).
    b. Salvaging building materials or salvage items at an off-site salvage or reuse center (i.e. lighting, fixtures).

c. Recycling source separated materials on site (i.e. crushing asphalt/ concrete for base course, or grinding for mulch).
d. Recycling source separated material at an offsite recycling center (i.e. scrap metal or green materials).
e. Use of material as Alternative Daily Cover (ADC) at landfills.
f. Delivery of soils or mixed inert material to an inert landfill for disposal (inert fill).
g. Disposal at a landfill or transfer station (where no recycling takes place).
h. Other (describe).

2. Contractor's Reuse, Recycling, and Disposal Report must quantify all materials generated in the Work, disposed in [Class III] landfills, or diverted from disposal through recycling. Indicate zero (0) if there is no quantity to report for a type of material.
3. As indicated on the form:
   a. Report disposal or recycling either in tons or in cubic yards: if scales are available at disposal or recycling facility, report in tons; otherwise, report in cubic yards. Report in units for salvage items when no tonnage or cubic yard measurement is feasible.
   b. Indicate locations to which materials are delivered for reuse, salvage, recycling, accepted as daily cover, inert backfill, or disposal in landfills or transfer stations.
   c. Provide legible copies of weigh tickets, receipts, or invoices that specifically identify the project generating the material. Said documents must be from recyclers and/or disposal site operators that can legally accept the materials for the purpose of re-use, recycling, or disposal.
4. Indicate project title, project number, progress payment number, name of the company completing the Contractor's Report and compiling backup documentation, the printed name, signature, and daytime phone number of the person completing the form, the beginning and ending dates of the period covered on the Contractor's Report, and the date that the Contractor's Report is completed.

C. For LEED Projects, complete the LEED Construction and Demolition Waste Management Calculator in format provided under the most current version of the U.S. Green Building Council's Leadership in Energy and Environmental Design (LEED) program. Include a signed cover letter with calculation summary on company letterhead.

1. Certify that the project has completed a waste management plan and diverted construction, demolition, and land clearing waste to uses other than landfill.
2. Provide quantities of diverted materials and means of diversion in accordance with the results table in the LEED Construction and Demolition Waste Management Calculator.
3. Indicate how and where waste was diverted.
4. Indicate quantities of waste diverted in tons [or cubic yards].
5. Letter will also include: Total quantity of diverted waste, total quantity of waste, and the percentage of waste diverted.
6. Include name, organization, and role in project. Provide signature and date completed.
7. Include legible copies of weigh tickets, receipts, or invoices that specifically identify the project generating the material. Said documents must be from recyclers and/or disposal site operators that can legally accept the materials for the purpose of re-use, recycling, or disposal.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 SALVAGE, RE-USE, RECYCLING AND PROCEDURES

A. Identify re-use, salvage, and recycling facilities.

B. Develop and implement procedures to re-use, salvage, and recycle new construction and excavation materials, based on the Contract Documents, the Contractor's Construction Waste and Recycling Plan, estimated quantities of available materials, and availability of recycling facilities. Procedures may include on-site recycling, source separated recycling, and/or mixed debris recycling efforts.

1. Identify materials that are feasible for salvage, determine requirements for site storage, and transportation of materials to a salvage facility.
2. Source separate new construction, excavation and demolition materials including, but not limited to the following types:
   a. Asphalt.
   b. Concrete, concrete block, slump stone (decorative concrete block), and rocks.
   c. Drywall.
   d. Green materials (i.e. tree trimmings and land clearing debris).
   e. Metal (ferrous and non-ferrous).
   f. Miscellaneous construction debris.
   g. Paper or cardboard.
   h. Red clay brick.
   i. Reuse or salvage materials
   j. Soils.
   k. Wire and cable.
   l. Wood.
   m. Other (describe)
3. Miscellaneous Construction Debris: Develop and implement a program to transport loads of mixed (commingled) new construction materials that cannot be feasibly source separated to a mixed materials recycling facility.

3.2 DISPOSAL OPERATIONS AND WASTE HAULING

A. Legally transport and dispose of materials that cannot be delivered to a source separated or mixed recycling facility to a transfer station or disposal facility that can legally accept the materials for the purpose of disposal.

B. Use a permitted waste hauler or Contractor's trucking services and personnel. To confirm valid permitted status of waste haulers, contact the local solid waste authority.

C. Become familiar with the conditions for acceptance of new construction, excavation and demolition materials at recycling facilities, and prior to delivering materials.

D. Deliver to facilities that can legally accept new construction, excavation and demolition materials for purpose of re-use, recycling, composting, or disposal.

E. Do not burn, bury or otherwise dispose of solid waste on the project job-site.

3.3 RE-USE AND DONATION OPTIONS

Implement a re-use program to the greatest extent feasible. Options may include:

California Materials Exchange (CAL-MAX) is a free program sponsored by CalRecycle and is designed to help connect businesses, organizations, manufacturers, schools, and individuals with the most effective online resources for exchanging materials. Go to http://www.calrecycle.ca.gov/CalMAX/. Public Surplus is a government agency surplus auction system used by many universities. Go to https://www.publicsurplus.com for more information.

3.4 REVENUE

Revenues or other savings obtained from recycled, re-used, or salvaged materials shall accrue to Contractor unless otherwise noted in the Contract Documents.

END OF SECTION 01 74 19

SECTION 01 74 19A

CONTRACTOR'S CONSTRUCTION WASTE AND RECYCLING PLAN

(Submit After Award of Contract and Prior to Start of Work)

Project Title:		
Contract or Work Order No.:		
Contractor's Name:		
Street Address:		
City:	State:	Zip:
Phone: ( )	Fax: ( )	
E-Mail Address:		
Prepared by: (Print Name)		

Date Submitted:		
Project Period:	From:	TO:

Reuse, Recycling or Disposal Processes To Be Used

Describe the types of recycling processes or disposal activities that will be used for material generated in the project. Indicate the type of process or activity by number, types of materials, and estimated quantities that will be recycled or disposed in the sections below:

01 - Reuse of building materials or salvage items on site (i.e. crushed base or red clay brick)
02 - Salvaging building materials or salvage items at an off site salvage or re-use center (i.e. lighting, fixtures)
03 - Recycling source separated materials on site (i.e. crushing asphalt/concrete for reuse or grinding for mulch)
04 - Recycling source separated materials at an off site recycling center (i.e. scrap metal or green matls)
05 - Recycling commingled loads of C&D matls at an off site mixed debris recycling center or transfer station
06 - Recycling material as Alternative Daily Cover at landfills
07 - Delivery of soils or mixed inerts to an inert landfill for disposal (inert fill).
08 - Disposal at a landfill or transfer station.
09 - Other (please describe) ____

Types of Material To Be Generated

Use these codes to indicate the types of material that will be generated on the project

A = Asphalt	C = Concrete	M = Metals	I = Mixed Inert	G = Green Matls
D = Drywall	P/C=Paper/Cardboard	W/C = Wire/Cable	S= Soils (Non Hazardous)	
M/C = Miscellaneous Construction Debris		R = Reuse/Salvage	W = Wood	O = Other (describe)

Facilities Used: Provide Name of Facility and Location (City)

Total Truck Loads: Provide Number of Trucks Hauled from Site During Reporting Period

Total Quantities: If scales are available at sites, report in tons. If not, quantify by cubic yards. For salvage/reuse items, quantify by estimated weight (or units).

SECTION I - RE-USED/RECYCLED MATERIALS

Include all recycling activities for source separated or mixed material recycling centers where recycling will occur.

Type of Material	Type of Activity	Facility to be Used, Location	Total Truck Loads	Total Quantities Tons	Cubic YD	Other Wt.
(ex.) M	04	ABC Metals, Los Angeles	24	355		
a. Total Diversion			-	-	-	-

SECTION 01 74 19A

CONTRACTOR'S CONSTRUCTION WASTE AND RECYCLING PLAN

Continued

SECTION II - DISPOSED MATERIALS						
Include all disposal activities for landfills, transfer stations, or inert landfills where no recycling will occur.						
Type of	Type	Facility to be Used,	Total Truck	Total Quantities		
Material	of Activity	Location	Loads	Tons	Cubic YD	Other Wt.
(ex.) D	08	DEF Landfill, Los Angeles	2	35		
b. Total Disposal				-	-	-

SECTION III - TOTAL MATERIALS GENERATED						
This section calculates the total materials to be generated during the project period (Reuse/Recycle + Disposal = Generation						
				Tons	Cubic YD	Other Wt.
a. Total Reused/Recycled				-	-	-
b. Total Disposed				-	-	-
c. Total Generated				-	-	-

SECTION IV - CONTRACTOR'S LANDFILL DIVERSION RATE CALCULATION			
Add totals from Section I + Section II			
	Tons	Cubic Yards	Other Wt.
a. Materials Re-Used and Recycled	-		
b. Materials Disposed	-		
c. Total Materials Generated (a. + b. = c.)	-	-	-
d. Landfill Diversion Rate (Tons Only)*	#DIV/0!		

** Use tons only to calculate recycling percentages: Tons Reused/Recycled/Tons Generated = % Recycled*

Contractor's Comments (Provide any additional information pertinent to planned reuse, recycling, or disposal activities):

Notes:

1. Suggested Conversion Factors: From Cubic Yards to Tons (Use when scales are not available)

Asphalt: .61 (ex. 1000 CY Asphalt = 610 tons. Applies to broken chunks of asphalt)

Concrete: .93 (ex. 1000 CY Concrete = 930 tons. Applies to broken chunks of concrete)

Ferrous Metals: .22 (ex. 1000 CY Ferrous Metal = 220 tons) Drywall Scrap: .20

Non-Ferrous Metals: .10 (ex. 1000 CY Non-Ferrous Metals = 100 tons) Wood Scrap: .16

SECTION 01 74 19B
CONTRACTOR'S REUSE, RECYCLING, AND DISPOSAL REPORT
(Submit With Each Progress Payment)

Project Title:		
Contract or Work Order No.:		
Contractor's Name:		
Street Address:		
City:	State:	Zip:
Phone: ( )	Fax: ( )	
E-Mail Address:		
Prepared by: (Print Name)		

Date Submitted:		
Period Covered:	From:	To:

Reuse, Recycling or Disposal Processes Used

Describe the types of recycling processes or disposal activities used for material generated in the project. Indicate the type of process or activity by number, types of materials, and quantities that were recycled or disposed in the sections below:

01 - Reuse of building materials or salvage items on site (i.e. crushed base or red clay brick)
02 - Salvaging building materials or salvage items at an off site salvage or re-use center (i.e. lighting, fixtures)
03 - Recycling source separated materials on site (i.e. crushing asphalt/concrete for reuse or grinding for mulch)
04 - Recycling source separated materials at an off site recycling center (i.e. scrap metal or green matls)
05 - Recycling commingled loads of C&D matls at an off site mixed debris recycling center or transfer station
06 - Recycling material as Alternative Daily Cover at landfills
07 - Delivery of soils or mixed inerts to an inert landfill for disposal (inert fill).
08 - Disposal at a landfill or transfer station.
09 - Other (please describe) ____

Types of Material Generated

Use these codes to indicate the types of material that were generated on the project

A = Asphalt	C = Concrete	M = Metals	I = Mixed Inert	G = Green Matls
D = Drywall	P/C=Paper/Cardboard	W/C = Wire/Cable	S= Soils (Non Hazardous)	
M/C = Miscellaneous Construction Debris		R = Reuse/Salvage	W = Wood	O = Other (describe)

Facilities Used: Provide Name of Facility and Location (City)

Total Truck Loads: Provide Number of Trucks Hauled from Site During Reporting Period

Total Quantities: If scales are available at sites, report in tons. If not, quantify by cubic yards. For salvage/reuse items, quantify by estimated weight (or units).

SECTION I - RE-USED/RECYCLED MATERIALS

Include all recycling activities for source separated or mixed material recycling centers where recycling occurred.

Type of Material	Type of Activity	Facilities Used, Location	Total Truck Loads	Total Quantities Tons	Cubic YD	Other Wt.
(ex.) M	04	ABC Metals, Los Angeles	24	355		
a. Total Diversion			-	-	-	-

SECTION 01 74 19B

CONTRACTOR'S REUSE, RECYCLING, AND DISPOSAL REPORT

Continued

SECTION II - DISPOSED MATERIALS						
Include all disposal activities for landfills, transfer stations, or inert landfills where no recycling occurred.						
Type of	Type	Facilities Used,	Total Truck	Total Quantities		
Material	of Activity	Location	Loads	Tons	Cubic YD	Other Wt.
(ex.) D	08	DEF Landfill, Los Angeles	2	35		
b. Total Disposal				-	-	-

SECTION III - TOTAL MATERIALS GENERATED						
This section calculates the total materials generated during the project period (Reuse/Recycle + Disposal = Generation						
				Tons	Cubic YD	Other Wt.
a. Total Reused/Recycled				-	-	-
b. Total Disposed				-	-	-
c. Total Generated				-	-	-

SECTION IV - CONTRACTOR'S LANDFILL DIVERSION RATE CALCULATION			
Add totals from Section I + Section II			
	Tons	Cubic Yards	Other Wt.
a. Materials Re-Used and Recycled	-		
b. Materials Disposed	-		
c. Total Materials Generated (a. + b. = c.)	-	-	-
d. Landfill Diversion Rate (Tons Only)*	#DIV/0!		

* *Use tons only to calculate recycling percentages: Tons Reused/Recycled/Tons Generated = % Recycled*

Contractor's Comments (*Provide any additional information pertinent to planned reuse, recycling, or disposal activities*):

Notes:

1. Suggested Conversion Factors: From Cubic Yards to Tons (Use when scales are not available)

Asphalt: .61 (ex. 1000 CY Asphalt = 610 tons. Applies to broken chunks of asphalt)

Concrete: .93 (ex. 1000 CY Concrete = 930 tons. Applies to broken chunks of concrete)

Ferrous Metals: .22 (ex. 1000 CY Ferrous Metal = 220 tons) Drywall Scrap: .20

Non-Ferrous Metals: .10 (ex. 1000 CY Non-Ferrous Metals = 100 tons) Wood Scrap: .16

SECTION 01 75 00

STARTING AND ADJUSTING PROCEDURES

****
THIS SECTION REQUIRES DEVELOPMENT, RELATED TO COMMISSIONING ACTIVITIES, REQUIREMENTS SPECIFIED IN DIVISIONS 13, 14, 23 AND 26, AND SECTION 01 79 00 – DEMONSTRATION AND TESTING (IF USED). MODIFY THIS SECTION TO COORDINATE WITH CAMPUS Cx AGENT REQUIREMENTS.
****

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Starting systems.

B. Demonstration and instructions.

C. Testing, adjusting, and balancing.

1.3 RELATED SECTIONS

A. Section 01 45 00 - Quality Control: Manufacturers field reports.

B. Section 01 78 03 - Operation and Maintenance Data: System operation and maintenance data and extra materials.

C. Section 01 79 00 – Demonstration and Testing

1.4 STARTING SYSTEMS

A. Contractor shall coordinate schedule for start-up of various equipment and systems.

B. Contractor shall notify University's Representative, Architect and Project Inspector in writing at least seven calendar days prior to start-up of each item.

C. Contractor shall verify that each piece of equipment or system has been checked for proper lubrication, drive rotation, belt tension, control sequence, and for conditions which may cause damage.

D. Contractor shall verify tests, meter readings, and specified electrical characteristics agree with those required by the equipment or system manufacturer.

E. Contractor shall verify that wiring and support components for equipment are complete and tested.

F. Contractor shall execute start-up under supervision of applicable manufacturer's representative and/or Contractor’s personnel in accordance with manufacturer’s instructions.

G. When specified in individual specification Sections, Contractor shall require manufacturer to provide authorized representative to be present at site to inspect, check, and approve equipment or system installation prior to start-up, and to supervise placing equipment or system in operation.

H. Contractor shall submit a written report in accordance with Section 01 33 00 - Submittals Procedures that equipment or system has been properly installed and is functioning correctly.

1.5 DEMONSTRATION AND INSTRUCTIONS

A. Contractor shall demonstrate operation and maintenance of Products to University's personnel at least two weeks prior to date of Contract Completion review.

B. Contractor shall demonstrate Project equipment and instruct in a classroom environment located at the University. The instruction shall be done by a qualified manufacturers' representative who is knowledgeable about the Project.

C. For equipment or systems requiring seasonal operation, Contractor shall perform demonstration for other season within six months of completion or, if possible, artificially create a load in the building.

D. Contractor shall utilize operation and maintenance manuals as basis for instruction. Contractor shall review contents of manual with University's personnel in detail to explain all aspects of operation and maintenance.

E. Contractor shall demonstrate start-up, operation, control, adjustment, trouble-shooting, servicing, maintenance, and shutdown of each item of equipment at scheduled agreed time and at equipment/designated location.

F. Contractor shall prepare and insert additional data in operations and maintenance manuals when need for additional data becomes apparent during instruction.

G. The amount of time required for instruction on each item of equipment and system is that specified in individual sections. If no time is specified in individual sections, Contractor shall include in his/her bid sum a reasonable sum to perform instruction to the satisfaction of the University.

1.6 TESTING, ADJUSTING, AND BALANCING

A. Testing Agency: Contractor shall appoint, employ, and pay for services of an independent firm to perform testing, adjusting and balancing.

B. Reports will be submitted by the independent firm to University's Representative, Architect and Project Inspector indicating observations and results of tests and indicating compliance or non-compliance with the requirements of the Contract Documents.

C. University reserves the right to hire its own independent testing and balancing company to check the work and the report submitted by the Contractor’s testing and balancing firm.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

Not Applicable to this Section.

END OF SECTION

SECTION 01 77 00

CONTRACT CLOSEOUT PROCEDURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Contract closeout procedures, including Contract Closeout meetings, correction ("punch") lists, submittals and final payment procedures.

1.3 RELATED SECTIONS

A. Section 01 33 00 - Submittals Procedures: General requirements for submittals.

B. Section 01 74 00 - Cleaning Requirements: Progress cleaning and cleaning as part of Contract closeout.

C. Section 01 78 33 - Warranties and Bonds: Documents to be submitted as part of Contract closeout.

D. Section 01 78 39 – Project Record Documents: Project record drawings and specifications to be submitted as part of Contract closeout; operation and maintenance data to be submitted as part of Contract closeout.

1.4 FINAL COMPLETION ACTIONS

A. Contractor Responsibility: Contractor shall be solely responsible for the timely completion of all required Contract closeout items except for filing of Notice of Completion by the Trustees.

B. Warranties, Bonds and Certificates: Contractor shall submit specific warranties, guarantees, workmanship bonds, maintenance agreements, final certifications and similar documents.

C. Locks and Keys: Contractor shall change temporary lock cylinders over to permanent keying and transmit keys to Trustees, unless otherwise directed or specified.

D. Tests and Instructions: Contractor shall complete start-up testing of systems, and instruction of the University's personnel. Contractor shall remove temporary facilities from the site, along with construction tools, mock-ups, and similar elements.

1.5 CONTRACT COMPLETION REVIEW

A. Contractor's Notification for Contract Completion Review Meeting: When the Contractor determines that the Work is complete in accordance with Contract Documents, the Contractor shall submit to University's Representative and Architect written certification that the Contract Documents have been reviewed, the Work has been inspected by the Contractor and by authorities having jurisdiction, and the facility is ready for the Contract Completion review.

B. Contract Completion Review Meeting: University's Representative and, as authorized by the Trustees, Architect and Architect's and Trustees' representatives and consultants, as appropriate, will attend a meeting at the Project site to review Contract closeout procedures and to review the items to be completed and corrected Punch List to make the Work ready for acceptance by the Trustees. This meeting shall be typically scheduled four to six weeks prior to scheduled completion date.

C. Punch List: Architect shall prepare subsequent to the Contract completion review meeting, a typewritten, comprehensive list of items to be completed and corrected (Punch List) to make the Work ready for acceptance by the Trustees.
1. The Punch List shall include all items to be completed or corrected prior to the Contractor's application for final payment.
2. The Punch List shall identify items by location (room number or name) and consecutive number. For example, 307-5 would identify item 5 in Room 307, Roof-4 would identify item 4 on Roof.
3. Architect and Architect’s consultants shall prepare separate lists according to categories used for Drawings. For example, provide lists for Architectural, Structural, Mechanical (HVAC), Plumbing, Fire Protection (sprinkler) system, Electrical and Equipment. But all lists shall be compiled by the Architect into the all inclusive Master Punch List.
4. Items to be considered shall include but not be limited to:
Corrections to construction.
Operation and maintenance data (manuals).
HVAC testing and balancing reports.
Spare parts and extra materials.
Keys, permanent keying and lock cylinders.
Warranties and guaranties.
Project record Drawings and Specifications.
Project record construction schedule.
State Fire Marshal Inspection.
Elevator Inspection (if applicable).
Other regulatory inspections.
Removal of construction facilities and temporary controls.
Final cleaning and pest control.
Landscape maintenance.
Commissioning/equipment startup.
Demonstration and training.
Acceptance.
Notice of Completion, filing by Trustees.
Final application for payment.
Occupancy by University.
Other closeout items specified.

D. Contract Completion Meeting: On a date mutually agreed by University's Representative, Architect and Contractor, a meeting shall be conducted at the Project site to determine whether the Work is satisfactory and has achieved Contract Completion.
1. Contractor shall provide a minimum seven calendar days written notice to the University's Representative and Architect for requested date of Contract Completion meeting.
2. Architect and the Architect's consultants will attend the Contract Completion meeting.
3. In addition to conducting a walk-through of the facility and reviewing the Punch List, the purpose of the meeting shall include submission of warranties, guarantees and bonds to University, submission of operation and maintenance data (manuals), provision of specified extra materials to University, and submission of other Contract closeout documents and materials as required and if not already submitted.
4. Architect and Architect's consultants, as appropriate, will conduct a walk-through of the facility with the University's Representative and Contractor to review the Punch List.
5. Architect shall update the Punch List and record additional items as may identified during the walk-through, including notations of corrective actions to be taken.
6. Architect shall retype the Punch List and distribute it within five calendar days to those attending the meeting.

E. Uncorrected Work: Refer to requirements specified in Section 01 45 00 - Quality Control regarding Contract adjustments for non-conforming Work.

F. Clearing and Cleaning: Prior to the Contract Completion review, Contractor shall conduct a thorough cleaning and clearing of the Project area, including removal of construction facilities and temporary controls. Refer to Section 01 74 00 - Cleaning Requirements.

G. Inspection and Testing: Prior to the Contract Completion review, Contractor shall complete inspection and testing required for the Work, including securing of approvals by authorities having jurisdiction.
   1. Complete all inspections, tests, balancing, sterilization and cleaning of plumbing and HVAC systems.
   2. Complete inspections and tests of electrical power and signal systems.
   3. Complete inspections and tests of conveying (elevator) systems.

H. Notice of Completion: University will record the Notice of Completion with County Recorder, when the Project is complete in all respects.

1.6 FINAL COMPLETION SUBMITTALS

A. Final Completion Submittals: Prior to application for Final Payment, Contractor shall submit the following.

B. Agency Document Submittals: Contractor shall submit to University all documents required by authorities having jurisdiction, including serving utilities and other agencies. Contractor shall submit original versions of all permit cards, with final sign-off by inspectors. Submit all certifications of inspections and tests.

C. Final Specifications Submittals: Contractor shall submit to University all documents and products required by Specifications to be submitted, including the following:
   1. Project record drawings and specifications.
   2. Operating and maintenance data.
   3. Guarantees, warranties and bonds.
   4. Keys and keying schedule.
   5. Spare parts and extra stock.
   6. Test reports and certificates of compliance.

D. Certificates of Compliance and Test Report Submittals: Contractor shall submit to University's Representative certificates and reports as specified and as required by authorities having jurisdiction, including but not limited to the following:

****

EDIT SUBPARAGRAPHS BELOW TO SUIT PROJECT REQUIREMENTS.

****

   1. Sterilization of water systems.
   2. Sanitary sewer system tests.
   3. Gas system tests.
   4. Lighting, power and signal system tests.
   5. Ventilation equipment and air balance tests.
   6. Fire sprinkler system tests.
   7. Roofing inspections and tests.

E. Subcontractors List: Contractor shall submit two copies of updated Subcontractor and Materials Supplier List to University's Representative and one copy to Architect.

F. Warranty Documents: Contractor shall prepare and submit to University all warranties and bonds as specified in Section 01 78 33 - Product Warranties and Bonds.

G. Service Agreements and Service Contracts: Contractor shall submit to University's Representative.

H. Contractor shall submit final electrical and water meter readings. Refer to section 01 51 00 – Temporary Utilities.

1.7 FINAL PAYMENT

A. Final Payment: After completion of all items listed for completion and correction and after submission of all documents and products and after final cleaning, Contractor shall submit final Application for Payment, identifying total adjusted Contract Sum, previous payments and sum remaining due. Payment will not be made until the following are accomplished:
   1. All Project Record Documents have been received and accepted by the Architect and the University.

2. All extra materials and maintenance stock have been transferred and accepted by University.
3. All warranty documents and operation, maintenance data, service agreements, maintenance contracts and salvage materials have been received and accepted by University's Representative.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

Not Applicable to this Section.

END OF SECTION

CSU The California State University

PROJECT ____________________ PROJECT NO. __________

CONTRACTOR ____________________ CONTRACT NO. __________

ARCHITECT ____________________ DATE __________

PROJECT CLOSEOUT CHECKLIST

Project Manager/Inspector of Record will complete this form and transmit it to the Construction Administrator with his/her recommendation to certify occupancy and completion, and release retention.

	Item	Verification: Date Completed	Verification: Initials
	Required for Occupancy		
1.	Final Inspection Punch List to GC		
2.	State Fire Marshal Inspection		
3.	Occupancy Change Order		
4.	Certification of Occupancy (Form 702.02-OCR) (certification by DBO)		
	Required for Notice of Completion		
4.	HVAC Balance Report		
5.	Keys/Keying		
6.	Training		
7.	Final Inspection Punch List Completed		
8.	Special Inspection Final Report		
9.	Elevator Inspection		
10.	Other Regulatory Inspection		
11.	Removal of Temporary Facilities		
12.	Final Cleaning		
13.	Commissioning		
14.	Cessation of Onsite Labor		
15.	Other 01 77 00 Requirements (Specify)		
16.	Certification of Completion (Form 702.02-OCR) (certification by A/E, PM, IOR, and DBO)		
	Required for Release of Retention		
17.	Spare Parts/Materials		
18.	Warranties		
19.	As Builts		
20.	As-Built Schedule		
21.	Landscape Maintenance Period Ends		
22.	O & M Manuals		
23.	Claims Resolved *or* Funds Held		
24.	Stop Notices Closed *or* Funds Held		
25.	Release from Labor Compliance Manager		
26.	Release of Retention (Form 702.02-OCR) (by PM and University Construction Administrator)		

The undersigned certifies that all of the above-listed items are complete.

____

{Name}, Construction Administrator Date

Distribution: *Copy to Architect/Engineer, Project Manager/Construction Inspector*
Original to Construction Administrator

SECTION 01 78 23

OPERATION AND MAINTENANCE DATA

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Format and content of operation and maintenance manuals.
   1. Data requirements for materials and finishes.
   2. Data requirements for equipment and operating systems.

B. Instruction of University's personnel.

C. Submission of operation and maintenance manuals.

1.3 RELATED SECTIONS

A. Section 01 31 00 - Coordination: Coordination documents and models prepared for performance of the Work, to be incorporated into operation and maintenance data submitted to University's Representative at Contract closeout.

B. Section 01 45 00 - Quality Control: Manufacturer's instructions; test and balance reports.

C. Section 01 61 00 - Basic Product Requirements: Systems demonstration.

D. Section 01 77 00 - Contract Closeout Procedures: Contract closeout procedures.

E. Section 01 78 33 - Product Warranties and Bonds: Requirements for warranties and bonds.

F. Section 01 78 39 - Project Record Documents: Submission of Project record documents.

E. Product Specifications Sections in Divisions 2 through 49: Specific requirements for operation and maintenance data.

1.4 QUALITY ASSURANCE

A. Contractor shall ensure that preparation of data shall be done by persons:
   1. Trained and experienced in maintenance and operation of the described products.
   2. Familiar with requirements of this Section.
   3. Skilled in technical writing to the extent required to communicate essential data.
   4. Skilled as drafters competent to prepare required drawings.

1.5 FORMAT AND CONTENT OF OPERATION AND MAINTENANCE MANUALS

A. Format for Operation and Maintenance Data Manuals: Contractor shall prepare data in the form of an instructional manual. Contractor shall comply with the general requirements specified below and comply with specific requirements for types of products in Articles following. See Article titled "SUBMISSION OF OPERATION AND MAINTENANCE MANUALS" for number of copies of manuals.

B. Operation and Maintenance Data Organization: Contractor shall organize operation and maintenance data in three-ring binders and organize the contents of each binder following the organization of the Contract Specifications. Contractor shall:

1. Organize the group of binders and the contents of individual binders in sequence according to the Section numbers and titles as listed in the Table of Contents of the Project Manual. Number the binders consecutively; coordinate with Paragraph below titled "Tables of Contents."
2. Organize each binder with color-coded tabbed dividers for each distinct product and system, with typed inserts in tabs identifying the product or system.
3. Organize the contents of each tabbed division according to the Article headings in PART 2 - PRODUCTS in each product Specification Section.
    a. Within each tabbed division, organize the information according to major component parts of equipment and systems, as applicable, and to facilitate locating information.
    b. Separate operation and maintenance data for each product under separate tabbed divisions, where feasible.
    c. Within each tabbed division, include a cover sheet identifying the specific products and component parts included in the tabbed division.
4. If the products of more than one Specification Section are included in the binder, provide separate, heavy cover stock dividers to separate information for each Section.

C. Binders: Contractor shall use 8-1/2 x 11 inch, standard three-ring binders with heavy duty vinyl covers with hard cardboard backing, black color, with provision on binder spine for inserting identification card; Maximum binder ring size shall be three inches. Contractor shall use multiple binders as necessary to avoid overfilling. When multiple binders are used, Contractor shall correlate data into related consistent groupings.

D. Cover: Contractor shall identify each binder with typed or printed card inserted on binder spine, stating OPERATION AND MAINTENANCE DATA, the Project name and the general subject matter of the contents of the binder.

E. Title Page: In each volume (binder) of operation and maintenance data, Contractor shall include a title page with the following:
1. Name of the Project.
2. Names, addresses and telephone numbers of the responsible design professionals (Architect and Architect's or University's consultant, as applicable).
3. Name, address and telephone numbers of Contractor, including names of contact persons.

F. Table of Contents: In each volume (binder) of operation and maintenance data, Contractor shall include a listing of the contents of the volume. In a separate, first binder, Contractor shall provide a master Table of Contents of operation and maintenance data, identifying the product and systems, the applicable Specification Section number and title, and the operation and maintenance data binder number.

G. Schedule of Products and Systems: In the first volume of the set of operation and maintenance data, Contractor shall include a schedule of products and systems, indexed to the Table of Contents of the volumes (binders) and cross-referenced to the Contract Drawings and Specifications.

H. Operation and Maintenance Data: In each tabbed division of operation and maintenance data for each product or system, Contractor shall provide the following:

1. On a cover page for each tabbed division, Contractor shall provide the following:
    a. Identify by name, address and telephone number, the manufacturer, supplier and installer. Include names of contact persons, if known.
    b. Identify by name, address and telephone number, local sources of supplies, replacement parts and factory-authorized service.
2. Within each tabbed division, Contractor shall include complete operation and maintenance data as published by the product manufacturer where feasible. Otherwise, present all data neatly typewritten on 20 pound, correspondence quality bond paper. Contractor shall strike-through information on printed literature where not applicable.
3. Contractor shall supplement the manufacturer's printed data with neatly typewritten text and professionally drafted diagrams as necessary to suit the particular installation for the Project and to fully

explain operation and maintenance procedures. Contractor shall provide logical sequence of instructions for each procedure.

I. Drawings: Contractor shall supplement operation and maintenance data to illustrate configurations and relationships of component parts of equipment and systems, and to show control and flow diagrams, as applicable.
   1. Contractor shall not use Project Record Documents as maintenance drawings.
   2. Contractor shall neatly fold drawings to size of text pages and provide reinforced, punched binding edge. Add binding strip as necessary to avoid punching through drawing content.

J. Additional Data: As specified in individual product Specification Sections.

K. Warranty and Guaranty: Contractor shall include copy of each warranty, and any guaranty, bond and service contract issued. Contractor shall provide information sheet identifying:
   1. Proper procedures in event of failure.
   2. Instances that might affect validity of warranties or bonds.

L. Material Safety Data Sheet (MSDS): For products requiring MSDS, according to CCR Title 8 and the University Contractor Safety Handbook, Contractor shall include copy of each applicable Material Safety Data Sheet (MSDS) for products delivered to the site and incorporated into the completed construction.

1.6 DATA REQUIREMENTS FOR MATERIALS AND FINISHES

A. Data for Building Products, Applied Materials and Finishes: Contractor shall include product data, with catalog number, size, composition, and color and texture designations. Contractor shall provide information for re-ordering custom manufactured Products.

B. Instructions for Care and Maintenance: Contractor shall include manufacturer's recommendations for cleaning agents and methods, precautions against detrimental agents and methods, and recommended schedule for cleaning and maintenance.

C. Data for Moisture Protection and Weather-Exposed Products: Contractor shall include product data listing applicable reference standards, chemical composition, and details of installation. Contractor shall provide recommendations for inspections, maintenance, and repair.

D. Additional Requirements: As specified in individual product Specification Sections.

1.7 DATA REQUIREMENTS FOR EQUIPMENT AND OPERATING SYSTEMS

A. Data for Equipment and Operating Systems: Contractor shall include description of each unit or system, and component parts.
   1. Include manufacturer's printed operation and maintenance instructions.
   2. Identify function, normal operating characteristics and limiting conditions.
   3. Include performance curves, with engineering data and tests.
   4. Include sequence of operation by controls manufacturer, as applicable.
   5. Provide diagrams by controls manufacturer for control systems, as applicable and as installed.

B. Piping Data: Contractor shall provide Contractor's coordination drawings, with piping diagrams as installed. Contractor shall provide charts of valve tag numbers, with location and function of each valve, keyed to flow and control diagrams. Contractor shall color code diagrams as necessary for clarity.

C. Reports: Contractor shall include test and balancing reports, as applicable and as specified in individual product Specification Sections.

D. Panelboard Circuit Directories: Contractor shall provide electrical service characteristics, controls and communications.

E. Wiring Diagrams: Contractor shall include diagrams of wiring as installed, with color coding as necessary for clarity.

F. Operating Procedures: Contractor shall include:
1. Start-up, break-in, and routine normal operating instructions and sequences.
2. Regulation, control, stopping, shut-down, and emergency instructions.
3. Summer and winter operating instructions.
4. Special operating instructions.

G. Maintenance Requirements: Contractor shall include:
1. Routine maintenance procedures and guide for trouble-shooting.
2. Disassembly, repair, and reassembly instructions.
3. Alignment, adjusting, balancing, and checking instructions.

H. Servicing and Lubrication: Contractor shall provide servicing and lubrication schedule, and list of lubricants required.

I. Parts Data: Contractor shall provide original manufacturer's parts list, illustrations, assembly drawings, and diagrams as necessary for service and maintenance. Contractor shall:
1. Include complete nomenclature and catalog numbers for consumable and replacement parts.
2. Provide list of original manufacturer's spare parts, current prices, and recommended quantities to be maintained in stock by the University or operator.

J. Software: Contractor shall provide all programming codes, access codes and other data necessary for operation, maintenance, future functioning and modifications of microprocessor-controlled products, independent of Original Equipment Manufacturer (OEM).

K. Additional Requirements: As specified in individual product Specification Sections.

1.8 DATA REQUIREMENTS FOR ELECTRIC AND ELECTRONIC SYSTEMS

A. Date Requirements for Electrical and Electronic Systems: Contractor shall provide description of each system and component parts, including:
1. Function, normal operating characteristics and limiting conditions.
2. Performance curves, engineering data and tests.
3. Complete nomenclature and commercial number of replaceable parts.

B. Circuit Directories of Panel Boards: Contractor shall include:
1. Electrical service.
2. Controls.
3. Communications.

C. Wiring Diagrams: As-installed, color-coded wiring diagrams.

D. Operating procedures: Contractor shall provide:
1. Routine and normal operating instructions.
2. Sequences required.
3. Special operating instructions.

E. Maintenance procedures: Contractor shall provide:
1. Routine operations.
2. Guide to "trouble-shooting."
3. Disassembly, repair and reassembly.
4. Adjustment and checking.

F. Contractor shall provide Manufacturer's printed operating and maintenance instructions.

G. Contractor shall provide list of original manufacturer's spare parts, manufacturer's current prices, and recommended quantities to be maintained in storage.

H. Contractor shall prepare and include additional data when the need for such data becomes apparent during instruction of Owner's personnel.

I. Additional requirements for operating and maintenance data: Respective sections of specifications.

1.9 INSTRUCTION OF UNIVERSITY'S PERSONNEL

A. Instruction of University's Personnel: Prior to Contract Completion review, Contractor shall complete instruction of University's designated personnel in the operation, adjustment and routine cleaning, service and maintenance of products, equipment, and systems. Contractor shall schedule indoctrination and training sessions at times acceptable to University. Contractor shall coordinate with requirements specified in Section 01 79 00 - Demonstration and Training.

B. Basis for Instruction: Contractor shall use operation and maintenance manuals as basis for instruction. Contractor shall review contents of manual with personnel in detail to explain all aspects of operation and maintenance.

C. Instructional Material: Contractor shall prepare and insert additional data in Operation and Maintenance Manual when need for such data becomes apparent during instruction.

1.10 SUBMISSION OF OPERATION AND MAINTENANCE MANUALS

A. Submittal: Contractor shall submit six copies to Architect for review and approval prior to submission of final Application for Payment.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

Not Applicable to this Section.

END OF SECTION

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SECTION 01 78 29

SURVEY AND LAYOUT DATA

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Administrative requirements for survey and layout data submittals.

1.3 RELATED SECTIONS

A. Section 01 45 00 - Quality Control: Test and inspection reports.

B. Section 01 72 00 - Preparation Requirements: Layout of the Work and other engineering services required for accomplishing the Work.

C. Section 01 77 00 - Contract Closeout Procedures: Submittals for occupancy, Acceptance and Final Payment.

1.4 LAYOUT OF THE WORK

A. Responsibility for Layout of the Work: Contractor shall be solely responsible for complete, timely and accurate layout of the Work including, but not necessarily limited to, horizontal and vertical control and dimensional coordination as necessary to construct the Work in accordance with the Contract Documents. Contractor shall:
   1. Employ a Land Surveyor or a Civil Engineer, registered in the State of California, to perform survey work.
   2. Employ a Professional Engineer, of the discipline required for the specific service on the Project, and licensed in the State of California where required in the specifications in Divisions 2 through 49.

B. Survey Reference Points: Existing basic horizontal and vertical control points are shown on the Contract Documents, or location of control points will be furnished by the University Representative. Contractor shall use the University Record of Survey, provided by the University Representative, as the Basis of Bearings for survey horizontal control, and shall tie at least one Project site control point to a point on the University Record of Survey. Contractor shall:
   1. Locate and protect control points prior to starting site work, and preserve all permanent reference points during construction.
   2. Make no changes or relocations without prior written notice to Architect.
   3. Report to University Representative and Architect when any reference point is lost or destroyed.
   4. Require a surveyor to replace project control points which may be lost or destroyed. Establish replacements based on original survey control.

1.5 LAYOUT RECORD SUBMITTALS

A. Land Surveyor: Contractor shall submit name, address and telephone number of land surveyor before starting survey work.

B. Survey Logs: On request, Contractor shall submit copies of field documents verifying accuracy of survey Work.

C. Submittal: Contractor shall submit a copy of registered site drawing and certificate signed by the land surveyor that the elevations and locations of the Work are in conformance with Contract Documents.

1.6 SURVEY RECORD DOCUMENTS

A. Survey Record Documents: Contractor shall maintain a complete and accurate log of control and survey work as Work progresses. Upon completion of foundation walls and major site improvements, Contractor shall prepare a certified survey illustrating dimensions, locations, angles and elevations of new construction and site work. Contractor shall submit survey record documents as specified in Section 01 77 00 - Contract Closeout Procedures.

B. Locations provided on the certified survey shall be coordinated with the control points tied to the University Record of Survey as per Section 1.4B.

C. For each new Project utility or improvement which is not to be owned and maintained by the University, Contractor shall provide a legal description and plot, stamped and signed by a properly licensed surveyor or Civil Engineer, and which will use the University Record of Survey as the Basis of Bearings and will provide a Point of Commencement shown on said Record of Survey.

1.7 CONTRACTOR'S REVIEW

A. Scope of Contractor's Review: Survey and layout data shall be reviewed by Contractor prior to submission for University's review or filing. Contractor shall sign each submittal copy certifying that:
   1. Field measurements have been determined and verified.
   2. Field construction criteria have been verified.
   3. Conformance with Drawings and Specifications requirements is confirmed.

B. Contractor's Review Action: Contractor shall indicate clearly on survey and layout data whether the dimensions and coordinates are in compliance with Contract requirements. Contractor shall note clearly and sign each submittal certifying that reported data "Conforms" or "Does Not Conform".

C. Changes and Deviations: Contractor shall identify all deviations from requirements of Drawings and Specifications. Changes in the Work shall not be authorized by submittals review actions. No review action, implicit or explicit, shall be interpreted to authorized changes in the Work. Changes shall only be authorized by separate written Change Order or Field Instruction, in accordance with the Contract General Conditions.

1.8 REVIEWS BY UNIVERSITY'S REPRESENTATIVE AND ARCHITECT

A. Reviews by University's Representative and Architect, General: Reviews of survey and layout data by University's Representative and Architect, or other responsible design professional, shall be only for general conformance with the design concept and requirements based on the information presented. Neither Architect nor other responsible design professional shall verify submitted survey and layout data.

B. Contract Requirements: Reviews by University's Representative, Architect or other responsible design professional shall not relieve the Contractor from compliance with requirements of the Drawings and Specifications. Changes shall only be authorized by separate written Change Order or Field Instruction, in accordance with the Contract General Conditions.

PART 2 - PRODUCTS

Not applicable to this Section.

PART 3 - EXECUTION

Not applicable to this Section.

END OF SECTION

SECTION 01 78 33

PRODUCT WARRANTIES AND BONDS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. General administrative and procedural requirements for preparation and submission of warranties and bonds required by the Contract Documents, including manufacturer's standard warranties on products and special Project warranties.
   1. Refer to the Contract General Conditions for terms of Contractor's special warranty of workmanship and materials.
   2. Certifications and other commitments and agreements for continuing services to University are specified elsewhere in the Contract Documents.

1.3 RELATED DOCUMENTS AND SECTIONS

A. Section 01 33 00 - Submittals Procedures: General administrative requirements for submittals, applicable to warranties and bonds.

B. Section 01 77 00 - Contract Closeout Procedures: General requirements for closeout of the Contract.

C. Section 01 78 23 - Operation and Maintenance Data: Operating and maintenance data binders, to include copies of warranties and bonds.

D. Product Specifications Sections in Divisions 2 through 49: Special Project warranty requirements for specific products or elements of the Work; commitments and agreements for continuing services to University.

1.4 DEFINITIONS

A. Warranty: Assurance to University by Contractor, installer, supplier, manufacturer or other party responsible as warrantor, for the quantity, quality, performance and other representations of a product, system service of the Work, in whole or in part, for the duration of the specified period of time. The University's standard warranty form shall be used for all warranties under this Contract unless otherwise agreed to in writing by the University Representative.

B. Guaranty: Assurance to University by Contractor or product manufacturer or other specified party, as guarantor, that the specified warranty will be fulfilled by the guarantor in the event of default by the warrantor.

C. Standard Product Warranty: Preprinted, written warranty published by product manufacturer for particular products and specifically endorsed by the manufacturer to the University.

D. Special Project Warranty: Written warranty required by or incorporated into Contract Documents, to extend time limits provided by standard warranty or to provide greater rights for University.

E. Guaranty Period: As defined in the Contract General Conditions, guaranty period shall be synonymous with "warranty period", "correction period" and similar terms used in the Contract Specifications. Warranty period shall be one year from the date of Project Completion unless otherwise agreed to in writing by the University Representative.

1.5 WARRANTIES AND GUARANTIES

A. Warranties and Guaranties, General: Contractor shall provide all warranties and guaranties with University named as beneficiary. For equipment and products, or components thereof, bearing a manufacturer's warranty or guaranty that extends for a period of time beyond the Contractor's warranty and guaranty, Contractor shall so state in the warranty or guaranty.

B. Provisions for Special Warranties: Contractor shall refer to Contract General Conditions for terms of the Contractor's special warranty of workmanship and materials.

C. General Warranty and Guaranty Requirements: Warranty shall be an agreement to repair or replace, without cost and undue hardship to University, Work performed under the Contract which is found to be defective during the guaranty period (warranty or guaranty) period. Repairs and replacements due to improper maintenance or operation, or due to normal wear, usage and weathering are excluded from warranty requirements unless otherwise specified.

D. Specific Warranty and Guaranty Requirements: Specific requirements are included in product Specifications Sections of Divisions 2 through 49, including content and limitations.

E. Disclaimers and Limitations: Manufacturer's disclaimers and limitations on product warranties and guaranties shall not relieve Contractor of responsibility for warranty and guaranty requirements for the Work that incorporates such products, nor shall they relieve suppliers, manufacturers, and installers required to countersign special warranties with Contractor.

F. Related Damages and Losses: When correcting warranted Work that has been found defective, Contractor shall remove and replace other Work that has been damaged as a result of such defect or that must be removed and replaced to provide access for correction of warranted Work.

G. Reinstatement of Warranty: When Work covered by a warranty has been found defective and has been corrected by replacement or rebuilding, Contractor shall reinstate the warranty by written endorsement.

H. Replacement Cost: Upon determination that Work covered by a warranty has been found to be defective, Contractor shall replace or reconstruct the Work to a condition acceptable to University's Representative, complying with applicable requirements of the Contract Documents. Contractor shall be responsible for all costs for replacing or reconstructing defective Work regardless of whether University has benefited from use of the Work through a portion of its anticipated useful service life.

I. University's Recourse: Written warranties made to University shall be in addition to implied warranties, and shall not limit the duties, obligations, rights and remedies otherwise available under law, nor shall warranty periods be interpreted as limitations on time in which University can enforce such other duties, obligations, rights, or remedies.

   1. Rejection of Warranties: University reserves the right to reject warranties and to limit selections to products with warranties not in conflict with requirements of the Contract Documents.

J. Warranty as Condition of Acceptance: University reserves the right to refuse to accept Work for the Project where a special warranty, certification, or similar commitment shall be required on such Work or part of the Work, until evidence is presented that entities required to countersign such commitments are willing to do so.

1.6 PREPARATION OF WARRANTY AND BOND SUBMITTALS

A. Project Warranty and Guaranty Forms: Forms for Project warranties and guaranties are included in the Contract Documents. Contractor shall submit the warranty package submittal to the Architect, with a copy to the University Representative, for review and approval. Contractor shall:
   1. Refer to product Specifications Sections of Divisions 2 through 49 for specific content requirements, and

particular requirements for submittal of special warranties.
2. Prepare standard warranties and guaranties, excepting manufacturers' standard printed warranties and guaranties, on Contractor's, subcontractor's, material suppliers, or manufacturer's own letterhead, addressed to University as directed by University's Representative.
3. Warranty and guaranty letters shall be signed by all responsible parties and by Contractor in every case, with modifications only as approved in advance by University's Representative to suit the conditions pertaining to the warranty or guaranty.

B. Manufacturer's Guaranty Form: Manufacturer's guaranty form may be used instead of special Project form included in the Contract Documents, if agreed to in writing by the University's Representative. Manufacturer's guaranty form shall contain appropriate terms and identification, ready for execution by the required parties.

1. If proposed terms and conditions restrict guaranty coverage or require actions by University beyond those specified, Contractor shall submit draft of guaranty to the Architect and the University's Representative for review and approval before performance of the Work.

2. In other cases, Contractor shall submit draft of guaranty to the Architect and the University's Representative for approval prior to final execution of guaranty.

C. Signatures: Signatures shall be by person authorized to sign warranties, guaranties and bonds on behalf of entity providing such warranty, guaranty or bond.

D. Co-Signature: All installer's warranties and bonds shall be co-signed by Contractor. Manufacturer's guaranties will not require co-signature.

1.7 FORM OF WARRANTY AND BOND SUBMITTALS

A. Form of Warranty and Bond Submittals: Prior to completion, Contractor shall collect and assemble all written warranties and guaranties into binders and deliver binders to the Architect, with a copy to the University Representative, for final review and acceptance. Contractor shall:
1. Prior to submission, verify that documents are in proper form and contain all required information and are properly signed by Contractor, subcontractor, supplier and manufacturer, as applicable.
2. Organize warranty and guaranty documents into an orderly sequence based on the Table of Contents of the Project Manual.
3. Include Table of Contents for binder, neatly typed, following order and section numbers and titles as used in the Project Manual.
4. Bind warranties, guaranties and bonds in heavy-duty, commercial quality, durable three-ring vinyl covered loose-leaf binders, thickness as necessary to accommodate contents, with clear front and spine to receive inserts, and sized to receive 8-1/2 inch by 11-inch paper.
5. Provide heavy paper dividers with celluloid or plastic covered tabs for each separate warranty. Mark tabs to identify products or installation, and section number and title.
6. Include on separate typed sheet, if information is not contained in warranty or guaranty form, a description of the product or installation, and the name, address, telephone number and responsible person for applicable installer, supplier and manufacturer.
7. Identify each binder on front and spine with typed or printed inserts with title "WARRANTIES AND BONDS", the Project title or name, and the name of the Contractor. If more than one volume of warranties, guaranties and bonds is produced, identify volume number on binder.
8. When operating and maintenance data manuals are required for warranted construction, include additional copies of each required warranty and guaranty in each required manual. Coordinate with requirements specified in Section 01783 - Operation and Maintenance Data.

1.8 TIME OF WARRANTY AND BOND SUBMITTALS

A. Submission of Preliminary Copies: Unless otherwise specified, Contractor shall obtain preliminary copies of warranties, guaranties and bonds within ten days of completion of applicable item or Work.

B. Submission of Final Copies: Contractor shall submit fully executed copies of warranties, guaranties and bonds prior to Notice of Completion.

C. Date of Warranties and Bonds: Unless otherwise directed or specified, commencement date of warranty, guaranty and bond periods shall be the date established in the Notice of Completion.

1. Warranties for Work accepted in advance of date stated in Notice of Completion: When a designated system, equipment, component parts or other portion of the Work is completed and occupied or put to beneficial use by University's Representative, by separate written agreement with Contractor, prior to completion date established in the Notice of Completion, Contractor shall submit properly executed warranties to University, as directed by University's Representative, within ten days of completion of that designated portion of the Work. Contractor shall list date of commencement of warranty, guaranty or bond period as the date established in the Notice of Completion.

D. Duration of Warranties and Guaranties: Unless otherwise specified or prescribed by law, warranty and guaranty periods shall be not less than the guaranty period required by the Contract General Conditions, but in no case less than one year from the date established for completion of the Project in the Notice of Completion. See product Specifications Sections in Divisions 2 through 49 of the Project Manual for extended warranty and guaranty beyond the minimum one-year duration.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

Not Applicable to this Section.

END OF SECTION

SECTION 01 78 39

PROJECT RECORD DOCUMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Requirements for Project Record Documents to be submitted for Contract closeout.

1.3 RELATED SECTIONS

A. Section 01 33 00 - Submittals Procedures: General requirements for submission for shop drawings, product data, samples and quality control reports.

1.4 PROJECT RECORD DOCUMENTS

A. Project Record Documents, General: Contractor shall not use Record Documents for construction purposes. Contractor shall protect from deterioration and loss in a secure, fire-resistive location; provide access to Record Documents for the Trustees' and the Architect's reference during normal working hours.

B. Record Drawings: Contractor shall record information continuously as Work progresses. Contractor shall not conceal Work permanently until all required information is recorded. Contractor shall**:**
   1. Maintain a clean, undamaged set of blue or black line white-prints of Contract Drawings and Shop Drawings. Mark the set to show the actual installation where the installation varies substantially from the Work as originally shown. Mark whichever drawing is most capable of showing conditions fully and accurately.
   2. Where Shop Drawings are used, record a cross-reference at the corresponding location on the Contract Drawings. Give particular attention to concealed elements that would be difficult to measure and record at a later date.
   3. Legibly and to scale, mark record sets with red erasable pencil. Use other colors to distinguish between variations in separate categories of the work.
   4. Mark new information that is important to the University, but was not shown on Contract Drawings or Shop Drawings. Record actual construction, including:
      a. Measured depths of foundations and footings encountered, measured in relation to finish First Floor datum.
      b. Measured horizontal and vertical locations of underground utilities and appurtenances, referenced to permanent ground improvements.
      c. Field changes of dimension and detail.
      d. Details not on original Contract Drawings. Application of copies of details produced and provided by Architect during construction will be accepted.
      e. Permanent Room names and Room numbers.
   5. Note related Change Order numbers where applicable.
   6. Organize record drawing sheets into manageable sets, bind with durable paper cover sheets, and print suitable titles, dates and other identification on the cover of each set.
   7. Store Record Documents separate from documents used for construction.

C. Record Specifications: Contractor shall record changes made by Addenda and Change Orders. In PART 2 - PRODUCTS in each Section, Contractor shall legibly mark and record in red ink actual Products installed or used, including:
   1. Manufacturer's name and product model or catalog number.
   2. Product substitutions or alternates utilized.

D. Submission:
   1. Contractor shall keep Project Record Documents current, as they will be reviewed for completeness by Architect, Inspector, and University's Representative as condition for certification of each Progress Payment Application.
   2. Prior to the date of the Notice of Completion, Contractor shall submit marked Record Documents to Architect for review, approval and further processing.

PART 2 - PRODUCTS

Not Applicable to this Section.

PART 3 - EXECUTION

Not Applicable to this Section.

END OF SECTION

SECTION 01 79 00

DEMONSTRATION AND TRAINING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Construction Drawings, Technical Specifications, Addenda, and general provisions of the Contract, including Contract General Conditions and Supplementary General Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. Administrative and procedural requirements for instructing University's personnel, including the following:
1. Demonstration of operation of systems, subsystems and equipment.
2. Training in proper operation and maintenance of systems, subsystems, and equipment installed under the Contract.

1.3 RELATED SECTIONS

A. Section 01 78 23 - Operation and Maintenance Data: Operating and maintenance instructions to be used during training and demonstration.

1.4 SUBMITTALS

A. Instruction Program: Contractor shall submit two copies of outline of instructional program for demonstration and training, including a schedule of proposed dates, times, length of instruction time, and instructors' names for each training module. Contractor shall include learning objective and outline for each training module. Contractor shall:
1. Make the operations and procedures manuals available for use during the training sessions.
2. Schedule submission of instruction program to allow sufficient time for receipt, review and acceptance of instruction program by the Architect and the University's Representative and shall be not less than three weeks prior to proposed date of first training session.
3. Submit, at completion of training, three complete training manuals for University's use.

B. Qualification Data: For firms and persons specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Contractor shall include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other information specified.

C. Attendance Record: For each training module, Contractor shall submit list of participants and length of instruction time.

D. Evaluations: For each participant and for each training module, Contractor shall submit results and documentation of performance-based test.

E. Demonstration and Training Video Record: Contractor shall submit two copies at end of each training session.

1.5 QUALITY ASSURANCE

A. Facilitator Qualifications: Contractor shall engage a qualified facilitator to prepare instruction program and training modules, to coordinate instructors, and to coordinate between Contractor and University's Representative for number of participants, instruction times, and location. Facilitator shall be firm or individual experienced in training or educating maintenance personnel in a training program similar in content and extent to that indicated for this Project, and whose work has resulted in training or education with a record of successful learning performance.

B. Instructor Qualifications: Contractor shall engage qualified instructors to instruct University's personnel how to adjust, operate, and maintain systems, subsystems, and equipment not part of a system. Instructors shall be factory-authorized service representatives, complying with requirements in Section 01450 - Quality Control, experienced in operation and maintenance procedures and training.
1. System manufacturers shall provide qualified instructor to describe system design, operational requirements, criteria, and regulatory requirements.
2. University's Representative will furnish Contractor with names and positions of participants.

C. Pre-Instruction Conference: Contractor shall conduct conference at Project site to comply with requirements in Section 01310 - Coordination. Contractor shall review methods and procedures related to demonstration and training including, but not limited to, the following:
1. Inspect and discuss locations and other facilities required for instruction.

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EDIT SUBPARAPH BELOW TO SUIT PROJECT REQUIREMENTS.

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2. Review and finalize instruction schedule and verify availability of educational materials, instructors' personnel, audiovisual equipment, and facilities needed to avoid delays.
3. Review required content of instruction.
4. For instruction that must occur outside, review weather and forecasted weather conditions and procedures to follow if conditions are unfavorable.

1.6 COORDINATION

A. Coordination of Instruction Schedule: Contractor shall coordinate instruction schedule with University's operations. Contractor shall adjust schedule as required to minimize disrupting University's operations.

B. Coordination of Instructors: Contractor shall coordinate instructors, including providing notification of dates, times, length of instruction time, and course content. Contractor shall allow for 30 days written notice to University's Representative.

C. Coordination with Operation and Maintenance Data: Contractor shall coordinate content of training modules with content of approved emergency, operation, and maintenance manuals.
1. Contractor shall not submit instruction program until operation and maintenance data have been reviewed and accepted by Architect and copies given to University's Representative.
2. Contractor shall coordinate review of operation and maintenance data to make operation and maintenance data available at least two weeks prior to date scheduled for initial training session.

PART 2 - PRODUCTS

2.1 INSTRUCTION PROGRAM

A. Program Structure: Contractor shall develop an instruction program that includes individual training sessions for each system and operating products not part of a system, as required by Division 2 through 48 Specification Sections. Contractor shall include instruction on operational interfaces between systems.

B. Schedule of Training Sessions: Contractor shall arrange to have training conducted on consecutive days, with no more than six hours of training scheduled for any one day. Concurrent classes will not be acceptable.

C. Training Sessions, General: Contractor shall develop a learning objective and teaching outline for each session. Contractor shall include a description of specific skills and knowledge that participant is expected to master. Training sessions shall progress logically. Each training session shall be comprised of time spent both in the classroom and at specific location of subject equipment or system. As a minimum, Contractor shall ensure that each training session covers the following subjects for each item of equipment and system:

1. Familiarization:
   a. Review catalog, parts lists, drawings, etc., which have been previously provided for the plant files and operation and maintenance manuals.

b. Check out the installation of the specific equipment items.
c. Demonstrate the unit and indicate how all parts of the specifications are met.
d. Answer questions.

2. Safety:
a. Using material previously provided, review safety references.
b. Discuss proper precautions around equipment.

3. Operation:
a. Using material previously provided, review reference literature.
b. Explain all modes of operation (including emergency).
c. Check out University's personnel on proper use of the equipment.

4. Preventive Maintenance:
a. Using material previously provided, review preventive maintenance (PM) lists including:
1) Reference material.
2) Daily, weekly, monthly, quarterly, semiannual, and annual jobs.
b. Demonstrate how to perform Preventive Maintenance tasks.
c. Demonstrate to University's personnel what to look for as indicators of equipment problems.

5. Corrective Maintenance:
a. List possible problems.
b. Discuss repairs--point out special problems.
c. Open up equipment and demonstrate procedures, where practical.

6. Parts:
a. Show how to use previously provided parts list and order parts.
b. Check over spare parts on hand. Make recommendations regarding additional parts that should be available.

7. Local Representatives:
a. Where to order parts: Name, address, telephone.
b. Service problems:
1) Who to call.
2) How to get emergency help.

8. Operation and Maintenance Manuals:
a. Review any other material submitted.
b. Update material, as required.

D. Classroom Training for Operations Personnel:
1. Using projected drawings and photographs, describe and discuss equipment locations in plant and present operational overview of systems. Thoroughly discuss operating and maintenance manuals.
2. Describe purpose and plant function of equipment and systems.
3. Describe operating theory of equipment.
4. Describe start-up, shutdown, normal operation and emergency operating procedures, including discussion of system integration and electrical interlocks, if any.
5. Identify and discuss safety items and procedures.
6. Describe routine preventive maintenance, including specific details on lubrication and maintenance of corrosion protection of the equipment and ancillary components.
7. Describe operator detection, without test instruments, of specific equipment trouble symptoms.
8. Describe required equipment performance test procedures and intervals.
9. Describe routine disassembly and assembly of equipment if applicable (as determined by University's Representative on case-by-case basis) for purposes such as operator inspection of equipment.

E. Classroom Training for Maintenance and Repair Personnel:
1. Theory of operation.
2. Description and function of equipment.
3. Start-up and shutdown procedures.

4. Normal and major repair procedures.
5. Equipment inspection and troubleshooting procedures including the use of applicable test instruments and the "pass" and "no pass" test instrument readings.
6. Routine and long-term calibration procedures.
7. Safety procedures.
8. Preventive maintenance such as lubrication; normal maintenance such as belt, seal, and bearing replacement; and up to major repairs such as replacement of major equipment part(s) with the use of special tools, bridge cranes, welding jigs, etc.

F. Field Training for Operations Personnel:
1. Identify locations of equipment components and controls.
2. Review of component functions and theory of operation.
3. Identifying piping and flow options.
4. Identifying valves and explain their functions at various settings.
5. Identifying instrumentation:
   a. Location of primary element.
   b. Location of instrument readout.
   c. Discuss purpose, basic operation, and information interpretation.
6. Discuss, demonstrate, and perform standard operating procedures and round checks, including system start-up and shutdown procedures.
7. Review and perform safety procedures.
8. Perform the required equipment exercise procedures.
9. Discuss and perform preventive maintenance activities.
10. Identify and review safety items and perform safety procedures, if feasible.

G. Field Training for Maintenance and Repair Personnel: In addition to field training specified above for operations personnel, include the following:
1. Describe normal repair procedures.
2. Perform routine disassembly and assembly of equipment, if applicable, for inspections and tests.
3. Perform routine maintenance and repair tasks, including mechanical and electrical operations for troubleshooting, adjustments and calibration.

H. Presentation Media:
1. Presentations may utilize computer-generated, projected graphics utilizing Microsoft PowerPoint software, including animation as appropriate to enhanced presentation and viewer interest. Graphics shall include text and still and moving images. PowerPoint presentation shall be suitable for incorporation into video record of instruction.
2. Each session shall include mock-ups, samples and other visual aids as appropriate.
3. Each session shall include printed handouts and notes for each participant.
4. Produce sufficient printed materials to provide minimum of five unused copies for University's use in subsequent training programs.

PART 3 - EXECUTION

3.1 INSTRUCTION

A. Preparation. Contractor shall:
5. Assemble educational materials necessary for instruction, including documentation and training module. Assemble training modules into a combined training manual.
6. Set up instructional equipment at instruction location.

B. Scheduling: Contractor shall provide instruction at mutually agreed on times. For equipment that requires seasonal operation, Contractor shall provide similar instruction at start of each season. Contractor shall:
1. Schedule training through University's Representative.
2. Schedule training at time and location convenient to University, with at least 14 calendar days' advance written notice to University's Representative.

C. Training Sessions: Contractor shall conduct classroom and field training sessions presenting content specified in Article 2.1, titled "Instruction Program," above.

D. Evaluation: At conclusion of each training session, Contractor shall assess and document each participant's mastery of module by use of written examination or performance-based demonstration test.

E. Cleanup. Contractor shall:
1. Collect used and leftover educational materials and deliver to University as directed by University's Representative.
2. Remove instructional equipment.
3. Restore systems and equipment to condition existing before initial training use.

END OF SECTION

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01 80 00 – Performance Requirements

Third Party Verified Certification - LEED Certification

New construction shall be designed and implemented to achieve LEED for New Construction Gold Certification. Design and construction teams are to provide all necessary services to complete the full certification process. All costs to achieve certification shall be the responsibility of the project team.

Performative Modeling

With each new construction or major renovation design submittal, the design / construction team shall complete the following iterative modeling at a minimum:

- Predictive Energy Modeling: Site EUI benchmarked against other projects of similar type and size.
- Predictive Daylight Modeling: The University prioritizes the use of daylighting in all regularly occupied spaces (unless indicated otherwise programmatically). While avoiding glare/contrast is necessary, spaces should meet the most current IESNA standards for daylight autonomy (300 – 3000 lux) at the work plane. Design Teams should demonstrate spatial daylight autonomy through the use of annual computer simulations.
- Predictive Water Use: For both indoor and outdoor water, the project should be benchmarked against other projects of similar size and type using a theoretical baseline.

THE CALIFORNIA STATE UNIVERSITY
OFFICE OF THE CHANCELLOR

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SAN DIEGO
SAN FRANCISCO
SAN JOSÉ
SAN LUIS OBISPO
SAN MARCOS
SONOMA
STANISLAUS

April 7, 2017

MEMORANDUM

TO: CSU Presidents

FROM: Timothy P. White
Chancellor

SUBJECT: Policy on Systemwide Smoke and Tobacco Free Environment
Executive Order 1108

Attached is a copy of Executive Order 1108 relating to a systemwide smoke and tobacco free environment. This executive order supersedes Executive Order 599, and all existing campus policies related to smoking and tobacco. Each campus president is asked to comply with the systemwide policy to create a smoke and tobacco-free campus, and to create a task force to lead the implementation of the new policy. Campus task forces will be responsible for developing an implementation plan, and ensuring all activities associated with implementation are carried out. Furthermore, a member of each campus task force will serve on the systemwide Smoke and Tobacco Free Policy Task Force to ensure all campuses are adequately moving forward with the implementation of this policy.

In accordance with policy of the California State University, the campus president has the responsibility for implementing executive orders where applicable and for maintaining the campus repository and index for all executive orders.

If you have questions regarding this executive order, please call the Chancellor's Office Labor Relations at (562) 951-4400.

TPW/jas

Attachment

c: CSU Office of the Chancellor Leadership
Provosts/Vice Presidents, Academic Affairs
Vice Presidents, Business and Administration
Vice Presidents, Student Affairs
Associate Vice Presidents, Academic and Faculty Affairs
Human Resources Officers

401 GOLDEN SHORE • LONG BEACH, CALIFORNIA 90802-4210 • (562) 951-4700 • Fax (562) 951-4986

THE CALIFORNIA STATE UNIVERSITY
Office of the Chancellor
401 Golden Shore
Long Beach, California 90802-4210
(562) 951-4400

Executive Order: 1108

Effective Date: April 7, 2017

Supersedes: Executive Order 599

Title: Policy on Systemwide Smoke and Tobacco Free Environment

I. AUTHORITY AND PURPOSE

This executive order is issued pursuant to Title 5, California Code of Regulations, Sections 42356, Government Code 7597.1, and the Standing Orders of the Board of Trustees.

A cornerstone of the California State University and higher education is the principle of one's individual freedom to learn, teach, work, think, and take part in their intellectual and career endeavors in a fulfilling, rewarding, safe, and healthy environment.

For decades, the health hazards of tobacco and second-hand smoke to individuals have been well studied and chronicled.

Further, studies have clearly demonstrated the acute health benefits, medical costs savings, and organizational costs savings when individuals quit smoking.

Thus, in order to provide the California State University's faculty, staff, students, guests and the public with campuses that support the principle of one's individual freedom to learn, teach, work, think and take part in their intellectual endeavors in a fulfilling, rewarding, safe and healthy environment, the creation and implementation of a "smoke and tobacco free" policy systemwide is necessary and welcome.

Campus Presidents or their designees shall have the responsibility for implementing the policy on their campuses with an implementation date of September 1, 2017.

II. DEFINITIONS

Members of the CSU Community: This includes all students, faculty, staff, alumni, university volunteers, contractors or vendors and visitors to any California State University campus or properties.

University Properties: These include the interior and exterior campus areas of any California State University campus. This definition includes buildings (including residence halls), structures (including parking structures), parking lots, and outdoor areas owned, leased or rented by the university or one of its auxiliaries. Also included are vehicles owned, leased or rented by the university or one of the university's auxiliaries. Private vehicles on university-owned, leased, or rented land or in university-owned, leased, or rented parking structures will also be subject to compliance with Executive Order 1108.

Smoke Free: "Smoke Free" means the use of cigarettes, pipes, cigars, and other "smoke" emanating products including e-cigarettes, vapor devices and other like products are prohibited on all University properties.

Smoke or Smoking: "Smoke" or "Smoking" means inhaling, exhaling, burning, or carrying any lighted or heated cigar, cigarette, cigarillo, pipe, hookah, or any other lighted or heated tobacco or plant product intended for inhalation, whether natural or synthetic, in any manner or in any form. "Smoke" or "Smoking" also includes the use of an electronic smoking device that creates an aerosol or vapor, in any manner or in any form, or the use of any oral smoking device for the purpose of circumventing the prohibition of smoking.

Tobacco Product:

(i) A product containing, made or derived from tobacco or nicotine that is intended for human consumption, whether smoked, heated, chewed, absorbed, dissolved, inhaled, snorted, sniffed, or ingested by any other means, including, but not limited to cigarettes, cigars, little cigars, chewing tobacco, pipe tobacco, and snuff.

(ii) An electronic device that delivers nicotine or other vaporized liquids to the person inhaling from the device, including, but not limited to, an electronic cigarette, cigar, pipe, or hookah.

(iii) Any component, part, accessory of a tobacco product, whether or not sold separately.

(iv) "Tobacco product" does not include a product that has been approved by the United States Food and Drug Administration for sale as a tobacco cessation product or for other therapeutic purposes where the product is market and sold solely for such an approved purpose.

Tobacco Free: "Tobacco Free" means the use of cigarettes, pipes, cigars, smokeless tobacco, snuffs, and other tobacco products are prohibited on all University properties.

III. POLICY TEXT

Campus Presidents or their designees shall have the responsibility of implementing this Executive Order on their campuses with an anticipated implementation date no later than September 1, 2017.

Scope of this Executive Order:

Effective September 1, 2017, all California State University campuses shall be 100% Smoke Free and Tobacco Free. Smoking, the use or sale of tobacco products, and the use of designated smoking areas are prohibited on all California State University properties. Members of the CSU community are expected to fully comply with the policy.

Any sponsorship and/or advertising in respect to any university activity or event by a tobacco product manufacturer is prohibited unless explicitly authorized by the University President or designee.

Exceptions:

(i) Smoking in university-sponsored theater and dance productions, student-authored or sponsored scenes, showcases or workshops produced as part of the department of theatre as well as ceremonial campus events may be authorized by the President or designee only when a required part of a specific performance. This includes smoking and/or tobacco use for traditional ceremonial activities of recognized cultural and/or religious groups.

(ii) The use of nicotine cessation products regulated by the United States Food and Drug Administration for treating nicotine or tobacco dependencies is permitted under the terms of this executive order.

(iii) Institutional Review Board approved research on tobacco or tobacco-related products.

Collective Bargaining:

Nothing in this executive order shall extend the existing grounds for employee discipline and, to the extent that any of these provisions are in conflict with a Collective Bargaining Agreement, the terms of the Collective Bargaining Agreement shall be controlling.

IV. COMPLIANCE, RESPONSIBILITIES AND ENFORCEMENT

Compliance is grounded in an informed and educated campus community. The success of this policy depends on the thoughtfulness, civility and cooperation of all members of the campus community, including visitors.

Members of the CSU community are individually responsible to comply with the creation of a systemwide smoke and tobacco free environment. While compliance with this executive order is an individual responsibility, members of the CSU community should be aware that enforcement of this policy may occur in the following instances:

(i) University Police shall reserve all enforcement authority with regards to any violation of existing state and federal law.

(ii) Individual agreements that prohibit smoking and proscribe penalties for breaches that are not impacted by this executive order (e.g. University Housing license agreements, other residential licenses, or existing leases).

Educational campaigns, outreach, communication and the promotion of tobacco cessation treatment options will be the primary means to promote compliance. A comprehensive education and outreach campaign, including resources and referrals for cessation will be made available as part of campus implementation programs.

The progress this policy represents in promoting the ability of students, faculty, staff and visitors to have a healthier and pleasant campus experience aligns well with the CSU's mission. Individual campus support and diligence in moving forward with the implementation and amendment of current policies is sincerely appreciated.

Hostile and/or violent interpersonal conduct directed against members of the CSU community requesting that an individual(s) comply regarding compliance with the terms of this executive order will not be tolerated, and will be enforced under systemwide or campus policies, including but not limited to workplace violence policies.

V. IMPLEMENTATION

The Vice Chancellor for Human Resources has overall responsibility for implementing this policy. This policy shall supersede all existing campus policies related to smoking and tobacco. Campus task forces will be responsible for developing an implementation plan, and ensuring all activities associated with implementation are carried out. Campus task forces shall include a student representative. A member of each campus task force will serve on the systemwide Smoke and Tobacco Free Policy Task Force to ensure all campuses are adequately moving forward with implementation of this policy. Exclusive Representatives may nominate an individual to serve on the Systemwide Task Force. To provide adequate time to create awareness, outreach, and educational programs, including smoking cessation and counseling programs, this policy is effective September 1, 2017.

Timothy P. White

Timothy P. White, Chancellor

Dated: April 7, 2017